B Sullivan

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION

APOLLO TRACKING STATION HONEYSUCKLE
COPY NO. 4
D/TLM
D/CMPTR
DOCUMENTATION SECTION

FINAL

APOLLO 11 FLIGHT PLAN

AS-506/CSM-107/LM-5

RECEIVED AT
11 JUL 1969
HONEYSUCKLE CREEK

JULY 1,1969

PREPARED BY
FLIGHT PLANNING BRANCH
FLIGHT CREW SUPPORT DIVISION

MANNED SPACECRAFT CENTER
HOUSTON, TEXAS

SECTION I
SECTION II
SECTION III
SECTION IV
SECTION V
SECTION VI

This final version of the Apollo 11 Flight Plan is number four of a set sent to Honeysuckle Creek Tracking Station, Canberra, Australia, for support of the mission.

It is dated July 1, 1969 and was received at the station on July 11, just five days before launch.

Honeysuckle Creek was one of NASA's three prime 26 metre dish tracking sites (the others being Goldstone, California and Madrid, Spain).

This copy was designated for the Telemetry (TLM) and Computer sections.

It has been preserved by Honeysuckle Creek's Bryan Sullivan, and includes his notations.

The Flight Plan was scanned, and this PDF file assembled, by Colin Mackellar for www.honeysucklecreek.net, May 2014.

For authenticity, the pages have been assembled without rotation of those pages where text and charts are sideways (they can be easily rotated in a PDF viewer). Page 1-7 is a foldout chart. Blank pages have also been retained.

www.honeysucklecreek.net

OPTIONAL FORM NO. 10
MAY 1962 EDITION
GSA FPMR (41 CFR) 101-11.6

UNITED STATES GOVERNMENT

Memorandum

TO : Distribution

DATE: July 8, 1969

In reply refer to:
CF342-9M-133

FROM : CF/Chief, Flight Crew Support Division

SUBJECT: Revision A to the Apollo 11 Final Flight Plan

Enclosed is Revision A to the Apollo 11 Final Flight Plan. Enclosure 1 contains the pen and ink changes to be made on the indicated pages. Enclosure 2 contains the new pages which replace the same numbered pages in the final edition of the flight plan. Changes on the new pages are made with prestige elite type rather than gothic type so they will be readily evident.

Warren J. North

Enclosures 2

CF342:TAGuillory:jat 7-8-69

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Buy U.S. Savings Bonds Regularly on the Payroll Savings Plan

APOLLO 11

APOLLO AS-506/CSM-107/LM-5

FINAL FLIGHT PLAN

JULY 1, 1969

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Any comments or questions on this document should be forwarded to T. A. Guillory, Flight Planning Branch, mail code CF34, extension 4271.

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Acknowledgment is made to Messrs. Richard Rogers, William Killian and Spencer Gardner for their technical support in the preparation of the Apollo 11 Flight Plan.

Views of the earth and the P52 stars shown in the Flight Plan were taken from the document, Views from the CM and LM During the Flight of Apollo 11 (Mission G).

The CSM and LM attitude information was taken from the document, Lunar Orbit Attitude Sequence for Mission G.

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ACCEL	Accelerometer
ACN	Ascension
ACT	Activation
ACQ	Acquisition
AEA	Abort Electronics Assembly
AGS	Abort Guidance Subsystem
AH	Ampere Hours
ALSCC	Apollo Lunar Surface Close-up Camera
ALT	Altitude
AMP or amp	Ampere
ANG	Antigua
ANT	Antenna
AOH	Apollo Operations Handbook
AOS	Acquisition of Signal or Acquisition of Site
AOT	Alignment Optical Telescope
APS	Ascent Propulsion Subsystem
ARS	Atmosphere Revitalization System
ATT	Attitude
AUX	Auxiliary
AZ	Azimuth
BAT	Battery
BDA	Bermuda
Bio	Bio-Medical Data on Voice Downlink
BP	Barber Pole
BT	Burn Time
BU	Backup
BW	Black & White
BRKT	Bracket
CAP COM	Capsule Communicator
CAL ∢	Calibration Angle
CAM	Camera
CB	Circuit Breaker
CDH	Constant Delta Altitude
CDR	Commander
CDU	Coupling Data Unit
CEX	Color External
CIN	Color Internal
CIRC	Circularization
CK	Check
CM	Command Module
CMC	Command Module Computer
CMD	Command
CMP	Command Module Pilot
CNTL	Control
C/O	Check out
COAS	Crew Optical Alignment Sight
COMM	Communications
CONFIG	Configuration

CONT	Continue
CP	Control Point
CRO	Carnarvon, Australia
CRYO	Cryogenic
CSC	Contingency Sample Collection
CSI	Coelliptic Sequence Initiation
CSM	Command Service Module
C&WS	Caution and Warning System
CYI	Grand Canary Island
DAP	Digital Auto Pilot
DB	Deadband
DCA	Digital Command Assembly
DEDA	Data Entry and Display Assembly
DEGS	Degrees
DEPL	Depletion
DET	Digital Event Timer
DIFF	Difference
DOI	Descent Orbit Insertion
DPS	Descent Propulsion System
DS	Documented Sample
DSE	Data Storage Equipment
DSKY	Display and Keyboard
DTO	Detailed Test Objective
DUA	Digital Uplink Assembly
DWN	Down
E	Erasable or Enter
EASEP	Early Apollo Scientific Experiment Package
ECS	Environmental Control System
ED	Explosive Device
EDT	Eastern Daylight Time
EFH	Earth Far Horizon
EI	Earth (atmosphere) Interface
EL	Elevation or Electric
EMS	Entry Monitor System
EMU	Extravehicular Mobility Unit
ENH	Earth Near Horizon
EPO	Earth Parking Orbit
EPS	Electrical Power Subsystem
EQUIP	Equipment
EST	Eastern Standard Time
EVA	Extravehicular Activity
EVAP	Evaporator
EVT	Extravehicular Transfer
EXT	External
f	F Stop
FC	Fuel Cell
FDAI	Flight Director Attitude Indicator

FLT	Flight
FM	Frequency Modulated
FOV	Field of View
fps or FPS	Feet per second
FT or ft	Feet
FTO	Flight Test Objective
FTP	Full Throttle Position
GBI	Grand Bahama Islands
GBM	Grand Bahama (MSFN)
GDC	Gyro Display Coupler
GDS	Goldstone, California
GET	Ground Elapsed Time
GETI	Ground Elapsed Time of Ignition
GLY	Glycol
GMT	Greenwich Mean Time
G&N	Guidance and Navigation
GNCS	Guidance Navigation Control System
GWM	Guam
GYM	Guaymas, Mexico
H2	Hydrogen
HA	Apogee Altitude
HAW	Hawaii
HBR	High Bit Rate (TLM)
HD	Highly Desirable
HGA	High Gain Antenna
HI	High
Hp	Perigee Altitude
HSK	Honeysuckle (Canberra, Australia)
HTR	Heater
HTV	USNS Huntsville
ICDU	Inertial Coupling Data Unit
ID	Identification
IGA	Inner Gimbal Angle
IGN	Ignition
IMU	Inertial Measurement Unit
INIT	Initialization
INT	Intervalometer
IP	Initial Point
ISA	Interim Storage Assembly
IU	Instrumentation Unit
IVC	Intervehicular Communications
IVT	Intravehicular Transfer
JETT	Jettison
KM	Kilometer
kwh	Kilowatt Hour

LA	Launch Azimuth
LAT	Latitude
LBR	Low Bit Rate (TLM)
LBS or lbs	Pounds
LCG	Liquid Cooled Garment
LDG	Landing
LDMK	Landmark
LEB	Lower Equipment Bay
LEC	Lunar Equipment Conveyor
LFH	Lunar Far Horizon
LGC	LM Guidance Computer
LH	Left-hand
L/H	Local Horizontal
LHEB	Left-hand Equipment Bay
LHFEB	Left-hand Forward Equipment Bay
LHSSC	Left Hand Side Storage Container
LiOH	Lithium Hydroxide
LLM	Lunar Landing Mission
LLOS	Landmark Line of Sight
LM	Lunar Module
LMP	Lunar Module Pilot
LNH	Lunar Near Horizon
LOI	Lunar Orbit Insertion
LONG	Longitude
LOS	Loss of Signal or Loss of Site
LPO	Lunar Parking Orbit
LR	Landing Radar
LRRR or LR3	Laser Ranging Retro-Reflector
LS	Landing Site
LT	Light
LTG	Lighting
LV	Launch Vehicle
L/V	Local Vertical
LVPD	Launch Vehicle Pressure Display
M	Mandatory
MAD	Madrid, Spain
MAN	Manual
MAX	Maximum
MAX Q	Maximum Dynamic Pressure
MCC	Midcourse Correction
MCC-H or MCC	Mission Control Center - Houston
MDC	Main Display Console
MEAS	Measurement
MER	USNS Mercury
MESA	Modularized Equipment Stowage Assembly
MET	Mission Event Timer

MGA	Middle Gimbal Angle
M/I	Minimum Impulse
MIN	Minimum
MLA	Merrit Island, Florida
MNVR	Maneuver
MPS	Main Propulsion System
MSFN	Manned Space Flight Network
MTVC	Manual Thrust Vector Control
N2	Nitrogen
NAV	Navigation
NM	Nautical Miles
NOM	Nominal
NXX	Noun XX
O2	Oxygen
OBS	Observation
O/F	Oxidizer to Fuel Ratio
OGA	Outer Gimbal Angle
OMNI	Omnidirectional Antenna
OPS	Oxygen Purge System
ORB	Orbital
ORDEAL	Orbit Rate Display Earth and Lunar
ORIENT	Orientation
OVHD	Overhead
P	Pitch or Program
PAD	Voice Update
PCM	Pulse Code Modulation
PC	Plane Change
PDI	Powered Descent Initiation
PGA	Pressure Garment Assembly
PGNCS	Primary Guidance Navigation Control Section
PIPA	Pulse Integrating Pendulous Accelerometer
PLSS	Personal Life Support Systems
PM	Phase Modulated
POL	Polarity or Polarizing
PRE	Pretoria, South Africa
PREF	Preferred
PREP	Preparation
PRESS	Pressure
PRIM	Primary
PROP	Proportional
PSE	Passive Seismic Experiment
PT	Point
PU	Propellant Utilization
PUGS	Propellant Utilization and Gaging System

PTC	Passive Thermal Control
PWR	Power
PXX	Program XX
Qty	Quantity
R	Roll or Range
R&B	Red & Blue
RAD	Radiator
RCDR	Recorder
RCS	Reaction Control System
RCU	Remote Control Unit
RCV	Receiver
RED	USNS Redstone
REFSMMAT	Reference Stable Member Matrix
REG	Regulator
REQD	Required
RH	Right-hand
RING	Ringsite
RLS	Radius of Landing Site
RNDZ	Rendezvous
RR	Rendezvous Radar
RSI	Roll Stability Indicator
RT	Real Time
RTC	Real Time Command
RXX	Routine XX
SA	Shaft Angle
S/C	Spacecraft
SCE	Signal Conditioning Equipment
SCS	Stabilization Control System
SCT	Scanning Telescope
SEC	Secondary
SECO	S-IVB Engine Cut-off
SECS	Sequential Events Control System
SEP	Separate
SEQ	Sequence
S-IVB	Saturn IV B(Third Stage)
SLA	Service Module LM Adapter
SLOS	Star Line-of-Sight
SM	Service Module
SPOT	Spot Meter
SPS	Service Propulsion System
SR	Sunrise
SRC	Sample Return Container
SRX	S-Band Receiver Mode No. X
SS	Sunset
STX	S-Band Transmit Mode No. X

S.V.	State Vector
SWC	Solar Wind Composition
Sw	Switch
SXT	Sextant
T EPHEM	Time of Ephemeris Update
TA	Trunnion Angle
TAN	Tananarive, Madagascar
TB	Time Base
TCA	Time of Closest Approach
TD&E	Transposition Docking & LM Ejection
TEC	Trans Earth Coast
TEI	Transearth Insertion
TEMP	Temperature
TERM	Terminate
TEX	Corpus Christi, Texas
TGT	Target
TIG	Time of Ignition
TLC	Trans Lunar Coast
TLI	Translunar Insertion
TLM or TM	Telemetry
TPF	Terminal Phase Final
TPI	Terminal Phase Initiation
TPM	Terminal Phase Midcourse
T/R	Transmitter/Receiver
TRANS	Translation
TV	Television
TVC	Thrust Vector Control
TWR	Tower
US	United States
V	Velocity
VAN	USNS Vanguard
VHF	Very High Frequency
VLV	Valve
VI	Inertial Velocity
VOX	Voice Keying
VXX	Verb XX
W/O	Without
WRT	With Respect to
WTN	USNS Watertown
XFER	Transfer
XMIT	Transmit or Transmitter
XPONDER	Transponder
Y	Yaw

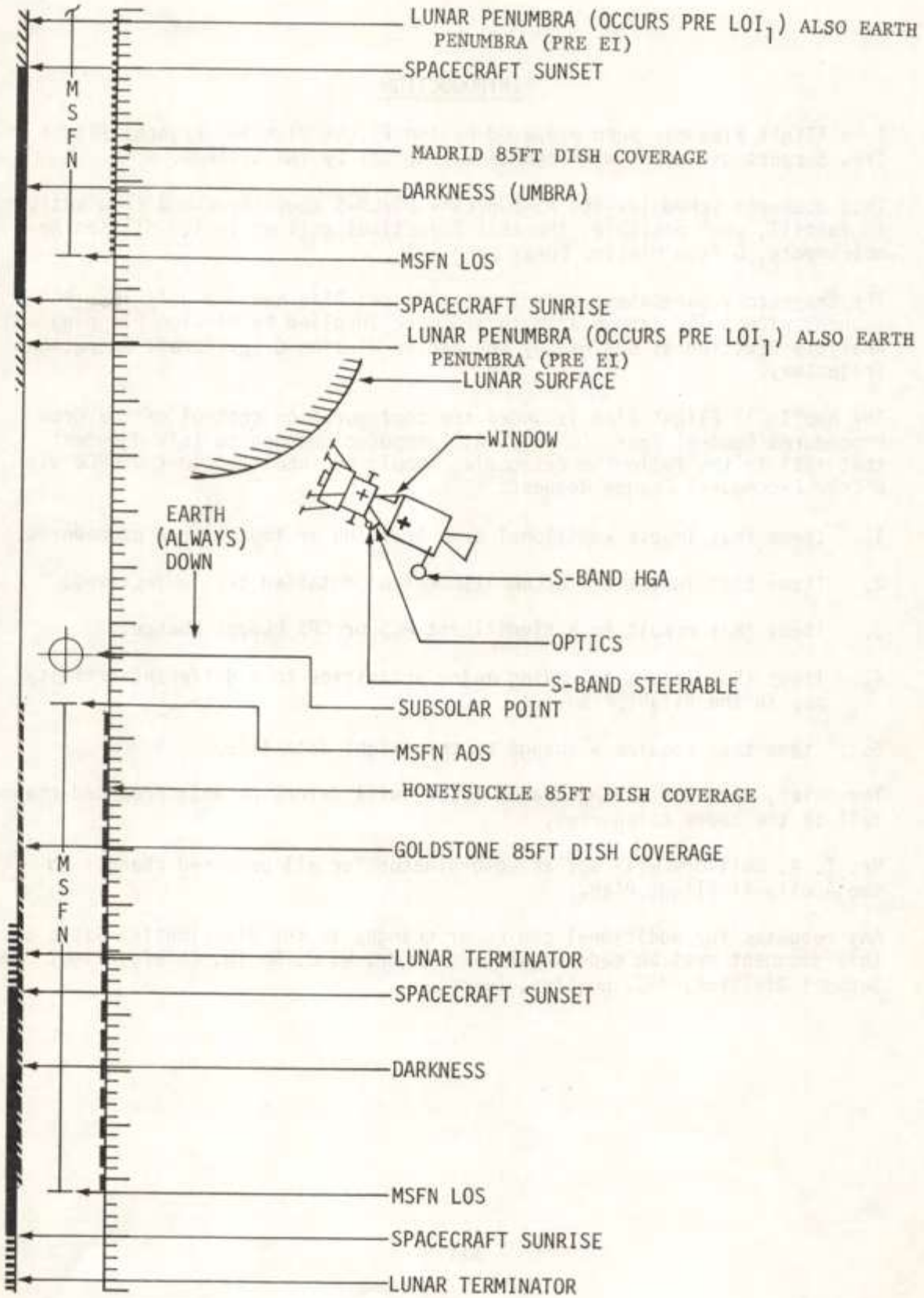
ΔV	Velocity Change (Differential)
ΔVC	Velocity Change at Engine Cutoff
ΔR	Position Change (Differential)
8-balls	Flight Director Attitude Indicator (FDAI)

CAMERA NOMENCLATURE

EL/250/BW-BRKT	Electric Hasselblad/250mm Lens/ Black & White film-Camera Bracket
INT (f5.6,250,INF)	Intervalometer ($\frac{1}{250}$ sec, f-stop 5.6, shutterspeed=$\frac{1}{250}$ sec, Infinity)
16mm/18/CEX-BRKT	16mm Camera/18mm Lens/Color Film External-Camera Bracket
MIR (f8,250,INF) 6fps	Mirror (f-stop 8, shutterspeed = $\frac{1}{250}$ sec, Infinity) 6 frames per sec

ENCLOSURE 2

SYMBOL NOMENCLATURE



INTRODUCTION

This Flight Plan has been prepared by the Flight Planning Branch, Flight Crew Support Division, with technical support by TRW Systems.

This document schedules the AS-506/CSM-107/LM-5 operations and crew activities to fulfill, when possible, the test objectives defined in the Mission Requirements, G Type Mission Lunar Landing.

The trajectory parameters used in this Flight Plan are for July 16, 1969 launch, with a 72° launch azimuth and were supplied by Mission Planning and Analysis Division as defined by the Apollo Mission G Spacecraft Operational Trajectory.

The Apollo 11 Flight Plan is under the configuration control of the Crew Procedures Control Board (CPCB). All proposed changes to this document that fall in the following categories should be submitted to the CPCB via a Crew Procedures Change Request:

1. Items that impose additional crew training or impact crew procedures.

2. Items that impact the accomplishment of detailed test objectives.

3. Items that result in a significant RCS or EPS budget change.

4. Items that result in moving major activities to a different activity day in the Flight Plan.

5. Items that require a change to the flight data file.

The Chief, Flight Planning Branch (FCSD) will determine what proposed changes fall in the above categories.

Mr. T. A. Guillory will act as co-ordinator for all proposed changes to the Apollo 11 Flight Plan.

Any requests for additional copies or changes to the distribution lists of this document must be made in writing to Mr. W. J. North, Chief, Flight Crew Support Division, MSC, Houston, Texas.

SECTION I - GENERAL

MISSION DESCRIPTION

1. Launch and EPO (Duration 2:44) LIFT OFF - 2:44 GET

   (a) Nominal launch time is 9:32 EDT, July 16, 1969, with a launch window duration of 4 hrs. 24 min.

   (b) Earth orbit insertion into a 100 nm circular orbit at 11 min. 43 sec. after lift-off

   (c) CSM systems C/O in earth orbit

   (d) Optional IMU realign (P52) to the pad REFSMMAT during the first night period

   (e) TLI occurs at 2:44:26 GET over the Pacific Ocean during the second revolution. (See Table 1-1 for burn data).

2. Translunar Coast (Duration 73:10) 2:44 - 75:54 GET

   After TLI, which places the spacecraft in a free lunar return trajectory, the following major events occur prior to LOI:

   (a) Transposition, docking and LM ejection, including SIVB photography

   (b) Separation from SIVB and a CSM evasive maneuver

   (c) SIVB propulsive venting of propellants (slingshot)

   (d) Two series of P23 cislunar navigation sightings, star/earth horizon, consisting of five sets at 06:00 GET and five sets at 24:30 GET

   (e) Four midcourse corrections which take place at TLI + 9, TLI + 24, LOI - 22 and LOI - 5 hours with ΔV nominally zero (See Table 1-1).

(f) Passive thermal control (PTC) will be conducted during all periods when other activities do not require different attitudes.

(g) LM inspection and housekeeping

(h) LOI_1, performed at 75:54:28 GET, ends the TLC phase.

3. Lunar Orbit (Duration 59:30) 75:54 - 135:24 GET

LOI Day (Duration 25:00) 69:00 - 94:00

(a) LOI_1

(b) Photos of targets of opportunity

(c) LOI_2

(d) Post LOI_2 LM entry and inspection. S-Band/VHF B Voice tests will be conducted.

(e) Post LOI_2 Pseudo landmark tracking (one set of sightings) (See Table 1-4)

(f) Rest period of 9 hours

DOI and EVA Day (Duration 28:00) 94:00 - 122:00 GET

(a) Docked LM activation and checkout

(b) Docked landing site landmark sighting (one set of sightings) (See Table 1-3)

(c) Undocking and separation

(d) DOI thru landing (See Figure 1-3 Powered Descent)

(e) LM post touchdown and simulated liftoff

(f) Rest period (LM) of 4 hours

(g) CSM plane change

(h) Rest period (CSM) of 4 hours

(i) EVA prep

(j) EVA for 2 hours 40 minutes

(k) Post EVA

(l) Rest period (LM) 4 hours 40 minutes

(m) Rest period (CSM) 4 hours 50 minutes

Ascent and TEI Day (Duration 25:00) 122:00 - 147:00 GET

(a) LM Lift-Off and Insertion

(b) LM active rendezvous

CSI

PC

CDH

TPI

Braking

(c) Docking

(d) LM jettison

(e) TEI

(f) Rest Period

4. Lunar Orbit Particulars (Average Values for a 60 x 60 nm orbit)

(a) Revolutions start at 180° longitude

(b) Revolution duration - 1 hr. 58.2 min.

(c) S/C night period duration - 47 min.

(d) MSFN coverage per rev. - 72 min.

(e) Orbit inclination - 1.25° for July 16, 1969 launch

(f) S/C orbital rate - 3°/min. (.05°/sec)

(g) Lighting change at fixed ground point - 1°West/Rev.

(h) Horizon visibility ± 20° selenocentric angle on the lunar surface

(i) One lunar degree on lunar surface is 16.35 nm

(j) Site 2 will be visible (3° sun angle) at REV. 7

(k) S/C subvehicle point to horizon 327 nm.

5. Transearth Coast and Entry (Duration 59:39) 131:52 - 195:03 GET

Transearth coast begins with TEI at 135:24:34 GET and consists of the following major events:

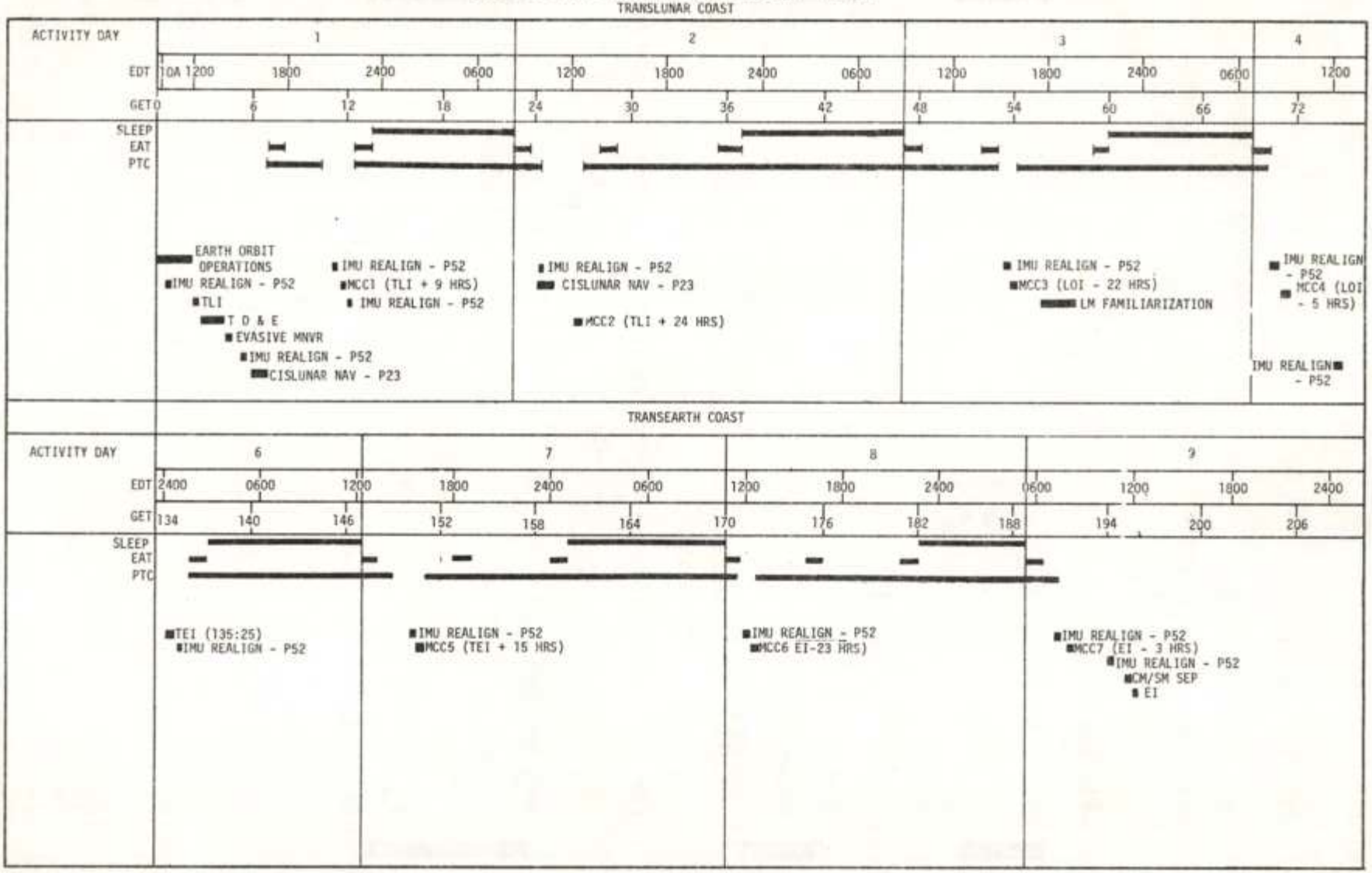
(a) Three midcourse corrections are scheduled at TEI + 15, EI - 23 and EI - 3 hours with ΔV nominally zero.

(b) CM/SM separation takes place at 194:51 GET and Entry Interface occurs at 195:03 GET.

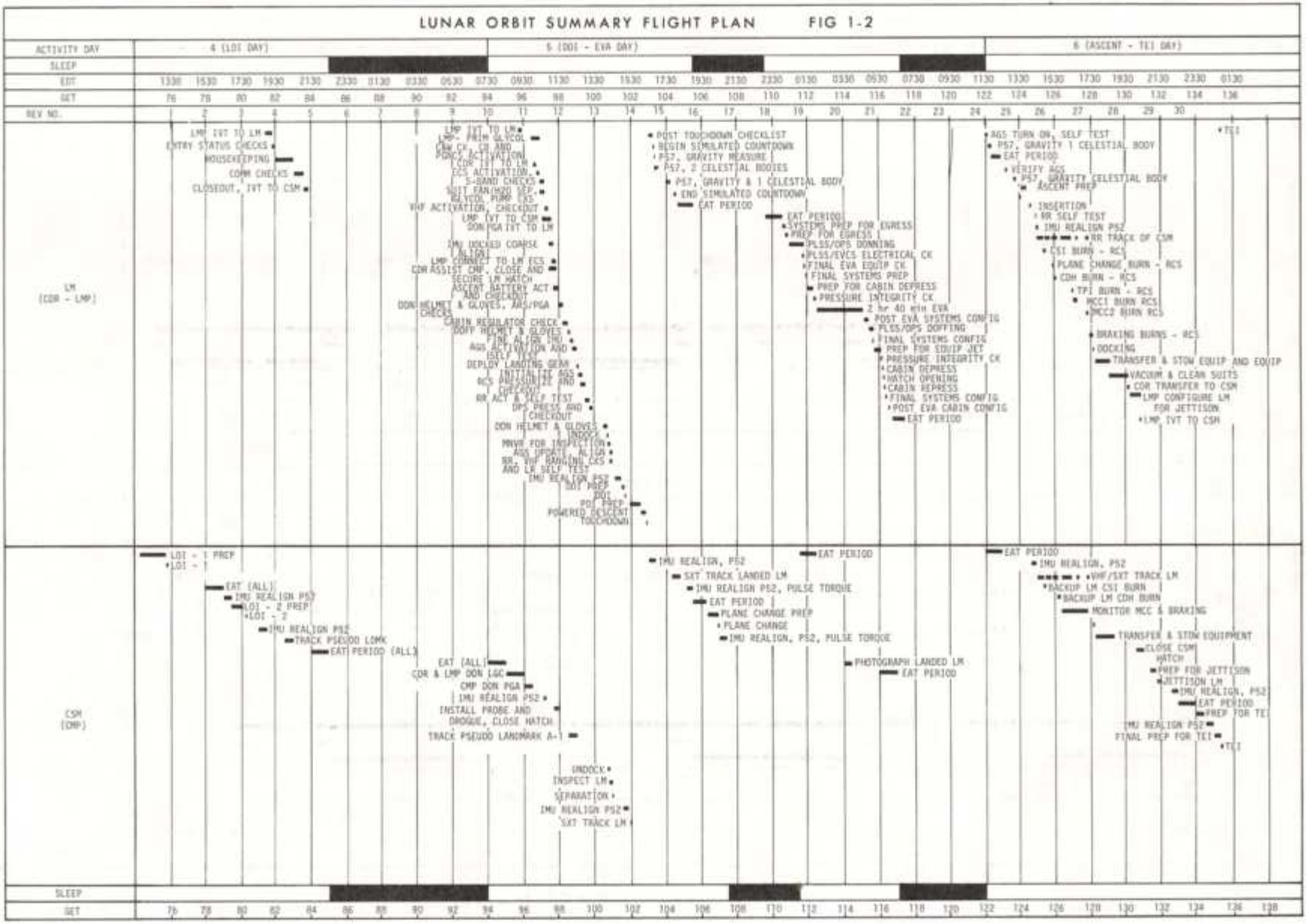
(c) Splashdown will occur in the Pacific Ocean at a longitude of about 172.4° West at 195:17 GET. This will occur approximately 25 minutes prior to sunrise local time.

FIGURE 1-1

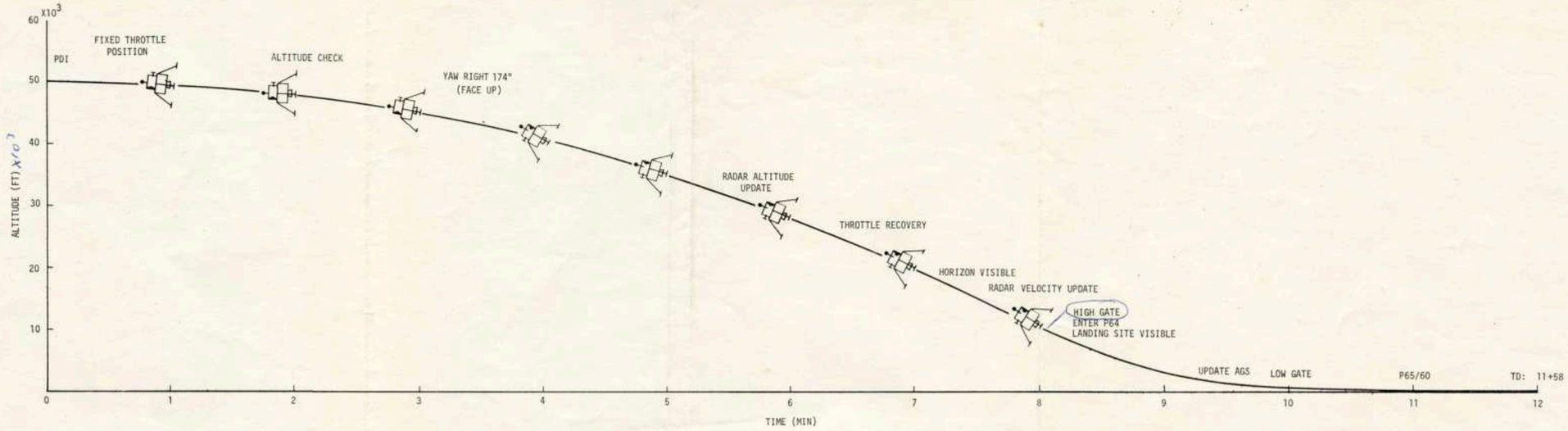
MISSION SUMMARY FLIGHT PLAN



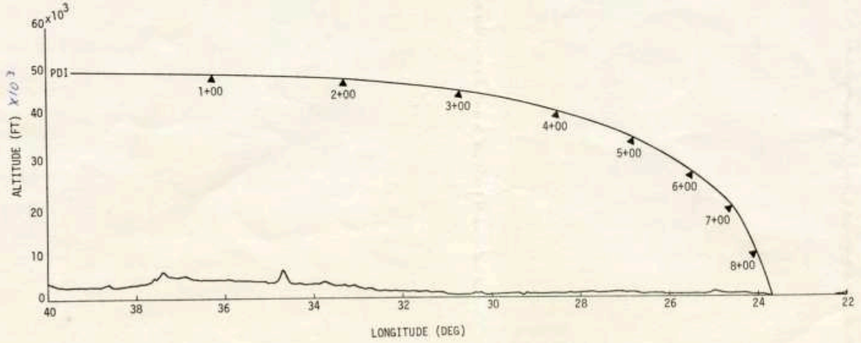
LUNAR ORBIT SUMMARY FLIGHT PLAN FIG 1-2



LM POWERED DESCENT



LM POWERED DESCENT/TERRAIN PROFILE



LM POWERED DESCENT/HIGH GATE TO TD

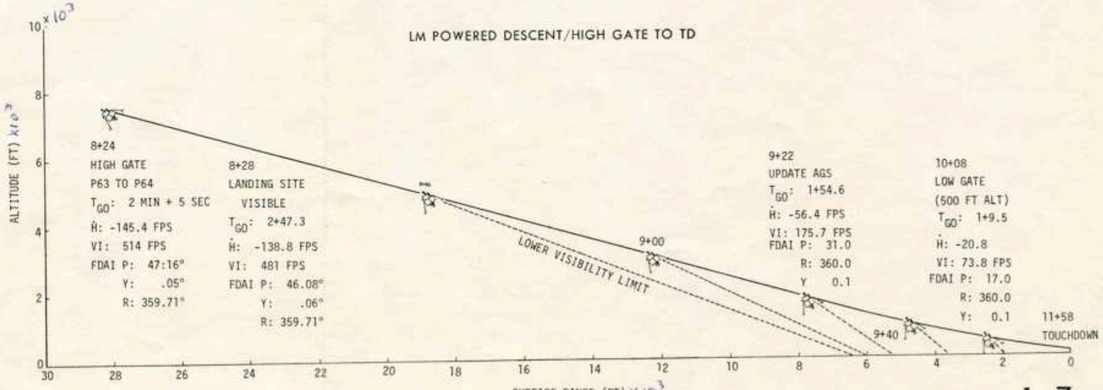


TABLE 1-1 CSM BURN SCHEDULE

BURN/MANEUVER	GETI BURN TIME ΔV_c	ATTITUDE (DEG) LH/LV	ATTITUDE (DEG) INERTIAL	LIGHTING	ΔV (FPS)	ULLAGE	TVC MODE	REFSMMAT	S/C WT. RESULTANT HA, HP	REMARKS
S-IVB TLI	2:44:26 5 MIN 20 SEC			BURNOUT AT SUNRISE	ΔVX: — ΔVY: — ΔVZ: — ΔV REQ: 10,451.2	———	———	PAD	WT: — HP: — HA: —	S-IVB BURN
CSM/LM S-IVB EVASIVE MNVR	04:39:44.9 2.8 SEC 15.6 FPS			DAYLIGHT	ΔVX: 5.1 ΔVY: 0.0 ΔVZ: 19.0 ΔV REQ: 19.7	NOT REQUIRED	G&N AUTO	PAD	WT: 96662.3 HP: 123.8 HA: 281953.9	SPS BURN
MIDCOURSE CORRECTIONS MCC_1 to MCC_4	11:45 26:45 53:55 70:55			———	ΔVX: NOMINALLY ΔVY: ZERO ΔVZ: ΔV REQ:	NOT REQUIRED	G&N AUTO	PAD PTC PTC LDG SITE	———	TLI + 9 TLI + 24 LOI - 22 LOI - 5
LOI_1	75:54:28.4 5 MIN 58.9 SEC 2914.8 FPS			DAYLIGHT (SS-1HR 7 MIN)	ΔVX: -2891.8 ΔVY: -433.1 ΔVZ: 20.4 ΔV REQ: 2924.1	NOT REQUIRED	G&N AUTO	LDG SITE	WT: 95207.4 HP: 59.2 HA: 169.8	SPS BURN
LOI_2	80:09:29.7 16.4 SEC			DAYLIGHT (SR + 9 MIN)	ΔVX: 138.3 ΔVY: 0.0 ΔVZ: 75.9 ΔV REQ: 157.8	2 JET 20 SEC	G&N AUTO	LDG SITE	WT: 71320.8lbs HP: 53.6 NM HA: 65.6 NM	SPS BURN
CSM/LM SEP	100:39:50.4 8 SEC			SUNLIGHT (SS-14 MIN)	ΔVX: 0.0 ΔVY: 0.0 ΔVZ: 2.5 ΔV REQ: 2.5	———	G&N AUTO	LDG SITE	WT: 36407.9 HP: 55.6 HA: 63.1	RCS BURN
*CSM PLANE CHANGE	107:05:33.4 .8 SEC 5.7			DARKNESS (SS + 17 MIN)	ΔVX: 0.0 ΔVY: 16.6 ΔVZ: 0.0 ΔV REQ: 16.6	2 JET 20 SEC	G&N AUTO	PLANE CHANGE	WT: 36325.4 HP: NO CHANGE HA: NO CHANGE	SPS BURN
LM JETTISON	131:53:04.7 3.1 SEC 0.8 FPS			DAYLIGHT (SR + 36 MIN)	ΔVX: -1.0 ΔVY: — ΔVZ: — ΔV REQ: 1.0	———	G&N AUTO	LIFT OFF	WT: 36154.7 HP: 58.5 HA: 59.4	RCS BURN
TEI	135:24:33.8 2 MIN 29.4 SEC NOT AVAILABLE			DAYLIGHT (SR + 10 MIN	ΔVX: 3213.3 ΔVY: 705.0 ΔVZ: -138.8 ΔV REQ: 3292.7	2 JET 16 SEC	G&N AUTO	LIFT OFF	WT: 36111.4 HP: — HA: —	SPS BURN
MIDCOURSE CORRECTIONS MCC_5 to MCC_7	150:24 172:00 192:06			———	ΔVX: ΔVY: NOMINALLY ΔVZ: ZERO ΔV REQ:	———	G&N AUTO	PTC PTC ENTRY	———	TEI + 15 EI -23 EI -3

* NEW DATA INDICATES THIS BURN MAY BE NOMINALLY ZERO

TABLE 1-2 LM BURN SCHEDULE

BURN/MANEUVER	GETI BURN TIME	ATTITUDE (DEG) LH	ATTITUDE (DEG) INERTIAL	LIGHTING	ΔV (FPS)	ULLAGE ΔV (FPS)	TVC MODE	REFSMMAT	S/C WT RESULTANT HA, HP	REMARKS
DOI	101:38:48 28.5 SEC			DARKNESS (SR-4 MIN)	ΔVX: 67.46 ΔVY: -26.68 ΔVZ: -12.51 ΔV REQ: 70 FPS	2 JET 7.5 SEC 1.3 FPS	PGNCS AUTO	LDG SITE	WT: 33,404 HP: 8.97 HA: 57.87NM	DPS BURN
PDI	102:35:13 11MIN 58SEC			DAYLIGHT	ΔVX: - ΔVY: - ΔVZ: - ΔV REQ: 6766	2 JET 7.5 SEC 1.3 FPS	PGNCS AUTO	LDG SITE	WT: 16,569 HP: 0 HA: 0	DPS BURN
ASCENT	124:23:26 7 MIN 18 SEC			DAYLIGHT	ΔVX: - ΔVY: - ΔVZ: - ΔV REQ: 6060	——	PGNCS AUTO	LIFT OFF	WT: 5,894 AT INS HP: 60,000 ft HA: 45 NM	APS BURN
CSI	125:21:19.1 45.0 SEC			DARKNESS (SR - 1 MIN)	ΔVX: 49.5 ΔVY: 0.0 ΔVZ: 0.0 ΔV REQ: 49.5	——	PGNCS AUTO	LIFT OFF	WT: 5875.0 HP: 44.9 HA: 45.0	RCS BURN
PLANE CHANGE	125:50:28 0			DAYLIGHT (SR + 25 MIN)	ΔVX: 0.0 ΔVY: 0.0 ΔVZ: 0.0 ΔV REQ: 0.0	——	PGNCS AUTO	LIFT OFF	WT: - HP: - HA: -	RCS + Y 2 JET BURN NOMINALLY ZERO
CDH	126:19:37.0 1.9 SEC			DAYLIGHT (SS - 19 MIN)	ΔVX: -1.1 ΔVY: 0.0 ΔVZ: 4.1 ΔV REQ: 4.3	——	PGNCS AUTO	LIFT OFF	WT: 5842.9 (TIG) HP: 43.8 HA: 45.3	RCS BURN
TPI	126:58:08.4 22.4 SEC			DARKNESS (SR - 23 MIN)	ΔVX: 22.0 ΔVY: 0.0 ΔVZ: -11.1 ΔV REQ: 24.8	——	PGNCS AUTO	LIFT OFF	WT: 5040.1 HP: 43.3 HA: 61.7	RCS BURN
MCC_1	127:13:08 0			DARKNESS (SR - 8 MIN)	ΔVX: 0.0 ΔVY: 0.0 ΔVZ: 0.0 ΔV REQ: 0.0	——	PGNCS AUTO	LIFT OFF	WT: - HP: - HA: -	RCS + Z 2 JET BURN NOMINALLY ZERO
MCC_2	127:28:08 0			DAYLIGHT (SR + 7 MIN)	ΔVX: 0.0 ΔVY: 0.0 ΔVZ: 0.0 ΔV REQ: 0.0	——	PGNCS AUTO	LIFT OFF	WT: - HP: - HA: -	RCS + Z 2 JET BURN NOMINALLY ZERO
1st BRAKING MNVR	127:36:57 0			DAYLIGHT (SR + 15 MIN)	ΔVX: 0.0 ΔVY: 0.0 ΔVZ: 0.0 ΔV REQ: 0.0	——	MANUAL	LIFT OFF	WT: - HP: - HA: -	RCS -Z 2 JET BURN NOMINALLY ZERO
2nd BRAKING MNVR	127:39:24.5 10.8 SEC			DAYLIGHT (SR + 18 MIN)	ΔVX: - ΔVY: - ΔVZ: - ΔV REQ: 12.0	——	PGNCS AUTO	LIFT OFF	WT: 5824.1 HP: 49.0 HA: 60.7	RCS -Z 2 JET
3rd BRAKING MNVR	127:40:32.8 8.8 SEC			DAYLIGHT (SR + 20 MIN)	ΔVX: - ΔVY: - ΔVZ: - ΔV REQ: 9.8	——	PGNCS AUTO	LIFT OFF	WT: 5816.4 HP: 53.7 HA: 60.3	RCS -Z 2 JET
4th BRAKING MNVR	127:42:16.1 4.3 SEC			DAYLIGHT (SR + 21 MIN)	ΔVX: - ΔVY: - ΔVZ: - ΔV REQ: 4.8	——	PGNCS AUTO	LIFT OFF	WT: 5810.1 HP: 56.2 HA: 60.1	RCS -Z 2 JET
5th BRAKING MNVR	127:43:35.7 4.2 SEC			DAYLIGHT (SR + 23 MIN)	ΔVX: - ΔVY: - ΔVZ: - ΔV REQ: 4.7	——	PGNCS AUTO	LIFT OFF	WT: 5807.0 HP: 59.9 HA: 58.9	RCS -Z 2 JET

TABLE 1-3 LUNAR LANDING SITE DATA

DAY	SITE DESIG	LATITUDE	LONGITUDE	[1]LAUNCH AZIMUTH/ SUN ELEVATION	[2]LAUNCH AZIMUTH/ SUN ELEVATION
JULY 16 0932 EDT	2(IIP6)	00°42'50"N 00.71388889°N	23°42'28"E 23.70777778°E	72°/10.5°	108°/13.5°
		(00.6914°N)	(23.7169°E)[3]		
JULY 18 1132 EDT	3(IIP8)	00°21'10"N 00.35277778°N	01°17'57"W 01.29916667°W	89.295°/11°	108°/13°
JULY 21 ~~1409~~ 1209 EDT	5(IIP13)	01°40'41"N 01.67805556°N	41°53'57"W 41.89916667°W	94.6775/9.7°	108°/11.7°

Data From TJ memo, Accuracy Estimates, Landing Site Landmarks, May 12, 1969, TJ-69-499.

1 Sun Elevation Angles Are For Approximately 27 Hours After LOI, 1st Opportunity TLI.
2 Includes 2nd Opportunity TLI.
3 Data From MPAD memo, landing site coordinates for G, June 12, 1969, 69-FM41-181.

TABLE 1-4 LANDMARK TRACKING DATA

July 16 Launch

LANDMARK DESIG.	LATITUDE	LONGITUDE	DELTA ALTITUDE (nm)	SUN EL
A1(Pseudo)	2°N 2.000°N	65° 30' 60.500°E	000.00	43°
IP(130)	1°53'N 1.885°N	28°42'E 28.726°E	000.00	-
130(Prime LDG SITE 2)	01°15'56"N 01.26555556°N	23°40'44" 23.67888889°E	-001.68	8.5°
	(01.24307°N)	(23.6880°E)[1]		
123(Alternate LDG SITE 2)	00°30'19"N 00.50527778°N	24°53'20°E 24.88888889°E	-001.71	-
129(Alternate LDG SITE 2)	01°17'06"N 01.28500000°N	23°44'37"E 23.74361111°E	-001.76	-
133(Alternate LDG SITE 2)	00°47'14"N 00.78722222°N	23°30'55"E 23.51527778°E	-001.68	-

1 Data from MPAD memo, landing site 2 position, June 20, 1969, 69-FM41-199.

TABLE 1-4 LANDMARK TRACKING DATA (CONT'D)

July 18 Launch

LANDMARK DESIG.	LATITUDE	LONGITUDE	DELTA ALTITUDE (nm)	SUN EL
IP(G1)	0°16'N 0.267°N	32°19'E 32.317°E	-	-
G1(129)	01°17'06"N 01.28500000°N	23°44'37"E 23.74361111°E	~~-001.97~~ -001.76	26°
IP(143)	00°18'N 00.300°N	3°23'E 3.383°E	-	-
143(Prime LDG SITE 3)	00°36'51"N 00.61416667°N	01°04'39"W 01.07750000°W	-001.01	9°
150(Alternate LDG SITE 3)	00°16'59"N 00.28305556°N	01°25'43"W 01.42861111°W	-001.01	-
147(Alternate LDG SITE 3)	00°03'42"N 00.06166667°N	01°16'36"W 01.27666667°W	-000.99	-

TABLE 1-4 LANDMARK TRACKING DATA (CONT'D)

July 21 Launch

LANDMARK DESIG.	LATITUDE	LONGITUDE	DELTA ALTITUDE (nm)	SUN EL
IP(G1)	0°30'S 0.500°S	26°33'W 26.550°W	-	-
G1	1°42'N 1.696°N	32°10'W 32.162°W	-001.77	8°
IP(180)	0°36'N 0.608°N	36°34'W 36.567°W	-	-
180(PRIME LDG SITE 5)	01°30'37"N 01.51027778°N	41°49'05"W 41.81805556°W	-001.25	8.9°
171(Alternate LDG SITE 5)	01°20'04"N 01.33444444°N	40°47'34"W 40.79277778°W	-001.29	-
178(Alternate LDG SITE 5)	01°45'33"N 01.75916667°N	41°34'12"W 41.57000000°W	-001.22	-
184(Alternate LDG SITE 5)	02°03'10"N 02.05277778°N	42°13'41"W 42.22805556°W	-001.23	-

FLIGHT PLAN NOTES

A. Crew

1. Crew designations are as follows:

Designation	Prime	Backup
Commander (CDR)	Armstrong	Lovell
Command Module Pilot (CMP)	Collins	Anders
Lunar Module Pilot (LMP)	Aldrin	Haise

2. Crew positions during the mission are as follows:

	CSM			LM	
	Left	Center	Right	Left	Right
Launch thru TLI	CDR	LMP	CMP		
T&D thru Entry	CMP	CDR	LMP		
Manned LM	CMP			CDR	LMP

3. The crew will eat and sleep simultaneously throughout the mission. Eat periods will be normally 1-hour duration, with additional activities held to a minimum during this time frame. Sleep periods will normally be 8 to 10 hour duration with two 4 to 5 hour sleep periods while the LM is on the lunar surface.

4.

Activity	PGA Configuration
Launch to insertion	PGA's with helmet & gloves (H&G)
Insertion to TLI	PGA's without H&G
TLI to evasive mnvr	PGA's with H&G
TLC & LOI 1&2	Constant wear garments
LM activation & checkout	PGA without H&G (CMP H&G donned for latch cocking & CDR/LMP H&G donned for pressure integrity check and cabin reg check)
Undocking through touchdown	PGA's with H&G except CMP without H&G after DOI
Touchdown through pre lift-off	PGA's without H&G except for CDR/LMP simulated countdown & EVA
Liftoff through LM jettison	PGA's with H&G (except H&G off after docking)
LM jettison through splashdown	Constant wear garmets

5. Two crew status reports via air-to-ground communications will be made by the flight crew during each activity day. The first report will be given after the first meal of the day and will concern the sleep obtained during the previous sleep period. The second report will be given following the final meal of the day and will concern the radiation dose received during the previous 24 hours and medication taken if any. The following information should be logged:
   a. Food Consumption
   b. Exercise
   c. Used fecal bags marked as to crewman and GET

6. Negative reporting will be used in reporting completion of each checklist.

7. Continuous CSM biomedical data are automatically transmitted to the ground.

8. LM biomedical switching is performed manually by the LMP from undocking to docking as scheduled in the timeline.

9. All onboard gage readings will be read directly from the gages. and will not be corrected by the appropriate calibration factors.

B. Photography

Photographic requirements were derived from the following:
   a. Lunar Surface Operations Plan
   b. Photographic Operations Plan

C. Procedures

1. CSM

Crew procedures called out in the flight plan may be found in the following documents:
   a. Apollo Operations Handbook - CSM-107 (AOH), Volume 2
   b. Crew Checklist
   c. CSM Rendezvous Procedure
   d. Abort Summary Document
   e. Apollo Entry Summary Document
   f. Photographic Operations Plan
   g. Descent Procedures Document
   h. Ascent Procedures Document
   i. Lunar Landmark Tracking Attitude Studies
   j. Lunar Orbit Attitude Sequence for Mission G
   k. Data Priority Documents

2. LM

Crew procedures called out in the flight plan may be found in the following documents:

a. Apollo Operations Handbook LM-5 Volume 2
b. Crew Checklist
c. LM Rendezvous Procedures
d. LM Descent/Ascent Summary Document
e. Lunar Landing Phase Photographic Operations Plan
f. Data Priority Documents
g. EVA Procedures
h. Apollo Lunar Surface Operations Plan

D. Communications

1. General

a. CSM and LM HBR data transmissions in lunar orbit will normally require the use of the high gain or steerable antennas
b. During communications, the spacecraft will be referred to by name (Apollo 11) and MCC-H will be referred to as Houston.
c. The preferred S-Band communications are:
   (1) CSM
      (a) Uplink Mode 6 (Voice, PRN, and Updata)
      (b) Downlink Mode 2 (Voice, PRN, TLM-HBR)
   (2) LM
      (a) Uplink Mode 7 (Voice, Updata)
      (b) Downlink Mode 1 (Voice, TLM-HBR)
d. LM voice recorder has a maximum utilization of 10 hours. This recorder will be used during LM operations to record all LM voice data during undocked operations (27 hours 42 minutes). This recorder will be operated in the VOX mode.
e. A small portable voice recorder will be carried in the CM to be used at the discretion of the crew as a voice recorder back-up. This recorder will not be transferred to the LM for use during undocked operations.
f. The S-band"squelch" will be on during the sleep periods in order to prevent MSFN fade-out noise from disturbing the crew.

2. DSE Operation

a. The DSE will normally be operated via ground command except for special cases where the operation is time limited. In these cases the crew may be asked to rewind the tape.
b. During the earth orbit period when the CSM is not over a MSFN station, CSM TLM-LBR data will be recorded on the DSE and will be dumped during the pass over the US and over CRO prior to TLI if possible.
c. DSE will be used for CSM HBR and voice recording during all CSM engine burns.
d. DSE data and voice recordings will be made in CSM LBR mode whenver possible in order to minimize the DSE dump time.

e. During PTC using the HGA REACQ communications mode the DSE will be used to record LBR data when the HGA is not in the MSFN field of view.
f. During lunar orbit LM operations, the DSE will be used to record LM-TLM-LBR data during all docked LM activites that occur on the lunar farside. For undocked LM activites only DOI will be recorded as VHF ranging is required.
g. DSE will be used to record all HBR entry data during the blackout region.

3. Launch - Earth Orbit Phase
a. OMNI B and VHF LEFT will be selected for lift off. OMNI D will be selected by the crew during boost phase if the launch azimuth is less than 96° or OMNI C if the launch azimuth is greater than 96°. OMNI D will probably be the best antenna for earth orbit.
b. VHF Duplex B will be used for launch, and Simplex A for earth orbit operations.
c. VHF Simplex A will be used for entry to be compatible with recovery forces communications.

4. Translunar and Transearth Coast Phase
The translunar and transearth sleep communications mode will be as follows. The CSM x-axis will be placed normal to the ecliptic plane. The CSM will be rolled at a rate of approximately three revolution per hour. During the near earth sleep periods prior to 30 hours GET (range less than 120Knm) omni antennas B and D will be used. During the other sleep periods (beyond 120Knm) the high gain antenna may be required (in the REACQ mode). The REACQ configuration will provide approximately 210 degrees of HGA coverage per CSM/LM revolution or 35 minutes of MSFN coverage per hour. The REACQ configuration will also allow MCC-H to use real time control to select TLM HBR or LBR and to dump the DSE during each spacecraft revolution.

5. Lunar Exploration Phase
a. Normal CSM communications between MSFN/LM will be by S-Band during the lunar exploration period.
b. If additional communications capability is required the S-Band erectable antenna will be deployed by the EVA crewman and will be utilized for all LM/MSFN/CSM communications.
c. During periods when both crewmen are EVA, the "AR" position (Relay Mode) will be the normal communication mode on each of the Extravehicular Communication System (EVCS). The CDR will relay the LMP VHF voice and data to the LM which in turn will relay to MCC-H via S-Band.

E. CSM Notes

1. Electrical Power System and Water Management

a. Spacecraft lift-off switch positions are listed in the Apollo Operations Handbook (Volume 2) for CSM 107.

b. The CSM will remain fully powered up throughout the mission (CMC, IMU and SCS in the "operate" configuration and optics power-up as required).

c. Fuel cell H2 and O2 purging is scheduled as follows H2 approximately every 48 hours and O2 approximately every 12 hours.

d. The hydrogen and oxygen VAC ION pumps will be inactive throughout the mission.

e. Potable water will be chlorinated once a day before each sleep period, starting with the First sleep period (GET 13:30). The POT H2O inlet valve will be opened prelaunch.

f. FC purges and waste water dumps will not be scheduled within one hour prior to optical sightings.

g. Waste H2O dumping will be managed to allow:

(1) Maximum QTY:85-90%
(2) Minimum QTY:25%
(3) At LOI:QTY = 75%
(4) At CM-SM SEP:QTY = 90% to 100%
(5) No dumping after MCC3 until after LOI
(6) Dumps will be performed (if required) within 2 hours preceding MCC maneuvers
(7) In lunar orbit if dumping is required, dumps will be performed immediately prior to sleep periods
(8) The water dump will not be operated in the automatic mode at anytime during the mission

h. The cryogenic heaters will be in AUTO during the mission and the fans will be operated manually. The fans will be cycled for one minute before and after each sleep cycle.

i. The batteries will be charged according to the following schedule:

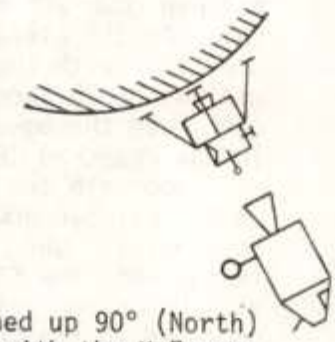
Time	Battery
5:20:00	B
12:20:00	A
48:10:00	B
80:25:00	A
107:15:00 ~~103:30:00~~	B
151:00:00 ~~148:00:00~~	A
172:00:00 ~~154:00:00~~	B

2. Environmental Control System and Cabin Pressurization

a. One CO2 odor absorber filter (LiOH canister) is changed approximately every 12 hours or if CO2 partial pressure is greater than 7.6mm Hg. There are 20 filters (2 in the canisters onboard and 18 stowed).

b. The coolant loop operation will be as follows:
   (1) Launch - primary loop operation
   (2) Earth Orbit - primary loop operation and secondary loop test
   (3) Post TLI - deactivate both evaporators
   (4) Pre LOI sec rad leak ck only.
   (5) At 112:30 activate primary evaporator
   (6) Post TEI - deactivate primary evaporator
   (7) Entry interface minus 1 hour - activate primary and secondary evaporator.

3. Guidance and Navigation



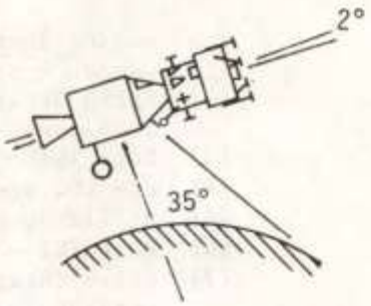
   a. During lunar orbit, the CSM and LM will utilize the same landing site REFSMMAT such that the gimbal angles would be 0,0,0 at landing with the LM sitting face forward on landing site number two and the CSM over the landing site pitched up 90° from local horizontal "heads up".
   b. During PTC the CSM/LM x-axis is pitched up 90° (North) for TLC and down 90° (South) for TEC with the Y-Z axes in the plane of the ecliptic. This change in x-axis pointing is to enable simultaneous viewing of the earth and moon through the side windows while maintaining a favorable high gain antenna position.
   c. The CSM tracking light will be on continuously from undocking to landing and from LM lift-off to docking.

4. Landmark Tracking

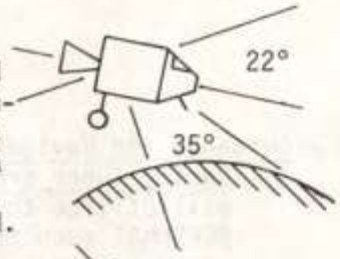
The following ground rules were used for landmark tracking.

   a. IMU to be realigned on the dark side preceding each tracking period.
   b. MSFN is reacquired after each tracking period. The tracking data will be acquired by MSFN after all the marks have been made and while N49 ($\Delta R, \Delta V$) is displayed. MSFN will give a GO when data acquisition has been verified.
   c. The pseudo landmark tracking (A1) will be used to determine the altitude of an area in which the LM will be making altitude checks after DOI. The data will be processed during the sleep period after the trackings and relayed to the LM prior to undocking.

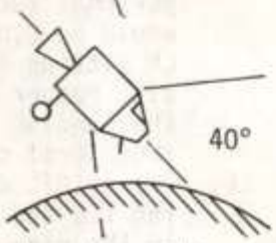
d. In the docked configuration the CSM/LM approaches the landmark in an inertial hold attitude. This inertial attitude places the spacecraft 2° below the local horizontal at the 35° elevation angle point. At 35° elevation angle a pitch down of 0.3°/sec is initiated. Five marks are then taken with the time between marks a minimum of 25 seconds. (See tracking profile)



e. In the undocked configuration the CSM approaches the landmark in ORB RATE and pitched down 22° from the local horizontal. At 35° elevation angle five marks are taken with the time between marks a minimum of 25 seconds. ORB RATE is continued throughout the marking period.



f. In the undocked COAS tracking the CSM will approach the LM in ORB RATE heads up and pitched down 40° from the local horizontal. When the LM is centered in the COAS the CSM will initiate a variable pitch rate to keep the LM centered in the COAS.



5. CSM/LM and CSM attitude maneuvers will normally be at a rate of 0.2°/sec or 0.5°/sec. unless other rates are required.
NOTE: At 0.2°/sec,15 minutes is required to maneuver 180°.
At 0.5°/sec, 6 minutes is required to maneuver 180°.

6. Passive thermal control mode will be initiated after MCC1 or as soon as MCC1 is scrubbed and maintained throughout the mission (except in lunar orbit) until at least three hours before entry except for interruptions for midcourse corrections, communications orientation (maximum interruption of three hours). PTC will not be initiated before approximately 7:00 GET.

7. Service Propulsion System All SPS burns will be initiated on Bank A except LOI1 which will be initiated on Bank B.

F. LM Notes

1. Entries into the LM

a. Three entries into the LM are scheduled in the timeline at 56:30, 81:30 and 95:52 GET respectively.

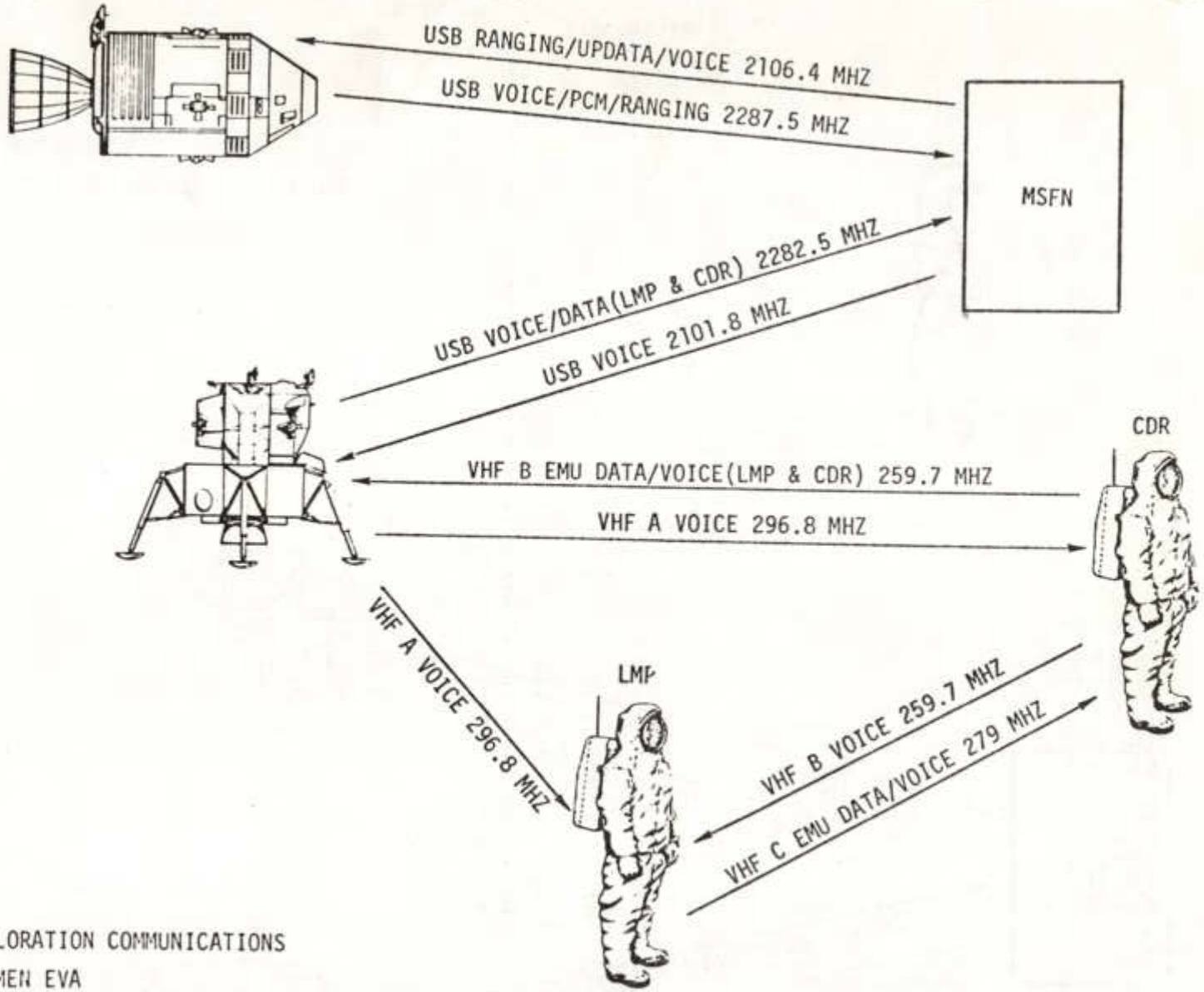
b. The first entry (56:30 GET) will be for LM familiarization and will be performed by the CDR and LMP in the constant wear garments. During this period there will be approximately 5 minutes of VHF-B LBR data which will be recorded by the DSE in the CSM. The LM will remain on CSM power during the crew familiarization period.
c. The second entry (81:30 GET) will be for LM housekeeping and will be performed by the LMP in constant wear garments. During this period the LM will go to internal power for the S-Band/VHF B voice activation.
d. The third entry into the LM (95:52 GET) will be performed by the LMP in LCG's to prepare the LM for undocking and descent to the lunar surface. During this period the LMP and CDR initially transfer to the LM in LCG's then return to the CSM for PGA donning.

2. Environmental Control System and Cabin Pressurization
   a. The LM cabin will contain ambient air at lift off and will bleed down to zero pressure psi during the launch.
   b. The LM will be pressurized for transposition and docking after which it will be isolated and the pressure periodically monitored.
   c. The LM will be pressurized prior to the first entry (LM familiarization) after which it will be isolated again for the remainder of the TLC period.
   d. Prior to the second entry (LM housekeeping) it will be pressurized again and will remain pressurized.

3. Guidance and Navigation
   a. Two LGC erasable memory dumps and MCC-H verifications will be accomplished prior to DOI. If a significant number of errors are found, memory correction and re-verification will be performed before DOI.
   b. The LM IMU will be manually aligned to the CSM IMU during the DOI Day LM activation and checkout. P52/AOT alignments will be performed as close to DOI as possible.
   c. All translations during the undocked manned LM operations will be under PGNCS control.
   d. The capability for MCC-H to update the LGC via uplink will normally be blocked by the LMP UP-DATA LINK switch (panel 12).

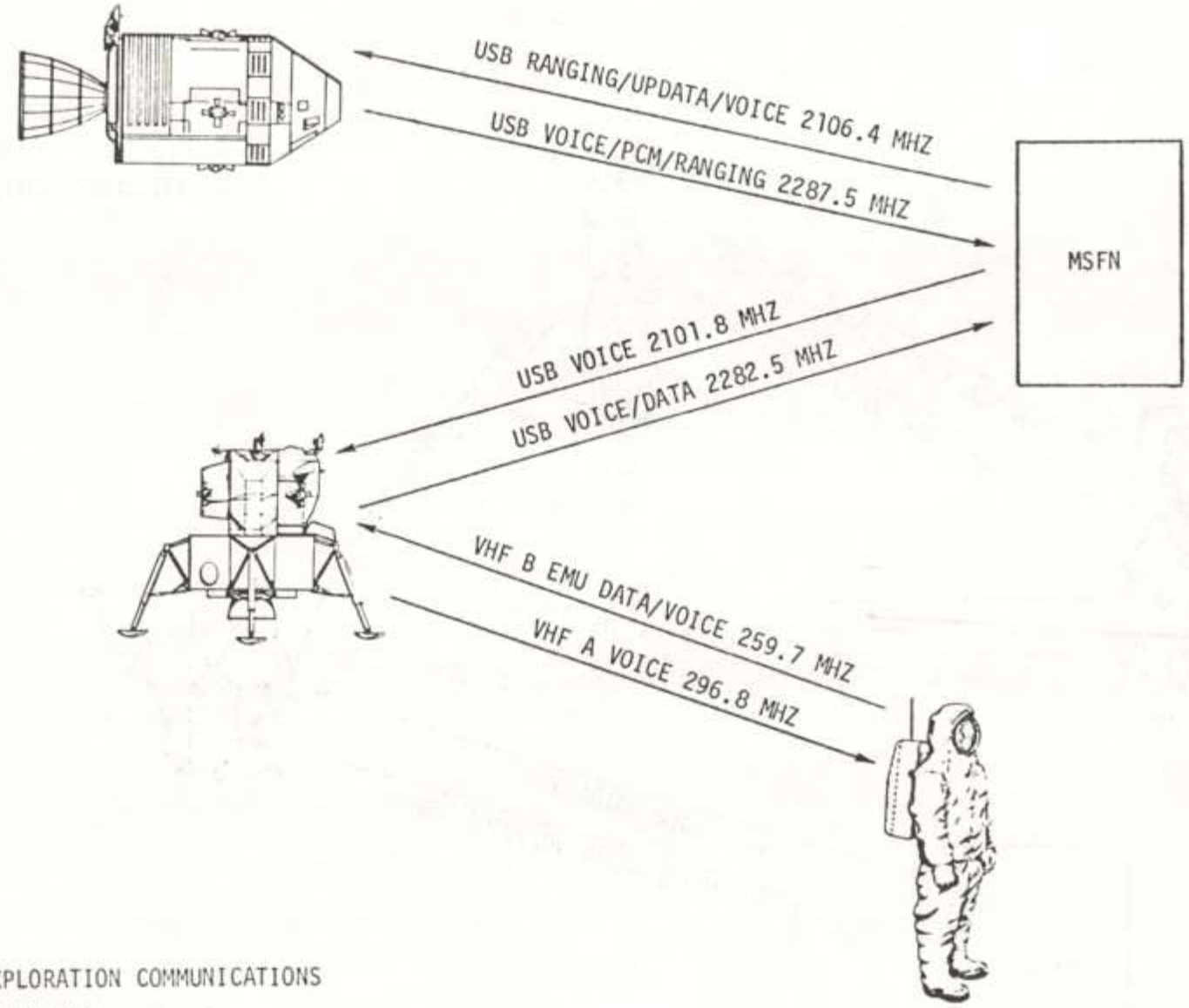
4. RCS Operation and Interface Constraints
   a. During CSM/LM docked checkout operations, the LM steerable and/or RR antennas will not be powered down once they have been activated. The SM B3 and C4 thrusters will be deactivated before the LM steerable and/or RR antennas have been unstowed in order to prevent SM-RCS impingement on these antennas.
   b. The CSM roll jets and LM yaw jets will be disabled when the probe is preloaded (docking latches are cocked) and the tunnel is pressurized prior to undocking. The jets will be activated after tunnel venting.
   c. LM RCS two jet ullage (System B) will be used for unstaged ullage maneuvers in order to prevent asymetrical RCS thrust caused by impingement on the descent stage.
   d. The RCS interconnect will be used during the APS lift-off and ascent, but will not be used during the rendezvous maneuvers.

5. Rendezvous
   a. The rendezvous radar will be pointed away from the sun and will be turned off when no functional use is required to prevent overheating of the antenna.
   b. The LM tracking light will be on continuously between separation and touchdown and between launch and docking except during PGNCS/AOT alignments. During PGNCS/AOT alignments (LM P52), the tracking light would interfere with the alignments. (dark adaption)



LUNAR EXPLORATION COMMUNICATIONS
BOTH CREWMEN EVA
EVCS DUAL MODE (RELAY)

Figure 1-4



LUNAR EXPLORATION COMMUNICATIONS
ONE CREWMAN EVA
PRIMARY MODE

Figure 1-5

SECTION II - UPDATE FORMS

UPDATE FORMS

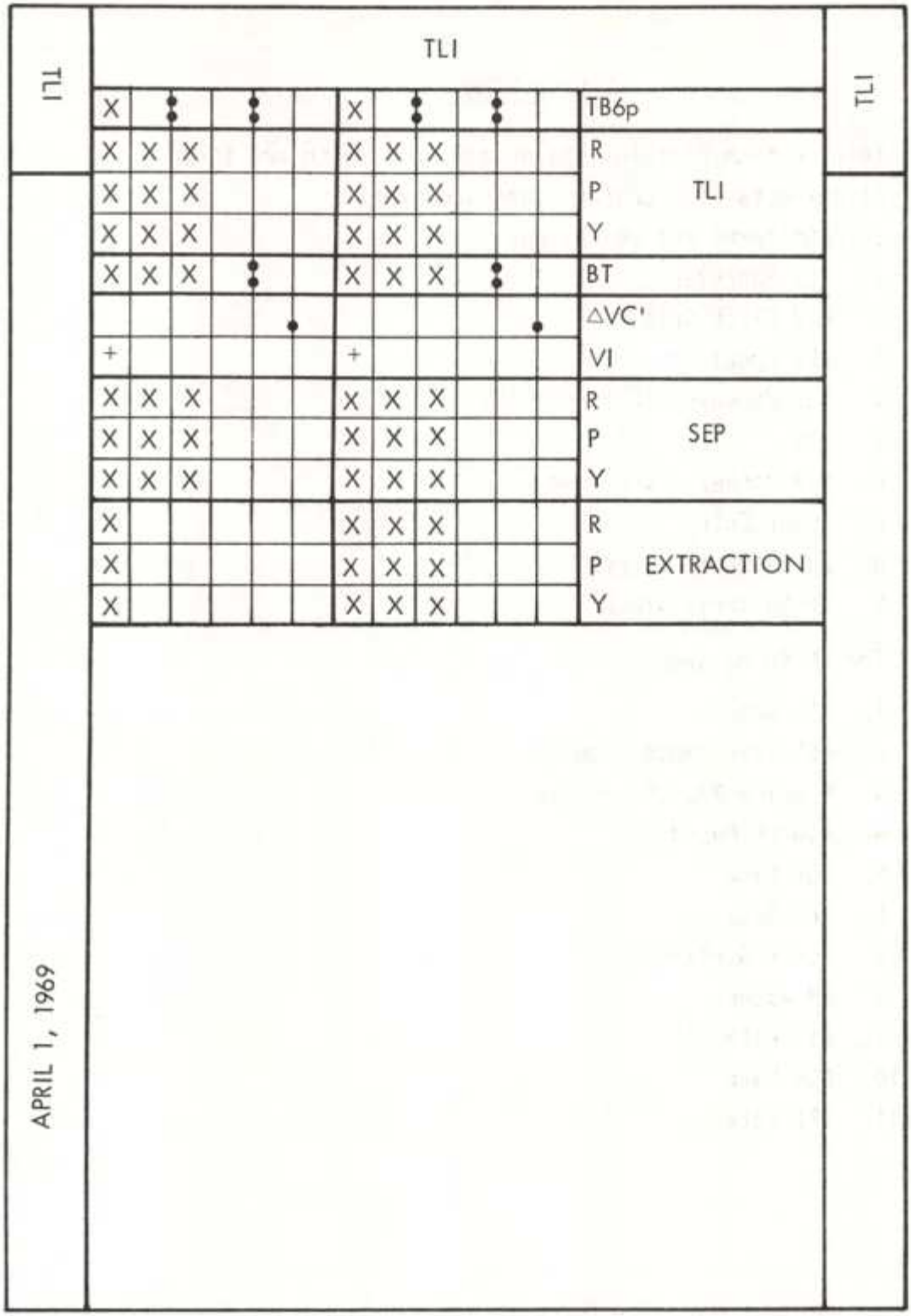
This section contains the update pads which are in the Flight Data File onboard the spacecraft.

The CSM forms are as follows:

1. TLI Maneuver
2. P37 Block Data
3. P27 Update
4. P30 Maneuver (External ΔV)
5. P76
6. CSM Rendezvous Rescue
7. Lunar Entry
8. Earth Orbit Entry
9. Earth Orbit Block Data

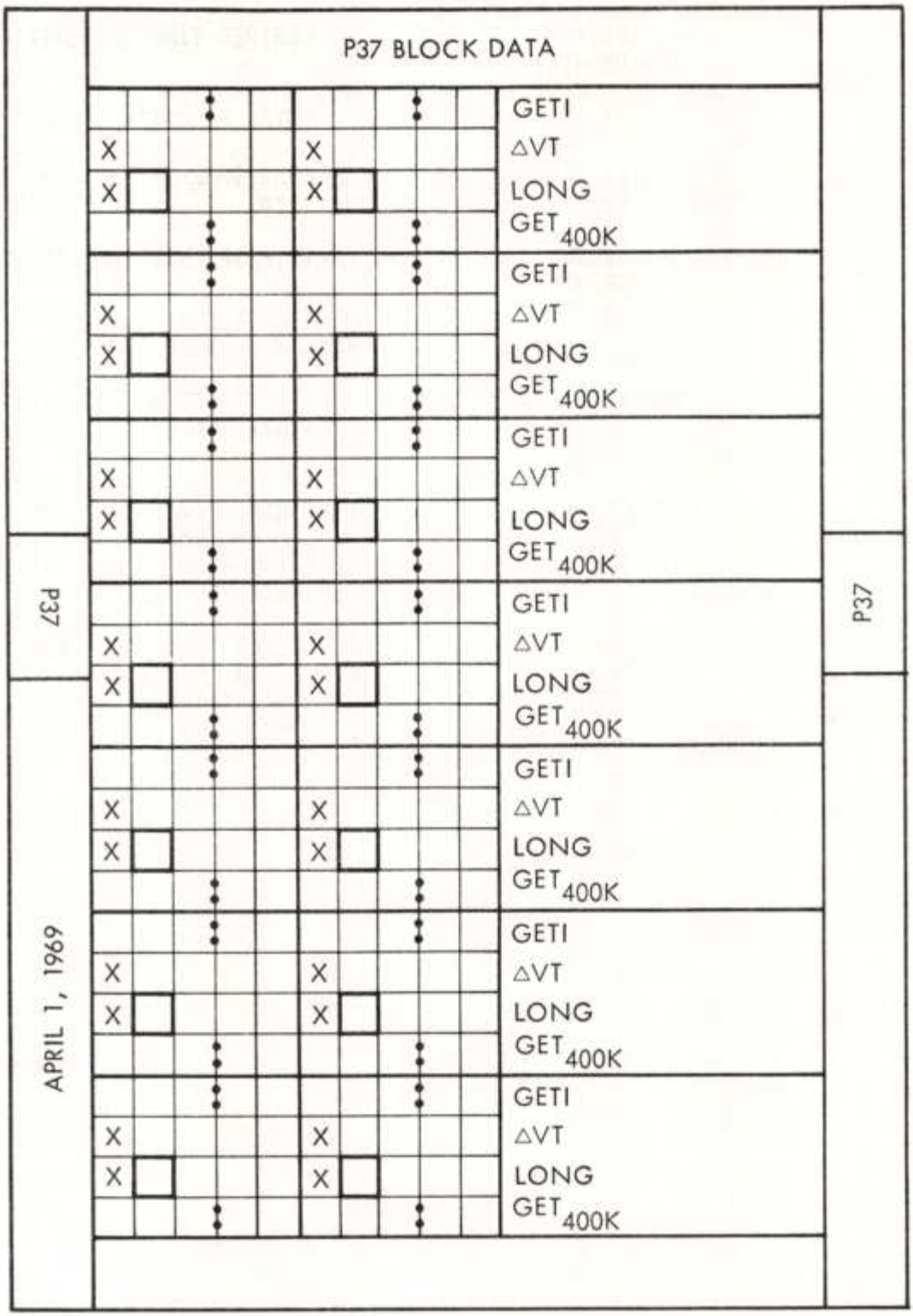
The LM forms are:

1. P27 Update
2. AGS State Vector Update
3. Phasing P30 LM Maneuver
4. P30 LM Maneuver
5. DOI Data
6. PDI Data
7. Lunar Surface
8. LM Ascent
9. CSI Data
10. CDH Data
11. TPI Data


TLI
TLI
TLI
TB6p
R
P
Y
TLI
BT
ΔVC'
VI
R
P
Y
SEP
R
P
Y
EXTRACTION
APRIL 1, 1969

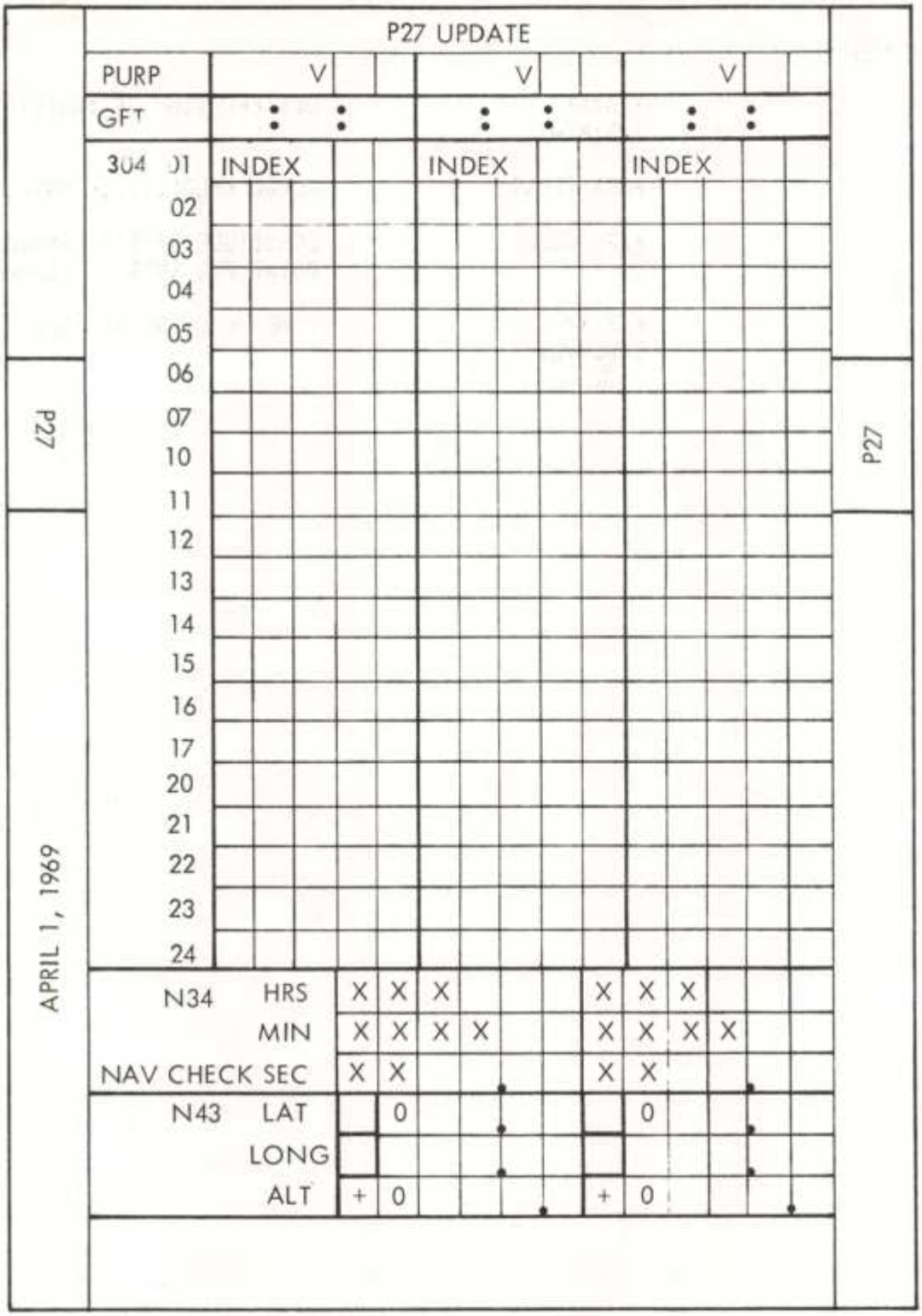
TLI PAD

TB 6p	X:XX:XX (HR:MIN:SEC)	PREDICTED TIME OF BEGINNING OF S-IVB RESTART PREPARATION FOR TLI (TB6 = TLI IGN -578.6 SEC)
R P Y	XXX (DEG) XXX (DEG) XXX (DEG)	PREDICTED SPACECRAFT IMU GIMBAL ANGLES AT TLI IGNITION
BT	X:XX (MIN:SEC)	DURATION OF TLI BURN
ΔVC	XXXX.X (FPS)	NOMINAL TLI ΔV SET INTO EMS ΔV COUNTER
VI	+XXXXX (FPS)	NOMINAL INERTIAL VELOCITY DISPLAYED ON DSKY AT TLI CUTOFF
R SEP P SEP Y SEP	XXX (DEG) XXX (DEG) XXX (DEG)	PREDICTED SPACECRAFT IMU GIMBAL ANGLES AT COMPLETION OF S-IVB MNVR TO CSM/S-IVB SEP ATTITUDE
R EXT P EXT Y EXT	XXX (DEG) XXX (DEG) XXX (DEG)	PREDICTED SPACECRAFT IMU GIMBAL ANGLES AT TIME OF CSM EXTRACTION OF LM FROM S-IVB


P37 BLOCK DATA
GETI
ΔVT
LONG
GET400K
P37
APRIL 1, 1969
P37

P37 BLOCK DATA

GETI	XXX:XX (HR:MIN)	DESIRED TIME OF IGNITION
ΔVT	XXXX (FPS)	TOTAL VELOCITY OF MNVR
LONG	±XXX (DEG)	LONGITUDE OF THE LANDING POINT FOR ENTRY GUIDANCE
GET 400K	XXX:XX (HR:MIN)	TIME OF ENTRY INTERFACE


P27 UPDATE
PURP
V
V
V
GFT
304 01
INDEX
INDEX
INDEX
02
03
04
05
06
07
10
11
12
13
14
15
16
17
20
21
22
23
24
N34
HRS
MIN
NAV CHECK SEC
N43
LAT
LONG
ALT
P27
APRIL 1, 1969
P27

P27 UPDATE - CSM

PURP	XXX	TYPE OF DATA TO BE RECEIVED (SUCH AS: CMC TIME)
V	XX (VERB)	TYPE OF COMMAND LOAD (70-71-72-73)
GET	XXX:XX:XX (HR:MIN:SEC)	TIME DATA RECORDED
304 01	XX (OCTAL)	INDEX NO. OF COMMAND WORDS IN LOAD
02-24	XX (OCTAL)	CORRECTION IDENTIFIERS
N34 NAV CHECK	XXX:XX:XX.XX (HR:MIN:SEC)	TIME FOR CONFIRMATION OF GROUND TRACK
N43		
LAT	XX.XX (DEG)	LATITUDE FOR GROUND TRACK CONFIRMATION
LONG	XXX.XX (DEG)	LONGITUDE FOR GROUND TRACK CONFIRMATION
ALT	XXX.X (DEG)	ALTITUDE FOR GROUND TRACK CONFIRMATION

TEI -P31

P30 MANEUVER

P30

APRIL 5, 1969

SET STARS

R ALIGN ___ ___ ___
P ALIGN ___ ___ ___
Y ALIGN ___ ___ ___

ULLAGE ___

HORIZON/WINDOW ___

OTHER ___

							PURPOSE
			1				PROP/GUID
+	3	6	6	9	1		WT N47
−	0	0	0	6	1		P_{TRIM} N48
+	0	0	0	6	6		Y_{TRIM}
+	0	0	1	3	7		HRS GETI
+	0	0	0	2	2		MIN N33
+	0	3	8	8	5		SEC
+	3	2	8	3	8		ΔV_X N81
+	0	6	8	4	5		ΔV_Y
−	0	2	4	8	7		ΔV_Z
X	X	X	N	A			R
X	X	X	0	5	2		P
X	X	X	N	/A			Y
+					•		H_A N42
					•		H_P
+					•		ΔVT
X	X	X	•				BT
X					•		ΔVC
X	X	X	X				SXTS
+				•		0	SFT
+			•		0	0	TRN
X	X	X					BSS
X	X				•		SPA
X	X	X			•		SXP
	0			•			LAT N61
				•			LONG
+					•		RTGO EMS
+							VIO
		•		•			GET 0.05G

MAL

P30 MANEUVER

PURPOSE	XXXXX	TYPE OF MNVR TO BE PERFORMED
PROP/GUID	XXX/XXX	PROPULSION SYSTEM (SPS/RCS)/ GUIDANCE (SCS/G&N)
WT	+XXXXX (lbs)	PREMANEUVER VEHICLE WEIGHT
P TRIM	±X.XX (DEG)	SPS PITCH GIMBAL OFFSET TO PLACE THRUST THROUGH THE CG
Y TRIM	±X.XX (DEG)	SPS YAW GIMBAL OFFSET TO PLACE THRUST THROUGH THE CG
GETI	XX:XX:XX.XX (HRS:MIN:SEC)	TIME OF MNVR IGNITION
ΔVX ΔVY ΔVZ	±XXXX.X (FPS) ±XXXX.X (FPS) ±XXXX.X (FPS)	P30 VELOCITY TO BE GAINED COMPONENTS IN LOCAL VERTICAL COORDINATES
R P Y	XXX (DEG) XXX (DEG) XXX (DEG)	IMU GIMBAL ANGLES OF MANEUVER ATTITUDE
HA	XXXX.X (NM)	PREDICTED APOGEE ALTITUDE AFTER MANEUVER
HP	±XXXX.X (NM)	PREDICTED PERIGEE ALTITUDE AFTER MANEUVER
ΔVT	+XXXX.X (FPS)	TOTAL VELOCITY OF MANEUVER
BT	X:XX (MIN:SEC)	MANEUVER DURATION
ΔVC	XXXX.X (FPS)	PREMANEUVER ΔV SETTING IN EMS ΔV COUNTER
SXTS	XX (OCTAL)	SEXTANT STAR FOR MANEUVER ATTITUDE CK
SFT	+XXX.X (DEG)	SEXTANT SHAFT SETTING FOR MANEUVER ATTITUDE CK
TRN	+XX.X (DEG)	SEXTANT TRUNNION SETTING FOR MANEUVER ATTITUDE CK
BSS	XX (OCTAL)	BORESIGHT STAR FOR MANEUVER ATTITUDE CK USING THE COAS

SPA	±XX.X (DEG)	BSS PITCH ANGLE ON COAS FOR MANEUVER ATTITUDE CK
SXP	±X.X (DEG)	BSS X POSITION ON COAS FOR MANEUVER ATTITUDE CK
LAT LONG	±XX.XX (DEG) ±XXX.XX (DEG)	LATITUDE AND LONGITUDE OF THE LANDING POINT FOR ENTRY GUIDANCE
RTGO	+XXXX.X (NM)	RANGE TO GO FOR EMS INITIALIZATION
VIO	+XXXXX (FPS)	INERTIAL VELOCITY AT .05G FOR EMS INITIALIZATION
GET (.05G)	XXX:XX:XX.XX (HRS:MIN:SEC)	TIME OF .05G
SET STARS	XX (OCTAL) XX (OCTAL)	STARS FOR BACKUP GDC ALIGN
R, P, Y (ALIGN)	XXX (DEG) XXX (DEG) XXX (DEG)	ATTITUDE TO BE SET IN ATTITUDE SET TW FOR BACKUP GDC ALIGN
ULLAGE	X (JETS) XX.X (SEC)	NO. OF SM RCS JETS USED AND LENGTH OF TIME OF ULLAGE
HORIZON/WINDOW	XX.X (DEG)	WINDOW MARKING AT WHICH HORIZON IS PLACED AT A SPECIFIED TIG (ATT CK)
OTHER		ADDITIONAL REMARKS VOICED UP BY MCC-H

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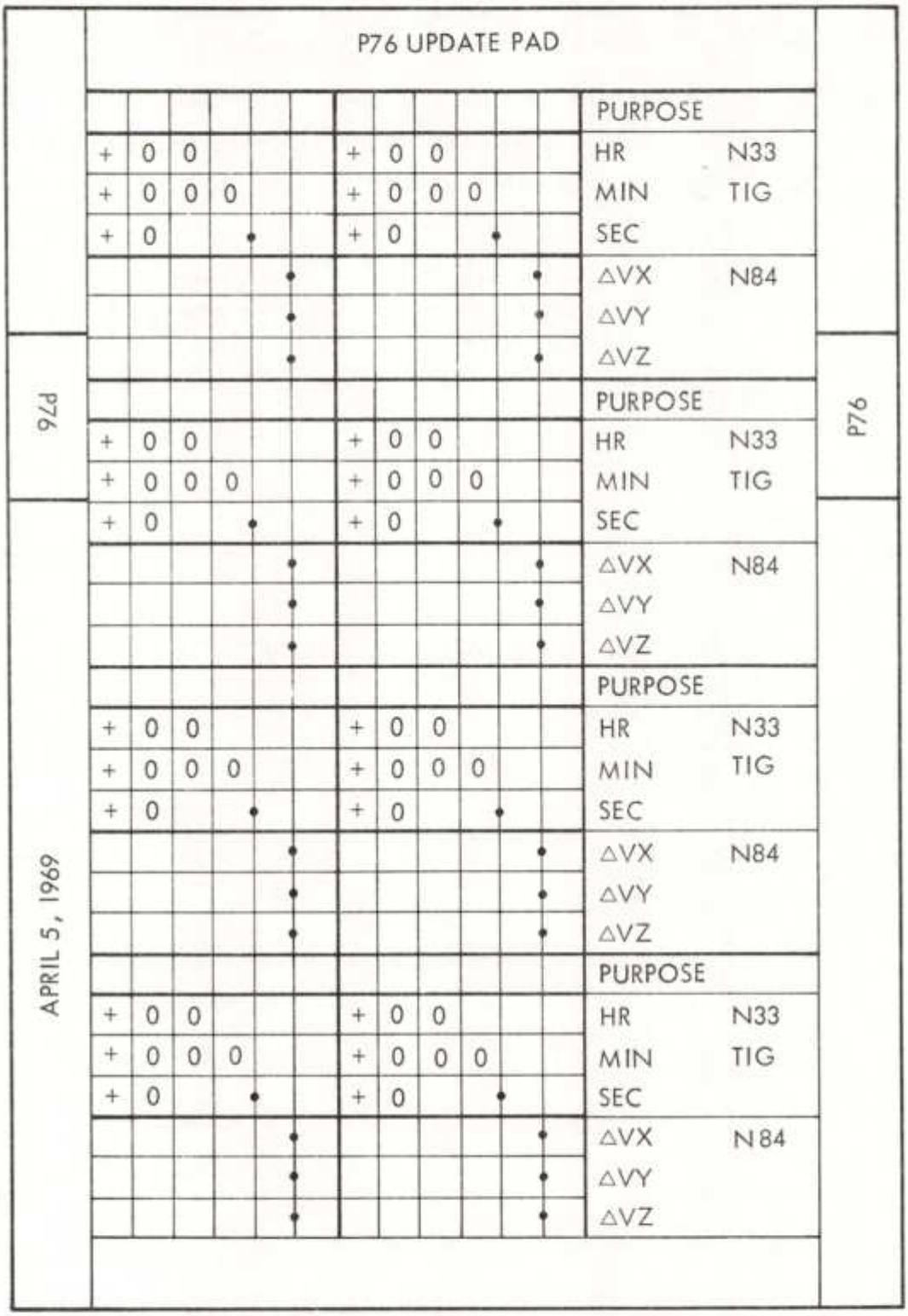
P76 UPDATE PAD
PURPOSE
HR N33
MIN TIG
SEC
ΔVX N84
ΔVY
ΔVZ
PURPOSE
HR N33
MIN TIG
SEC
ΔVX N84
ΔVY
ΔVZ
PURPOSE
HR N33
MIN TIG
SEC
ΔVX N84
ΔVY
ΔVZ
PURPOSE
HR N33
MIN TIG
SEC
ΔVX N84
ΔVY
ΔVZ
P76
P76
APRIL 5, 1969

P76 UPDATE PAD

PURPOSE	XXXXX	PURPOSE OF MANEUVER
N33 TIG	XX:XX:XX.XX (HR:MIN:SEC)	TIME OF IGNITION
N84		
ΔVX ΔVY ΔVZ	XXXX.X (FPS) XXXX.X (FPS) XXXX.X (FPS)	COMPONENTS OF ΔV APPLIED ALONG LOCAL VERTICAL AXIS AT TIG (LM)

CSM RENDEZVOUS RESCUE PADS

CSM SEP PAD

33	00		000		0 .	
81	+	0000.0	+	0000.0	−	0002.5
22	XXX		XXX		XXX	

DOI PAD

84	•	•	•
33			

PDI_1 +12 ABORT PAD

84	•	•	•
33			

"CSM RESCUE" PAD

PHAS	33	00	000	0	.
TPI(PDI < 10)	37	00	000	0	.
TPI(PDI > 10)	37	00	000	0	.

"CSM RESCUE UPDATE" PAD

PHAS	33	00	000	0	.
TPI(PDI < 145)	37	00	000	0	.
$TPI(T_2)$	37	00	000		.

RESCUE TWO PAD

47	+ .	+	00000.	
48	.		.	
33	00	000	0	
81	.	.	.	
22	XXX	XXX	XXX	
ΔV_C	X .			
11	00	000	0 .	
37	00	000	0 .	
N				

CSI ONE

11		000	0 .
81	.	.	.
N			

P22 PAD

T1	•	•	•	(HOR)
T2	•	•	•	(LMK)
	•	NM (N OR S)		
89	•	•	•	
	LAT	LONG/2	ALT	

NOMINAL LM IGNITION TIMES

CSI	11	00	000	0 .
PC	33	00	000	0 .
TPI	37	00	000	0 .

CSI TWO

11	00	000	0 .
81	.	.	.
N			

CSI THREE

11	00	000	0 .
81	.	.	.
N			

CSI FOUR

11	00	000	0 .
81	.	.	.
N			

CDH

13	00	000	0 .
81	.	.	.

TPI

37	00	000	0 .	
81	.	.	.	
59	.	.	.	
LOS BT	XX	XX	XX	

CSM SEP PAD

33	GETI	XXX:XX:XX.XX (HRS:MIN:SEC)	GET OF CSM/LM SEPARATION BURN
81	DELTA VX	+XXXX.X (FPS)	LOCAL VERTICAL VELOCITY COMPONENTS OF SEP BURN
	DELTA VY	+XXXX.X (FPS)	
	DELTA VZ	+XXXX.X (FPS)	
22	R	XXX (DEG)	SEPARATION BURN INERTIAL GIMBAL ANGLES
	P	XXX (DEG)	
	Y	XXX (DEG)	

DOI PAD

84	DELTA VX	XXXX.X (FPS)	LM LOCAL VERTICAL VELOCITY COMPONENTS FOR DOI BURN
	DELTA VY	XXXX.X (FPS)	
	DELTA VZ	XXXX.X (FPS)	
33	GETI	XXX:XX:XX.XX (HRS:MIN:SEC)	GET OF DOI BURN

PDI + 12 ABORT PAD

84	DELTA VX	XXXX.X (FPS)	LM LOCAL VERTICAL VELOCITY COMPONENTS FOR FIRST OPPORTUNITY PDI PLUS 12 MIN ABORT
	DELTA VY	XXXX.X (FPS)	
	DELTA VZ	XXXX.X (FPS)	
33	GETI	XXX:XX:XX.XX (HRS:MIN:SEC)	GET OF PDI + 12 MIN ABORT BURN

"CSM RESCUE" PAD

PHAS	33	GETI	XXX:XX:XX.XX (HRS:MIN:SEC)	GET OF CSM ABORT PHASING BURN
TPI (PDI 10)	37	GETI	XXX:XX:XX.XX (HRS:MIN:SEC)	GET OF TPI FOR LM ABORTS BETWEEN PDI AND PDI + 10 MIN
TPI (PDI 10)	37	GETI	XXX:XX:XX.XX (HRS:MIN:SEC)	GET OF TPI FOR LM ABORTS AFTER PDI + 10 MIN

"CSM RESCUE UPDATE" PAD

PHAS	33	GETI	XXX:XX:XX.XX (HRS:MIN:SEC)	GET OF CSM ABORT PHASING BURN FOR 2ND OPPORTUNITY (1 REV DELAY)

TPI (PDI 14.5)		GETI	XXX:XX:XX.XX (HRS:MIN:SEC)	GET OF TPI FOR LM ABORTS BETWEEN PDI AND PDI + 14.5 MIN FOR 2ND OPPORTUNITY
	37			
TPI (T2)		GETI	XXX:XX:XX.XX (HRS:MIN:SEC)	GET OF PREFERRED LM LIFTOFF TIME
	37			
RESCUE TWO PAD				
47	WT		XXXX.X (lbs)	PREMANEUVER CSM WEIGHT
48	P TRIM		X.XX (DEG)	SPS PITCH & YAW GIMBAL OFFSET TO PLACE THRUST THROUGH THE CG
	Y TRIM		X.XX (DEG)	
33	GETI		XXX:XX:XX.XX (HRS:MIN:SEC)	GET OF RESCUE BURN
81	DELTA VX		XXXX.X (FPS)	LOCAL VERTICAL VELOCITY COMPONENTS OF RESCUE BURN
	DELTA VY		XXXX.X (FPS)	
	DELTA VZ		XXXX.X (FPS)	
22	R		XXX (DEG)	RESCUE BURN GIMBAL ANGLES
	P		XXX (DEG)	
	Y		XXX (DEG)	
ΔVc	ΔVc		XX.X (FPS)	VELOCITY TO BE SET IN EMS COUNTER FOR RESCUE BURN
11	GETI		XXX:XX:XX.XX (HRS:MIN:SEC)	GET OF CSI BURN BASED ON RESCUE BURN
37	GETI		XXX:XX:XX.XX (HRS:MIN:SEC)	GET OF TPI BURN BASED ON RESCUE BURN
N			X	THE FUTURE APSIDAL CROSSING (APOLUNE OR PERILUNE) OF THE ACTIVE VEHICLE AT WHICH CDH SHOULD OCCUR

CSI ONE

11	GETI	XXX:XX:XX.X (HRS:MIN:SEC)	GET OF CSI ONE BURN
81	DELTA VX DELTA VY DELTA VZ	XXXX.X (FPS) XXXX.X (FPS) XXXX.X (FPS)	LOCAL VERTICAL VELOCITY COMPONENTS OF CSI ONE BURN
N		X	THE FUTURE APSIDAL CROSSING (APOLUNE OR PERILUNE) OF THE ACTIVE VEHICLE AT WHICH CDH SHOULD OCCUR

CSI TWO, THREE, FOUR

SAME AS ABOVE EXCEPT CSI TWO, THREE, FOUR

CDH

13	GETI	XXX:XX:XX.X (HRS:MIN:SEC)	GET OF CDH BURN
81	DELTA VX DELTA VY DELTA VZ	XXXX.X (FPS) XXXX.X (FPS) XXXX.X (FPS)	LOCAL VERTICAL VELOCITY COMPONENTS OF CDH BURN

TPI

37		XXX:XX:XX.X (HRS:MIN:SEC)	GET OF LM TPI BURN
81	DELTA VX DELTA VY DELTA VZ	XXX (FPS) XXX (FPS) XXX (FPS)	LOCAL VERTICAL VELOCITY COMPONENTS OF TPI BURN
59	ΔV (LOS)	XXX (FPS)	VELOCITY COMPONENTS ALONG THE LINE OF SIGHT TO TARGET
LOS BT		X:XX MIN:SEC	BURN DURATION ALONG THE LINE OF SIGHT

P22 PAD

T1		XXX:XX:XX.XX (HRS:MIN:SEC)	GET AT WHICH LANDMARK APPEARS ON HORIZON

T2		XXX:XX:XX.XX (HR:MIN:SEC)	GET AT WHICH LANDMARK LOS IS 35° ABOVE LOCAL HORIZONTAL
NM (N OR S)		XX.X (NM)	DISTANCE OF LANDMARK NORTH OR SOUTH OF ORBITAL TRACK
89	LAT LONG ALT	±XX.X (DEG) ±XX (DEG)	LATITUDE OF LANDMARK LONGITUDE OF LANDMARK ALTITUDE OF LANDMARK ABOVE OR BELOW MEAN LUNAR RADIUS

NOMINAL LM IGNITION TIMES

CSI 11	GETI	XXX:XX:XX.X (HRS:MIN:SEC)	NOMINAL GET OF LM CSI BURN
PC 33	GETI	XXX:XX:XX.XX (HRS:MIN:SEC)	NOMINAL GET OF LM PLANE CHANGE BURN
TPI 37	GETI	XXX:XX:XX.XX (HRS:MIN:SEC)	NOMINAL GET OF LM TPI BURN

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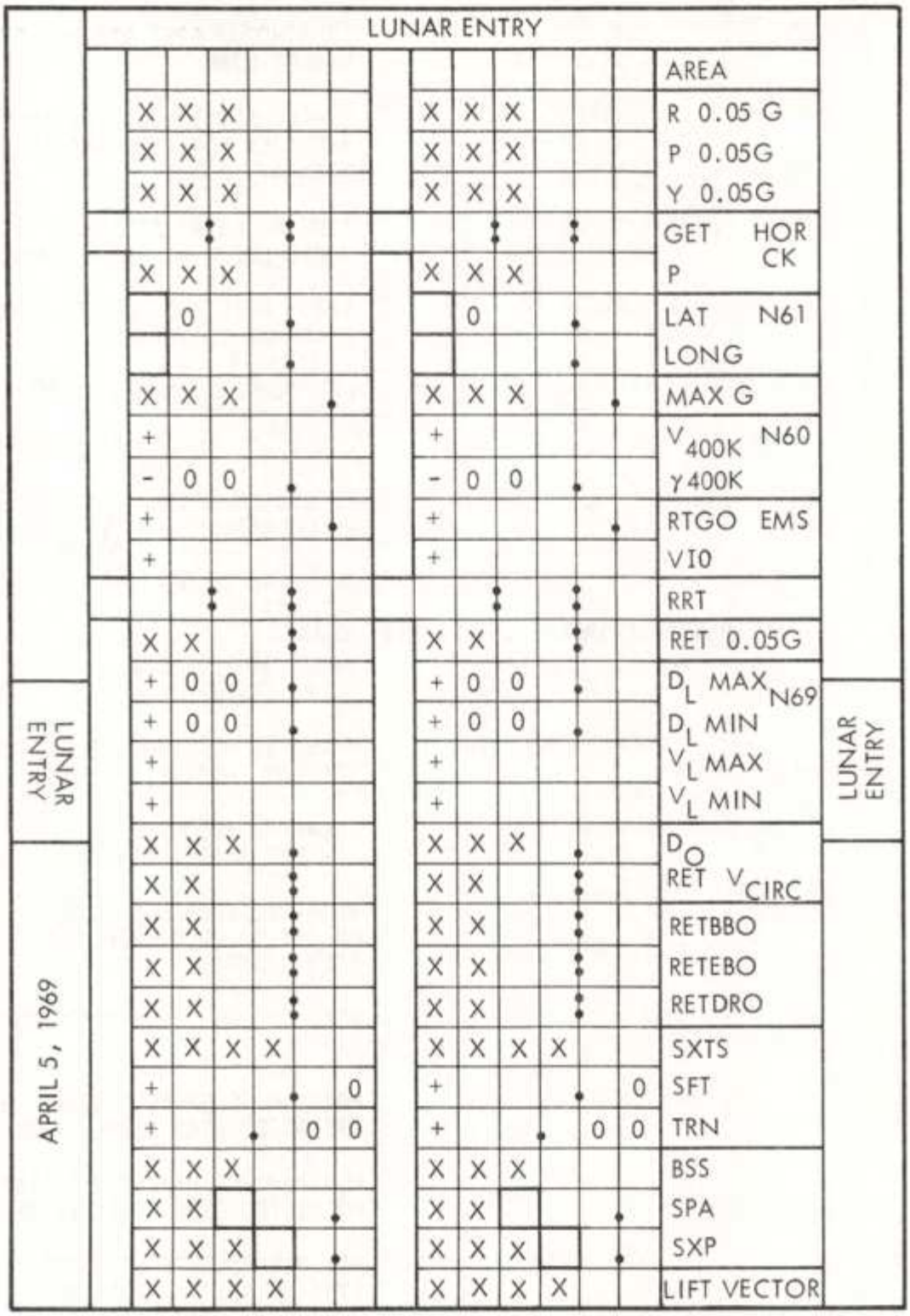
LUNAR ENTRY
AREA
R 0.05 G
P 0.05G
Y 0.05G
GET HOR CK
P
LAT N61
LONG
MAX G
V 400K N60
γ 400K
RTGO EMS
VIO
RRT
RET 0.05G
D L MAX N69
D L MIN
V L MAX
V L MIN
D O
RET V CIRC
RETBBO
RETEBO
RETDRO
SXTS
SFT
TRN
BSS
SPA
SXP
LIFT VECTOR
LUNAR ENTRY
LUNAR ENTRY
APRIL 5, 1969

LUNAR ENTRY PAD

AREA	XXXXX	SPLASHDOWN AREA DEFINED BY TARGET LINE
R .05G P .05G Y .05G	XXX (DEG) XXX (DEG) XXX (DEG)	SPACECRAFT IMU GIMBAL ANGLES REQUIRED FOR AERODYNAMIC TRIM AT .05G
GET (HOR CK)	XXX:XX:XX (HRS:MIN:SEC)	TIME OF ENTRY ATTITUDE HORIZ CHECK AT EI -17 MIN.
P (HOR CK)	XXX (DEG)	PITCH ATTITUDE FOR HORIZON CHECK AT EI -17 MIN.
LAT	±XX.XX (DEG)	LATITUDE OF TARGET POINT
LONG	±XXX.XX (DEG)	LONGITUDE OF TARGET POINT
MAX G	XX.X (G's)	PREDICTED MAXIMUM REENTRY ACCELERATION
V400K	+XXXXX (FPS)	INERTIAL VELOCITY AT ENTRY INTERFACE
400K	-X.XX (DEG)	INERTIAL FLIGHT PATH ANGLE AT ENTRY INTERFACE
RTGO	+XXXX.X (NM)	RANGE TO GO FROM .05G TO TARGET FOR EMS INITIALIZATION
VIO	+XXXXX (fps)	INERTIAL VELOCITY AT .05G FOR EMS INITIALIZATION
RRT	XXX:XX:XX (HRS:MIN:SEC)	REENTRY REFERENCE TIME BASED ON GET OF PREDICTED 400K (DET START)
RET .05G	XX:XX (MIN:SEC)	TIME OF .05G FROM 400K (RRT)
DL MAX	+X.XX (G's)	MAXIMUM ACCEPTABLE VALUE OF PREDICTED DRAG LEVEL (FROM CMC)
DL MIN	+X.XX (G's)	MINIMUM ACCEPTABLE VALUE OF PREDICTED DRAG LEVEL (FROM CMC)
VL MAX	+XXXXX (FPS)	MAXIMUM ACCEPTABLE VALUE OF EXIT VELOCITY (FROM CMC)

VL MIN	+XXXXX (FPS)	MINIMUM ACCEPTABLE VALUE OF EXIT VELOCITY (FROM CMC)
DO	X.XX (G's)	PLANNED DRAG LEVEL DURING CONSTANT G
RET VCIRC	XX:XX (MIN:SEC)	TIME FROM EI THAT S/C VELOCITY BECOMES CIRCULAR
RETBBO	XX:XX (MIN:SEC)	TIME FROM EI TO THE BEGINNING OF BLACKOUT
RETEBO	XX:XX (MIN:SEC)	TIME FROM EI TO THE END OF BLACKOUT
RETDRO	XX:XX (MIN:SEC)	TIME FROM EI TO DROGUE DEPLOY
SXTS	XX (OCTAL)	SEXTANT STAR FOR ENTRY ATTITUDE CHECK
SFT	+XXX.X (DEG)	SEXTANT SHAFT SETTING FOR ENTRY ATTITUDE CHECK
TRN	+XX.X (DEG)	SEXTANT TRUNNION SETTING FOR ENTRY ATTITUDE CHECK
BSS	XXX (OCTAL)	BORESIGHT STAR FOR ENTRY ATTITUDE CHECK USING THE COAS
SPA	±XX.X (DEG)	BSS PITCH ANGLE ON COAS FOR ENTRY ATTITUDE CHECK
SXP	±X.X (DEG)	BSS X POSITION ON COAS FOR ENTRY ATTITUDE CHECK
LIFT VECTOR	XX (UP/DN)	LIFT VECTOR DESIRED AT .05G's BASED ON ENTRY CORRIDOR

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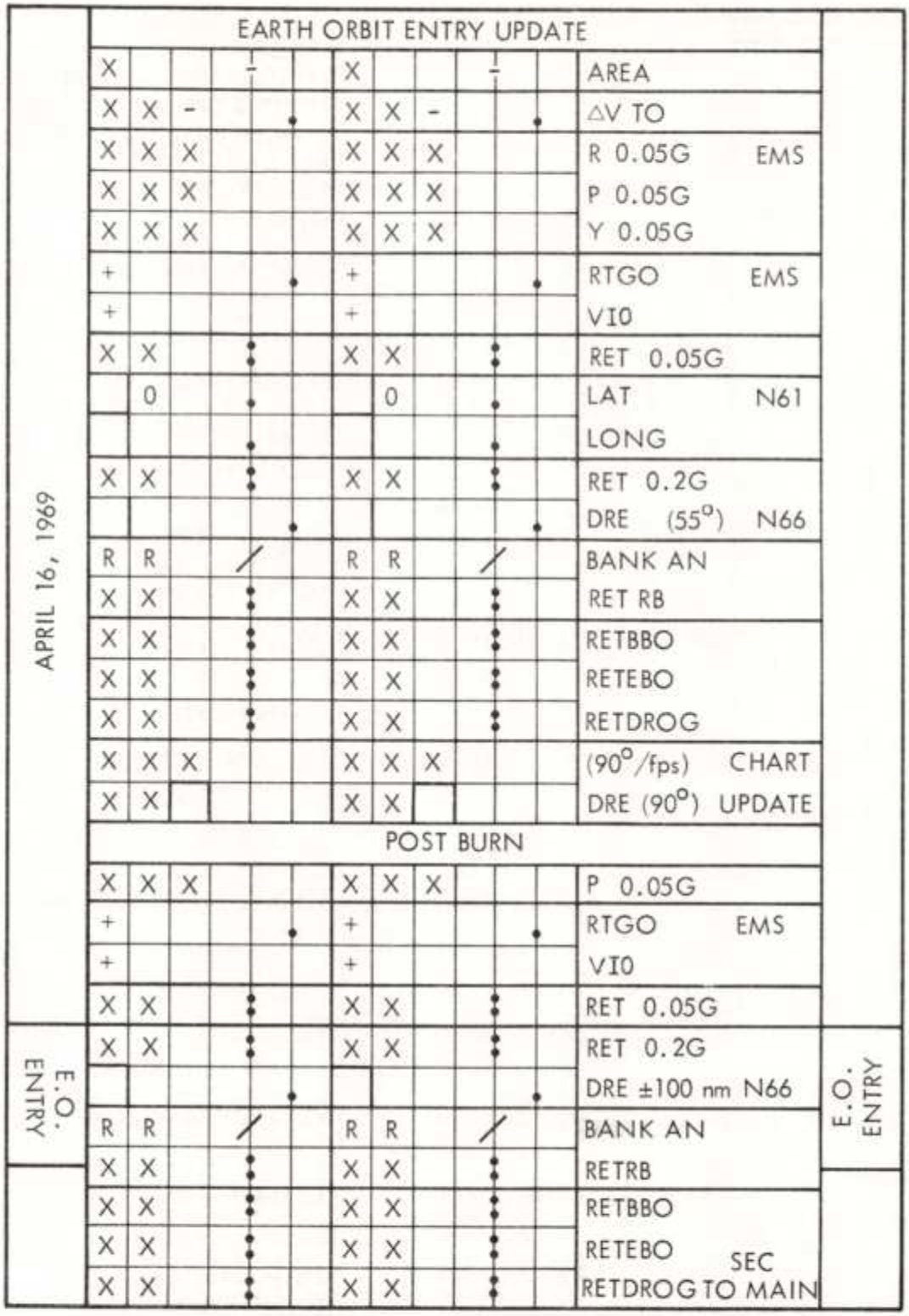
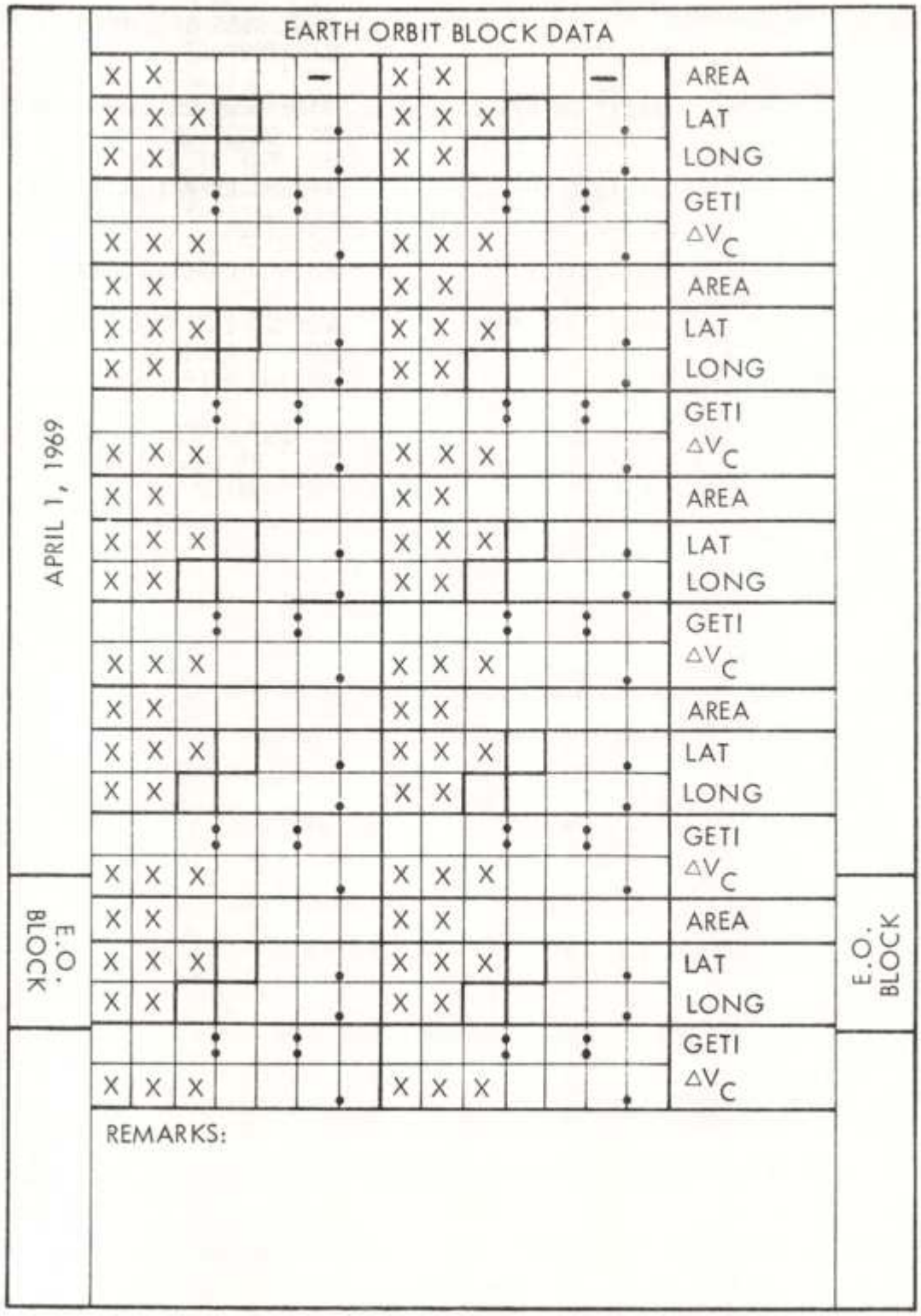

APRIL 16, 1969
EARTH ORBIT ENTRY UPDATE
AREA
ΔV TO
R 0.05G EMS
P 0.05G
Y 0.05G
RTGO EMS
VIO
RET 0.05G
LAT N61
LONG
RET 0.2G
DRE (55°) N66
BANK AN
RET RB
RETBBO
RETEBO
RETDROG
(90°/fps) CHART
DRE (90°) UPDATE
POST BURN
P 0.05G
RTGO EMS
VIO
RET 0.05G
RET 0.2G
DRE ±100 nm N66
BANK AN
RETRB
RETBBO
RETEBO SEC
RETDROG TO MAIN
E.O. ENTRY
E.O. ENTRY

EARTH ORBIT ENTRY UPDATE

AREA	XXX-X	RECOVERY AREA - FIRST 3 DIGITS DENOTES REV IN WHICH LANDING OCCURS. LAST DIGIT DENOTES RECOVERY AREA AND SUPPORT CAPABILITIES
ΔV TO	XX.X (FPS)	ΔV DUE TO ENGINE TAILOFF
EMS		
R 0.05G P 0.05G Y 0.05G	XXX (DEG) XXX (DEG) XXX (DEG)	SPACECRAFT IMU GIMBAL ANGLES REQUIRED FOR AERODYNAMIC TRIM AT 0.05G.
EMS		
RTGO	XXXX.X (NM)	RANGE TO GO FROM .05G TO TARGET
VIO	XXXXX (FPS)	INERTIAL VELOCITY AT .05G FOR EMS INITIALIZATION
RET 0.05G	XX:XX (MIN:SEC)	TIME FROM RETROFIRE TO .05G
N61		
LAT	±XX.XX (DEG)	LATITUDE OF IMPACT LANDING POINT
LONG	±XXX.XX (DEG)	LONGITUDE OF IMPACT LANDING POINT
N66		
RET .2G	XX:XX (MIN:SEC)	TIME FROM RETROFIRE TO .2G
DRE (55°)	±XXXX.X (NM)	DOWNRANGE ERROR AT .2G
BANK AN	XX/XX (DEG/DEG)	BACKUP BANK ANGLE FOR SCS ENTRY: ROLL RIGHT/ROLL LEFT

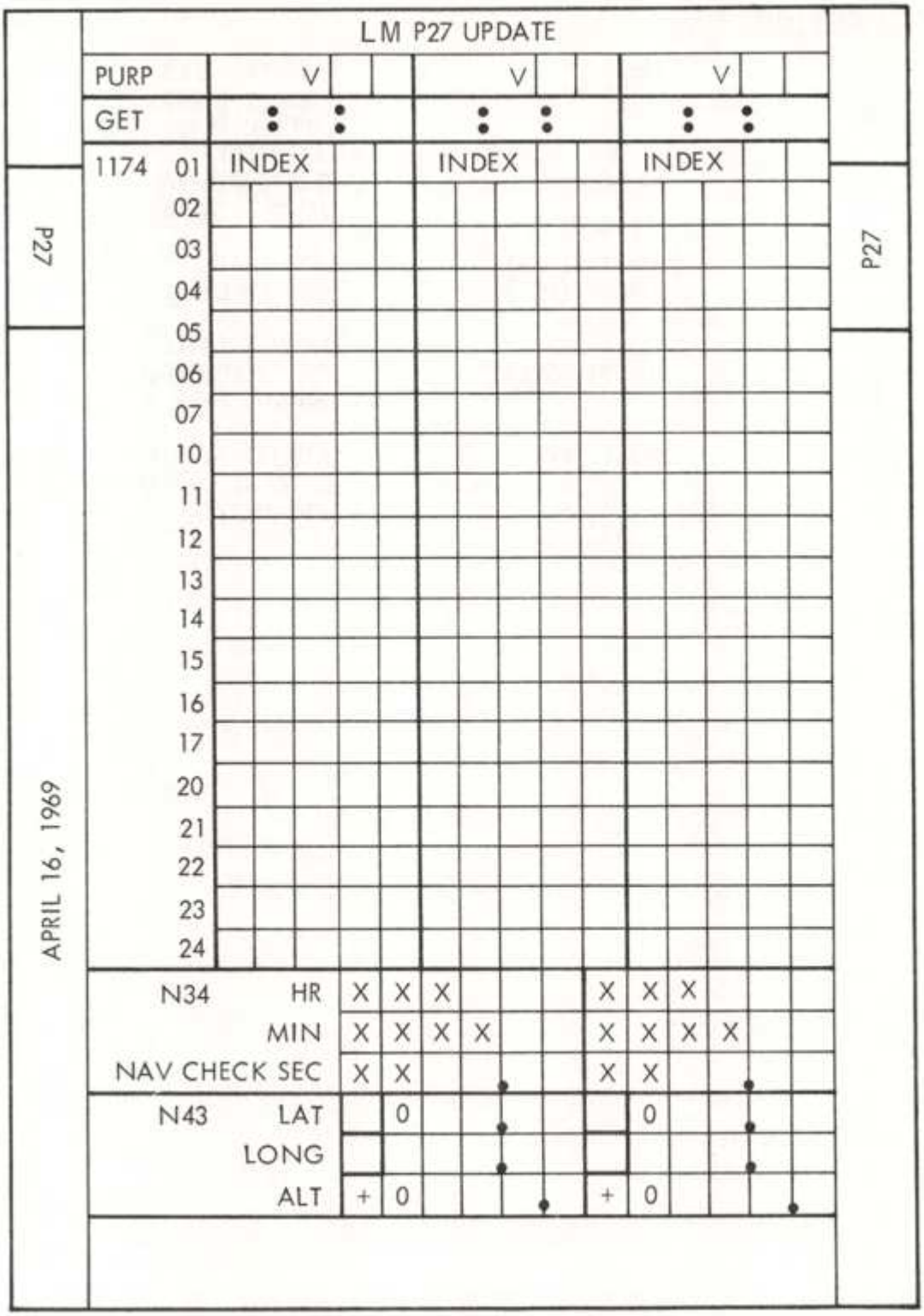
RETRB	XX:XX (MIN:SEC)	TIME FROM RETROFIRE TO REVERSE BACKUP BANK ANGLE
RETBBO	XX:XX (MIN:SEC)	TIME FROM RETROFIRE TO BEGINNING OF COMMUNICATIONS BLACKOUT
RETEBO	XX:XX (MIN:SEC)	TIME FROM RETROFIRE TO END OF COMMUNICATIONS BLACKOUT
RETDROG	XX:XX (MIN:SEC)	TIME FROM RETROFIRE TO DROGUE CHUTE DEPLOYMENT
CHART UPDATE		
90°/FPS DRE (90°)	±XX ±XXX	VALUES USED TO RE-PLOT BACKUP ENTRY CHART - ΔV AND DOWN RANGE ERROR (DRE) @ 90° BANK ANGLE
POST BURN		
P 0.05G	XXX (DEG)	PITCH ANGLE @ ENTRY INTERFACE
EMS		
RTGO	+XXXX.X (NM)	RANGE TO GO FROM 0.05G TO TARGET FOR EMS COUNTER
VIO	+XXXXX (FPS)	INERTIAL VELOCITY @ 0.05G
RET 0.05G	XX:XX (MIN:SEC)	TIME FROM RETROFIRE TO 0.5G
RET 02G	XX:XX (MIN:SEC)	TIME FROM RETROFIRE TO 0.2G
DRE	±XXXX.X (NM)	DOWN RANGE ERROR
BANK AN	XX/XX (DEG/DEG)	BACKUP BANK ANGLE FOR SCS ENTRY: ROLL RIGHT/ROLL LEFT
RETRB	XX:XX (MIN:SEC)	TIME FROM RETROFIRE TO REVERSE BACKUP BANK ANGLE

RETBBO	XX:XX (MIN:SEC)	TIME FROM RETROFIRE TO BEGINNING OF COMMUNICATIONS BLACKOUT
RETEBO	XX:XX (MIN:SEC)	TIME FROM RETROFIRE TO END OF COMMUNICATIONS BLACKOUT
RETDROG	XX:XX (MIN:SEC)	TIME FROM RETROFIRE TO DROGUE CHUTE DEPLOYMENT


EARTH ORBIT BLOCK DATA
AREA
LAT
LONG
GETI
ΔV_C
AREA
LAT
LONG
GETI
ΔV_C
AREA
LAT
LONG
GETI
ΔV_C
AREA
LAT
LONG
GETI
ΔV_C
AREA
LAT
LONG
GETI
ΔV_C
REMARKS:
APRIL 1, 1969
E.O. BLOCK
E.O. BLOCK

EARTH ORBIT BLOCK DATA

AREA	XXX-X	RECOVERY AREA FIRST 3 DIGITS - LANDING REVOLUTION LAST DIGIT - RECOVERY AREA AND SUPPORT CAPABILITIES
LAT LONG	±XX.XX (DEG) ±XXX.X (DEG)	COORDINATES OF THE DESIRED LANDING AREA
GETI	XXX:XX:XX.XX (HR:MIN:SEC)	DEORBIT IGNITION TIME FOR THE DESIRED LANDING AREA
ΔVC	XXX.X (FPS)	DEORBIT MANEUVER ΔV TO BE LOADED INTO THE EMS COUNTER.


LM P27 UPDATE
PURP
V
V
V
GET
1174
01
INDEX
INDEX
INDEX
02
03
04
05
06
07
10
11
12
13
14
15
16
17
20
21
22
23
24
N34
HR
MIN
NAV CHECK SEC
N43
LAT
LONG
ALT
P27
APRIL 16, 1969
P27

P27 UPDATE-LM

PURP	XXX	TYPE OF DATA TO BE RECEIVED (SUCH AS: LDG TIME)
V	XX (VERB)	TYPE OF COMMAND LOAD (70-71-72-73)
GET	XXX:XX:XX (HR:MIN:SEC)	TIME DATA RECORDED
1174 01	XX (OCTAL)	INDEX NO. OF COMMAND WORDS IN LOAD
02-24	XX (OCTAL)	CORRECTION WORD IDENTIFIERS
N34 NAV CHECK TIME	XXX:XX:XX.XX (HR:MIN:SEC)	TIME FOR CONFIRMATION OF GROUND TRACK
N43		
LAT	XX.XX (DEG)	LATITUDE FOR GROUND TRACK CONFIRMATION
LONG	XXX.XX (DEG)	LONGITUDE FOR GROUND TRACK CONFIRMATION
ALT	XXX.X (NM)	ALTITUDE FOR GROUND TRACK CONFIRMATION

AGS STATE VECTOR UPDATE

												PURP	
												240	
												241	
												242	
												260	
												261	
												262	
+						+						254	
												244	
												245	
												246	
												264	
												265	
												266	
+						+						272	

REMARKS:

APRIL 5, 1969

AGS SV

AGS SV

AGS STATE VECTOR UPDATE

PURP		PURPOSE FOR AGS STATE VECTOR UPDATE
240	±XXXXX (100 FT)	LM STATE VECTOR-POSITION COMPONENTS
241	±XXXXX (100 FT)	
242	±XXXXX (100 FT)	
260	±XXXX.X (FPS)	LM STATE VECTOR-VELOCITY COMPONENTS
261	±XXXX.X (FPS)	
262	±XXXX.X (FPS)	
254	+XXXX.X (MIN)	LM TIME FOR WHICH THE STATE VECTOR IS ACCURATE
244	±XXXXX (100 FT)	CSM STATE VECTOR-POSITION COMPONENTS
245	±XXXXX (100 FT)	
246	±XXXXX (100 FT)	
264	±XXXX.X (FPS)	CSM STATE VECTOR-VELOCITY COMPONENTS
265	±XXXX.X (FPS)	
266	±XXXX.X (FPS)	
272	+XXXX.X (MIN)	CSM TIME FOR WHICH THE STATE VECTOR IS ACCURATE

	ASCENT PHASING						P30 LM MANEUVER					
HR N33	+	0	0				+	0	0			
MIN TIG	+	0	0	0			+	0	0	0		
SEC	+	0			•		+	0			•	
ΔVX N81					•							•
ΔVY LOCAL					•							•
ΔVZ VERT					•							•
H_A N42	+				•		+					•
H_P					•							•
ΔVR	+				•		+					•
BT	X	X	X	•			X	X	X	•		
R FDAI	X	X	X				X	X	X			
P INER	X	X	X				X	X	X			
ΔVX AGS N86					•							•
ΔVY AGS					•							•
ΔVZ AGS					•							•
BSS	X	X	X				X	X	X			
SPA	X	X			•		X	X				•
SXP	X	X	X		•		X	X	X			•

ASCENT
PHASING

N33 PHASING TIG	XXX:XX:XX.XX (HR:MIN:SEC)	IGNITION TIME OF LM MANEUVER
N81 LOCAL VERTICAL ΔV		
ΔVX	±XXXX.X (FPS)	LOCAL VERTICAL ΔV COMPONENTS OF THE MANEUVER
ΔVY	±XXXX.X (FPS)	
ΔVZ	±XXXX.X (FPS)	
N42 ORBITAL PARAMETERS		
HA	+XXXX.X (NM)	PREDICTED APOGEE RESULTING FROM MANEUVER
HP	±XXXX.X (NM)	PREDICTED PERIGEE RESULTING FROM MANEUVER
ΔVR	+XXXX.X (FPS)	TOTAL ΔV REQUIRED FOR THE MANEUVER
BT	X:XX (MIN:SEC)	DURATION OF THE MANEUVER
FDAI		
R	XXX (DEG)	INERTIAL FDAI ANGLES AT THE BURN ATTITUDE
P	XXX (DEG)	
AGS ΔV		
ΔVX AGS ΔVY AGS ΔVZ AGS	±XXXX.X (FPS) ±XXXX.X (FPS) ±XXXX.X (FPS)	LOCAL VERTICAL ΔV COMPONENTS OF THE MANEUVER TO TARGET THE AGS
BSS	XX (OCTAL)	BSS STAR FOR MANEUVER ATTITUDE CHECK
SPA SXP	±XX.X (DEG) ±XX.X (DEG)	BSS PITCH ANGLE ON COAS, & BSS X POSITION ON COAS FOR MANEUVER ATTITUDE CHECK

P30 LM MANEUVER

P30

												PURPOSE	
+	0	0				+	0	0				HR	N33
+	0	0	0			+	0	0	0			MIN TIG	
+	0					+	0					SEC	
												ΔVX	N81
												ΔVY	LOCAL
												ΔVZ	VERT
+						+						Ha	N42
												Hp	
+						+						ΔVR	
X	X	X				X	X	X				BT	
X	X	X				X	X	X				R	FDAI
X	X	X				X	X	X				P	INER
												ΔVX AGS	N86
												ΔVY AGS	
												ΔVZ AGS	
X	X	X				X	X	X				BSS	
X	X					X	X					SPA	
X	X	X				X	X	X				SXP	

REMARKS:

P30

P30 LM MANEUVER

PURPOSE	XXXXX	PURPOSE OF MANEUVER (SUCH AS DOI TARGETING)
N33 TIG OF MANEUVER	XXX:XX:XX.XX (HR:MIN:SEC)	IGNITION TIME FOR THE MANEUVER
N81 LOCAL VERTICAL ΔV		
ΔVX	±XXXX.X (FPS)	LOCAL VERTICAL ΔV COMPONENTS OF THE MANEUVER
ΔVY	±XXXX.X (FPS)	
ΔVZ	±XXXX.X (FPS)	
N42 ORBITAL PARAMETERS		
HA	+XXXX.X (NM)	PREDICTED APOGEE AND PERIGEE RESULTING FROM THE MANEUVER
HP	±XXXX.X (NM)	
ΔVR	+XXXX.X (FPS)	TOTAL ΔV REQUIRED FOR THE MANEUVER
BT	X:XX(MIN:SEC)	DURATION OF THE MANEUVER
FDAI		
R	XXX (DEG)	INERTIAL FDAI ANGLES AT THE BURN ATTITUDE
P	XXX (DEG)	
N86 AGS ΔV		
ΔVX AGS	±XXXX.X (FPS)	LOCAL VERTICAL ΔV COMPONENTS OF THE MANEUVER USED TO TARGET THE AGS
ΔVY AGS	±XXXX.X (FPS)	
ΔVZ AGS	±XXXX.X (FPS)	
BSS	XX (OCTAL)	BSS STAR FOR BURN ATTITUDE CHECK
SPA	±XX.X (DEG)	BSS PITCH ANGLE ON COAS, & BSS X POSITION ON COAS FOR MANEUVER ATTITUDE CHECK
SXP	±XX.X (DEG)	

LOI DATA CARD

P30

HR	N33	+	0	0				+	0	0			
MIN	TIG	+	0	0	0			+	0	0	0		
SEC		+	0			•		+	0			•	
ΔVX	N81						•						•
ΔVY	LOCAL						•						•
ΔVZ	VERT						•						•
H_A	N42	+					•						•
H_P							•						•
ΔVR		+					•	+					•
BT		X	X	X	•			X	X	X	•		
R	FDAI	X	X	X				X	X	X			
P	INER	X	X	X				X	X	X			
ΔVX AGS	N86						•						•
ΔVY AGS							•						•
ΔVZ AGS							•						•
BSS		X	X	X				X	X	X			
SPA		X	X				•	X	X				•
SXP		X	X	X			•	X	X	X			•

LR SELF TEST

H TM (+7994±30) _____

Ḣ TM (-480±6) _____

N66 SLANTRNG (+08275.±5.0) _____

N67 VX (-00494. ±2.0) _____

VY (+01858. ±2.0) _____

VZ (+01329. ±2.0) _____

RR/TM/VHF

R_1 R_2 Ṙ

N78 _____ _____ _____

TM _____ _____ _____

CMC _____ _____ _____

VHF _____ _____

P52

STAR 1 _____ 2 _____ 3 _____

N05 (STAR ∢ DIFF) _____

N93 (TORQUING ∢) X _____

Y _____

Z _____

GET ____ : ____ : ____

RESIDUALS

		PGNS				AGS			
ΔVX	N35				•	500			•
ΔVY					•	501			•
ΔVZ					•	502			•

MANUAL SHUT-DOWN

A. ΔVG NEGATIVE (PGNS)

OR

B. V_T: 2 SECONDS OVER BURN

-AND-

AGS VGX 2 FPS OVER

MANUAL TAKEOVER

ATT±5° RATE±5°/sec

DOI DATA CARD

N33 DOI TIG	XXX:XX:XX.XX (HR:MIN:SEC)	IGNITION TIME OF LM MANEUVER
N81 LOCAL VERTICAL ΔV		LOCAL VERTICAL ΔV COMPONENTS OF THE MANEUVER
ΔVX	±XXXX.X (FPS)	
ΔVY	±XXXX.X (FPS)	
ΔVZ	±XXXX.X (FPS)	
N42 ORBITAL PARAMETERS		
HA	+XXXX.X (NM)	PREDICTED APOGEE RESULTING FROM MANEUVER
HP	±XXXX.X (NM)	PREDICTED PERIGEE RESULTING FROM MANEUVER
ΔVR	+XXXX.X (FPS)	TOTAL ΔV REQUIRED FOR THE MANEUVER
BT	X:XX (MIN:SEC)	DURATION OF THE MANEUVER
FDAI		
R	XXX (DEG)	INERTIAL FDAI ANGLES AT THE BURN ATTITUDE
P	XXX (DEG)	
N86 AGS ΔV		
ΔVX AGS ΔVY AGS ΔVZ AGS	+XXXX.X (FPS) ±XXXX.X (FPS) ±XXXX.X (FPS)	LOCAL VERTICAL ΔV COMPONENTS OF THE MANEUVER TO TARGET THE AGS
BSS	XXX (OCTAL)	BSS STAR FOR MANEUVER ATTITUDE CHECK
SPA SXP	+XX.X (DEG) ±XX.X (DEG)	BSS PITCH ANGLE ON COAS, & BSS X POSITION ON COAS FOR MANEUVER ATTITUDE CHECK

PDI PAD			
HRS	TIG	+ 0 0	+ 0 0
MIN	PDI	+ 0 0 0	+ 0 0 0
SEC		+ 0	+ 0
TGO	N61	X X	X X
CROSSRANGE			
R	FDAI	X X X	X X X
P	AT TIG	X X X	X X X
Y		X X X	X X X
DEDA 231 IF RQD			

PDI ABORT < 10 MIN			
LOG INSERTION GET=		__ __ __ : __ __ : __ __	
+		5 0 : 0 0	
CSI TIG=		__ __ __ : __ __ : __ __	
HRS	N37	+ 0 0	+ 0 0
MIN	TPI	+ 0 0 0	+ 0 0 0
SEC		+ 0	+ 0

PDI ABORT > 10 MIN			
HRS		+ 0 0	+ 0 0
MIN		+ 0 0 0	+ 0 0 0
SEC	PHASING TIG	+ 0	+ 0
HRS	N37	+ 0 0	+ 0 0
MIN	TPI	+ 0 0 0	+ 0 0 0
SEC		+ 0	+ 0

NO PDI + 12 ABORT			
HR	N33	+ 0 0	+ 0 0
MIN	TIG	+ 0 0 0	+ 0 0 0
SEC		+ 0	+ 0
ΔVX	N81		
ΔVY	LOCAL		
ΔVZ	VERT		
H_A	N42	+	
H_P			
ΔVR		+	+
BT		X X X	X X X
R	FDAI	X X X	X X X
P	INER	X X X	X X X
ΔVX AGS	N86		
ΔVY AGS			
ΔVZ AGS			
HRS	N11	+ 0 0	+ 0 0
MIN	CSI	+ 0 0 0	+ 0 0 0
SEC		+ 0	+ 0
HRS	N37	+ 0 0	+ 0 0
MIN	TPI	+ 0 0 0	+ 0 0 0
SEC		+ 0	+ 0

R2 SUN CHECK
N22 _____ N20 _____
_____
_____

PDI DATA CARD

PDI PAD

TIG PDI	XXX:XX:XX.XX (HR:MIN:SEC)	PDI IGNITION TIME
TGO	XX:XX (MIN:SEC)	TIME TO HIGH GATE
CROSS RANGE	±XXXX.X (NM)	OUT-OF-PLANE DISTANCE BETWEEN THE INITIAL LM ORBITAL PLANE AND THE LANDING SITE (POSITIVE INDICATES LANDING SITE IS NORTH OF ORBITAL PLANE)
FDAI AT TIG		
R	XXX (DEG)	INERTIAL FDAI ANGLES AT IGNITION
P	XXX (DEG)	
Y	XXX (DEG)	
DEDA 231 (IF REQ'D)	XXXXX (100 FT)	LUNAR RADIUS AT THE LANDING SITE

PDI ABORT <10 MIN

TPI TIG	XXX:XX:XX.XX (HR:MIN:SEC)	TPI IGNITION TIME

PDI ABORT >10 MIN

PHASING TIG	XXX:XX:XX.XX (HR:MIN:SEC)	TIME OF IGNITION OF LM PHASING MANEUVER
TPI TIG	XXX:XX:XX.XX (HR:MIN:SEC)	TPI IGNITION TIME

NO PDI +12 ABORT

N33 ABORT TIG	XXX:XX:XX.XX (HR:MIN:SEC)	IGNITION TIME FOR ABORT BURN
N81 LOCAL VERTICAL ΔV		
ΔVX	±XXXX.X (FPS)	LOCAL VERTICAL ΔV COMPONENTS OF THE PHASING MANEUVER
ΔVY	±XXXX.X (FPS)	
ΔVZ	±XXXX.X (FPS)	
N42 ORBITAL PARAMETERS		
HA	+XXXX.X (NM)	PREDICTED APOGEE RESULTING FROM MANEUVER
HP	±XXXX.X (NM)	PREDICTED PERIGEE RESULTING FROM MANEUVER
ΔVR	XXXX.X (FPS)	TOTAL ΔV REQUIRED FOR THE MANEUVER
BT	X:XX (MIN:SEC)	DURATION OF MANEUVER
FDAI		
R	XXX (DEG)	INERTIAL FDAI ANGLES AT THE BURN ATTITUDE
P	XXX (DEG)	
N86 AGS ΔV		
ΔVX AGS	±XXXX.X (FPS)	LOCAL VERTICAL ΔV COMPONENTS OF THE MANEUVER TO TARGET THE AGS
ΔVY AGS	±XXXX.X (FPS)	
ΔVZ AGS	±XXXX.X (FPS)	
N11 CSI TIG	XXX:XX:XX.XX (HR:MIN:SEC)	TIME OF IGNITION FOR CSI BURN
N37 TPI TIG	XXX:XX:XX.XX (HR:MIN:SEC)	TIME OF IGNITION FOR TPI BURN

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T 2 ABORT			
HR	T2	+ 0 0	+ 0 0
MIN	TIG	+ 0 0 0	+ 0 0 0
SEC		+ 0	+ 0
HR	N33	+ 0 0	+ 0 0
MIN	PHASING	+ 0 0 0	+ 0 0 0
SEC	TIG	+ 0	+ 0
HRS	N11	+ 0 0	+ 0 0
MIN	CSI_1	+ 0 0 0	+ 0 0 0
SEC		+ 0	+ 0
HRS	N37	+ 0 0	+ 0 0
MIN	TPI	+ 0 0 0	+ 0 0 0
SEC		+ 0	+ 0

P68

N43 _____ LAT

_____ LONG

_____ ALT

P12

N76 _____ V (HOR) (5515.2)

_____ V (VERT) (19.5)

_____ CROSSRANGE (0.0)

NOTE: IF CROSSRANGE >8 N.M., LOAD 8 N.M.

N74 _____ YAW

_____ PITCH

T 3 ABORT			
HRS	T3	+ 0 0	+ 0 0
MIN	TIG	+ 0 0 0	+ 0 0 0
SEC		+ 0	+ 0
HRS	CSM	+ 0 0	+ 0 0
MIN	PERIOD	+ 0 0 0	+ 0 0 0
SEC		+ 0	+ 0
HRS		+ 0 0	+ 0 0
MIN	P+ΔT	+ 0 0 0	+ 0 0 0
SEC		+ 0	+ 0
HRS	N11	+ 0 0	+ 0 0
MIN	CSI TIG	+ 0 0 0	+ 0 0 0
SEC		+ 0	+ 0
HRS	N37	+ 0 0	+ 0 0
MIN	TPI	+ 0 0 0	+ 0 0 0
SEC		+ 0	+ 0

P12

N76 _____ V (HOR) (5535.6)

_____ V (VERT) (32.0)

_____ CROSSRANGE (0.0)

NOTE: IF CROSSRANGE >8 N.M., LOAD 8 N.M.

N74 _____ YAW

_____ PITCH

LUNAR SURFACE DATA CARD

T2 ABORT

T2 TIG	XXX:XX:XX.XX (HR:MIN:SEC)	LIFTOFF TIME- SECOND PREFERRED TIME AFTER TOUCH- DOWN (∿T.D. +12 MIN.)
N33 PHASING TIG	XXX:XX:XX.XX (HR:MIN:SEC)	TIME OF IGNITION FOR PHASING BURN
N11 CSI TIG	XXX:XX:XX.XX (HR:MIN:SEC)	TIME OF IGNITION FOR CSI BURN
N37 TPI TIG	XXX:XX:XX.XX (HR:MIN:SEC)	TIME OF IGNITION FOR TPI BURN

T3 ABORT

T3 TIG	XXX:XX:XX.XX (HR:MIN:SEC)	LIFT OFF TIME AFTER FIRST CSM REVOLUTION
CSM PERIOD	XXX:XX:XX.XX (HR:MIN:SEC)	CSM ORBITAL PERIOD
P + ΔT	XXX:XX:XX.XX (HR:MIN:SEC)	CSM PERIOD PLUS THE TIME INTERVAL BETWEEN CLOSEST APPROACH AND LIFTOFF TIMES
N11 CSI TIG	XXX:XX:XX.XX (HR:MIN:SEC)	TIME OF IGNITION FOR CSI BURN
N37 TPI TIG	XXX:XX:XX.XX (HR:MIN:SEC)	TIME OF IGNITION FOR TPI BURN

LM ASCENT PAD

+	0	0				+	0	0				HRS
+	0	0	0			+	0	0	0			MIN TIG
+	0					+	0					SEC
+						+						V (HOR)
+						+						V (VERT) N76
	0						0					*CROSSRANGE
												DEDA 047
												DEDA 053
												DEDA 225/226
												DEDA 231

*NOTE: LOAD 8 NM IF CROSSRANGE IS GREATER THAN 8 NM
COMMENTS:

LM ASCENT PAD

ASCENT TIG	XXX:XX:XX.XX (HR:MIN:SEC)	TIME OF APS IGNITION FOR LM ASCENT
N76 INSERTION TARGET		
V(HOR)	XXXX.X (FPS)	HORIZONTAL VELOCITY AT ORBIT INSERTION
V(VERT)	XXXX.X (FPS)	VERTICAL VELOCITY AT ORBIT INSERTION
CROSSRANGE	±XXX.X (NM)	CROSSRANGE DISTANCE AT ORBITAL INSERTION
DEDA 047	XXXXXX (OCTAL)	SINE OF LANDING AZIMUTH ANGLE
DEDA 053	XXXXXX (OCTAL)	COSINE OF LANDING AZIMUTH ANGLE
DEDA 225	XXXXXX (100 FT)	LOWER LIMIT OF $\alpha\}$ AT ORBIT INSERTION
DEDA 226	XXXXXX (100 FT)	UPPER LIMIT OF $\alpha\}$ AT ORBIT INSERTION
DEDA 231	XXXXXX (100 FT)	RADIAL DISTANCE OF LAUNCH SITE FROM CENTER OF MOON

CSI DATA CARD June 19, 1969

HR	TIG N11	+	0	0				+	0	0			
MIN	CSI	+	0	0	0			+	0	0	0		
SEC		+	0			•		+	0			•	
N55			(+00001)					(+02660)					(+13000)
HR	TIG N37	+	0	0				+	0	0			
MIN	TPI	+	0	0	0			+	0	0	0		
SEC		+	0			•		+	0			•	
ΔVX	N81	+	0			•		+	0			•	
ΔVY			0			•			0			•	
FDAI PITCH		+	X	X				+	X	X			
373 (+0321.3)		+				•		+				•	
275 ~~(+0418.1)~~ +418.5		+				•		+				•	
410+1, 605+00777, 416+1													
ΔVX	AGS N86		0			•			0			•	
ΔVY	AGS		0			•			0			•	
ΔVZ	AGS		0			•			0			•	

RESIDUALS

	PGNS	AGS
ΔVX		500
ΔVY	N85	501
ΔVZ		502

P52

STAR 1 ____ 2 ____ 3 ____

N05 (STAR ∤ DIFF) _____

N93 (TORQUING ∤) X _____

Y _____

Z _____

PGNCS

N75: ΔH (15.0) _____ CSI/CDH (57:68) _____ CDH/TPI (38:21) _____

N81: ΔVX CSI (+50.5) _____ YDOT CSI (+0.0) _____

N82: ΔVX CDH (-0.4) _____ ΔVY CDH (+0.0) _____ ΔVZ CDH (+0.0) _____

AGS

402 ΔH _____ 372 ΔT(CSI-CDH) _____ 267 ΔVG (CSI) _____

CSM SOLUTION: ΔVX CSI _____ ΔVY CSI _____

N86 (AGS): ΔVX _____ ΔVY _____ ΔVZ _____

371 ΔV CDH _____

CSI DATA CARD (P32 LM MANEUVER)

N11 CSI TIG	XXX:XX:XX.XX (HR:MIN:SEC)	CSI IGNITION TIME
N37 TPI TIG	XXX:XX:XX.XX (HR:MIN:SEC)	TPI IGNITION TIME
N81		
ΔVX	XXX.X (FPS)	LOCAL VERTICAL ΔV COMPONENTS OF THE CSI MANEUVER
ΔVY	XXX.X (FPS)	
FDAI PITCH	XXX (DEG)	FDAI INERTIAL PITCH ANGLE AT THE CSI BURN ATTITUDE
DEDA 373	XXXX.X (MIN)	AGS IGNITION TIME OF NEXT MANEUVER
DEDA 275	XXXX.X (MIN)	DESIRED TPI TIG (FOR CSI CALCULATION ONLY)
N86 AGS ΔV		
ΔVX AGS	XX.XX (FPS)	LOCAL VERTICAL ΔV COMPONENTS OF CSI USED TO TARGET AGS EXT ΔV
ΔVY AGS	XX.XX (FPS)	
ΔVZ AGS	XX.XX (FPS)	

CDH PAD

HR	N13	+	0	0				+	0	0			
MIN	TIG	+	0	0	0			+	0	0	0		
SEC	CDH	+	0					+	0				
ΔVX			0						0				
ΔVY	N81		0						0				
ΔVZ			0						0				
PLM	FDAI	X	X	X				X	X	X			
373 (+0379.6)		+						+					
ΔVX	N86		0						0				
ΔVY	AGS		0						0				
ΔVZ			0						0				

CDH DATA CARD

PLANE CHANGE P30,V90,410+5

TIG CDH -3 0:0 0.0 0

TIG PC

YDOT

CSM N90	PGNS N90	AGS 263
(-)	(-)	(-)
(-)	(-)	(-)

RESIDUALS

	PGNS	AGS
ΔVX	N85	500
ΔVY		501
ΔVZ		502

RESIDUALS

	PGNS	AGS
ΔVX	N85	500
ΔVY		501
ΔVZ		502

PGNCS

N75

ΔH (15.0)

ΔT TPI/CDH (37:31)

TPI SLIP (0:00)

N81

ΔVX (-1.1)

YDOT

CDH N90 (+0.0)

ΔVZ (+4.1)

(-)

(-)

(-)

PGNCS

AGS

402 ΔH

450 ΔVX

452 ΔVZ

CSM SOLUTION

ΔVX ΔVY ΔVZ

N86 (AGS)

ΔVX ΔVY ΔVZ

AGS

CDH DATA CARD

N13 CDH TIG	XXX:XX:XX.XX (HR:MIN:SEC)	IGNITION TIME FOR CDH MANEUVER
N81 LOCAL VERTICAL ΔV		
ΔVX	±XXX.X (FPS)	LOCAL VERTICAL ΔV COMPONENTS OF CDH MANEUVER
ΔVY	±XXX.X (FPS)	
ΔVZ	±XXX.X (FPS)	
PLM FDAI	XXX (DEG)	FDAI INERTIAL PITCH ANGLE AT CDH BURN ATTITUDE
DEDA 373	XXXX.X (MIN)	AGS IGNITION TIME OF NEXT MANEUVER
N86 AGS ΔV		
ΔVX AGS	±XXX.X (FPS)	LOCAL VERTICAL ΔV COMPONENTS OF CDH USED TO TARGET AGS EXT ΔV
ΔVY AGS	±XXX.X (FPS)	
ΔVZ AGS	±XXX.X (FPS)	

TPI DATA CARD June 19, 1969

HR N37	+	0	0				+	0	0			
MIN TIG	+	0	0	0			+	0	0	0		
SEC TPI	+	0			•		+	0			•	
N55	(BLANK)			(+026.60)			(+130.00)					
ΔVX N81		0				•		0				•
ΔVY		0				•		0				•
ΔVZ		0				•		0				•
ΔVR N42	+	0				•	+	0				•
RLM	X	X	X				X	X	X			
PLM	X	X	X				X	X	X			
R TPI N54	+	0			•		+	0				
R TPI TIG-5		0				•		0				
F/A (+/-) N59		0						0				
R/L (+/-) ΔV		0						0				
D/U (+/-) LOS		0						0				
BT	X	X		:			X	X		:		

307+043.00, 314+0

RESIDUALS

		PGNS				AGS			
ΔVX	N85				•	500			•
ΔVY					•	501			•
ΔVZ					•	502			•

PGNCS

N58: HP ( ) ΔV TPI ( ) ΔV TPF ( )

N81: ΔVX ( ) ΔVY ( ) ΔVZ ( )

ΔVF/A ( ) ΔVR/L ( ) ΔVD/U ( )

AGS

267 ΔVTPI | 371 ΔV TPI+TPF

CSM SOLUTION: ΔVX ΔVY ΔVZ

N86 (AGS): ΔVX ΔVY ΔVZ

TPI DATA CARD

N37 TPI TIG	XXX:XX:XX.XX (HR:MIN:SEC)	IGNITION TIME FOR THE TPI MANEUVER

N81 LOCAL VERTICAL ΔV

ΔVX	±XX.X (FPS)	LOCAL VERTICAL ΔV
ΔVY	±XX.X (FPS)	COMPONENTS OF THE
ΔVZ	±XX.X (FPS)	TPI MANEUVER
N42 Δ VR	+XX.X (FPS)	TOTAL ΔV REQUIRED FOR THE MANEUVER
RLM	XXX (DEG)	ROLL AND PITCH
PLM	XXX (DEG)	FDAI ANGLES AT TPI BURN ATTITUDE

N54 TIG-5

R TPI	XX.XX (FT)	RANGE AT TPI TIG - 5 MIN
$\dot{R}$ TPI	±XXX.X (FPS)	RANGE RATE AT TPI TIG - 5 MIN

N59 Δ V LOS

F/A	+XX.X (FPS)	LINE-OF-SIGHT ΔV
R/L	±XX.X (FPS)	COMPONENTS OF THE
D/U	±XX.X (FPS)	TPI MANEUVER
BT	XX:XX (MIN:SEC)	DURATION OF THE MANEUVER

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SECTION III - DETAILED TIMELINE

FLIGHT PLAN

TIME	EVENT	REMARKS
- 00:09	LCC: REPORT IGNITION	FIRST OPPORTUNITY LIFT-OFF JULY 16, 0932 EDT, 72° LA, TARGETED FOR LANDING SITE 2. LIFT-OFF: 1332 GMT
00:00	LCC: CDR: REPORT LIFT-OFF	
00:02	CDR: REPORT YAW MNVR	
00:10	LCC: REPORT CLEAR OF TOWER	
00:15	CDR: REPORT ROLL AND PITCH PROGRAM INITIATE	
00:32	CDR: REPORT ROLL COMPLETE	
00:42	MCC: REPORT MARK MODE IB	PROP DUMP TO RCS CMD
00:51	LMP: REPORT CABIN PRESS DECREASING	ALTITUDE 14,00 ft
01:21	MAX Q	
01:56	MCC: REPORT MARK MODE IC	ALTITUDE 100,000 ft
02:00	MCC: CDR: REPORT GO/NO GO FOR STAGING	
02:15	CDR: REPORT INBOARD OUT	
02:41	CDR: REPORT OUTBOARD OUT	
02:42	CDR: REPORT STAGING/SII IGNITION	
03:12	CDR: REPORT S-II SEP LIGHT OUT	
03:17	CDR: REPORT TWR JETT AND MODE II	
03:21	CDR: REPORT GUIDANCE	

FLIGHT PLAN

TIME	EVENT	REMARKS
04:00	MCC: REPORT TRAJECTORY AND GUIDANCE GO/NO GO	
04:00	CDR: REPORT S/C GO/NO GO	
05:00	CDR: REPORT S/C GO/NO GO	
05:25	MCC: REPORT ~~S-IVB TO ORBIT CAPABILITY~~ COI CAPABILITY	
06:00	CDR: REPORT S/C GO/NO GO	
07:00	CDR: REPORT S/C GO/NO GO	
08:00	CDR: REPORT S/C GO/NO GO	
08:30	MCC: CDR: REPORT GO/NO GO FOR STAGING	
08:57	MCC: REPORT MODE IV	
	CDR: REPORT S/C GO/NO GO	
	MCC: REPORT TRAJECTORY AND GUIDANCE GO/NO GO	
09:11	CDR: REPORT S-II CUTOFF	
09:15	CDR: REPORT S-IVB IGNITION	
10:00	MCC: CDR: REPORT GO/NO GO FOR ORBIT	
	MCC: REPORT PREDICTED SECO	
11:40	CDR: REPORT SECO $TB_5 = 0$	IMU GIMBAL ANGLES @ INSERTION R 180° P 340° Y 0°
	S-IVB MAINTAINS COMMANDED CUTOFF INERTIAL ATTITUDE	H pad 103.3 NM

MISSON G	EDITION FINAL	DATE JULY 1, 1969	PAGE 3-ii

FLIGHT PLAN

MCC-H

0930 EDT

00:00 LIFTOFF

US

:10 SECO-INSERTION CHECKLIST
LMP - SM RCS MON CK, CM RCS MON CK,
C&W OPERATIONAL CK
REMOVE AND STOW HELMETS AND GLOVES
UNSTOW CAMERAS

TCYI

:20 CMP - TRANS TO LEB - O2 MAIN REG CK
CMP/LMP - SEC RAD LEAK CK
CDR/CMP - ECS POST INSERTION CONFIG
LMP - FUEL CELL PURGE CK, EPS MON. CK,
FUEL CELL POWER PLANT CK, DC VOLT -
AMP CK, ECS MON CK, SPS MON CK
GDC ALIGN TO IMU - RECORD DRIFT
CDR - UNSTOW SEQ CAMERA BRACKET AND ORDEAL
INSTALL ORDEAL & COAS

:30 MOUNT AND INITIALIZE ORDEAL
CMP - ~~ECS REDUNDANT COMPONENT CHECK~~ SEC GLYCOL LOOP CK

TAN

:40 JETTISON OPTICS COVER (DIRECT, HIGH, SHAFT RIGHT)
~~RECORD ΔAZ CORRECTION~~
RECORD Z TORQUE

:50 IMU REALIGN - P52
(OPTION 3 - REFSMMAT)
(OPTIONAL)

TCROT

S-BAND VOL - UP FOR HSK

01:00 TWO-WAY USB VOICE CK

UPDATE
~~ΔAZ CORRECTION~~
Z TORQUE

SET UP CAMERA EQUIP(T&D)
16mm/18/CEX-BRKT-MIR
(f8,250,7) 6 FPS
1 MAG (FOR T&D)
EL/80/CEX
(f8,250,INF)

NOTES

LIFTOFF CREW POSITIONS
LEFT COUCH - CDR
CENTER COUCH - LMP
RIGHT COUCH - CMP
INSERTION IMU GIMBAL ANGLES P 340 R 180 Y 0
AT SECO +20 SEC, SIV-B MNVRS TO LH AND INITIALIZES ORB RATE (HEADS DOWN)

COOLANT CONTROL ATTENUATION PANEL NOT OPENED

REPORT
P52 - (PAD REFSMMAT)
N71: __ __.__ __
N05: __ __ __.__ __
N93:
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __
GET __ __ __:__ __:__

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	00:00 - 01:00	1/1	3-1

MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

MCC-H

1000 EDT

01:00

:10

:20

:30

:40

:50

02:00

THSK

USCYI

P52 - CONT'D

GDC ALIGN TO IMU
STOW OPTICS

UPLINK CMC
CSM STATE VECTOR

SCS ATT REF COMPARISON CK
EXTEND DOCKING PROBE

UPDATE
PAD DATA

V66 - TRANS CSM STATE VECTOR TO LM SLOT
RECORD PAD DATA
(TLI, TLI + 90 MIN ABORT, AND
P37 - TLI + 4 HR ABORT)
SM RCS HOT FIRE
(MIN IMPULSE - ALL JETS)

GO/NO GO

GO/NO GO FOR PYRO ARM
BEGIN TLI PREPARATION (CHECKLIST PG L-2-19)
DON HELMETS & GLOVES - ALL

EMS ΔV TEST

NOTES

AS A GENERAL RULE MSFN WILL ALWAYS UP-LINK THE STATE VECTOR TO THE CSM SLOT AND THE CREW WILL TRANSFER IT VIA V66 TO THE LM SLOT

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	01:00 - 02:00	1/1	3-2

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TLI
BURN CHART

	P OR Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
TLI	10°/SEC SHUTDOWN	+45° SHUTDOWN	BT + 6 SEC & VI = PAD VALUE	DO NOT TRIM

FLIGHT PLAN

MCC-H

1130 EDT

02:00

:10 TAN

:20

:30 CRO

:40 MER

:50 MSFN

03:00

GO/NO GO

GDC ALIGN TO IMU
PYRO LOGIC ARM
CMP - TRANS TO COUCH

SET ORDEAL TO ORB RATE

GO/NO GO FOR TLI

TB-6 (02:34:48)

THRUST MON - P47

TLI

GETI = 02:44:26
BT = 5:20
ΔV_T =10,451.2FPS
POSIGRADE

POO - CMC IDLING
V66 - TRANS CSM SV TO LM SLOT
TLI BURN STATUS REPORT
CDR - TRANS TO CENTER COUCH, CMP - LEFT COUCH,
LMP - RIGHT COUCH

NOTES

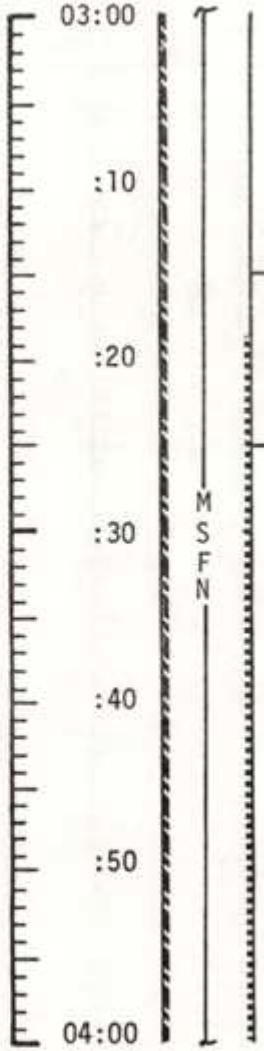
REPORT:
TLI BURN STATUS
VI _ _ _ _ _
ΔVC _ _ _ _ _

AT SECO: SIVB INERTIAL
AT SECO+20 SEC: SIVB
TO LOCAL HORIZONTAL
ORB RATE, HEADS DOWN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	02:00 - 03:00	1/2	3-3

MCC-H

FLIGHT PLAN



WASTE STOWAGE VENT VLV - CLOSED
PRESS CABIN TO 5.7 PSIA
GDC ALIGN TO IMU
VERIFY AND ACTIVATE RCS DAP
R1=11103, R2=01111
CSM SEP PREPARATION
LOAD DOCKING GIMBAL ANGLES

START 16mm CAMERA

CSM/SIV-B SEP
GET = 03:15

DOCKING PHOTOGRAPHY

T&D MNVR
+X 0.8 FPS
AFTER 15 SEC, -X 0.3 FPS
AUTO MNVR TO DOCKING ATT
NULL TRANSLATION & RATES
+X TO CLOSE AT .25 TO .5 FPS

DOCK GET = 03:25

POST DOCKING
S-BAND HGA ANGLES = P_____ Y _____

STOP 16mm CAMERA

CMP - INITIATE CM/LM PRESS EQUALIZATION

CONFIGURE FOR EXTRACTION
CHECK LM PRESS STABILIZATION
REMOVE AND TEMP STOW TUNNEL HATCH
CHECK DOCKING LATCHES
VENT DOCKING PROBE
CONNECT UMBILICALS - VERIFY PWR TO LM
REINSTALL HATCH
LM/CM ΔP VALVE - LM/CM ΔP
VERIFY TUNNEL EQUALIZATION - CLOSED
WASTE STOWAGE VENT VALVE - OPEN (FOR 8 HOURS)

NOTES

DAP LOAD FOR SEPARATION
CSM, 0.5°DB, 2.0°/SEC,
B/D ROLL, 4 JETS

EL PHOTOS AS
CONVENIENT

DECISION TO END CM
CABIN PURGE WILL BE
MADE REAL TIME BASED
ON LM LEAK RATE

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	03:00 - 04:00	1/TLC	3-4

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EVASIVE MANEUVER
BURN CHART

	P OR Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
EVASIVE MNVR	10°/SEC TAKEOVER	±10° TAKEOVER	BT + 1 SEC	DO NOT TRIM

FLIGHT PLAN

MCC-H

1330 EDT

UPDATE
EVASIVE MNVR
GO/NO GO-GO

04:00

RECORD EVASIVE MNVR PAD
GO/NO-GO FOR PYRO ARM & LM EJECTION
DAP - R1 = 21101, R2 = 11111
PYRO LOGIC ARM
~~THRUST MONITOR - P47~~

DAP CONFIGURATION FOR LM EJECTION: CSM & LM 0.5° DB, .05°/SEC, A/C ROLL, 4 JETS

:10

LM EJECTION

GETI = 04:09:45
BT = 3 SEC
ΔV_T =

EXT ΔV = P30
DAP- R1 = 21111 R2 = 11111
DAP CONFIGURATION AFTER LM EJECTION SAME AS ABOVE EXCEPT CHANGE TO 5° DB

:20

:30

SPS THRUST - P40
PITCH DOWN 75° WRT LOCAL HORIZONTAL
ROLL TO VISUALLY ACQ SIV-B

SM RCS MON CK

GDC ALIGN TO IMU

TLI CUT OFF
+ 1 HR 50 MIN

:40

EVASIVE MNVR

GETI = 04:39:45
NO ULLAGE
BT = 2.8 SEC
ΔV_T = 19.7
IN PLANE

SM RCS MON CK
SPS MON CK
BURN STATUS REPORT
V66 - TRANS CSM STATE VECTOR TO LM SLOT

:50

~~MANEUVER TO OBSERVE SLINGSHOT~~

05:00

MSFN

NOTE:

WITH RT TLM, ONLY ITEMS NORMALLY REQUIRED IN BURN STATUS REPORT ARE ΔVC, FUEL, OX, AND UNBAL

NOTES

FOR LM EJECTION RELATIVE ΔV FROM SPRINGS ≈1 FPS

BURN STATUS REPORT

X	X	:	ΔTIG
X	X	:	BT
		•	V_{gx}
	TRIM		
X	X	X	R
X	X	X	P
X	X	X	Y
☐		•	V_{gx}
☐		•	V_{gy}
☐		•	V_{gz}
☐		•	ΔV_c
X	X	X	FUEL
X	X	X	OX
X	X	X	UNBAL

FIRST SPS BURN WILL ALWAYS START ON BANK A AND BANK B WILL BE ACTIVATED IF THE THE BURN > 5 SEC

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	04:00 - 05:00	1/TLC	3-5

FLIGHT PLAN

MCC-H

1430 EDT

05:00

S-IVB SLINGSHOT

REPORT LM/CM ΔP

:10

UPLINK CMC

ZERO TRUNNION BIAS

DOFF & STOW
HELMET, GLOVES
AND PGA'S

:20

BATTERY CHARGE, BATTERY B

M
S
F
N

:30

:40

IMU REALIGN - P52
OPTION 3 - REFSMMAT

:50

UPDATE

BLOCK DATA

GDC ALIGN TO IMU
RECORD BLOCK DATA
(P37 - TLI + 11 HRS ABORT)

MNVR TO SIGHTING ATT

06:00

NOTES

REPORT

P52 - (PAD REFSMMAT)

N71: __ __,__ __

N05: __ __ __.__ __

N93:

X __ __.__ __ __

Y __ __.__ __ __

Z __ __.__ __ __

GET __ __ __:__ __:__ __

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	05:00 - 06:00	1/TLC	3-6

MCC-H

FLIGHT PLAN

1500 EDT
06:00
:10
:20
:30
:40
:50
07:00

MSFN

CISLUNAR NAVIGATION - P23
OPTICS CALIBRATION

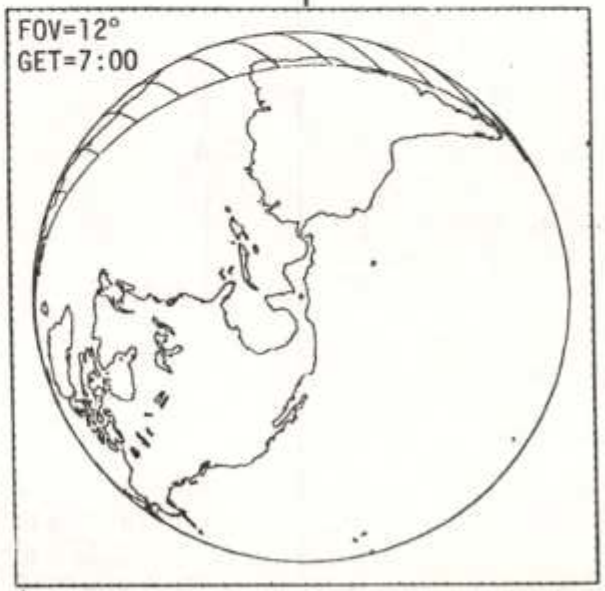
1. STAR 02 ENH (R3=00110)
2. STAR 40 EFH (R3=00120)
3. STAR 44 ENH (R3=00110)
4. STAR 44 ENH (R3-00110)
5. STAR 45 ENH (R3-00110)

NOTES

3 MARKS ON EACH STAR

INCORPORATE P23 MARK DATA AND UPDATE ONBOARD STATE VECTOR

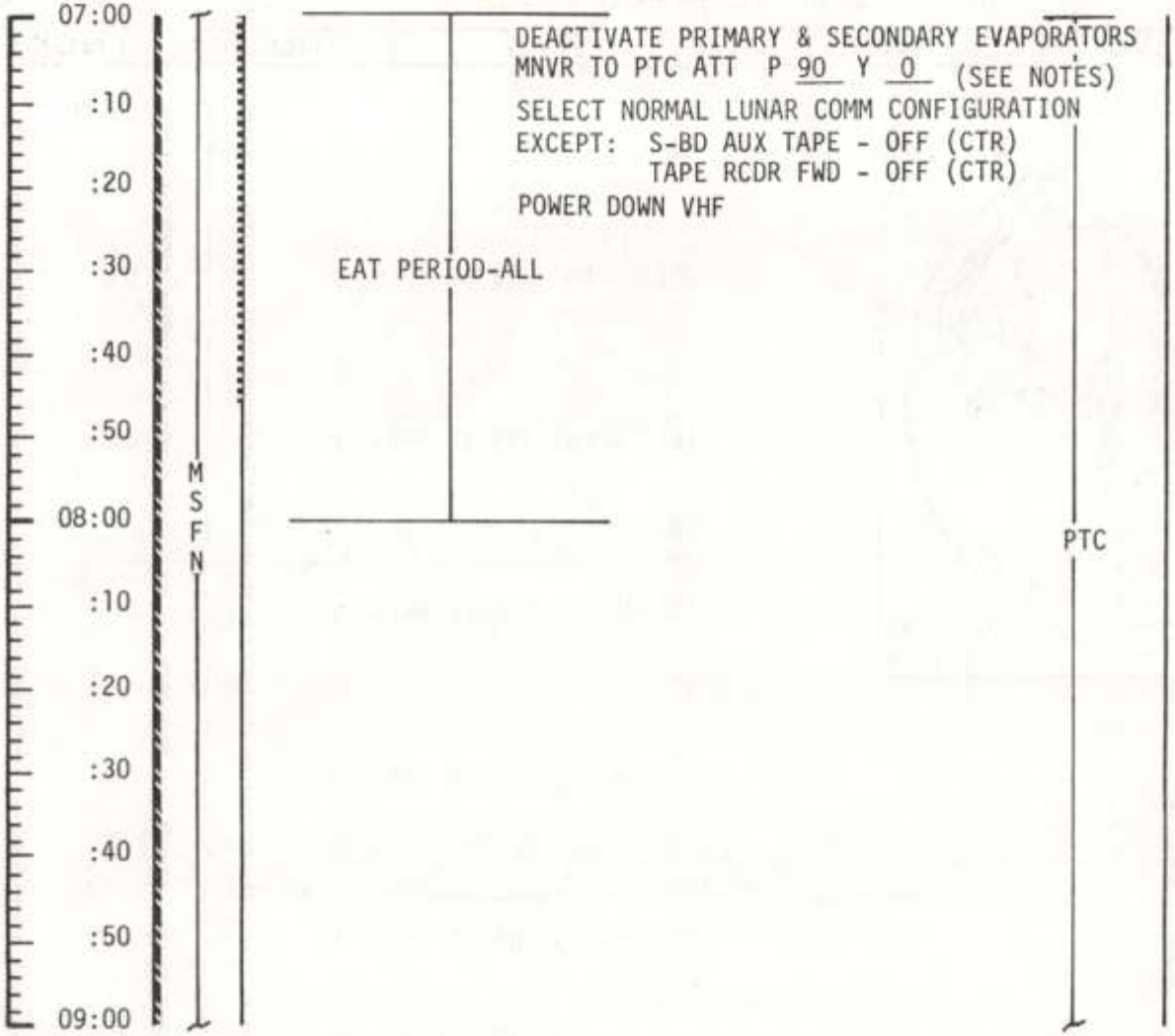
TRN BIAS CALIBRATION REPEATED UNTIL 2 CKS AGREE TO WITHIN 0.003° REPEAT CKS EVERY 30 MIN DURING P23



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	06:00 - 07:00	1/TLC	3-7

MCC-H

FLIGHT PLAN



NOTES

PTC WILL BE INITIATED AFTER MCC1, OR AFTER MCC1 IS SCRUBBED

PHOTOS OF EARTH AS CONVENIENT
EL/250/CEX-RING
(f11,250)

LUNAR
EL/250/CEX-RING
(f5.6,250,INF)

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	07:00 - 09:00	1/TLC	3-8

MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

MCC-H

1830 EDT

09:00
:10
:20
:30
:40
:50
10:00
:10
:20
:30
:40
:50
11:00

MSFN

UPLINK CMC
- EARTH HORIZON BIAS (ΔH) (IF REQUIRED)
- CSM STATE VECTOR
- MCC1 TGT LOAD

UPDATE
- MCC1 MNVR PAD

CO_2 FILTER CHANGE NO. 1
(3 INTO A, STORE 1 IN B5)

V66 - TRANS CSM STATE VECTOR TO LM SLOT

O_2 FUEL CELL PURGE

RECORD MCC1 MNVR PAD

CONTINUE PTC IF MCC1 IS SCRUBBED

IMU REALIGN - P52
OPTION 3 - REFSMMAT
(OPTIONAL)

PTC

NOTES

THE EARTH HORIZON BIAS (ΔH) WILL BE UPDATED TO THE CMC IF THE DIFFERENCE BETWEEN THE SIGHTING ΔH & THE E-MEMORY ΔH IS ≥8.3KM

REPORT:

P52 (PAD REFSMMAT)
N71: __ __,__ __
N05: __ __ __.__ __
N93:
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __
GET __ __ __:__ __:__ __

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	09:00 - 11:00	1/TLC	3-9

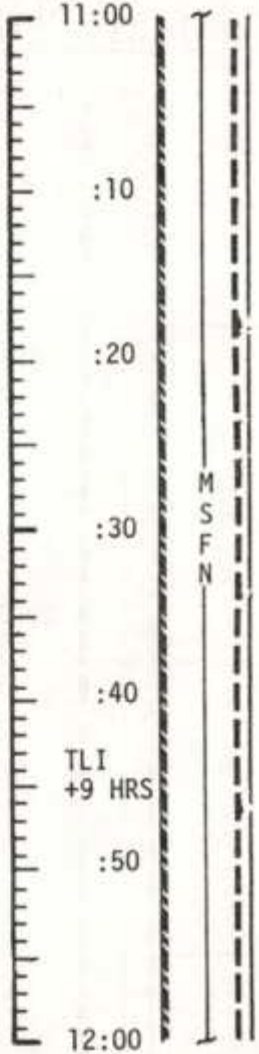
MCC
BURN CHART

	P OR Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
MCC1	10°/SEC TAKEOVER	±10° TAKEOVER	BT + 1 SEC	TRIM X AXIS ONLY (UNLESS X > 2 FPS)

FLIGHT PLAN

MCC-H

2030 EDT



EXT ΔV - P30

SPS/RCS THRUST - P40/41

MNVR TO BURN ATT

SXT STAR CK

EMS ΔV TEST

SM RCS MON CK

GDC ALIGN TO IMU

MCC1 ΔV = NOMINALLY ZERO

SM RCS MON CK
SPS MON CK
MCC1 BURN STATUS REPORT
V66 - TRANS CSM STATE VECTOR TO LM SLOT

RECORD BLOCK DATA - (P37 - TLI + 25, 35, 44, AND 53 HR ABORTS

UPLINK CMC
DESIRED ORIENTATION (PTC)
UPDATE
BLOCK DATA

NOTES

BURN STATUS REPORT

X	X	□	•			ΔTIG
X	X		•			BT
□				•		V_{gx}
TRIM						
X	X	X				R
X	X	X				P
X	X	X				Y
□				•		V_{gx}
□				•		V_{gy}
□				•		V_{gz}
□				•		ΔV_c
X	X	X				FUEL
X	X	X				OX
X	X	X				UNBAL

~~MCC1 WIL BE PERFORMED IF ΔV WOULD EXCEED 25 FPS IF DELAYED TO MCC3 (LOI-22 HRS)~~

MCC1 WILL BE DELAYED TO MCC2 IF PROPELLANT COST IS NOT PROHIBITIVE

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	11:00 - 12:00	1/TLC	3-10

MCC-H

FLIGHT PLAN

2100 EDT

12:00
:10
:20
:30
:40
:50
13:00
:10
:20
:30
:40
:50
14:00

M S F N

IMU REALIGN - P52
(OPTION 1 - PREFERRED)
& STAR CHECK
(IF MCC1 WAS PERFORMED)

REPORT LM/CM ΔP
BATTERY CHARGE, BATTERY A

EAT PERIOD - ALL

REST PERIOD
(9 HOURS)

WASTE STOWAGE VENT VLV - CLOSED

VENT BATTS UNTIL SYSTEM TEST METER (4A) = 0

START PTC
P 90° Y 0°

PRESLEEP CHECKLIST

CREW STATUS REPORT (RADIATION, MEDICATION)
CYCLE O_2 & H_2 FANS
CHLORINATE POTABLE WATER
SELECT NORMAL LUNAR CONFIGURATION
EXCEPT
- S-BD NORMAL MODE VOICE - OFF
- S-BD SQUELCH - ENABLE
- S-BD AUX TAPE - OFF
- S-BD ANT - OMNI
- S-BD ANT OMNI - B
- TAPE RCDR FWD - OFF

VERIFY:
- WASTE MNGT OVBD DRAIN - OFF
- WASTE STOW VENT VLV - CLOSED
- EMERG CABIN PRESS VLV - BOTH
- SURGE TK O_2 VLV - ON
- REPRESS PACK O_2 VLV - ON
- LM TUNNEL VENT VLV - LM/CM ΔP
- POT H_2O HTR - OFF

AUTO RCS JET SELECT (16) - OFF

NOTES

P52 - PULSE TORQUE TO PTC REFSMMAT. ALIGNMENT CHECKED WITH OPTICS

PTC ESTABLISHED IN G&N P, Y ±30°DB, R RATE OF 0.3°/SEC

ON BOARD READOUT

BAT C _____
PYRO BAT A _____
PYRO BAT B _____
RCS A _____
B _____
C _____
D _____

DC IND SEL TO MNA OR MNB

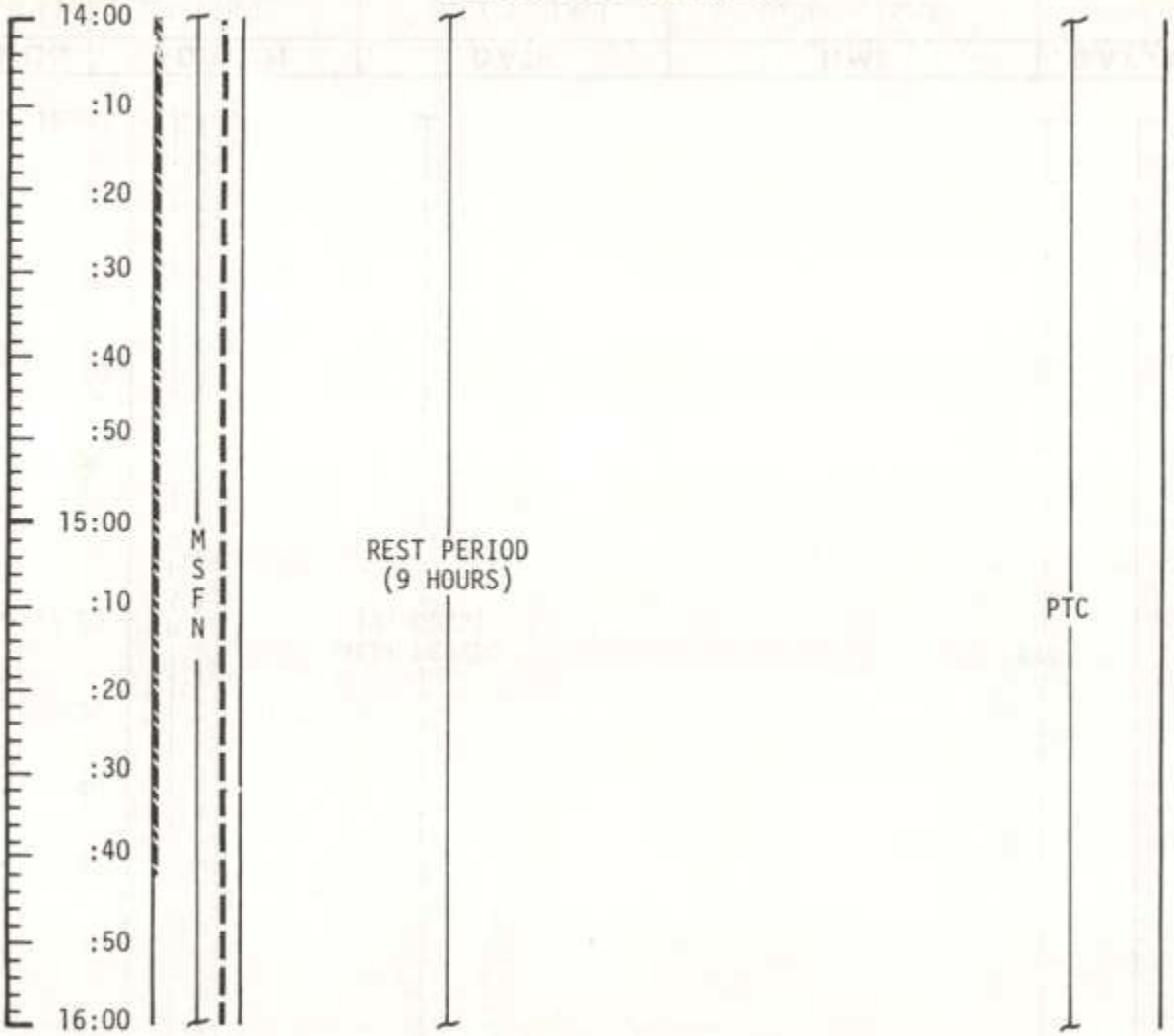
DURING REST PERIOD, 2 CREWMEN IN REST STATION, 1 IN LEFT COUCH

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	12:00 - 14:00	1/TLC	3-11

MCC-H

FLIGHT PLAN

NOTES

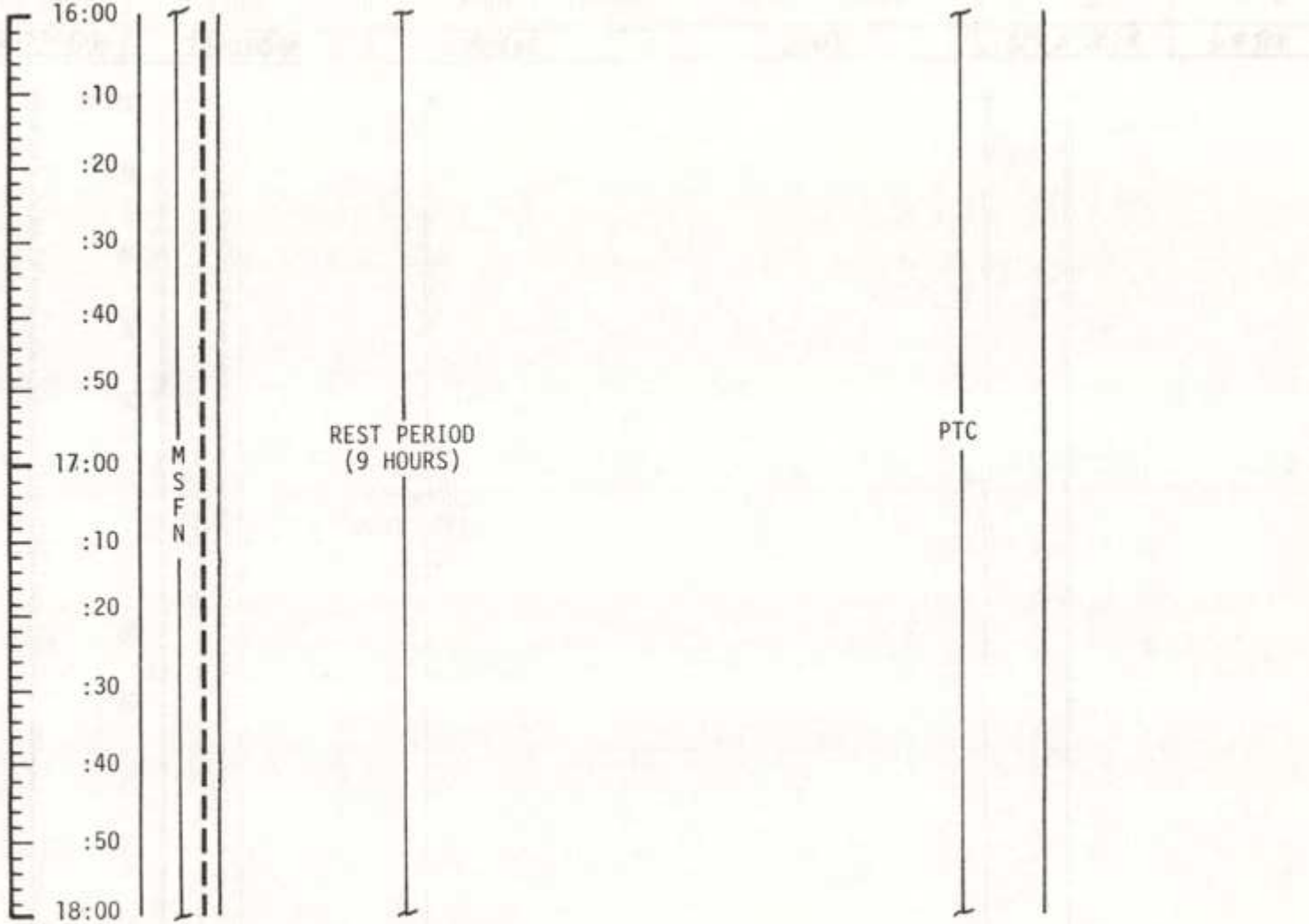


MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	14:00 - 16:00	1/TLC	3-12

MCC-H

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	16:00 - 18:00	1/TLC	3-13

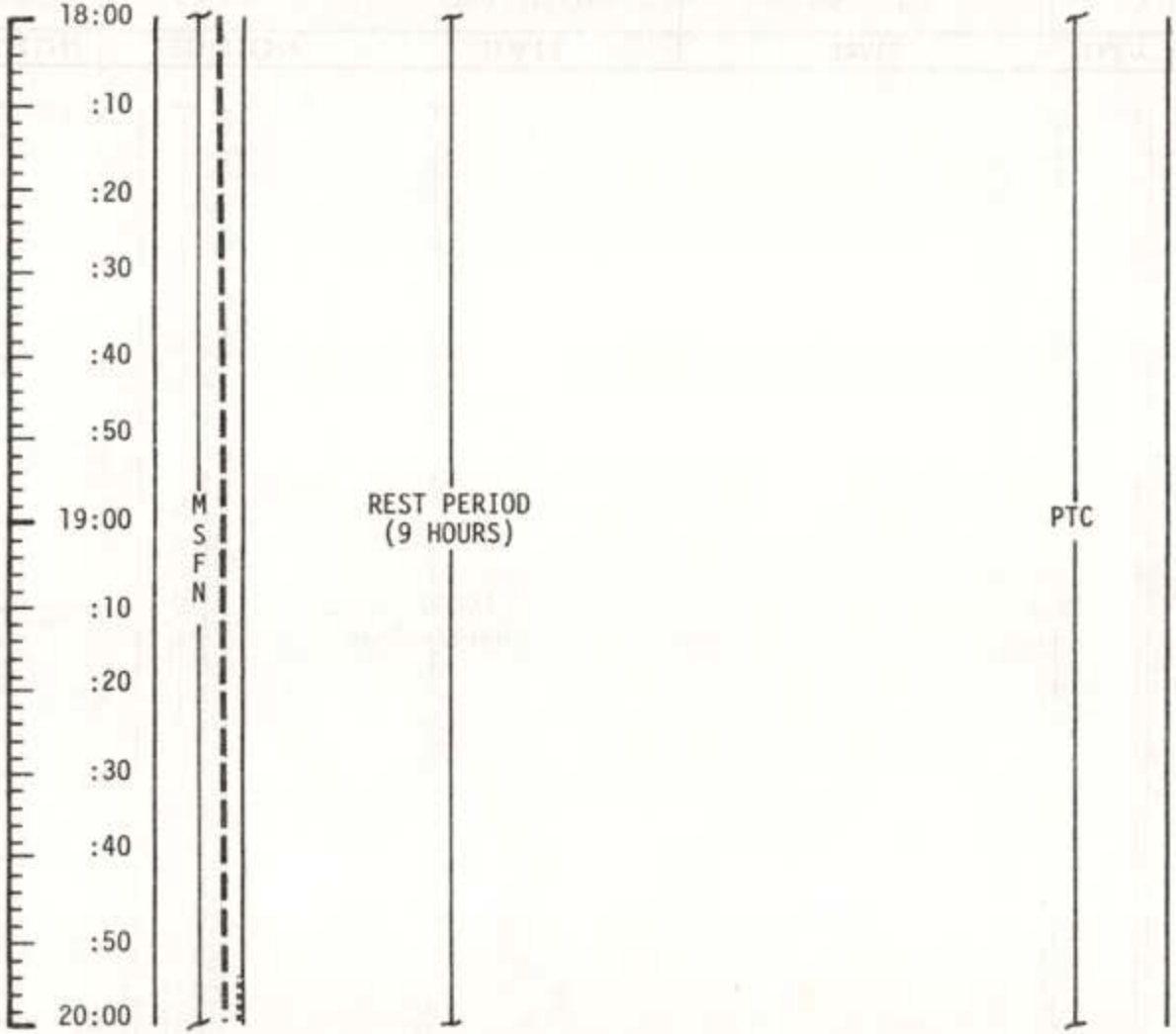
MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

MCC-H

FLIGHT PLAN

NOTES

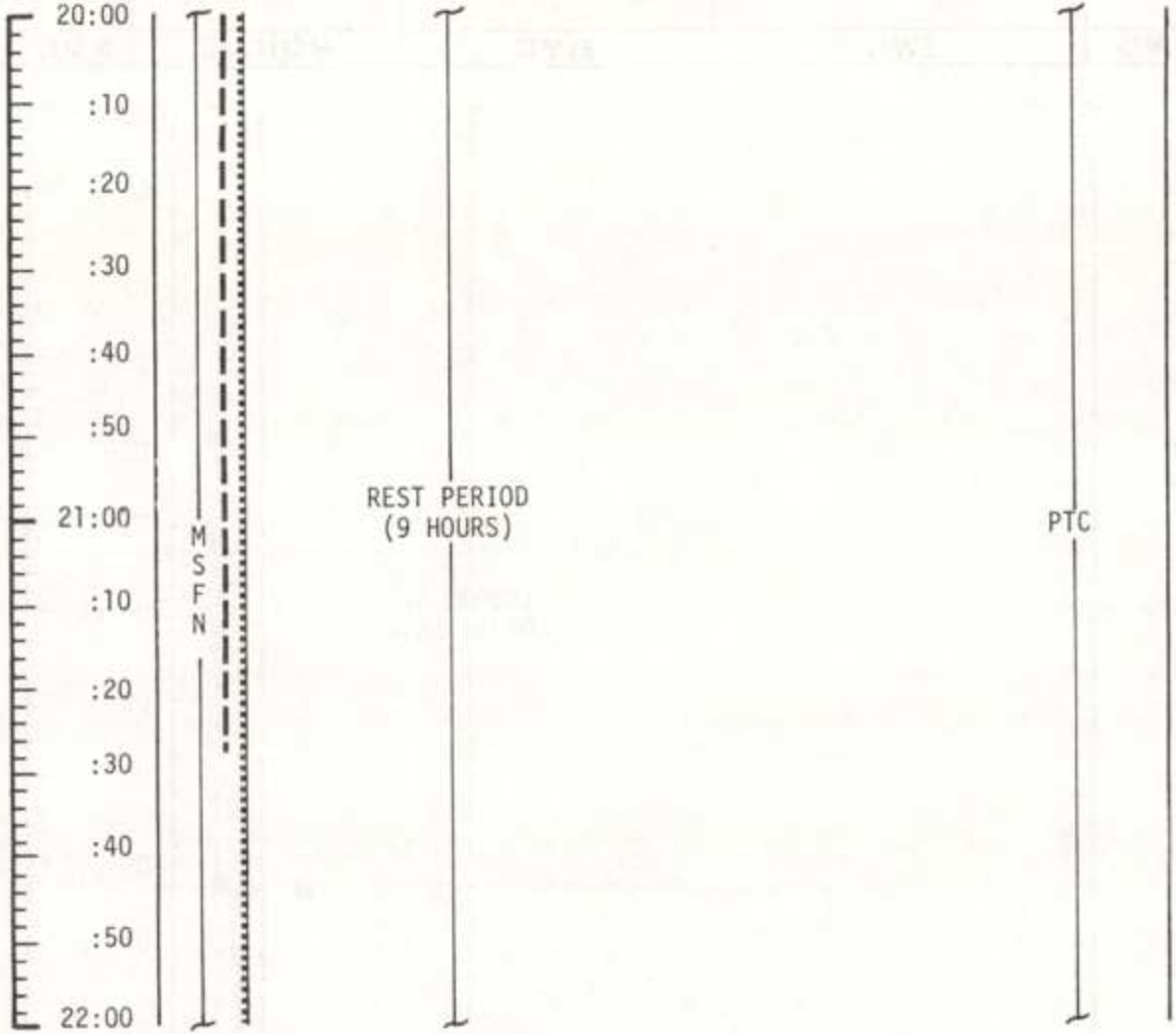


MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	18:00 - 20:00	1/TLC	3-14

MCC-H

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	20:00-22:00	1/TLC	3-15

MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

MCC-H

0730 EDT

NOTES

22:00
:10
:20
:30
:40
:50
23:00
:10
:20
:30
:40
:50
24:00

MSFN

REST PERIOD
(9 HOURS)

CO_2 FILTER CHANGE NO.2
(4 INTO B, STORE 2 IN B5)

EAT PERIOD

UPDATE
CONSUMABLES

POST SLEEP CHECKLIST
CREW STATUS REPORT (SLEEP)
CYCLE O_2 & H_2 FANS
GDC ALIGN TO IMU
REPORT LM/CM ΔP
CONSUMABLES UPDATE
SELECT NORMAL LUNAR
CONFIGURATION EXCEPT:
S-BD AUX TAPE - OFF
TAPE RCDR FWD - OFF
POT H_2O HTR - ON
AUTO RCS JET SELECT (16) - ON

PTC

CONSUMABLE UPDATE
(Δ FROM NOMINAL)
GET: _____
RCS TOT _____
A _____
B _____
C _____
D _____
H_2 TOT _____
O_2 TOT _____

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	22:00 - 24:00	2/TLC	3-16

MCC-H

FLIGHT PLAN

0930 EDT

24:00

:10

:20

:30

:40

:50

25:00

MSFN

O_2 FUEL CELL PURGE

IMU REALIGN - P52
OPTION 3 - REFSMMAT

GDC ALIGN TO IMU

MNVR TO SIGHTING ATT

CISLUNAR NAVIGATION P23

OPTICS CALIBRATION

1. STAR 01 ENH (R3=00110)

2. STAR 02 ENH (R3 00110)

PTC

REPORT:

P52 - PTC REFSMMAT
N71: __ __,__ __
N05: __ __ __.__ __
N93:
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __
GET __ __ __:__ __:__ __

NOTES

3 MARKS ON EACH STAR
INCORPORATE P23 MARK
DATA AND UPDATE
ONBOARD STATE VECTOR
TRN BIAS CALIBRATION
REPEATED UNTIL 2 CKS
AGREE TO WITHIN 0.003°
REPEAT CKS EVERY 30
MIN DURING P23'S

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	24:00 - 25:00	2/TLC	3-17

FLIGHT PLAN

MCC-H

1030 EDT

25:00

:10

:20

:30

:40

:50

26:00

MSFN

UPLINK CMC
- CSM STATE VECTOR
- MCC2 TGT LOAD

UPDATE
- MCC2 PAD DATA

3. STAR 44 EFH (R3=00120)

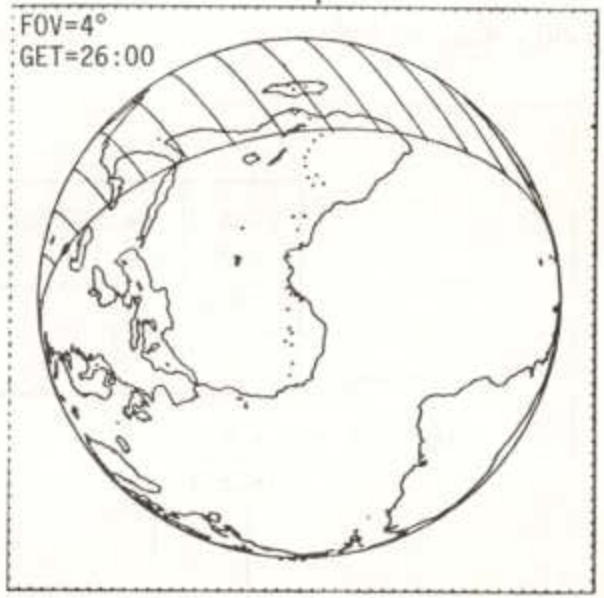
4. STAR 44 EFH (R3=00120)

5. STAR 45 EFH (R3=00120)

V66 - TRANS CSM STATE VECTOR
TO LM SLOT

RECORD MCC2 MNVR PAD

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	25:00 - 26:00	2/TLC	3-18

MCC
BURN CHART

	P OR Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
MCC2	10°/SEC TAKEOVER	±10° TAKEOVER	BT + 1 SEC	TRIM X AXIS ONLY (UNLESS X > 2 FPS)

MCC-H

FLIGHT PLAN

1130 EDT

26:00

:10

:20

26:30

:40

TLI + 24 HRS

:50

27:00

MSFN

EXT ΔV - P30

SPS/RCS THRUST - P40/41

MNVR TO BURN ATT

SXT STAR CK

EMS ΔV TEST

SM RCS MON CK

GDC ALIGN TO IMU

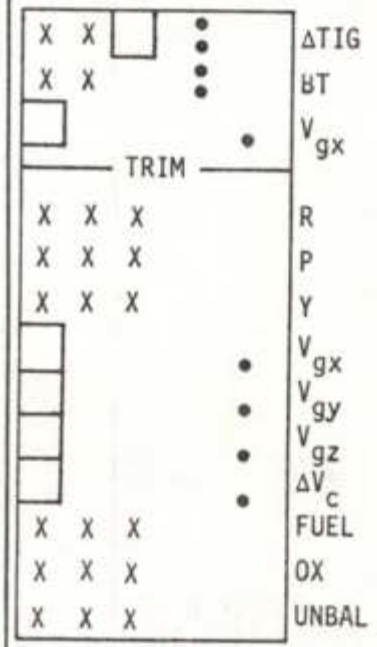
MCC2 ΔV=NOMINALLY ZERO

SM RCS MON
SPS MON CK
MCC2 BURN STATUS REPORT

V66 - TRANS CSM STATE VECTOR TO LM SLOT

NOTES

BURN STATUS REPORT



~~MCC2 WILL BE PERFORMED IF ΔV WOULD EXCEED 25 FPS IF DELAYED TO MCC3 (LOI-22 HRS)~~

TO AVOID EXECUTION OF MCC3, A NON FREE MCC2 WILL BE EXECUTED (SPS BURN IF THE BT >3 SEC)

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	26:00 - 27:00	2/TLC	3-19

FLIGHT PLAN

MCC-H

UPLINK
EARTH HORIZON BIAS (ΔH) (IF REQUIRED)

1230 EDT

27:00
:10
:20
:30
:40
:50
28:00
:10
:20
:30
:40
:50
29:00

MSFN

EAT PERIOD

START PTC
P 90 Y 0

PTC

NOTES

THE EARTH HORIZON BIAS WILL BE UPDATED TO THE CMC IF THE DIFFERENCE BETWEEN THE SIGHTING ΔH IS ≥8.3KM

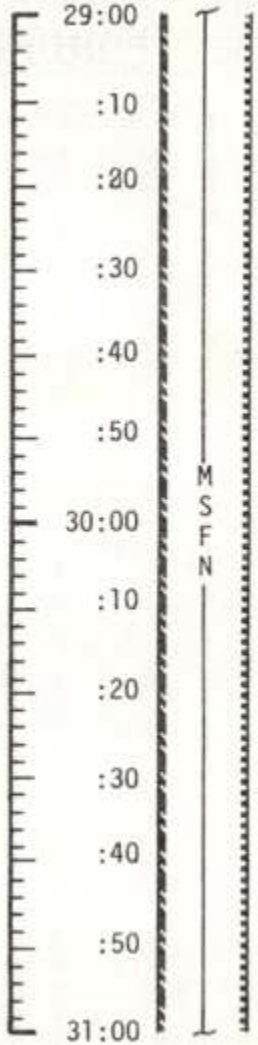
PTC ESTABLISHED IN G&N P, Y +30° DB, R RATE OF 0.3°/SEC

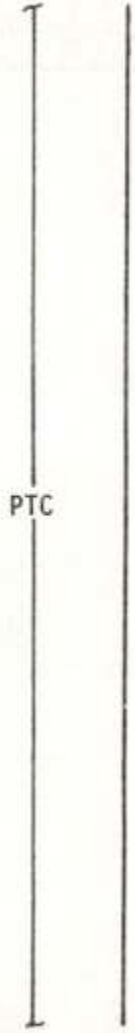
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	27:00 - 29:00	2/TLC	3-20

MCC-H

FLIGHT PLAN

NOTES



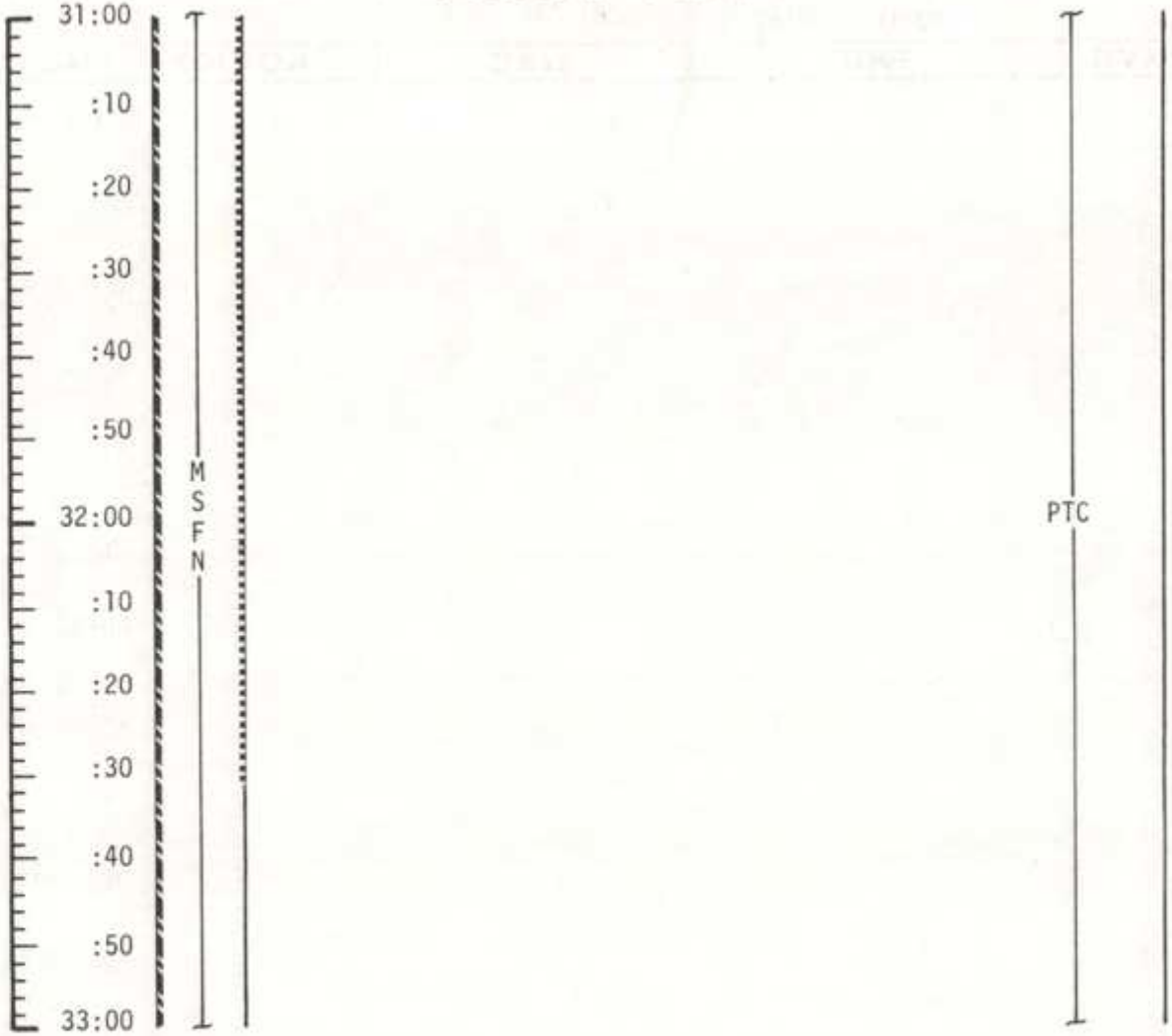


MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	29:00 - 31:00	2/TLC	3-21

MCC-H

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	31:00 - 33:00	2/TLC	3-22

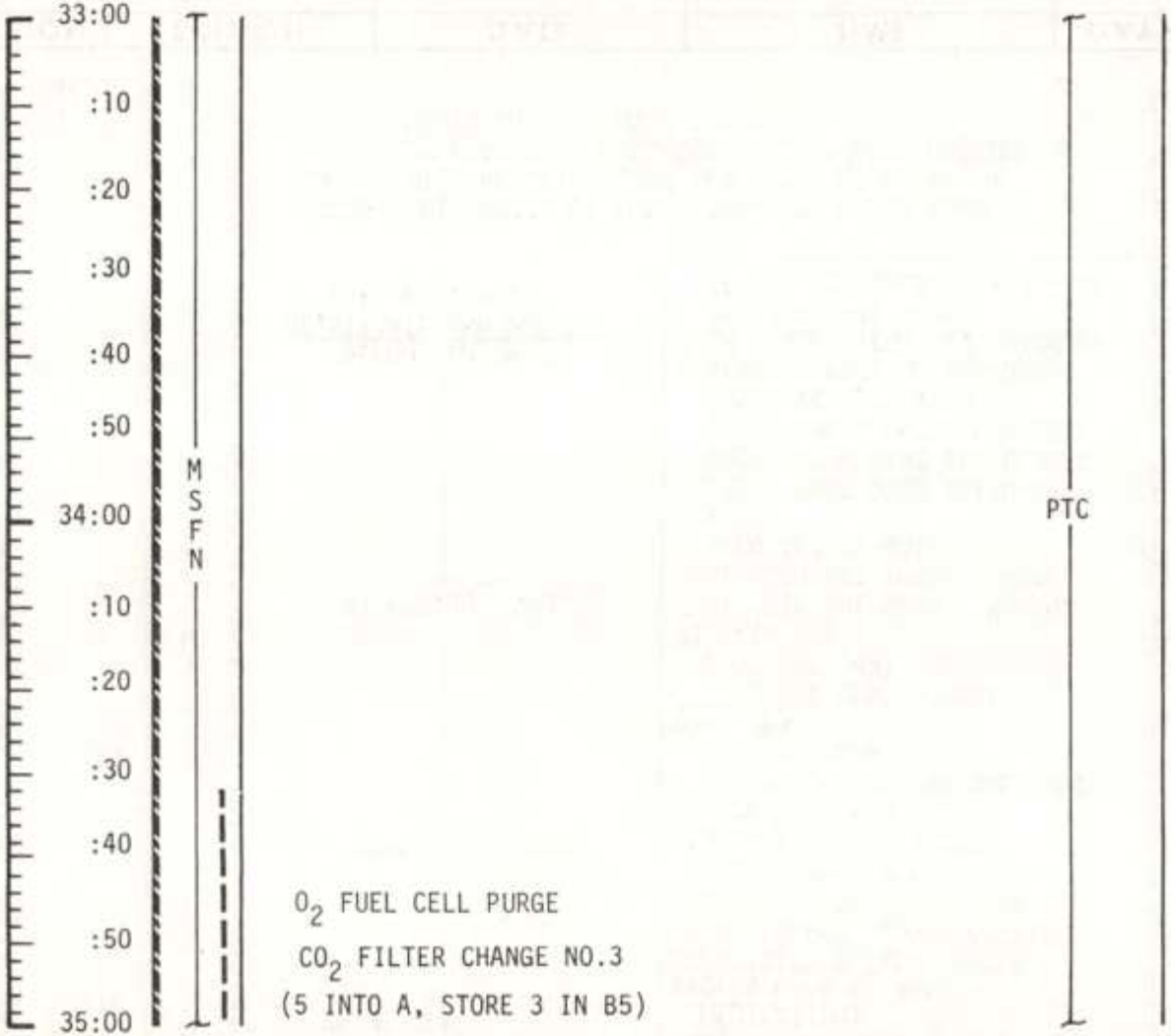
MSC Form 29 (May 68)

FLIGHT PLANNING BRANCH

MCC-H

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	33:00 - 35:00	2/TLC	3-23

FLIGHT PLAN

MCC-H

NOTES

(2030 EDT)

UPDATE BLOCK DATA

35:00 :10 :20 :30 :40 :50 36:00 :10 :20 :30 :40 :50 37:00

MSFN

RECORD BLOCK DATA-
LOI-5 FLYBY TO
PRIME CLA

EAT PERIOD - ALL

REPORT LM/CM ΔP
REINITIATE CSM PURGE
(IF REQUIRED)

PRESLEEP CHECKLIST

CREW STATUS REPORT (RADIATION, MEDICATION)
CYCLES O_2 & H_2 FANS
CHLORINATE POTABLE WATER
SELECT NORMAL LUNAR CONFIG
EXCEPT: (FOR COAST ASLEEP)
- S-BD NORMAL MODE VOICE - OFF
- S-BD AUX TAPE - OFF
- TAPE RCDR FWD - OFF
- GO TO HGA OR CONTINUE OMNI OPS PER MSFN

OMNI OPS
- S-BD ANT OMNI - OMNI
- S-BD ANT OMNI - B

HI GAIN OPS
- HI GAIN ANT BEAM - NARROW
- HI GAIN ANT TRACK - REACQ
- S-BD ANT-HI GAIN

VERIFY:
- WASTE MNGT OVBD DRAIN-OFF
- WASTE STOW VENT VLV-CLOSED
- EMERG CABIN PRESS VLV-BOTH
- SURGE TK O_2 VLV-ON
- REPRESS PACK O_2 VLV-OFF
- LM TUNNEL VENT VLV-LM/CM ΔP
- POT H_2O HTR - OFF

AUTO RCS JET SELECT (16) - OFF

NOTE: THE LENGTH OF THE SECOND CSM CABIN PURGE WILL BE DETERMINED REAL TIME BASED ON THE LM LEAK RATE ENSURING LM O_2 PURITY REQUIREMENTS ON THE LUNAR SURFACE

ONBOARD READOUT

BAT C _____
PYRO BAT A _____
PYRO BAT B _____

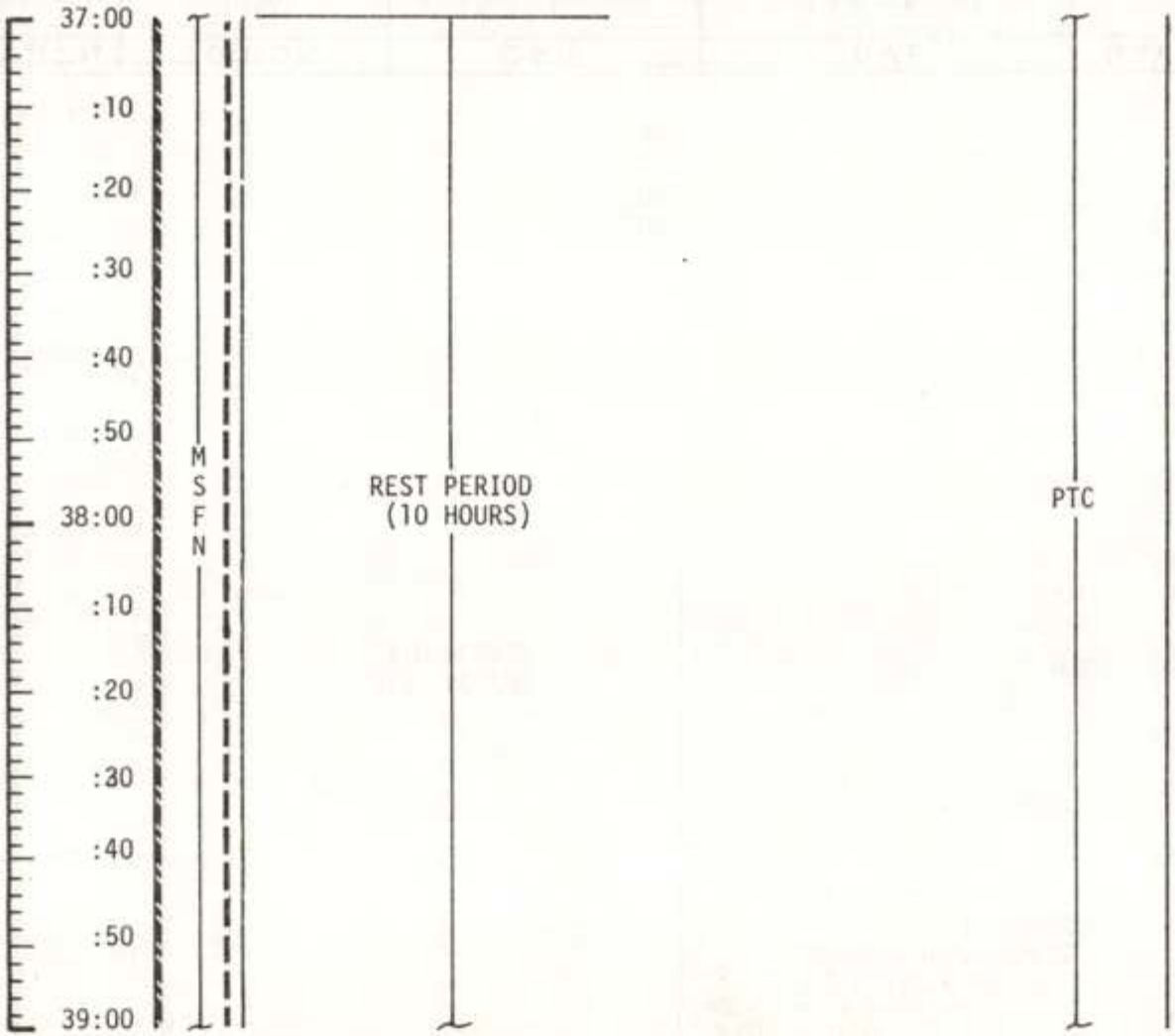
RCS A _____
B _____
C _____
D _____

DC IND SEL TO MNA OR MNB

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	35:00 - 37:00	2/TLC	3-24

MCC-H

FLIGHT PLAN



NOTES

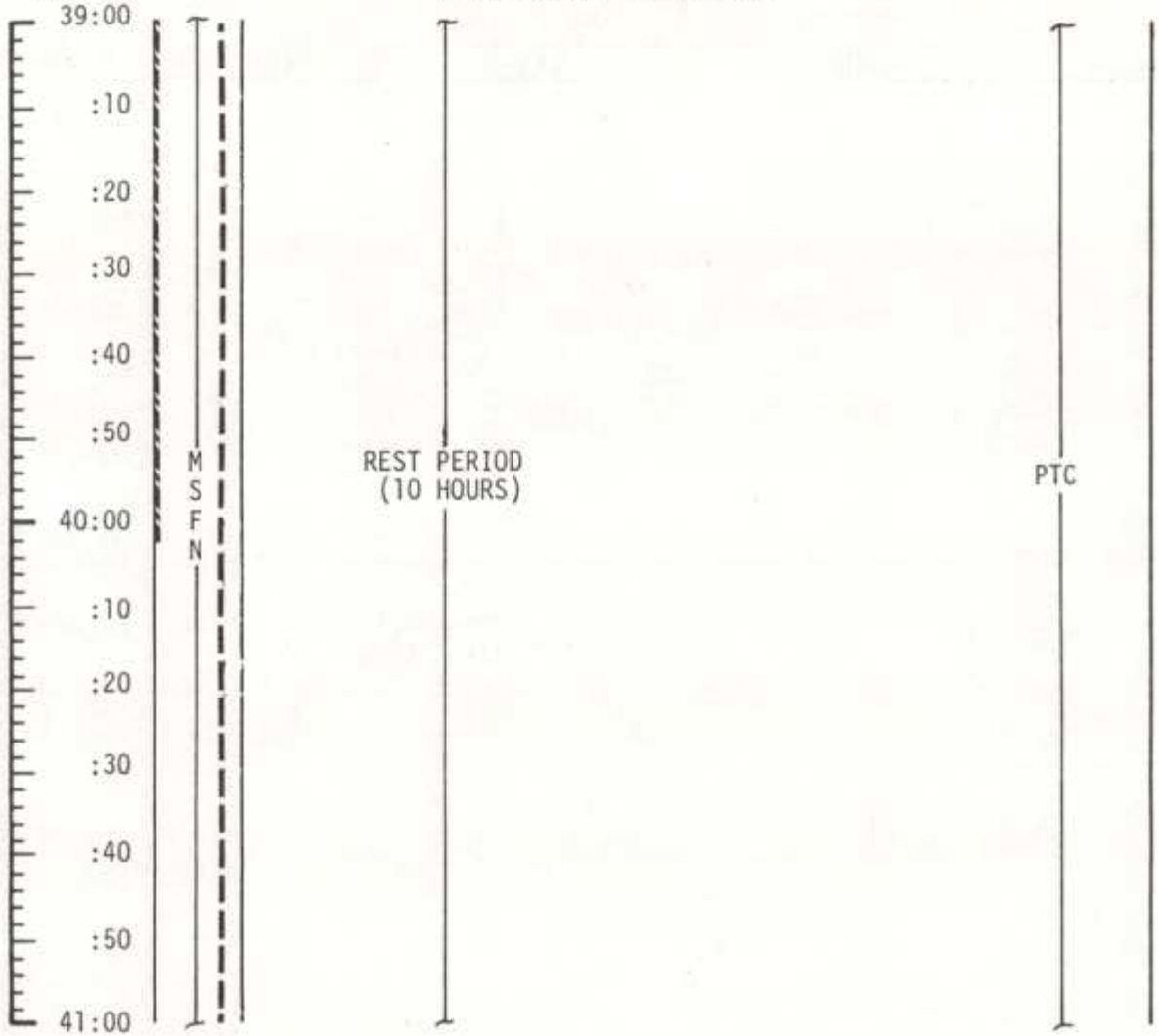
DURING REST PERIOD 2 CREWMEN IN REST STATION, 1 IN LEFT COUCH

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	37:00 - 39:00	2/TLC	3-25

MCC-H

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	39:00 - 41:00	2/TLC	3-26

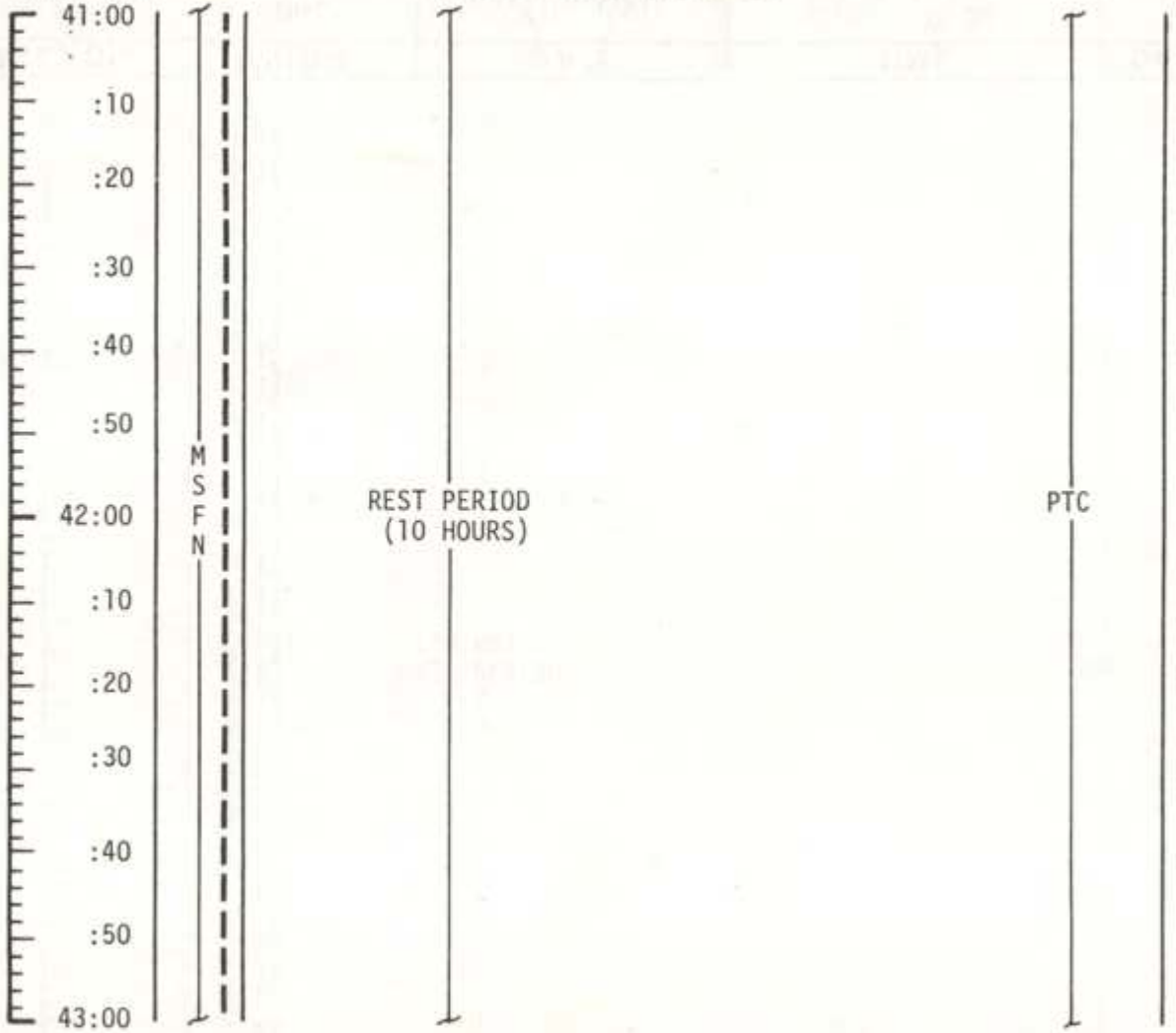
MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

MCC-H

FLIGHT PLAN

NOTES



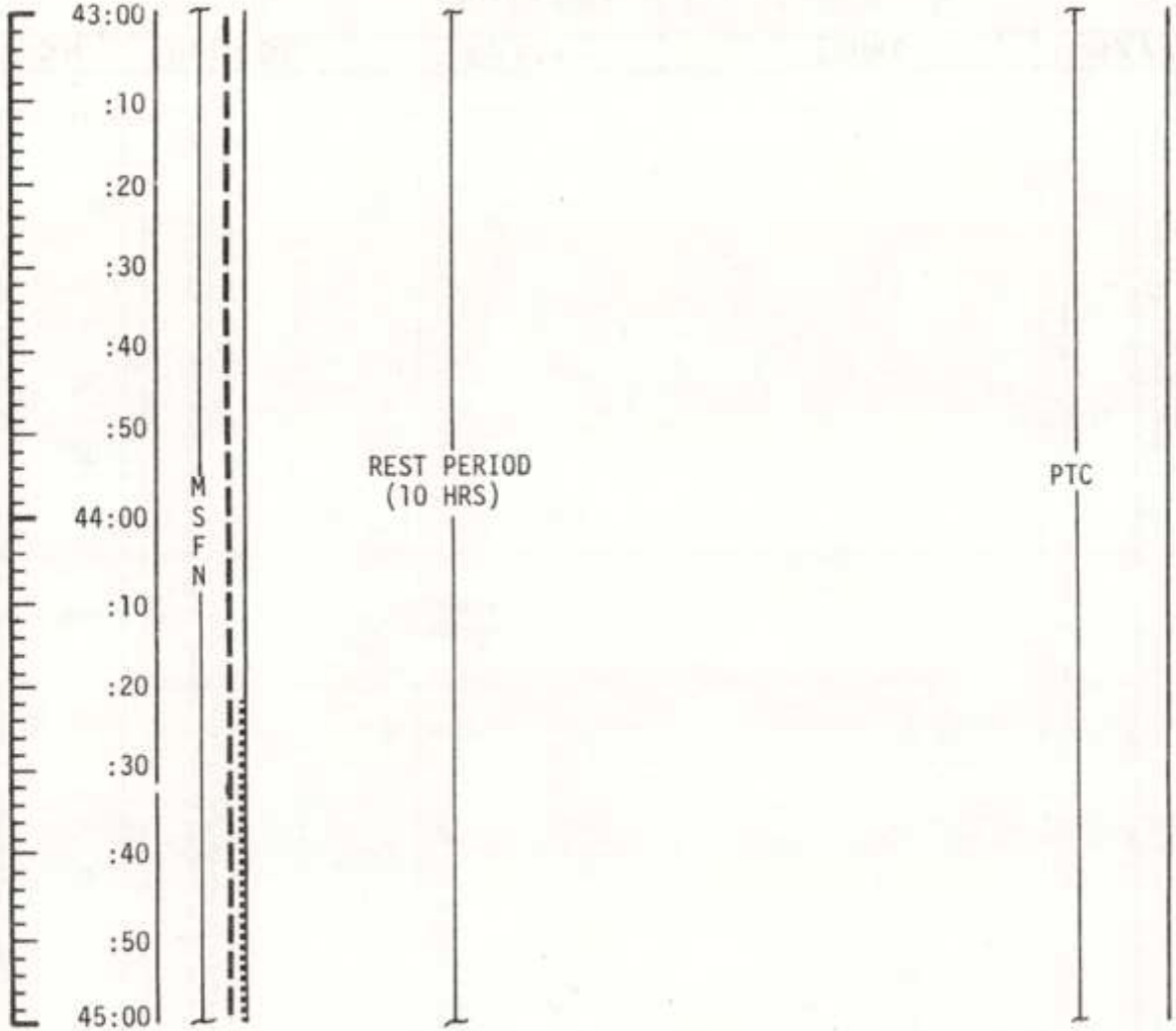
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	41:00 - 43:00	2/ILC	3-27

MCC-H

0430 EDT

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	43:00 - 45:00	2/TLC	3-28

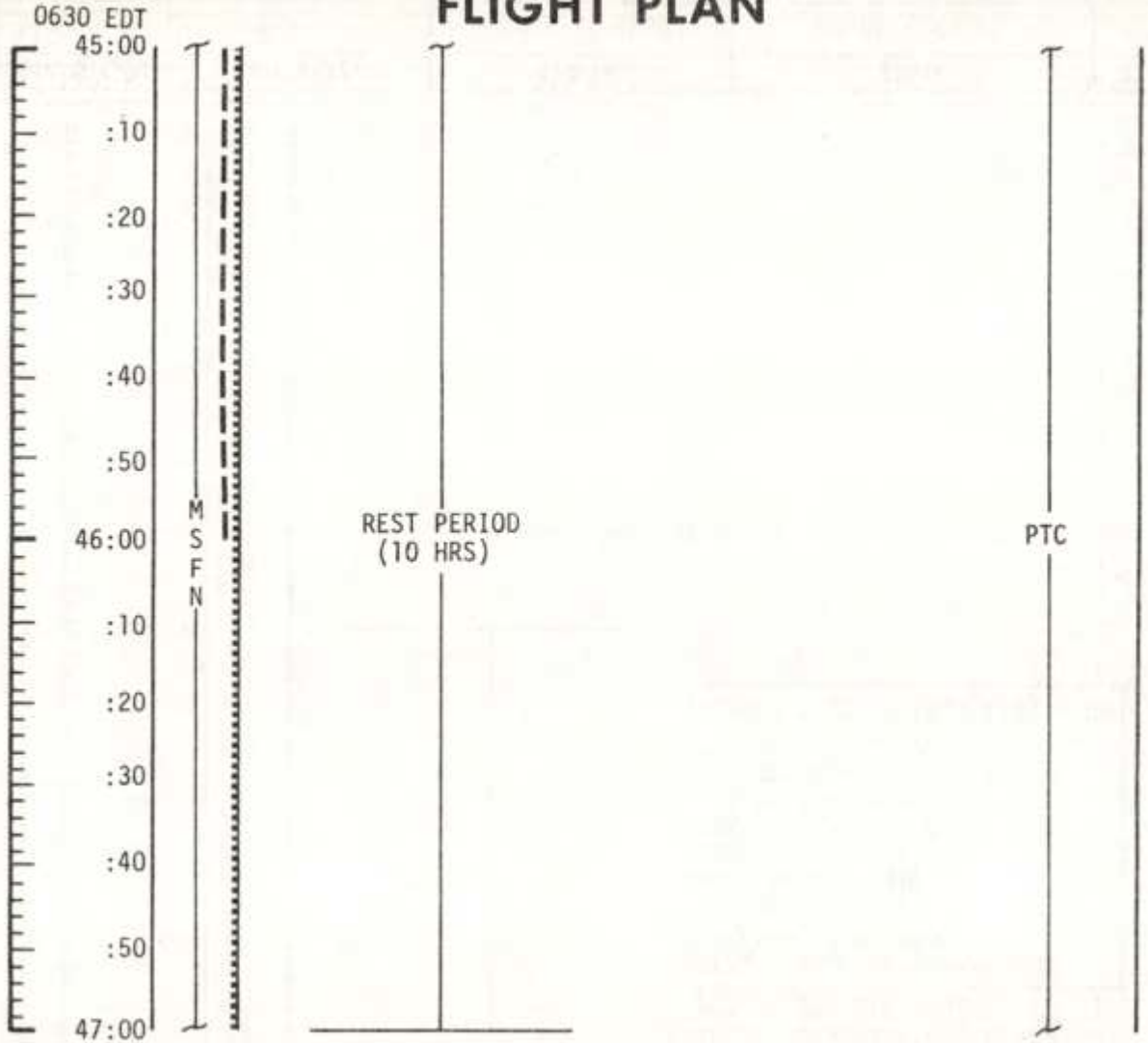
MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

MCC-H

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	45:00 - 47:00	2/TLC	3-29

FLIGHT PLAN

MCC-H

0830 EDT

47:00
:10
:20
:30
:40
:50
48:00
:10
:20
:30
:40
:50
49:00

MSFN

UPDATE
CONSUMABLES

EAT PERIOD-ALL

CO_2 FILTER CHANGE NO.4
(6 INTO B, STORE 4 IN B5)
POST SLEEP CHECKLIST

CREW STATUS REPORT(SLEEP)
CYCLE O_2 & H_2 FANS
GDC ALIGN TO IMU
CONSUMABLES UPDATE
SELECT NORMAL LUNAR
CONFIGURATION EXCEPT:
S-BD AUX TAPE - OFF
TAPE RCDR FWD - OFF
POT H_2O HTR - ON
AUTO RCS JET SELECT (16) - ON

BATTERY CHARGE, BATTERY B

PTC

NOTES

CONSUMABLES UPDATE
(Δ FROM NOMINAL)

GET: _____

RCS TOT _____
A _____
B _____
C _____
D _____

H_2 TOT _____

O_2 TOT _____

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	47:00 - 49:00	3/TLC	3-30

MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

MCC-H

FLIGHT PLAN

NOTES

1030 EDT

49:00
:10
:20
:30
:40
:50
50:00
:10
:20
:30
:40
:50
51:00

MSFN

PTC

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	49:00 - 51:00	3/TLC	3-31

FLIGHT PLAN

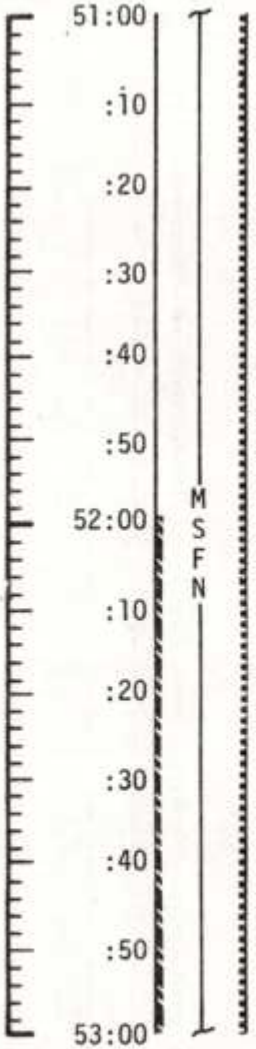
MCC-H

1230 EDT

NOTES

UPLINK
- CSM STATE VECTOR
- MCC 3 TGT LOAD

UPDATE
- MCC 3 MNVR PAD



V66 TRANSFER CSM STATE VECTOR TO LM SLOT

RECORD MCC3 MNVR PAD
H_2 PURGE LINE HTRS-ON

O_2 & H_2 FUEL CELL PURGE

EAT PERIOD-ALL

PTC

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	51:00 - 53:00	3/TLC	3-32

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MCC
BURN CHART

	P OR Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
MCC3	10°/SEC TAKEOVER	+10° TAKEOVER	BT + 1 SEC	TRIM X AXIS ONLY (UNLESS X > 2 FPS)

MCC-H

FLIGHT PLAN

1430 EDT

53:00
:10
:20
53:30
:40
:50
LOI -22 HRS
54:00

MSFN

IMU REALIGN - P52
OPTION 3 - REFSMMAT

EXT ΔV - P30

SPS/RCS THRUST - P40/41

MNVR TO BURN ATT

SXT STAR CK (STOW OPTICS)

EMS ΔV TEST

SM RCS MON CK

GDC ALIGN TO IMU

MCC3 ΔV=NOMINALLY ZERO

SM RCS MON CK
SPS MON CK

BURN STATUS REPORT

X X ☐ ●●●	ΔTIG
X X ●●●	BT
☐ ●	V_{gx}
TRIM	
X X X	R
X X X	P
X X X	Y
☐ ●	V_{gx}
☐ ●	V_{gy}
☐ ●	V_{gz}
☐ ●	ΔV_c
X X X	FUEL
X X X	OX
X X X	UNBAL

P52 (PTC REFSMMAT)
N71: __ __,__ __
N05: __ __ __.__ __
N93:
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __
GET __ __ __:__ __:__ __

NOTES

MCC3 WILL BE DELAYED TO MCC4 IF PROPELLANT COST IS NOT PROHIBITIVE

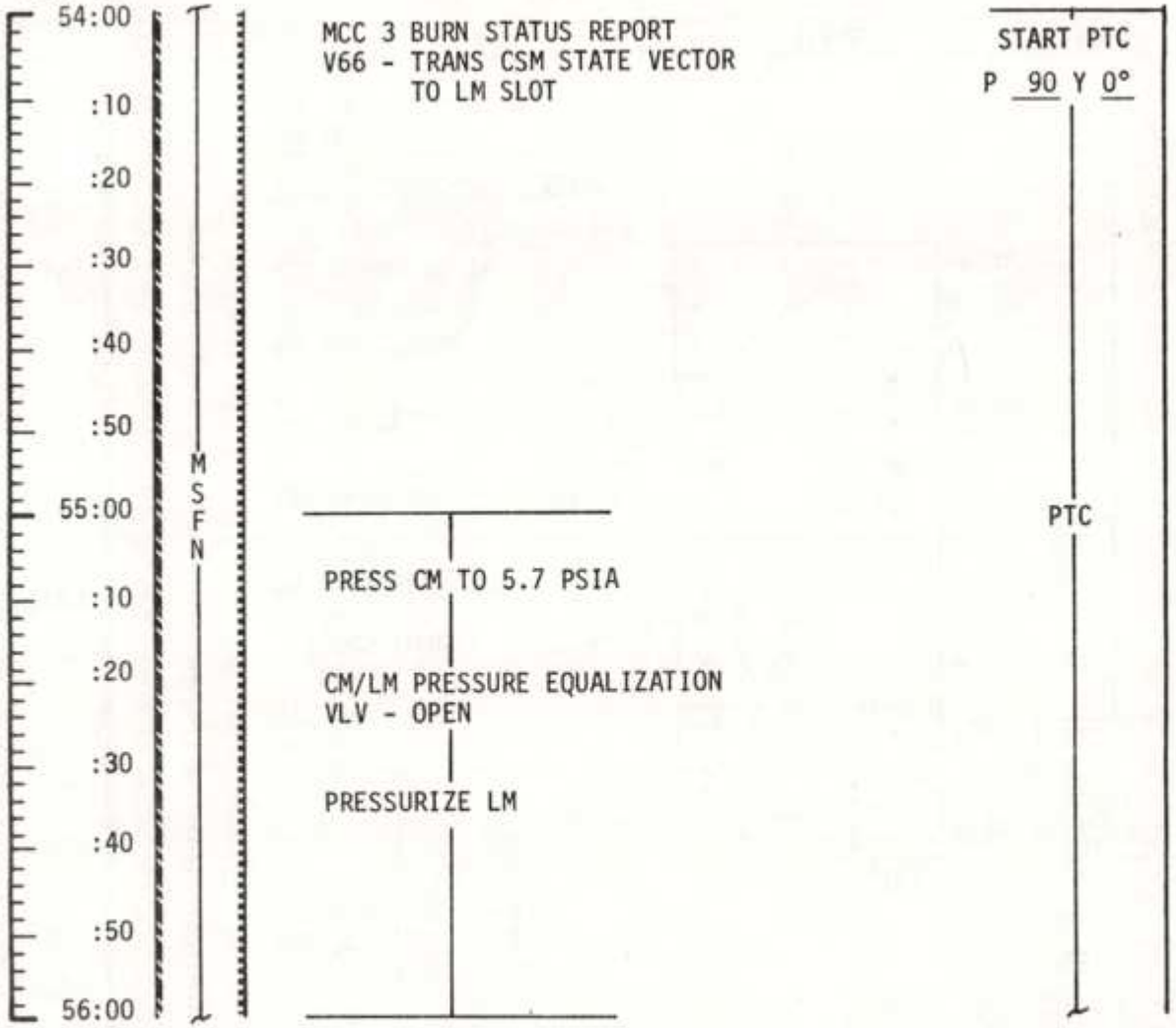
~~MCC3 WILL BE EXECUTED IF ΔV > 3 FPS AND IF LOI1 CANNOT BE TARGETED TO CORRECT THE TLC DISPERSIONS.~~

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	53:00 - 54:00	3/TLC	3-33

MCC-H

FLIGHT PLAN

NOTES

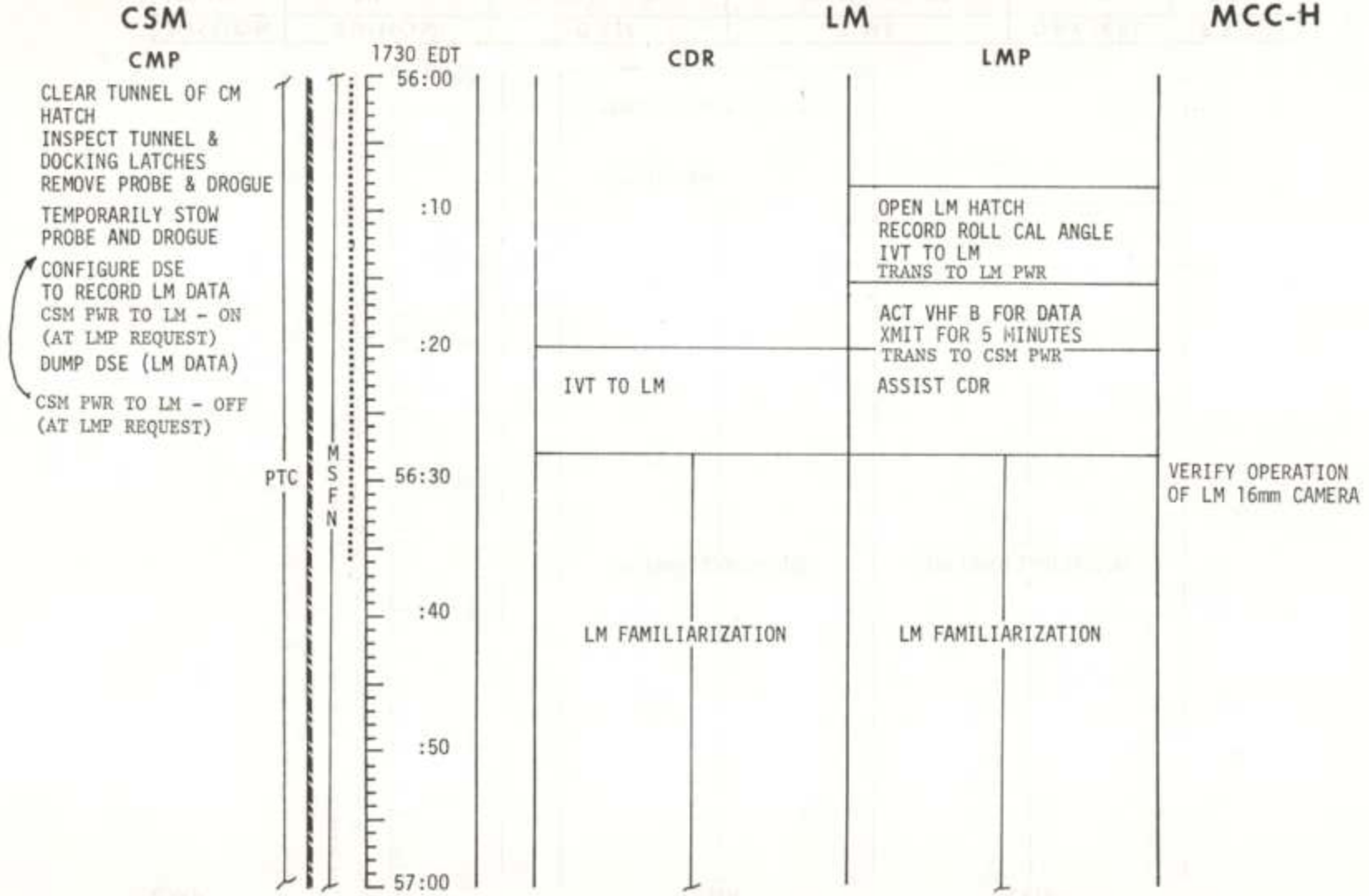


MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	54:00 - 56:00	3/TLC	3-34

MSC Form 29 (May 69)

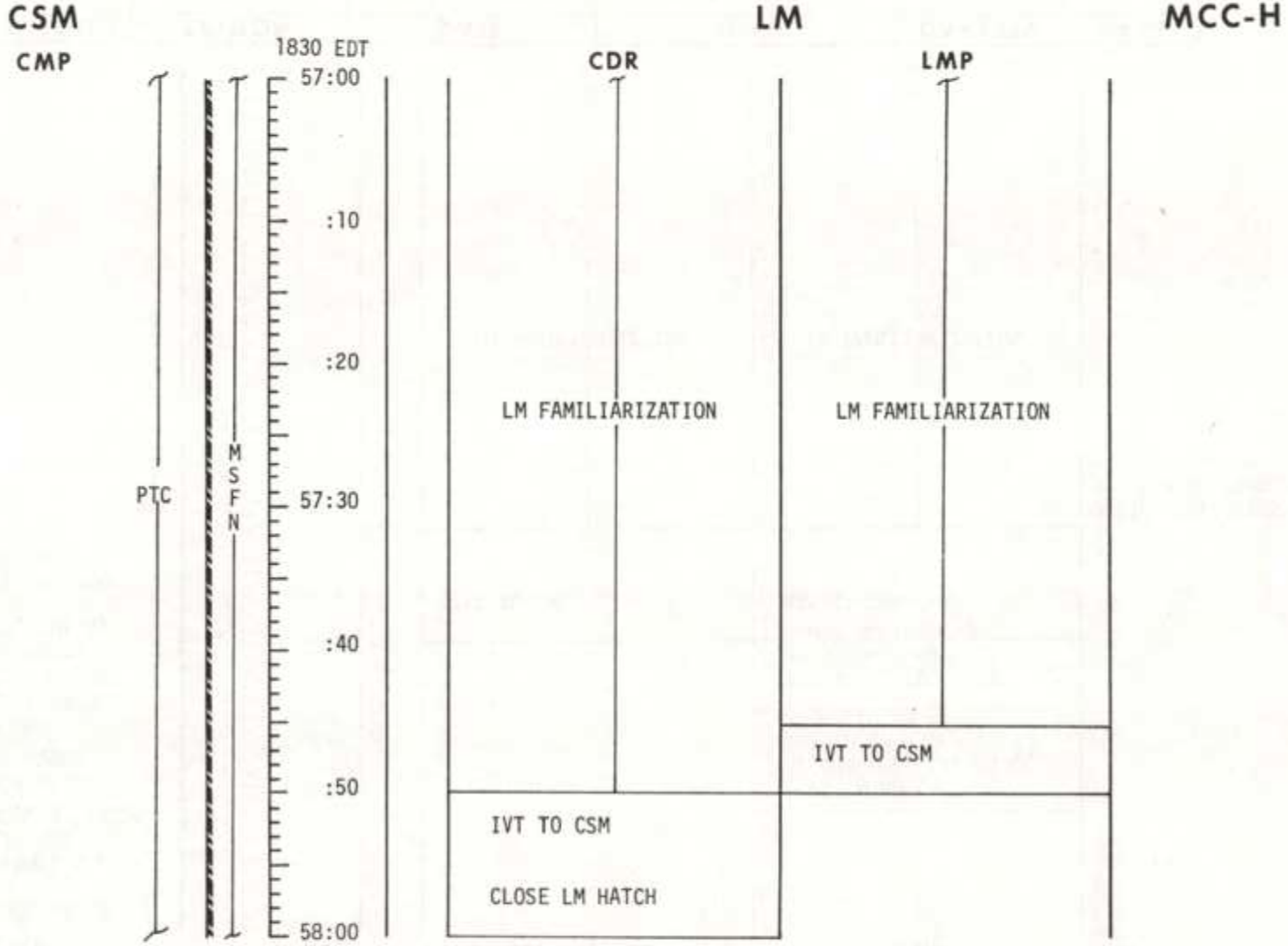
FLIGHT PLANNING BRANCH

FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	56:00 - 57:00	3/TLC	3-35

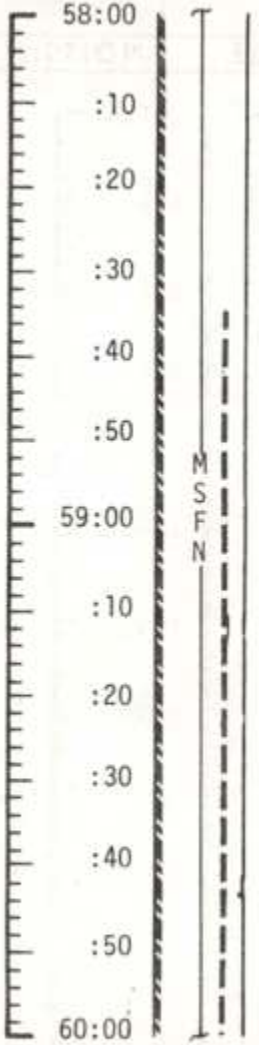
FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	57:00 - 58:00	3/TLC	3-36

MCC-H

FLIGHT PLAN



MSFN

INSTALL PROBE AND DROGUE
INSTALL CM HATCH
LM TUNNEL VENT VALVE - LM/CM ΔP

CO_2 FILTER CHANGE NO.5
(7 INTO A, STORE 5 IN B6)

EAT PERIOD-ALL

PTC

NOTES

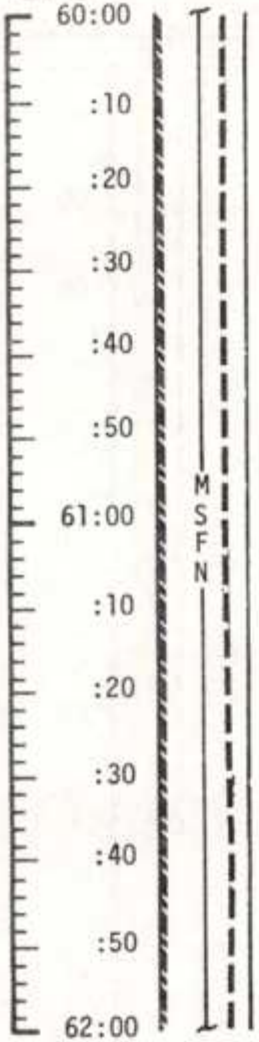
180,000 NM from EARTH

ONBOARD READOUT
BAT C _____
PYRO BAT A _____
PYRO BAT B _____
RCS A _____
B _____
C _____
D _____
DC IND SEL TO MNA
OR MNB

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	58:00 - 60:00	3/TLC	3-37

MCC-H

FLIGHT PLAN



PRESLEEP CHECKLIST

CREW STATUS REPORT (RADIATION & MEDICATION)
CYCLE O_2 & H_2 FANS

CHLORINATE POTABLE WATER
SELECT NORMAL LUNAR CONFIGURATION EXCEPT:
- S-BD NORMAL MODE
- VOICE - OFF
- S-BD SQUELCH - ENABLE
- S-BD AUX TAPE OFF
- TAPE RCDR FWD - OFF
- GO TO HGA OR CONTINUE OMNI OPS PER MSFN

OMNI OPS
- S-BD ANT OMNI - OMNI
- S-BD ANT OMNI - B

HI GAIN OPS
- HI GAIN ANT BEAM - NARROW
- HI GAIN ANT TRACK - REAQ
- S-BD ANT - HI GAIN

VERIFY:
- WASTE MNGT OVBD DRAIN - OFF
- WASTE STOW VENT VLV - CLOSED
- EMERG CABIN PRESS VLV - ON
- REPRESS PACK O_2 VLV - OFF
- LM TUNNEL VENT VLV - LM/CM ΔP
- POT H_2O HTR - OFF

AUTO RCS JET SELECT (16) - OFF

PTC

NOTES

DURING REST PERIOD 2 CREWMAN IN REST STATIONS, 1 IN LEFT COUCH.

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	60:00 - 62:00	3/TLC	3-38

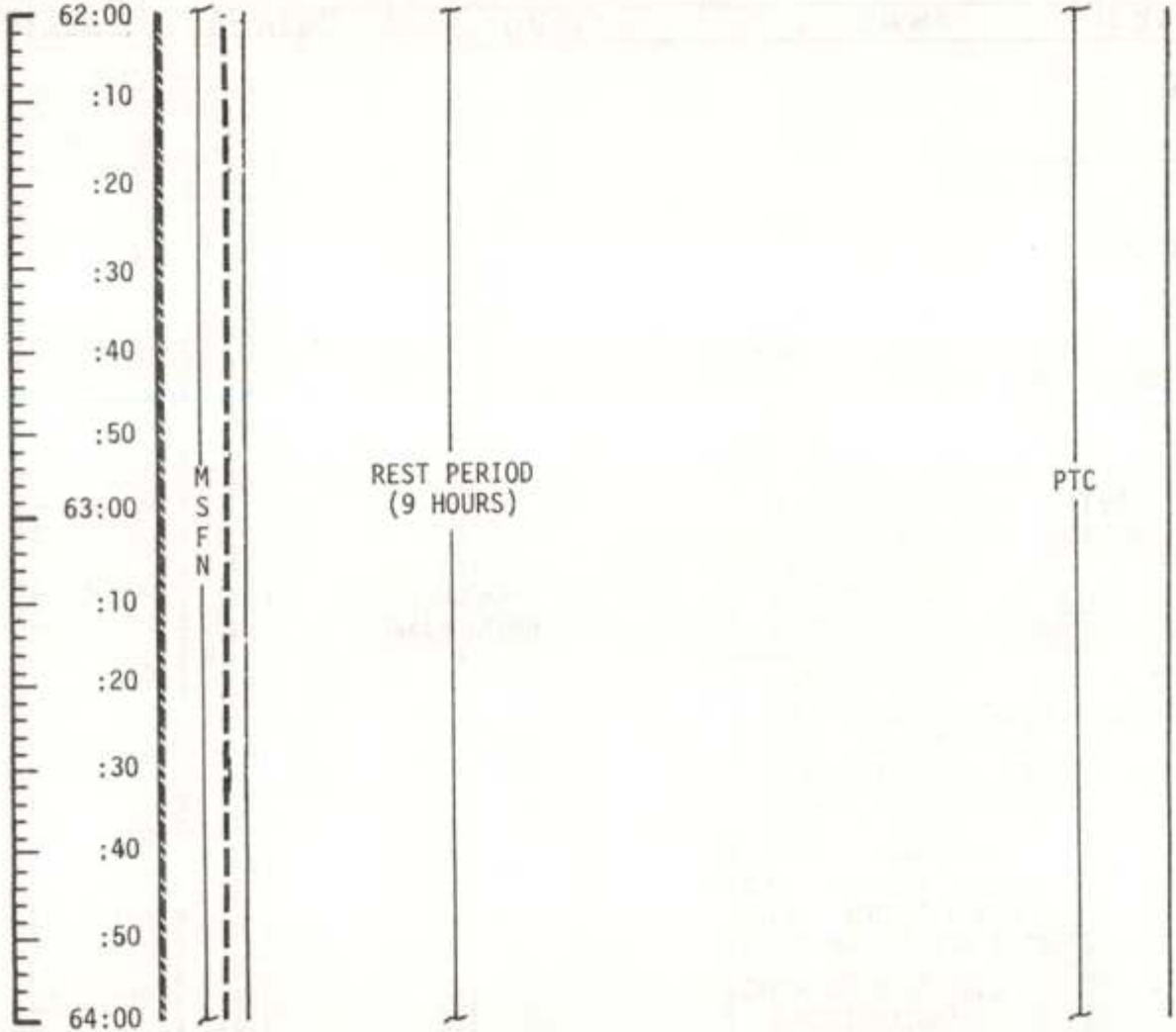
MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

MCC-H

FLIGHT PLAN

NOTES



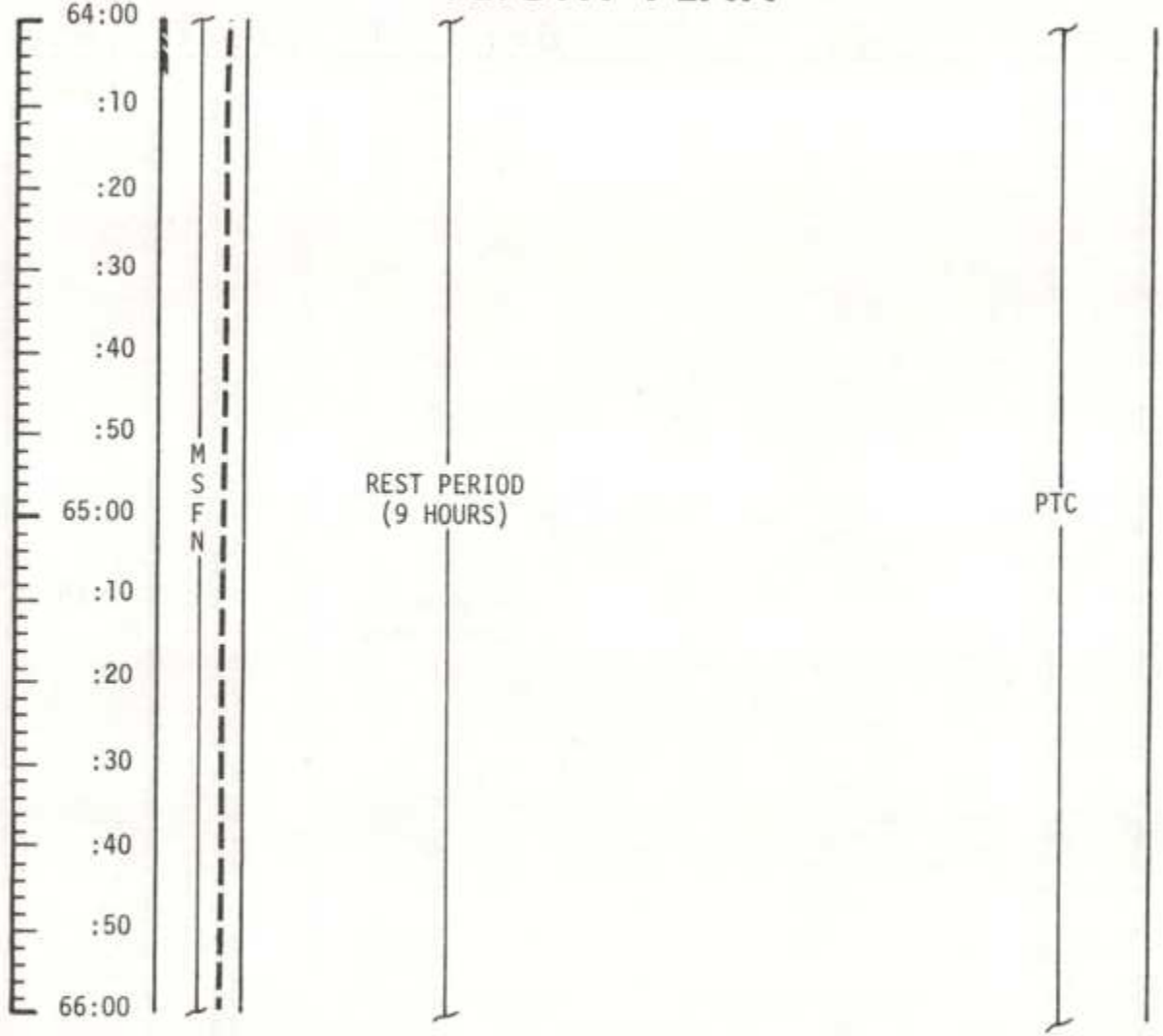
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	62:00 - 64:00	3/TLC	3-39

MCC-H

0130 EDT

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	64:00 - 66:00	3/TLC	3-40

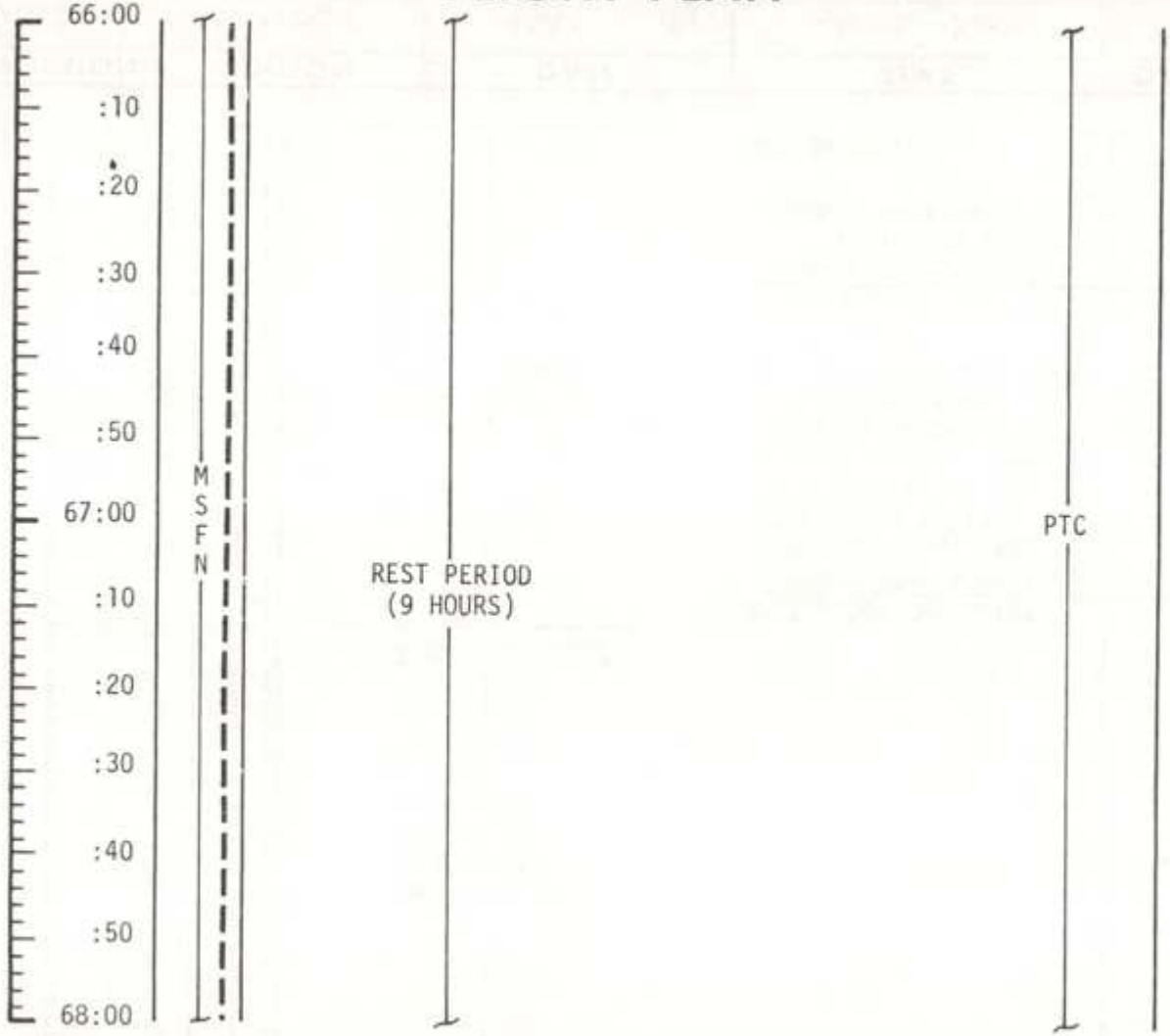
MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

MCC-H

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	66:00 - 68:00	3/TLC	3-41

MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

MCC-H

0530 EDT

NOTES

68:00
:10
:20
:30
:40
:50
69:00
:10
:20
:30
:40
:50
70:00

MSFN

REST PERIOD
(9 HOURS)

O_2 FUEL CELL PURGE

EAT PERIOD-ALL

UPLINK CMC
- CSM STATE VECTOR
- MCC4 TGT LOAD
- DESIRED ORIENTATION (LDG SITE)

UPDATE
- MCC4 MNVR PAD

PTC

POST SLEEP CHECKLIST
- CREW STATUS REPORT (SLEEP)
- CYCLE O_2 & H_2 FANS
- GDC ALIGN TO IMU
- CONSUMABLES UPDATE
- SELECT NORMAL LUNAR CONFIGURATION EXCEPT:
  - S-BD AUX TAPE - OFF
  - TAPE RCDR FWD - OFF
- POT H_2O HTR - ON
- AUTO RCS JET SELECT (16) - ON

V66-TRANS CSM STATE VECTOR TO LM SLOT
RECORD MCC4 MNVR PAD

PTC

UNSTOW OPTICS

CONSUMABLES REPORT
(Δ FROM NOMINAL)

GET: _____

RCS TOT _____
- A _____
- B _____
- C _____
- D _____

H_2 TOT _____

O_2 TOT _____

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	68:00 - 70:00	3/TLC	3-42

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MCC
BURN CHART

	P OR Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
MCC4	10°/SEC TAKEOVER	+10° TAKEOVER	BT + 1 SEC	TRIM X AXIS ONLY

FLIGHT PLAN

MCC-H

UPDATE
BLOCK DATA

0730 EDT
70:00
:10
:20
70:30
:40
:50
LOI
-5 HRS
70:55
71:00

MSFN

IMU REALIGN - P52
(OPTION 1 - PREFERRED)

EXT ΔV - P30

SPS/RCS THRUST P40/41

MNVR TO BURN ATT

SXT STAR CK

EMS ΔV TEST

SM RCS MON CK

GDC ALIGN TO IMU

MCC 4 ΔV=NOMINALLY ZERO

SM RCS MON CK

RECORD BLOCK DATA-
PC + 2 HRS FAST
RETURN TO ANY CLA

BURN STATUS REPORT

X	X		•	ΔTIG
X	X		•	BT
			•	V_{gx}
		TRIM		
X	X	X		R
X	X	X		P
X	X	X		Y
			• •	V_{gx}
			•	V_{gy}
			•	V_{gz}
			•	ΔV_c
X	X	X		FUEL
X	X	X		OX
X	X	X		UNBAL

NOTES

REPORT:

P52 (LDG SITE REFSMMAT)
N71: __ __,__ __
N05: __ __ __.__ __
N93:
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __
GET __ __ __:__ __:__ __

MCC3 WILL BE EXECUTED WITH THE SPS IF THE BT >3 SEC

~~MCC 4 WILL BE EXECUTED ONLY IF LOI, CANNOT BE TARGETED TO CONNECT MCC 3 DISPERSIONS~~

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	70:00 - 71:00	3/TLC	3-43

MCC-H

FLIGHT PLAN

NOTES

0830 EDT

71:00
:10
:20
:30
:40
:50
72:00
:10
:20
:30
:40
:50
73:00

MSFN

SPS MON CK
MCC 4 BURN STATUS REPORT
V66 - TRANS CSM STATE VECTOR TO LM SLOT

CO_2 FILTER CHANGE NO.6
(8 INTO B, STORE 6 IN B6)

~~PRE-LOI ECS REDUNDANT COMPONENT CK~~ SECONDARY RADIATOR LEAK CHECK

ACTIVATE PRIMARY EVAPORATOR

FOV=3°
GET=72:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	71:00 - 73:00	4/TLC	3-44

FLIGHT PLAN

MCC-H

1030 EDT

UPLINK CMC
CSM STATE VECTOR
LOI_1 TGT LOAD

UPDATE CSM
LOI_1 MNVR PAD

73:00

V66 TRANSFER CSM STATE VECTOR TO LM SLOT

COPY LOI_1 P30 MANEUVER PAD

73:30

MSFN

IMU REALIGN - P52
AND DRIFT CK
OPTION 3 - REFSMMAT

73:52

UPDATE CSM
BLOCK DATA

74:00

COPY BLOCK DATA (TEI_1 & TEI_4)

NOTES

TEI_1 BLOCK DATA ASSUM
LOI_1 ACCOMPLISHED
TEI_4 ASSUMES LOI_1
ACCOMPLISHED BUT
NO LOI_2

REPORT:

P52 (LDG SITE REFSMMAT)
N71: __ __,__ __
N05: __ __ __.__ __
N93:
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __
GET __ __ __:__ __:__

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	73:00 - 74:00	4/TLC	3-45

MCC-H

FLIGHT PLAN

NOTES

P30 EXTERNAL ΔV
EXT ΔV - P30
P00. V49
MANEUVER TO BURN ATTITUDE R 357.9, P 225.4, Y 346.2
SEXTANT STAR CHECK

ROLL TO ACQUIRE MSFN
SPS PRETHRUST - P40 (TVC TEST)
ROLL TO BURN ATTITUDE

S-BAND SQUELCH - OFF

PITCH UP 360° AT 0.2°/SEC
TO OBSERVE LUNAR SURFACE

GO INERTIAL
V64 REACQUIRE MSFN

EMS ΔV TEST

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	74:00 - 75:00	4/TLC	3-46

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LOI_1
BURN CHART

	P OR Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
LOI_1	10°/SEC TAKEOVER	±10° TAKEOVER	BT + 10 SEC	DO NOT TRIM

LOI_1 ABORT MODES			
LOI_1 V_{GO}	BT	TRAJECTORY	ABORT MODE
2924.0 - 2129.0	0-01:50	HYPERBOLIC	MODE I - COAST 2 HR - DPS - P37 (P37 BEYOND SPHERE FOR VGO >2279 AND BT <01:30
2129.0 - 1589.0	01:50 - 03:00	UNSTABLE	MODE II - COAST 2 HR - 2 DPS BURNS FOR STABILIZATION AND WATER or CLA LANDING
1589.0 - 0	03:00 - 06:05	LUNAR ORBIT	MODE III - DPS BURN AFTER ONE REV

FLIGHT PLAN

MCC-H | 1230 EDT | NOTES

75:00

CMP - PRE LOI_1 SYSTEMS CKS

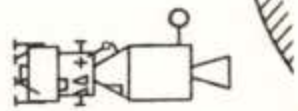
C&W CK
CM RCS CK
SM RCS CK
SPS PERIODIC MON
EPS PERIODIC MON
ECS PERIODIC MON

COPY UPDATE: LOS __ __:__ __:__ __
AOS WITH LOI_1 __ __:__ __:__ __
AOS W/O LOI_1 __ __:__ __:__ __

UPDATE CSM
LOS AND AOS
(WITH & WITHOUT
LOI_1)

M S F N

EXT ΔV - P30 (RELOAD N81 WITH PAD VALUES)
SPS THRUST - P40
MNVR TO BURN ATTITUDE



75:30

R357.9, P225.4, Y346.2

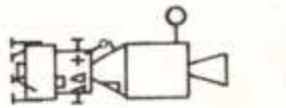
SEXTANT STAR CHECK

GO/NO GO

GO/NO GO FOR LOI_1
PCM-LO
S-BAND AUX-DOWN VOICE BACKUP

75:46

GDC ALIGN TO IMU



LOI_1

GETI: 75:54:28
NO ULLAGE
BT: 5 MIN 59.9 SEC
ΔV_T: 2924.1 FPS
ORBIT: 59.2 X 169.8
RETROGRADE
DO NOT TRIM

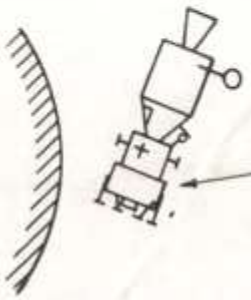
NOTE: INITIATE LOI_1
WITH BANK B BALL VALVES

76:00

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	75:00 - 76:00	4/1	3-47

FLIGHT PLAN

MCC-H



R180, P315/295 Y0
HGA P-20, Y355
DUMP DSE

1330 EDT

76:00
REV 1

76:19

V66 - TRANSFER CSM STATE
VECTOR TO LM SLOT

SM RCS AND SPS MON CK
ROLL 180°, PITCH DOWN 70°,
YAW LEFT 14°
V64 REACQUIRE MSFN
ORB RATE

LOI_1 BURN STATUS REPORT

76:49

MSFN

77:00
77:02

77:44

77:49

77:56
78:00

BURN STATUS REPORT

X	X	□	•	ΔTIG
X	X		•	BT
□			•	V_{gx}
		TRIM		
X	X	X		R
X	X	X		P
X	X	X		Y
□			•	V_{gx}
□			•	V_{gy}
□			•	V_{gz}
□			•	ΔV_c
X	X	X		FUEL
X	X	X		OX
X	X	X		UNBAL

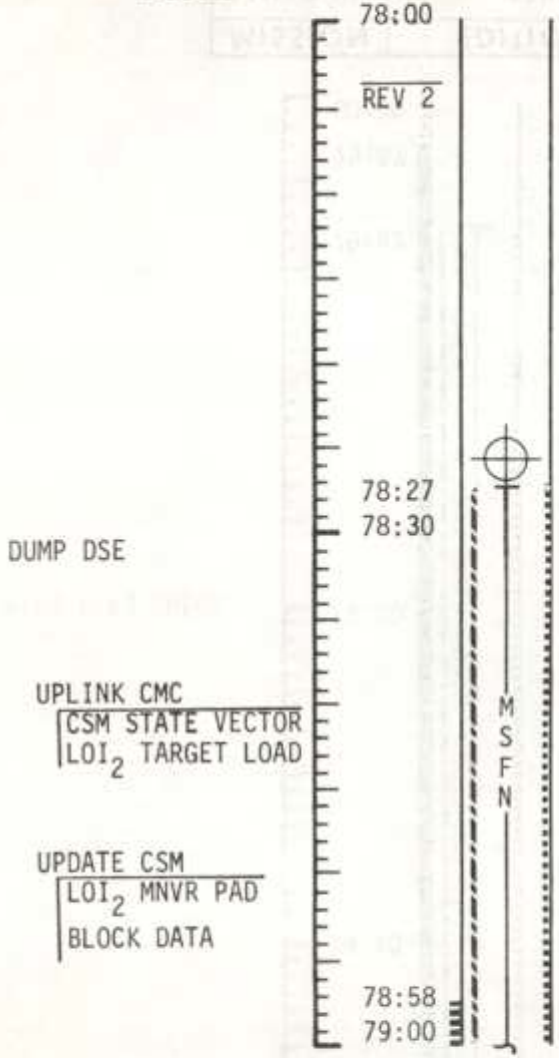
NOTES

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	76:00 - 78:00	4/1	3-48

MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

FLIGHT PLAN



EAT PERIOD

V64 REACQUIRE MSFN
HGA P-20, Y359

V66 TRANSFER CSM STATE VECTOR TO LM SLOT

RECORD LOI_2 MNVR PAD AND BLOCK DATA (TEI_5)

NOTES

TEI_5 BLOCK DATA
ASSUMES LOI_1 & LOI_2
ACCOMPLISHED

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	78:00 - 79:00	4/2	3-49

FLIGHT PLAN

MCC-H

1630 EDT

79:00

79:10

PIPA BIAS CHECK

79:30

79:52

79:57

80:00

MSFN

CMP - PRE LOI_2 SYSTEMS MONITOR

GO INERTIAL

IMU REALIGN - P52
OPTION 3 - REFSMMAT
DRIFT CHECK

R 180, P 315/182, Y 0
HGA P -45, Y 180

P30 EXTERNAL ΔV
MANEUVER TO LOI_2 BURN ATTITUDE

P40 SPS THRUST
GO INERTIAL

R 0, P 50.1/212.3, Y 359.6

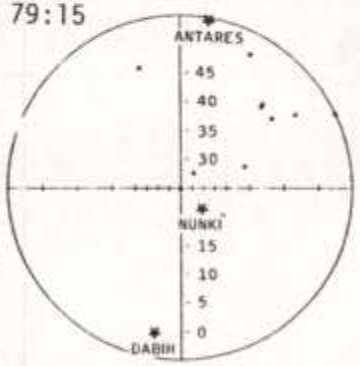
SEXTANT STAR CHECK
EMS ΔV TEST
SM RCS CHECK
LOAD DAP FOR 2 JET ULLAGE
R1 = 20111
R2 = 11111

NOTES

REPORT:

P52 (LDG SITE REFSMMAT)
N71: __ __.__ __
N05: __ __ __.__ __
N93:
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __
GET __ __ __:__ __:__ __

79:15



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	79:00 - 80:00	4/2	3-50

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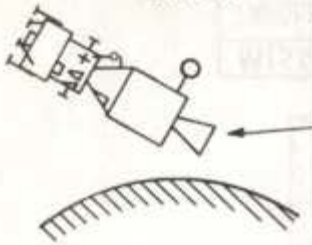
LOI_2
BURN CHART

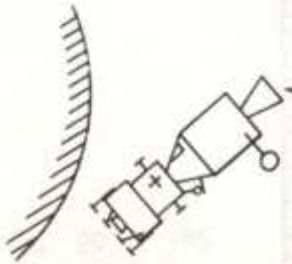
	P OR Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
LOI_2	10°/SEC TAKEOVER	±10° TAKEOVER	BT + 1 SEC	TRIM X AXIS TO 1 FPS

FLIGHT PLAN

MCC-H



R 0, P 151.5/212.3, Y 359.6



R180, P315/293, Y0
HGA P -34, Y 358

DUMP DSE

1730 EDT

80:00

80:04

REV 3

80:30

80:37

81:00

MSFN

LOI_2

GDC ALIGN TO IMU

GETI: 80:09:30
ULLAGE: 2 JET, 19 SEC
BT: 16.4 SEC
ΔV_T: 157.8 FPS
RETROGRADE
ORBIT 53.6 X 65.6
~~DO NOT TRIM~~

TRIM X AXIS TO 1 FPS

SM RCS CHECK
SPS MONITOR CHECK
V66 - TRANSFER CSM STATE VECTOR TO LM SLOT
BATTERY CHARGE, BATTERY A

ROLL 180°, PITCH DOWN 81°, ORB RATE

OBSERVE LUNAR SURFACE

V64 ACQUIRE MSFN

LOI_2 BURN STATUS REPORT

PRESSURIZE THE LM
CM/LM PRESSURE EQUALIZATION
VALVE - OPEN

NOTES

BURN STATUS REPORT

X	X	□	●	ΔTIG
X	X		●	BT
□				V_{gx}
			●	
		TRIM		
X	X	X		R
X	X	X		P
X	X	X		Y
□			●	V_{gx}
□			●	V_{gy}
□			●	V_{gz}
□			●	ΔV_c
X	X	X		FUEL
X	X	X		OX
X	X	X		UNBAL

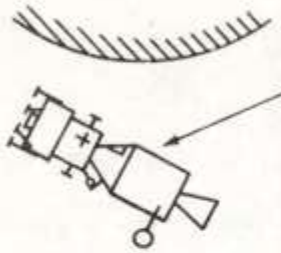
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	80:00 - 81:00	4/3	3-51

FLIGHT PLAN

MCC-H

1830 EDT

UPDATE CSM
P22 AUTO OPTICS
ALT LMK (A-1)
TRACKING
(SEE GET 82:35)



R 180, P 318/196, Y 0

HGA P -60, Y 182

81:00
81:04
81:10
81:30
81:49
81:56
82:00

MSFN

RECORD UPDATE (SEE GET 82:40)

GO INERTIAL

IMU REALIGN - P52
OPTION 3 REFSMMAT

STOW OPTICS

TUNNEL VENT VALVE - LM/CM ΔP
VERIFY LM/CM ΔP <0.2

PREPARE FOR LM INGRESS

CMP - CLEAR TUNNEL OF CM HATCH
INSPECT TUNNEL & DOCKING LATCHES
REMOVE PROBE & DROGUE
STOW CM HATCH, PROBE & DROGUE

LMP - OPEN LM HATCH
VERIFY DOCKING TUNNEL INDEX ANGLE
IVT TO LM

LMP - LM ENTRY STATUS CHECKS

NOTES

REPORT:

P52 (LDG SITE REFSMMAT)
N71: __ __,__ __
N05: __ __ __.__ __
N93:
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __
GET __ __ __:__ __:__ __

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	81:00 - 82:00	4/3	3-52

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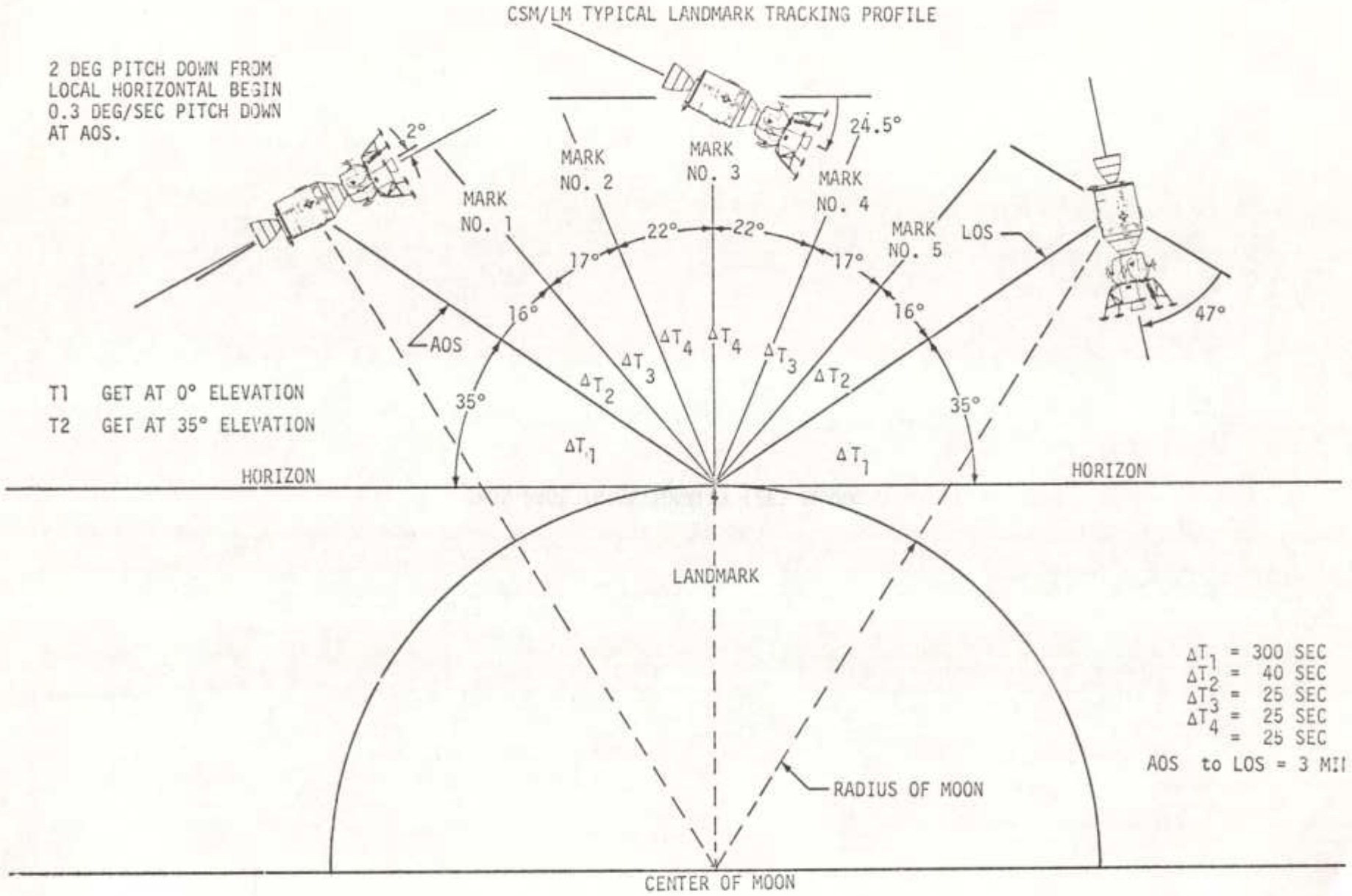

CSM/LM TYPICAL LANDMARK TRACKING PROFILE
2 DEG PITCH DOWN FROM LOCAL HORIZONTAL BEGIN 0.3 DEG/SEC PITCH DOWN AT AOS.
2°
24.5°
MARK NO. 1
MARK NO. 2
MARK NO. 3
MARK NO. 4
MARK NO. 5
LOS
22°
22°
17°
17°
16°
16°
47°
AOS
35°
35°
ΔT4
ΔT4
ΔT3
ΔT3
ΔT2
ΔT2
ΔT1
ΔT1
T1 GET AT 0° ELEVATION
T2 GET AT 35° ELEVATION
HORIZON
HORIZON
LANDMARK
RADIUS OF MOON
CENTER OF MOON
ΔT1 = 300 SEC
ΔT2 = 40 SEC
ΔT3 = 25 SEC
ΔT4 = 25 SEC
= 25 SEC
AOS to LOS = 3 MI

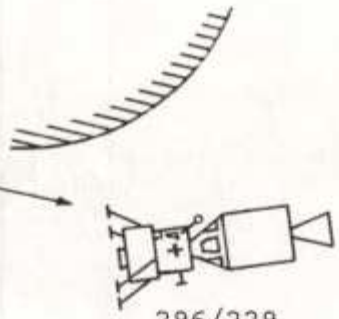
FLIGHT PLAN

CSM			LM	MCC-H
1930 EDT	CMP	CDR	LMP	
82:00				
82:02			PERFORM HOUSEKEEPING CHORES 1 STOW HELMET STOWAGE BAGS 2 UNSTOW MIRROR, CHECKLIST AND DISPOSAL ASSEMBLY 3 STOW INTERIM STOWAGE ASSEMBLY 4 UNSTOW AND CONFIGURE FOR USE: 16mm/HCEX (f4,500,INF) 6 fps	
REV 4				
	R 0, ~~P 257/297~~, Y 0 P ~~296~~/303 MANEUVER TO LANDMARK TRACK ATTITUDE GO INERTIAL SELECT OMNI B P22 ORBITAL NAVIGATION UNSTOW OPTICS	AID LMP AS REQUIRED		
82:30				
82:35			P22 AUTO OPTICS LMK ID A-1 T1 82:41:06 (HOR) T2 82:46:17 (35°) 14 NM (N) N 89 LAT 02.000 LONG/2 32.750 ALTITUDE- 000.00 NM	
	TRACK LANDMARK ALT LMK (A-1) (5 MARKS ON LMK) PITCH DOWN 0.3°/SEC DO NOT INCORPORATE MARKS			
	STOW OPTICS			
83:00				

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	82:00 - 83:00	4/4	3-53

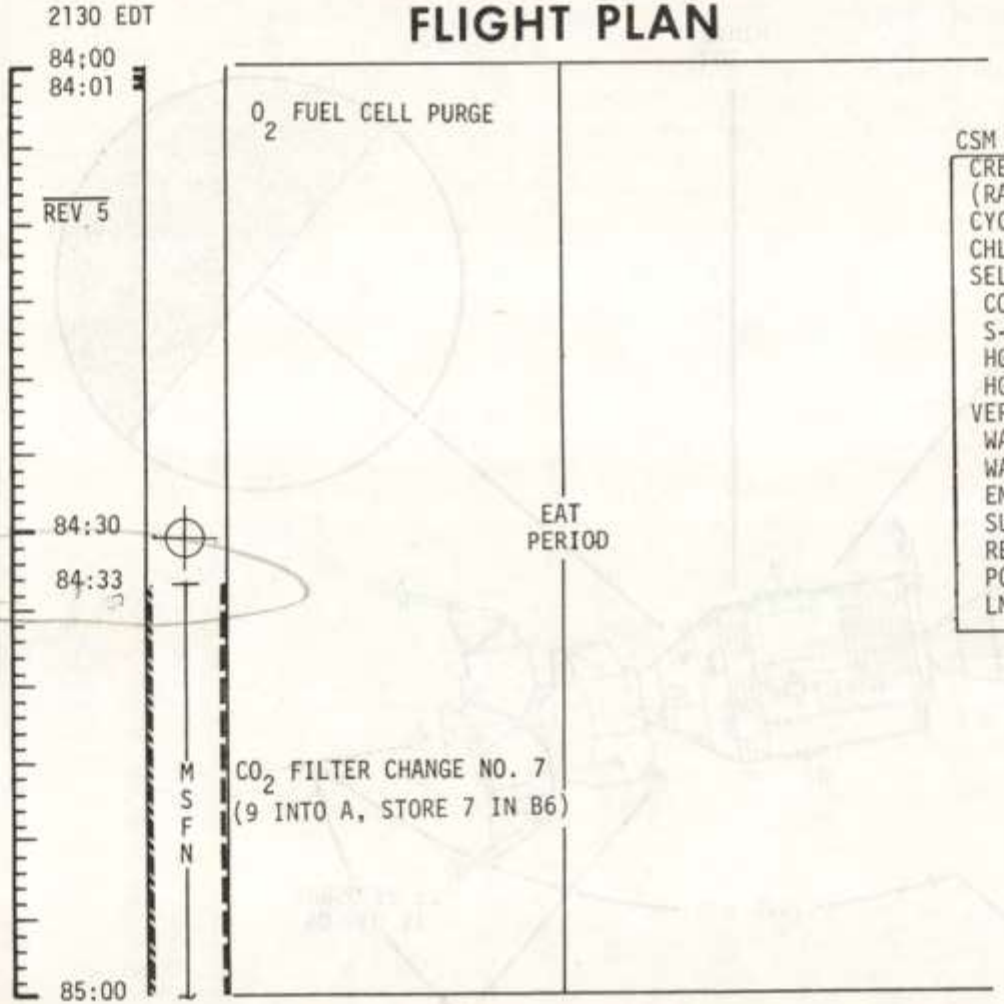
FLIGHT PLAN

CSM			LM	MCC-H
CMP	2030 EDT	CDR	LMP	
CSM POWER TO LM - OFF (AT LMP REQUEST) STOP PITCH RATE, ROLL TO SLEEP ATTITUDE GO INERTIAL V64 ACQUIRE MSFN	83:00 83:02 83:08	 296/229 R82, ~~350/229~~, Y0 HGA P 8 Y270	TRANSFER TO LM POWER COMM ACTIVATION	
	MSFN		S-BAND/VHF B VOICE & TM TEST	DUMP DSE
			REPORT OPS SOURCE PRESSURE	RECORD OPS SOURCE PRESSURE UPDATE CSM BLOCK DATA
RECORD BLOCK DATA (TEI 11) V66-TRANSFER STATE VECTOR TO LM SLOT CSM POWER TO LM-ON (AT LMP REQUEST) INSTALL CM HATCH TUNNEL VENT VALVE-LM PRESS	83:30		COMM DEACTIVATION TRANSFER TO CSM POWER	UPLINK CMC CSM STATE VECTOR
	83:47		LMP IVT TO CSM CLOSE LM HATCH	
	83:55			
	84:00			

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	83:00 - 84:00	4/4	3-54

FLIGHT PLAN

MCC-H



0202Z

LO 5

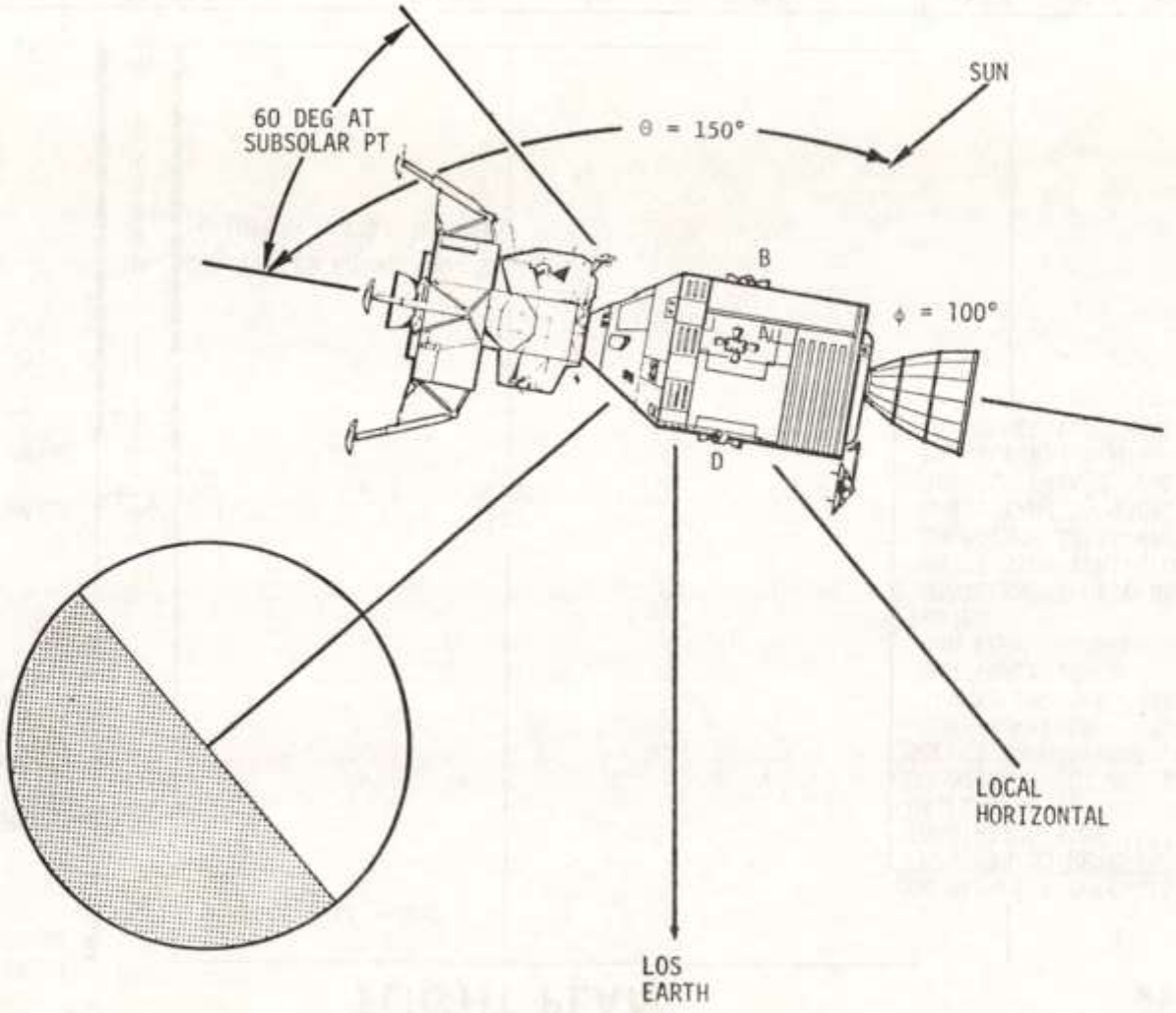
NOTES

CSM PRESLEEP CHECKLIST
- CREW STATUS REPORT
- (RADIATION & MEDICATION)
- CYCLE O_2 & H_2 FANS
- CHLORINATE POTABLE WATER
- SELECT NORMAL LUNAR COMM
  - CONFIGURATION - EXCEPT:
  - S-BAND SQUELCH-ENABLE
  - HGA TRACK-REACQ
  - HGA BEAM - NARROW
- VERIFY:
  - WASTE MNGMT OVBD DRAIN-OFF
  - WASTE STOW VENT-CLOSED
  - EMERGENCY CABIN PRESS-BOTH
  - SURGE TANK O_2 VALVE-ON
  - REPRESS PACK O_2 VALVE - OFF
  - POTABLE WATER HTR-OFF
  - LM TUNNEL VENT VALVE-LM PRESS

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	84:00 - 85:00	4/5	3-55

MSC Form 29 (May 69) FLIGHT PLANNING BRANCH

LUNAR ORBIT REST PERIOD ATTITUDE

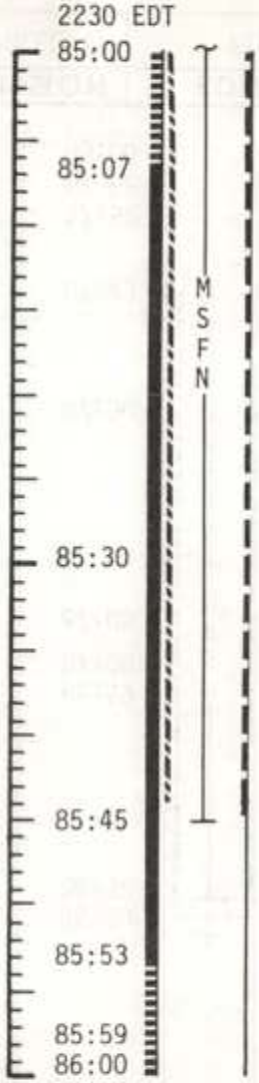


Revision A

FLIGHT PLAN

MCC-H

DUMP DSE



REST PERIOD
(9 HOURS)

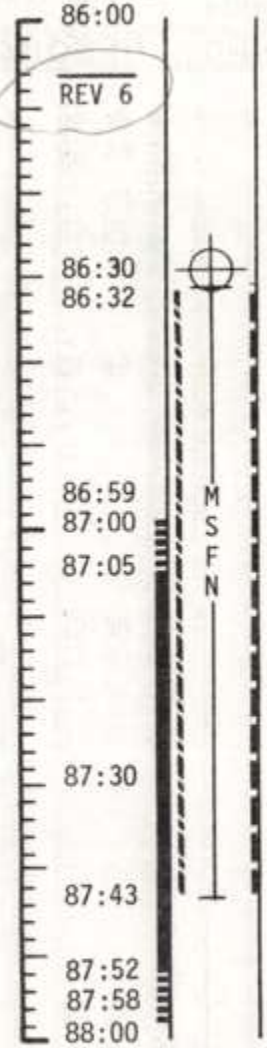
NOTES

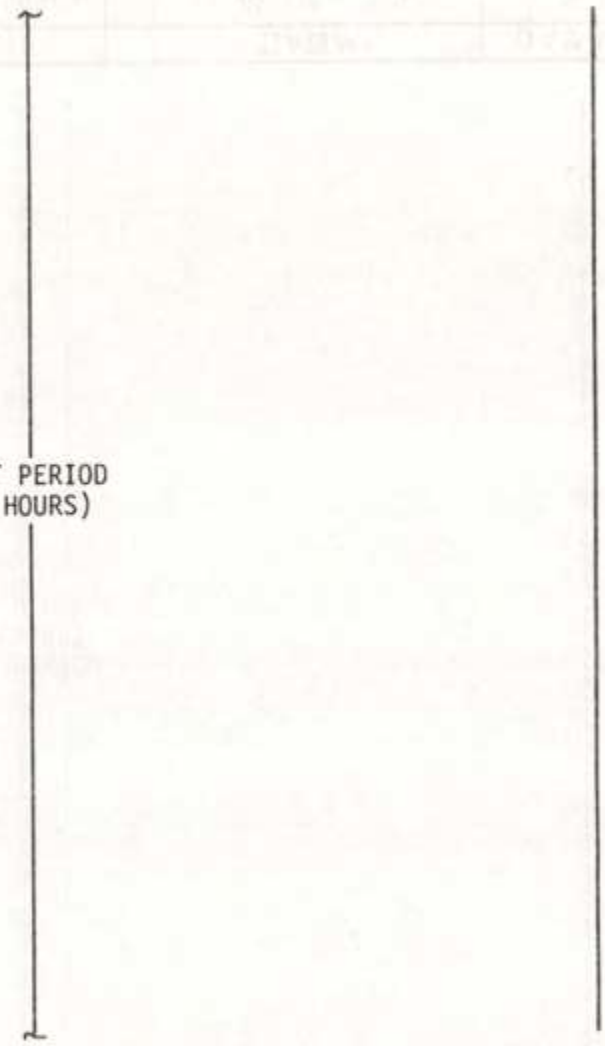
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	85:00 - 86:00	4/5	3-56

FLIGHT PLAN

MCC-H

NOTES





MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	86:00 - 88:00	4/6	3-57

MSC Form 29 (May 68)

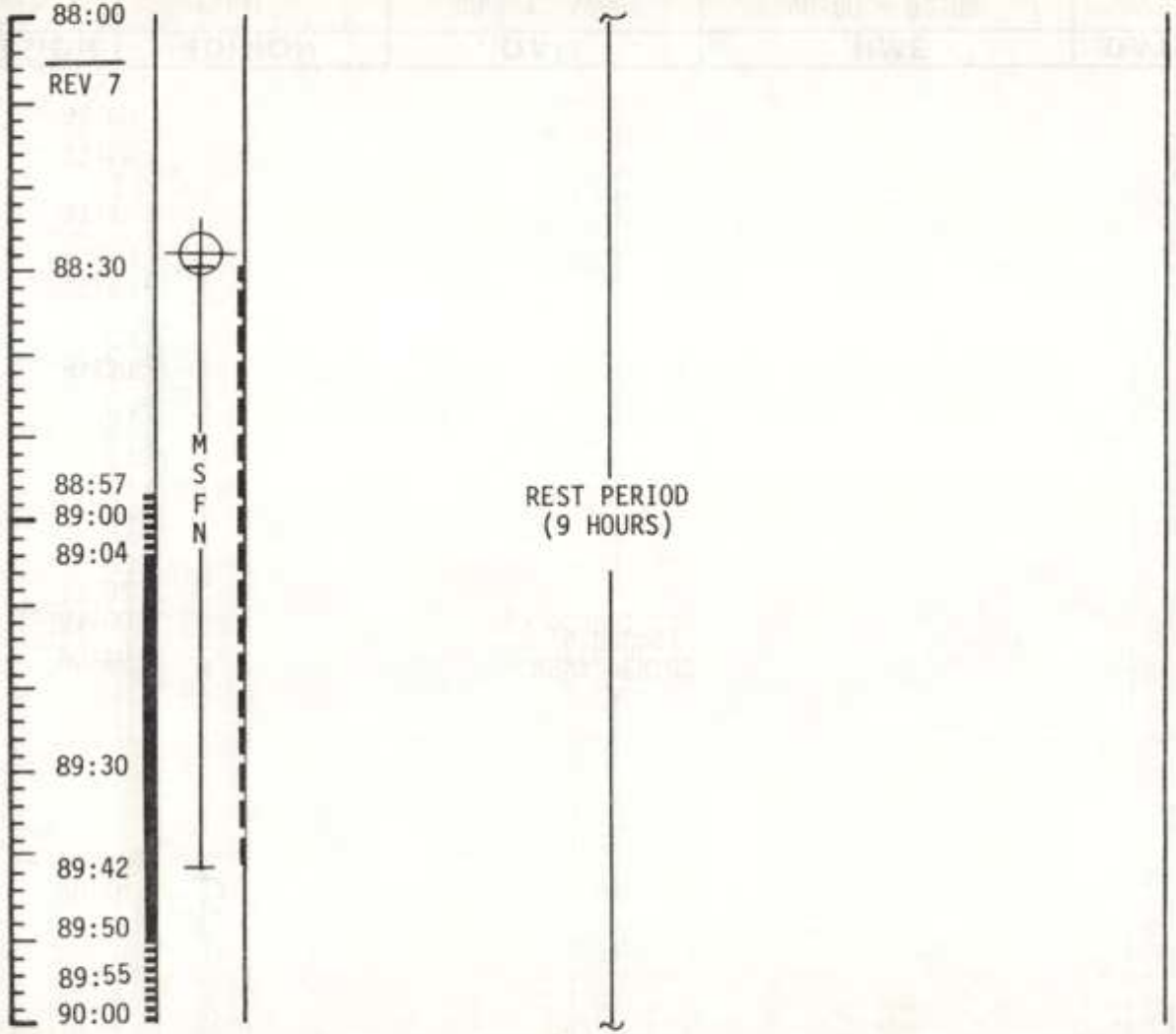
FLIGHT PLANNING BRANCH

FLIGHT PLAN

MCC-H

NOTES

DUMP DSE

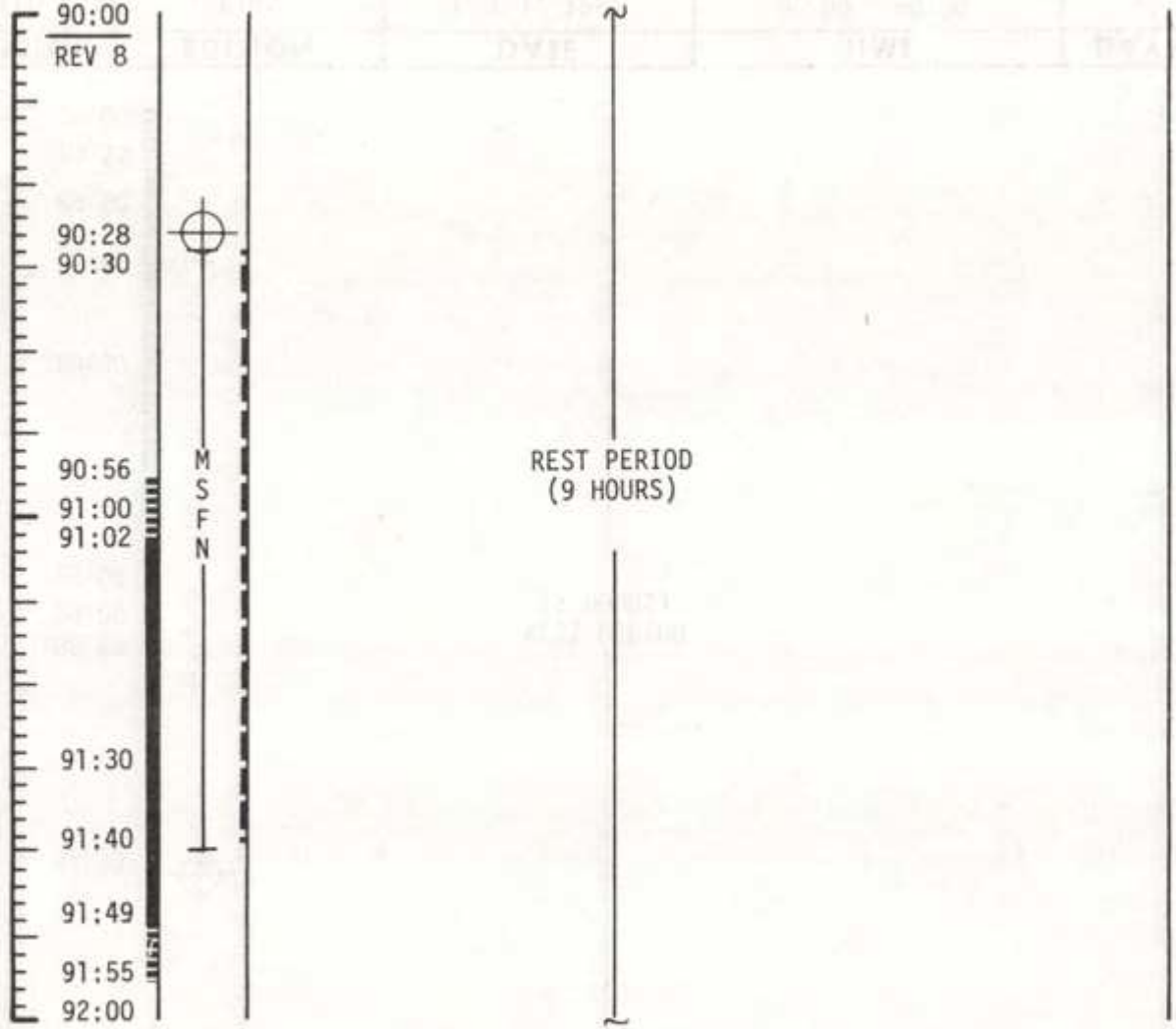


MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	88:00 - 90:00	4/7	3-58

MCC-H

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	90:00 - 92:00	4/8	3-59

MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

MCC-H

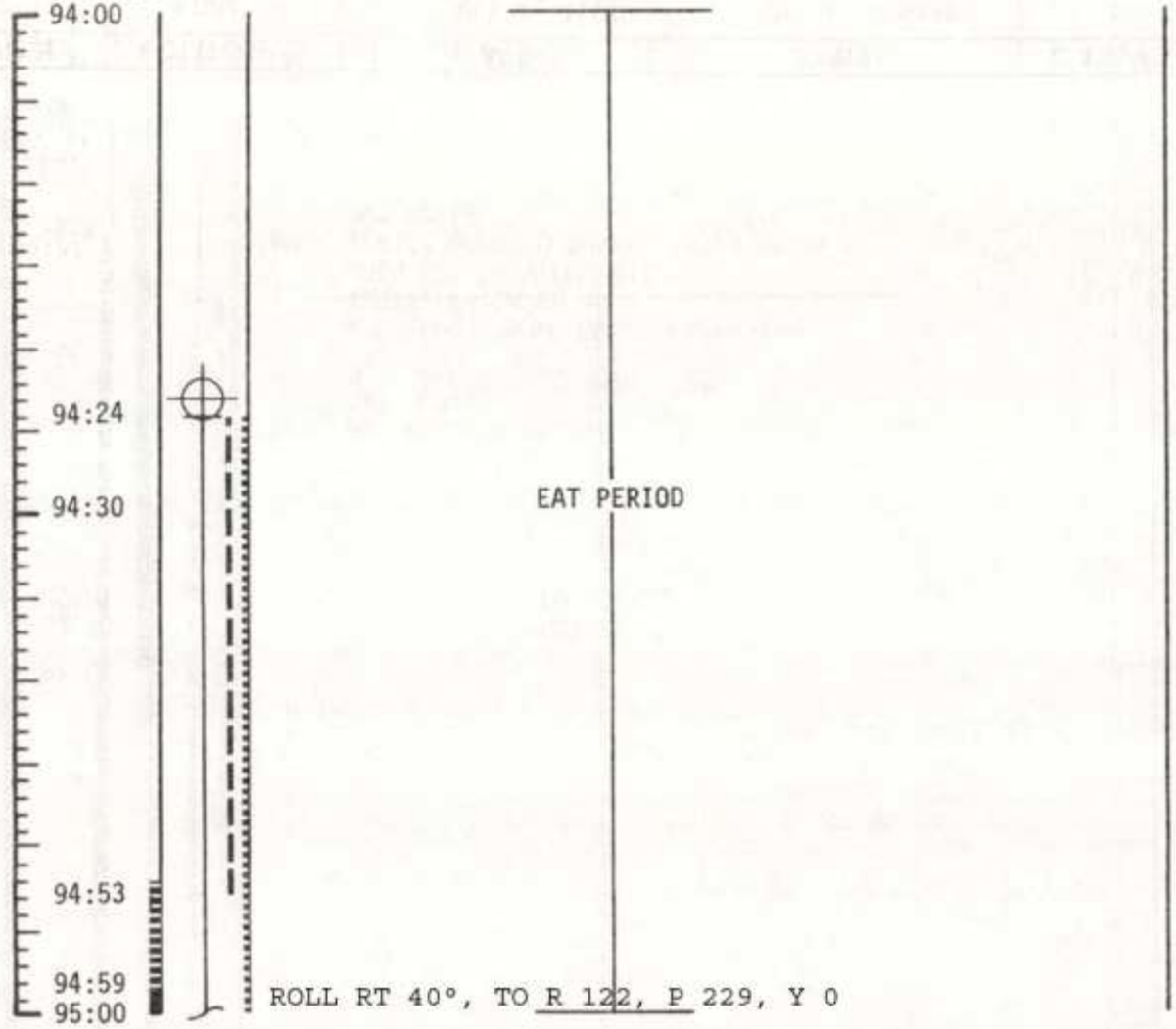
NOTES

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	92:00 - 94:00	4/9 - 10	3-60

MCC-H

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	94:00 - 95:00	5/10	3-61

MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

MCC-H

0830 EDT

UPDATE
BLOCK DATA

UPDATE
BASELINE
ALTITUDE FOR
DESCENT ALTITUDE
SIGHTINGS

95:00

MSFN

CDR & LMP DON LCG'S
CMP-RECORD BLOCK DATA - TEI_{30}

LMP-COPY BASELINE ALTITUDE

POST SLEEP CHECKLIST
CREW STATUS REPORT (SLEEP)
CYCLE H2, O2 FANS
GDC ALIGN TO IMU
CONSUMABLES UPDATE
SELECT COMM NORMAL
LUNAR CONFIGURATION

CO_2 FILTER CHANGE NO. 8
(10 INTO B, STORE 8 IN B6)

95:30

95:36

CMP: DON PGA W/O HELMET
AND GLOVES
H_2 - PURGE LINE HTRS - ON

95:46

LM TUNNEL VENT VALVE - LM/CM ΔP
VERIFY LM/CM ΔP <0.2
OPEN AND STOW CM HATCH

95:52

LMP: VERIFY DOCKING TUNNEL INDEX ANGLE
IVT TO LM

REV 11

96:00

NOTES

CONSUMABLES UPDATE
(Δ FROM NOMINAL)

GET: _____
RCS TOT _____
A _____
B _____
C _____
D _____
H_2TOT _____
O_2TOT _____

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	95:00 - 96:00	5/10-11	3-62

FLIGHT PLAN

CSM		LM		MCC-H
CMP	0930 EDT	CDR	LMP	
UNDOCKING PHOTO 16mm/18/CEX-BRKT-MIR (f8,250,7) 6 fps O_2 & H_2 FUEL CELL PURGE	96:00	DON PGA W/O HELMET AND GLOVES	LM FAMILIARIZATION	
V64 ACQUIRE MSFN CREW STATUS REPORT REPORT DOCKING TUNNEL INDEX ANGLE TO MSFN DEACTIVATE B3 & C4 JETS CONFIGURE DAP 21112 WIDE DB 11001 (FOR LM STEERABLE ANTENNA ACTIVATION)	96:22	CSM POWER TO LM-OFF (AT LMP REQUEST)	LM POWER-ON	DUMP DSE
	96:30	DISCONNECT AND STOW LM POWER UMBILICAL	EPS ACTIVATION MISSION TIMER ACTIVATION PRIMARY GLYCOL LOOP ACT	UPLINK CMC CSM STATE VECTOR DESIRED ORIENT (LS REFSMMAT)
RECORD LMK 130 PAD DATA (SEE GET 98:35) AND CSM DAP DATA AND LOAD			CAUTION/WARNING CHECKOUT CB ACTIVATION TB VERIFICATION	UPDATE CSM LMK 130 PAD ~~BLOCK DATA~~ CSM DAP DATA
	96:51	IVT TO LM TRANSFER HELMET & GLOVES	PGNCS TURN - ON AND SELF TEST	
UNSTOW OPTICS P52-IMU REALIGN OPTION 1 PREFERRED	96:58	ECS ACTIVATION AND C/O CONNECT TO LM ECS	BIO MED SWITCH - LEFT	
	97:00			

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	96:00 - 97:00	5/11	3-63

FLIGHT PLAN

CSM

REPORT: CMP

P52 (LDG SITE REFSMMAT)
N71: __ __,__ __
N05: __ __ __.__ __
N93:
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __
GET __ __ __:__ __:__ __

VHF CHECKOUT
CSM TIME MARK TO LM
STOW OPTICS

VO6N20E
(ON MARK FROM CDR)

RECORD LM PCM DATA

DON HELMET AND GLOVES
PGA PRESSURE INTEGRITY CHECK
INSTALL DROGUE & PROBE, PRELOAD PROBE
INHIBIT ROLL COMMANDS UNTIL LM/CM ΔP>3.5psia
COCK LATCHES (12)
INSTALL HATCH
VENT TUNNEL
HATCH INTEGRITY CHECK
INSTALL AND ALIGN DOCKING TARGET

MSFN

1030 EDT
97:00
97:30
97:34
97:44
97:49
REV 12
98:00

LM

CDR	LMP
SUIT FAN/H2O SEP CHECK	SEC S-BAND T/R AND POWER AMPLIFIER CHECK
GLYCOL PUMP CHECK	S-BAND STEERABLE ANTENNA ACTIVATION P 152, Y -9
VHF-B ACTIVATION	
E MEMORY DUMP	IVT TO CSM
VHF CHECKOUT (COMM CHECK WITH CSM)	
LGC/CMC CLOCK SYNC T EPHEM UPDATE	DON PGA
DOCKED IMU COARSE ALIGN REPORT GIMBAL ANGLES AND TIME TO MSFN	
AFT OMNI - LBR SLEW STEERABLE ANTENNA P 187, Y 70	IVT TO LM TRANSFER HELMET & GLOVES
VERIFY DROGUE AND PROBE INSTALLATION CLOSE AND SECURE HATCH	CONNECT TO LM ECS AND COMM
	ASCENT BATTERY ACTIVATION AND CHECKOUT RECORD ED BAT VOLTS

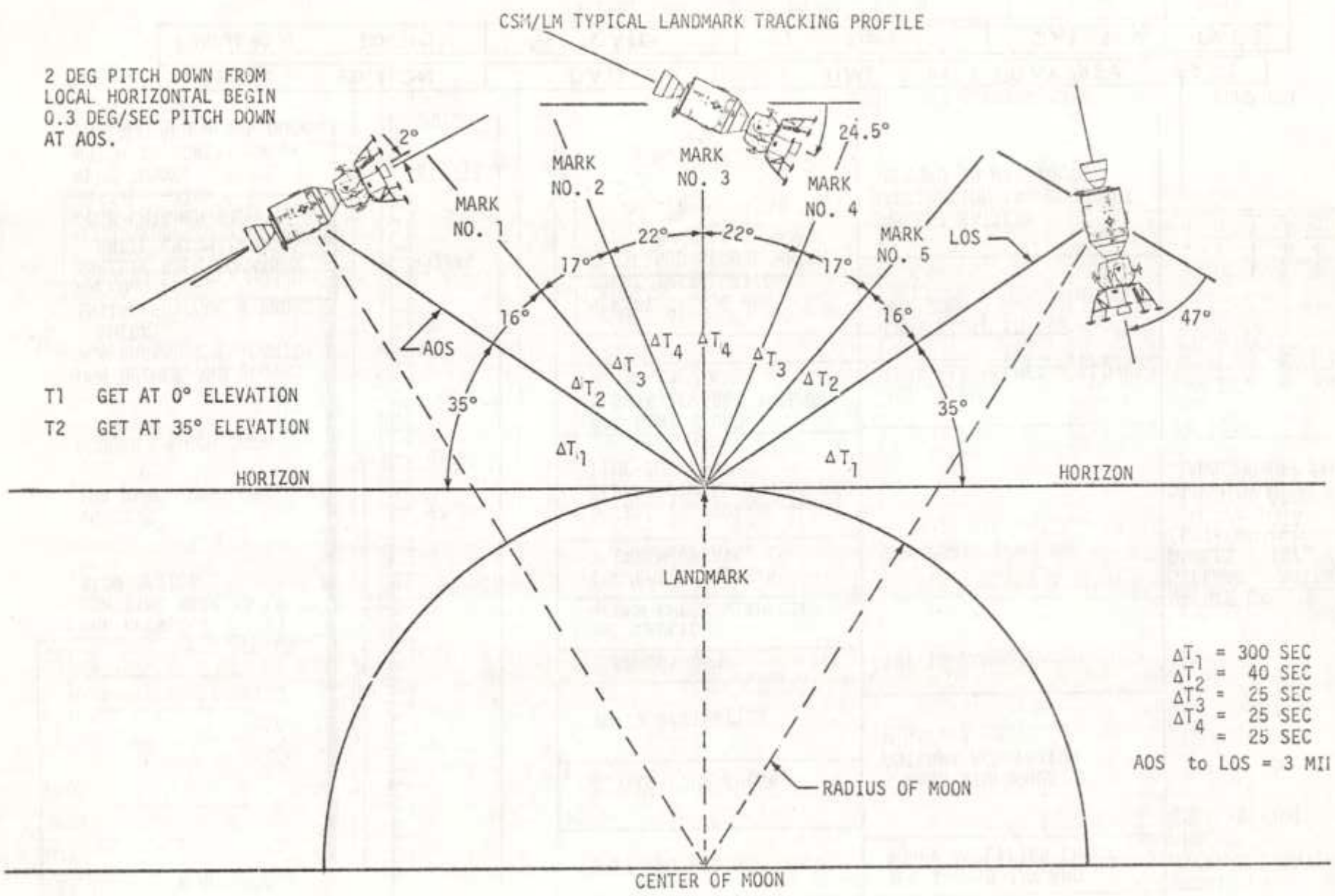
MCC-H

UPDATE LM STEERABLE ANTENNA ANGLES (GET :97:10)

UPDATE LM STEERABLE ANTENNA ANGLES P 187, Y 70 (GET: 98:55)

COPY GIMBAL ANGLES AND TIME

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	97:00 - 98:00	5/11-12	3-64


CSM/LM TYPICAL LANDMARK TRACKING PROFILE
2 DEG PITCH DOWN FROM LOCAL HORIZONTAL BEGIN 0.3 DEG/SEC PITCH DOWN AT AOS.
2°
MARK NO. 1
MARK NO. 2
MARK NO. 3
MARK NO. 4
MARK NO. 5
24.5°
LOS
22°
22°
17°
17°
16°
16°
47°
AOS
35°
35°
ΔT4
ΔT4
ΔT3
ΔT3
ΔT2
ΔT2
ΔT1
ΔT1
T1 GET AT 0° ELEVATION
T2 GET AT 35° ELEVATION
HORIZON
HORIZON
LANDMARK
RADIUS OF MOON
CENTER OF MOON
ΔT1 = 300 SEC
ΔT2 = 40 SEC
ΔT3 = 25 SEC
ΔT4 = 25 SEC
AOS to LOS = 3 MII

FLIGHT PLAN

CSM		LM		MCC-H
CMP	1130 EDT	CDR	LMP	
	98:00	DON HELMET & GLOVES	DON HELMET & GLOVES	
		ARS/PGA PRESSURE INTEGRITY CHECK	ARS/PGA PRESSURE INTEGRITY CHECK	
MANEUVER TO TRACKING ATTITUDE R 0 , P 297/270, Y 0 DOFF HELMET & GLOVES		CABIN REGULATOR CHECK	CABIN REGULATOR CHECK	
SELECT OMNI C	98:21	DOFF HELMET & GLOVES COPY DAP DATA COPY GYRO TORQUE ANGLES AND FINE ALIGN IMU X ___, Y ___, Z ___	DOFF HELMET & GLOVES SELECT OMNI FWD BIO MED SWITCH - RIGHT	UPDATE LM DAP DATA GYRO TORQUE ANGLES
GO INERTIAL UNSTOW OPTICS P22 ORBITAL NAVIGATION	98:30			P22 AUTO OPTICS LMK ID 130 T1 98:40:02 (HOR) T2 98:45:08 (35°) 9.6 NM (N) OR S) N 89 LAT + 01.243 LONG/2 + 11.844 ALTITUDE- 001.46 NM
TRACK LDG SITE LANDMARK (5 MARKS ON LDMK 130) PITCH DOWN 0.3°/SEC AT T2 DO NOT INCORPORATE MARKS		RATE GYRO CHECK	AGS ACT & SELF TEST	
	98:50	R0, P126.5/22.5, Y0 HGA P-45, Y3		
CONTINUE PITCH TO AGS CAL PITCH ATT (22.5)	98:56			
V64 ACQUIRE MSFN	99:00		V64 ACQUIRE MSFN ANT P 187, Y 70	DUMP DSE

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	98:00 - 99:00	5/12	3-65

FLIGHT PLAN

CSM

CMP

STOW FLIGHT PLAN
UNSTOW SOLO BOOK
COPY PADS

DON HELMET & GLOVES
SC CONT - SCS
MIN/MAX DB, LOW/HIGH RATE
(AT REQUEST OF CDR)
GO/NO-GO FOR UNDOCKING
DISABLE ROLL JETS FOR RCS HOT FIRE

VERIFY TUNNEL VENT VALVE - OFF

RECORD LM PCM DATA
SC CONT - CMC AUTO
MANEUVER TO
AGS CALIBRATION ATTITUDE

SC CONT - SCS
RATES <0.1°/SEC
DISABLE THRUSTERS FOR 32 SEC
(AT REQUEST OF LMP)
MANEUVER TO UNDOCKING ATTITUDE
R0, P320/14, Y0

1230 EDT
99:00

M S F N

99:30
99:32

99:43

99:49
REV 13

100:00

LM

CDR

DRIFT CHECK-REPORT GIMBAL ANGLES & TIME TO MSFN
DEPLOY LANDING GEAR

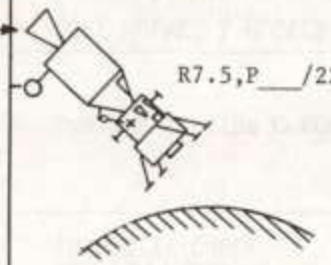
ORDEAL INITIALIZATION

LOAD DAP DATA - 32012
CSM WT _____
P TRIM _____
Y TRIM _____
DPS GIMBAL DRIVE AND THROTTLE TEST

RCS PRESSURIZATION

RCS CHECKOUT

RR ACT & SELF TEST



DPS PRESS & CHECKOUT

LMP

V47 INITIALIZE AGS

COPY AGS ABORT CONSTANT AND K FACTOR

RCS PRESSURIZATION

GO/NO-GO FOR UNDOCKING
RCS CHECKOUT

AFT OMNI - LBR

SLEW STEERABLE ANTENNA
ANT P 123, Y -37

AGS ACCEL & GYRO CALIBRATION

MCC-H

UPLINK LGC
- LS REFSMMAT
- LM & CSM STATE VECTORS
- LGC/CMC CLOCK SYNC
- PIPA BIAS
- LGC ABORT CONSTANT

UPDATE LM
- AGS ABORT CONSTANT
- AGS K FACTOR

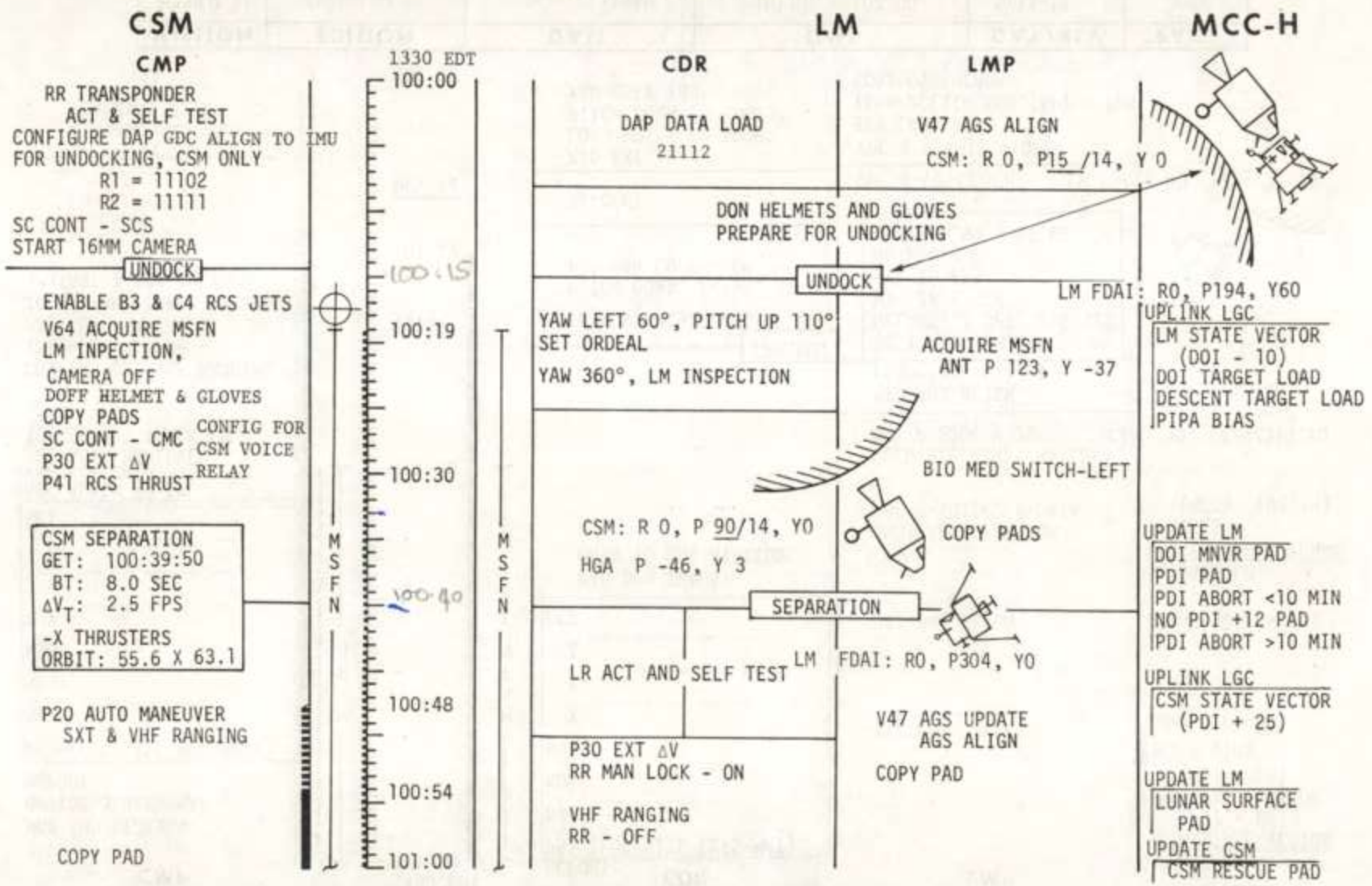
UPLINK CMC
- LM & CSM STATE VECTORS

UPDATE CSM
- P30 MNVR PAD (SEPARATION)
- GO/NO-GO

UPDATE LM
- STEERABLE ANTENNA ANGLES(GET:100:25)

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	99:00 - 100:00	5/12-13	3-66

FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	100:00 - 101:00	5/13	3-67

FLIGHT PLAN

CSM

CMP

P52 IMU REALIGN
OPTION 3 REFSMMAT
REPORT:

P52 (LDG SITE REFSMMAT)	
N71:	__ __,__ __
N05:	__ __ __.__ __
N93:	
X	__ __.__ __ __
Y	__ __.__ __ __
Z	__ __.__ __ __
GET	__ __ __:__ __:__ __

GDC ALIGN TO IMU
VHF B - DATA
P20 AUTO MANEUVER
TO SEXTANT TRACK LM

CONFIRM DOI-VHF RANGING
INCORPORATE P76
P20 AUTO MANEUVER
TO SEXTANT TRACK LM
SEXTANT TRACK ONLY

SEXTANT AND VHF
TRACKING OF LM

MSFN

1430 EDT
101:00
101:30
101:41
101:47
REV 14
102:00

MSFN

LM

CDR

REPORT:

P52 (LDG SITE REFSMMAT)	
N71:	__ __,__ __
N05:	__ __ __.__ __
N93:	
X	__ __.__ __ __
Y	__ __.__ __ __
Z	__ __.__ __ __
GET	__ __ __:__ __:__ __

P40 DPS THRUST
MNVR TO DOI ATTITUDE

DPS,DOI

TRIM V_X RESIDUALS
PITCH DOWN, P=195, RR-ON
P20 MAN LOCK - ON

RR-OFF

P30 EXT ΔV
LOAD PDI + 12 ABORT
PITCH DOWN TO 106.5°
YAW LEFT 180°

LMP

SYSTEMS CHECKS

V47 AGS ALIGN

OMNI AFT, PCM LBR
VHF A-VOICE, B-DATA

SLEW STEERABLE ANTENNA
ANT P 220, Y 28

N20 AGS ALIGN
LOADS AGS EXT ΔV

GETI: 101:38:19.6
ULLAGE: 2 JET, 7.5 SEC
BT: 28.5 SEC
ΔV: 70 FPS
RETROGRADE
ORBIT 8.97 X 57.87

VHF A-VOICE/RNG

VHF B - XMTR - OFF
SET CAMERA
16mm/HCEX(4,500,INF) 6 fps
COAS OVERHEAD

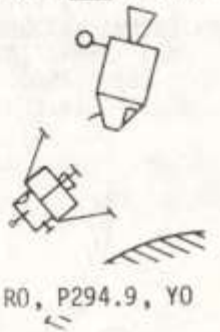
MCC-H

UPLINK CMC
CSM STATE VECTOR
(PDI + 25)
LM STATE VECTOR
(DOI - 10)
PIPA BIAS
DUMP DSE

GO/NO GO FOR DOI

UPDATE LM
STEERABLE ANTENNA
ANGLES
(GET: 102:19)

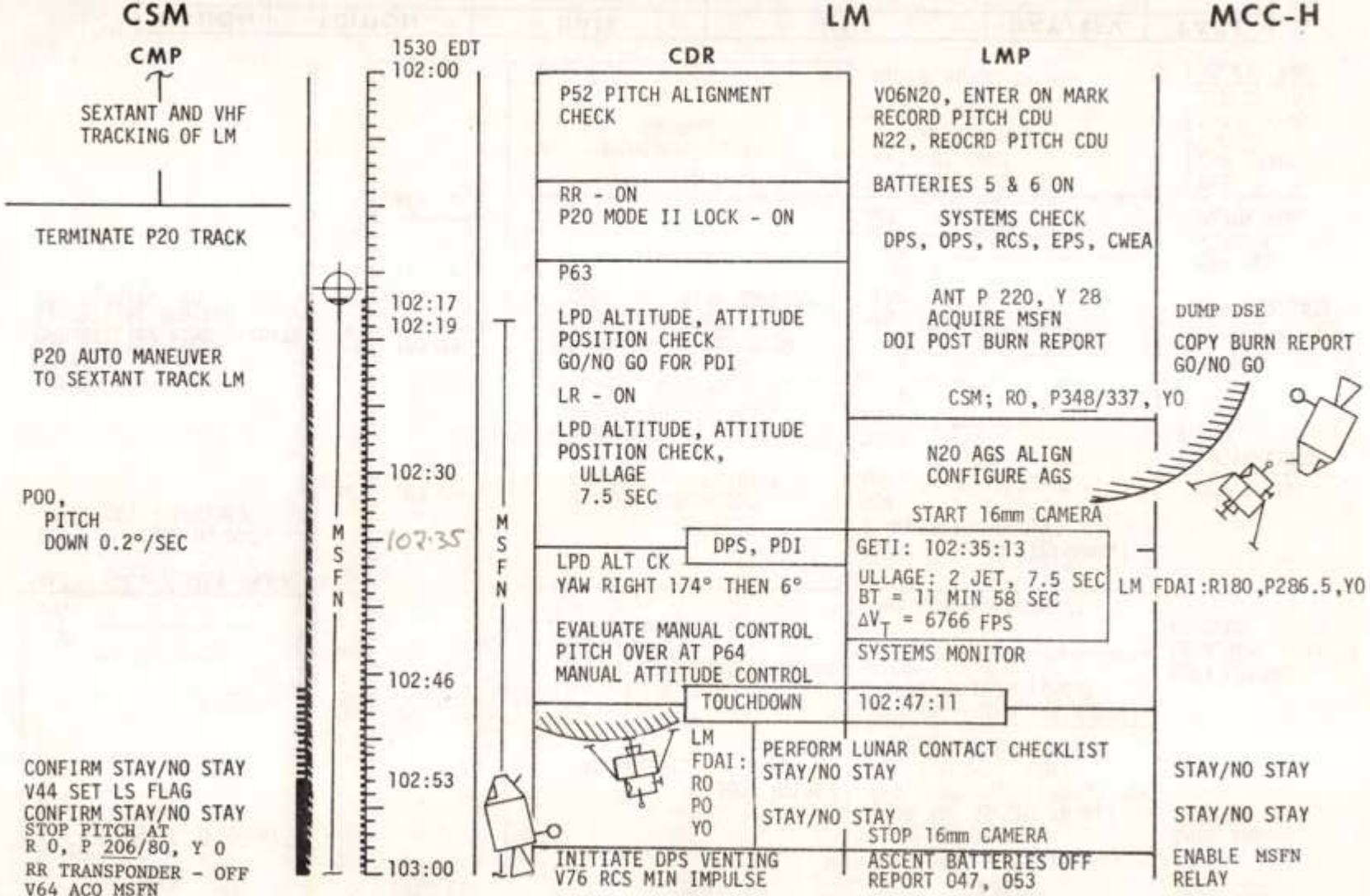
CSM: RO, P215/319,YO



LM FDAI: RO, P294.9, YO

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	101:00 - 102:00	5/13-14	3-68

FLIGHT PLAN



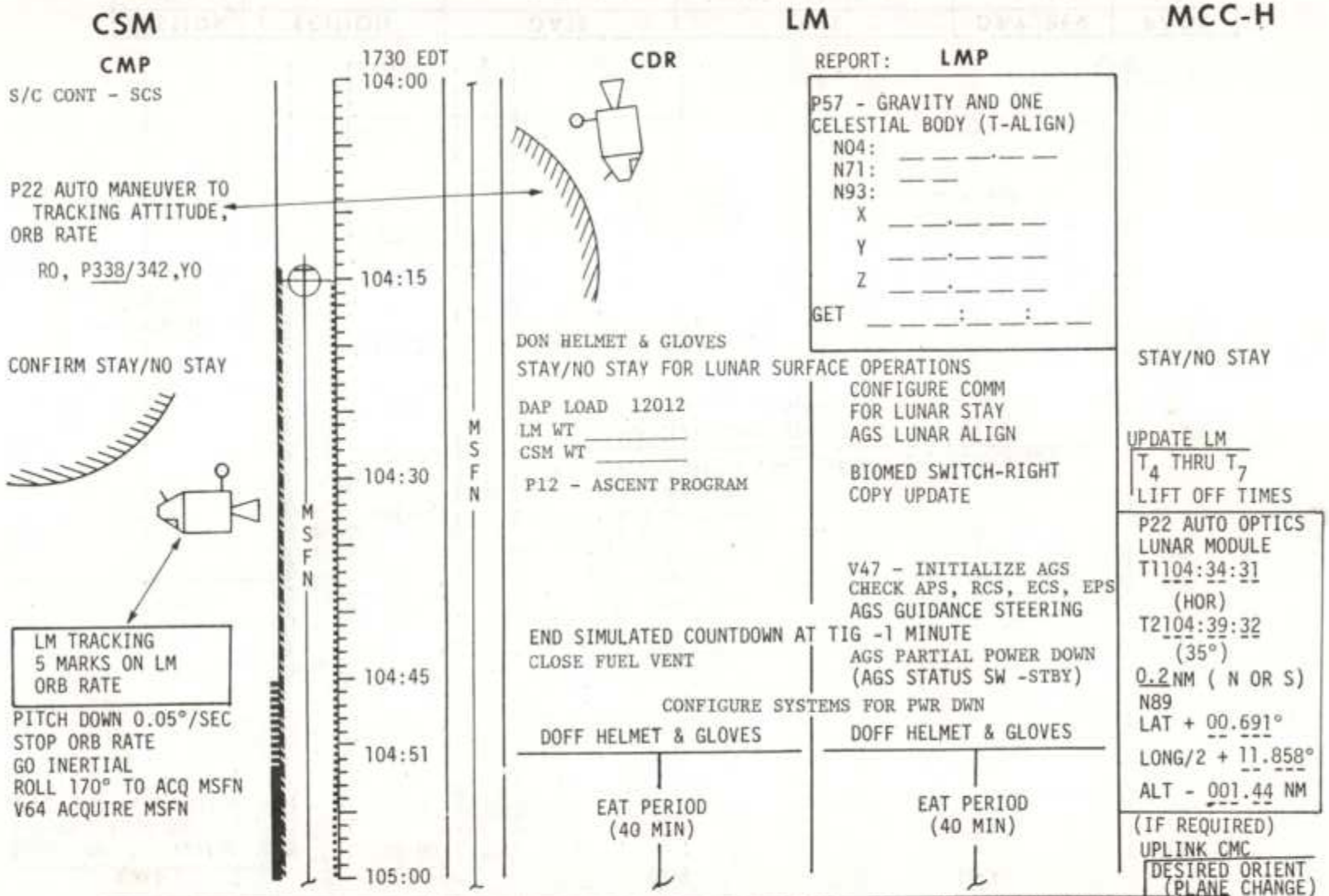
ON	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	102:00 - 103:00	5/14	3-69

FLIGHT PLAN

CSM		LM		MCC-H
CMP	1630 EDT	CDR	LMP	
REPORT: P52 (LDG SITE REFSMMAT) OPTION 3 N71: __ __,__ __ N05: __ __ __.__ __ N93: X __ __.__ __ __ Y __ __.__ __ __ Z __ __.__ __ __ GET: __ __ : __ : __ ALIGN GDC, VERIFY ORDEAL COPY LM TRACKING PAD (SEE GET - 104:35) P22 AUTO MNVR TO TRACKING ATTITUDE ORB RATE(IF TIME PERMITS IF NOT, AT 104:10) RO P338/76, YO	103:00 MSFN 103:29 103:30 103:39 103:45 REV 15 104:00	DOFF HELMET AND GLOVES RR TO STANDBY REPORT ESTIMATE OF LANDED LOCATION CLOSE SHADES 80mm/BW/CHECKLIST 60mm/HCEX/CHECKLIST 6 FRAMES FAR FIELD (FOCUS 50') 6 FRAMES NEAR FIELD (FOCUS 20') WITH EACH CAMERA REMOVE MAGS AND STOW INSTALL PROTECTIVE COVER AND STOW CAMERAS PHOTOGRAPH LUNAR SURFACE	AGS LUNAR SURFACE GYRO CALIBRATION LOAD AGS ASCENT TARGET H=60,000 FT, H DOT=32 FPS CLOSE SHADES, DOFF HELMET AND GLOVES P57-IMU ALIGN (REFSMMAT) GRAVITY MEASUREMENT N04: __ __ __ __ __ __ AGS LUNAR ALIGNMENT P57-IMU ALIGN (REFSMMAT) 2 CELESTIAL BODIES N04: __ __ __.__ __ N05: __ __ __.__ __ N71: __ __,__ __ N93: X __ __.__ __ __ Y __ __.__ __ __ Z __ __.__ __ __ N89: LAT __ __.__ __ __ LONG/2 __ __.__ __ __ ALT __ __.__ __ __ GET __ __ __:__ __:__ __ INITIALIZE AGS COPY AND LOAD ASCENT PAD DATA INSTALL WINDOW SHADES COARSE ALIGN RR CDU	COPY AGS AZIMUTH COPY LANDED LOCATION, GRAVITY MEASURE UPDATE CSM LM TRACKING PAD COPY P57 DATA COPY AGS AZIMUTH UPLINK LGC RLS CSM STATE VECTOR (TD + 1:40) UPDATE LM ASCENT PAD

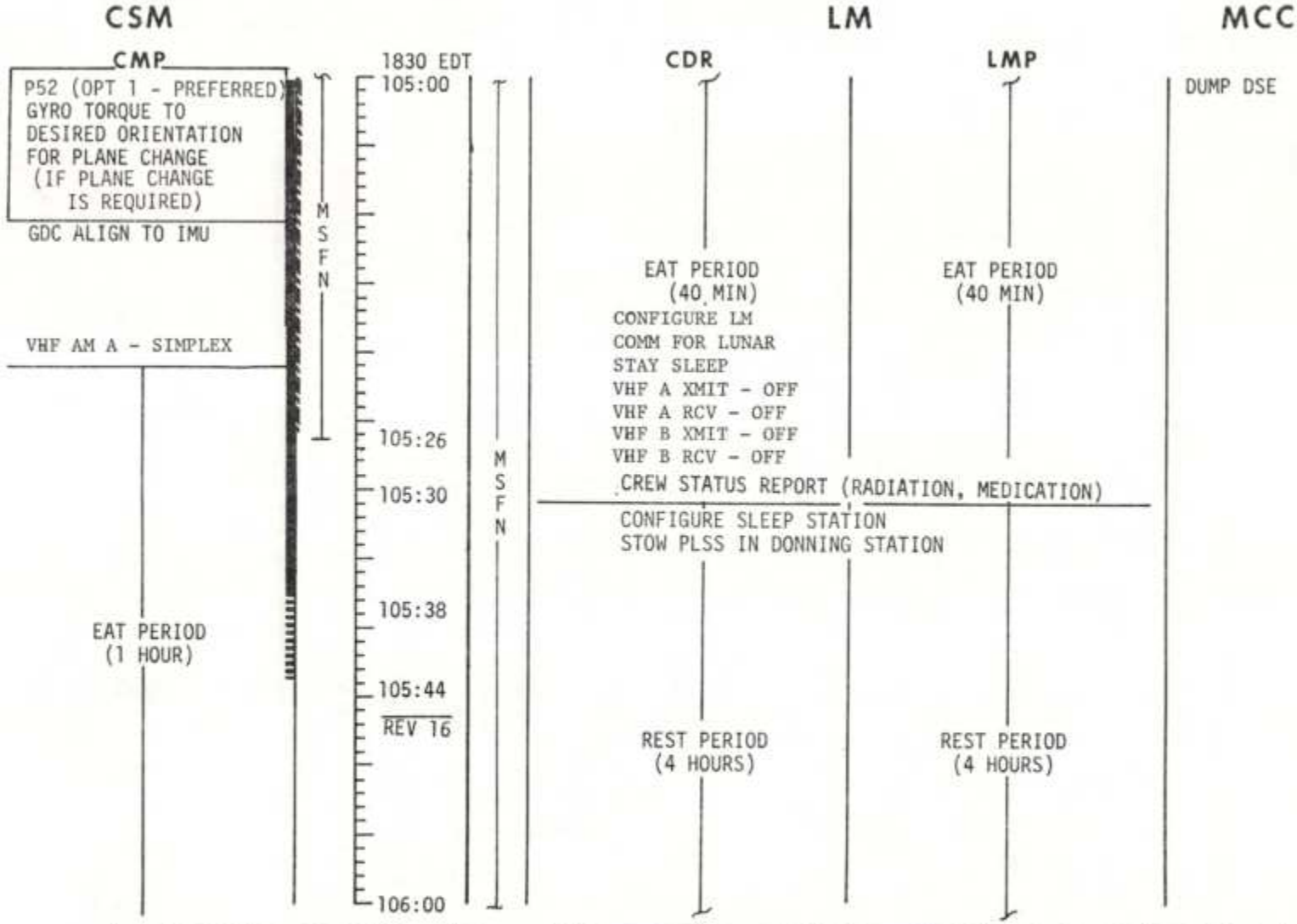
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	103:00 - 104:00	5/14-15	3-70

FLIGHT PLAN



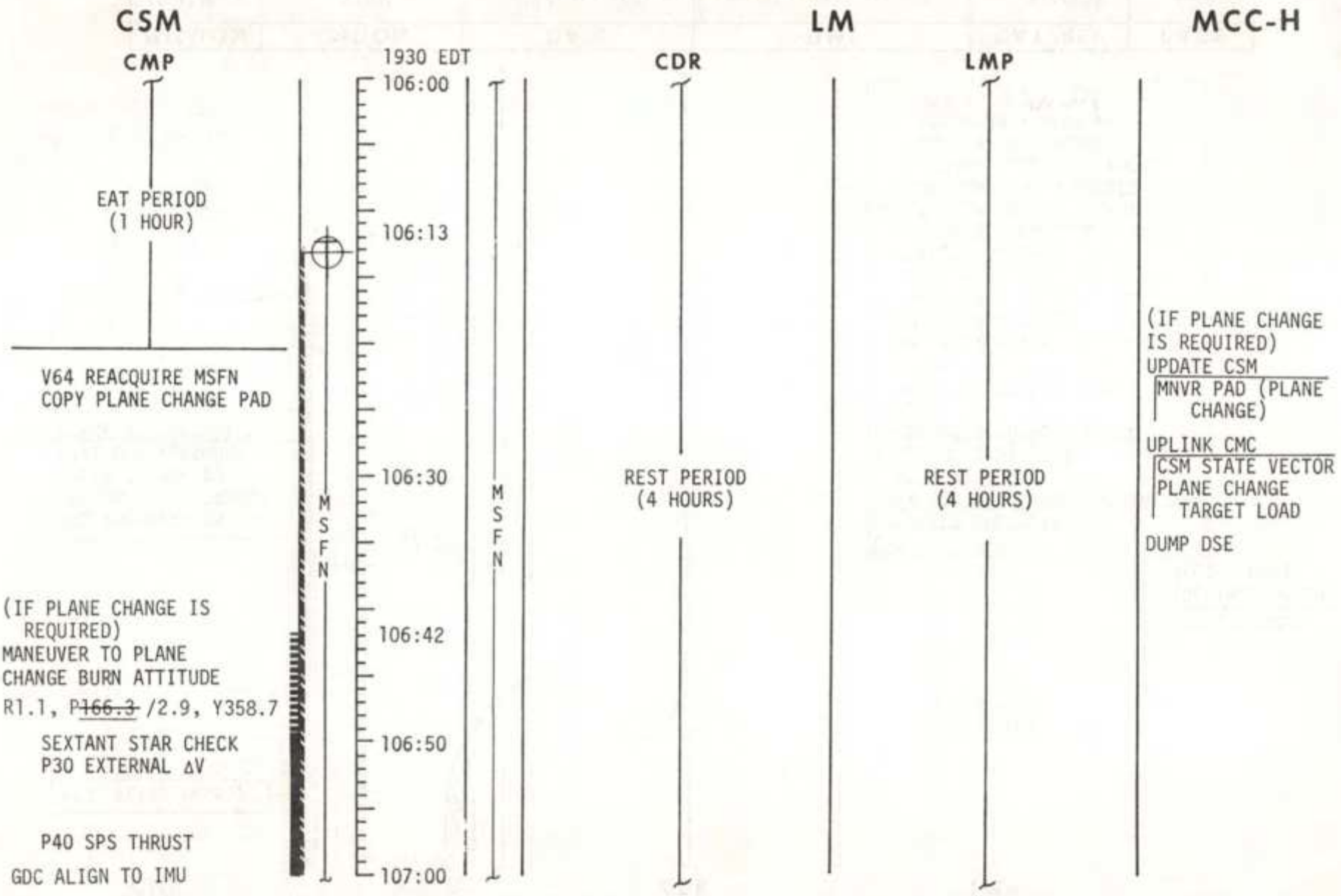
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	104:00 - 105:00	5/15	3-71

FLIGHT PLAN



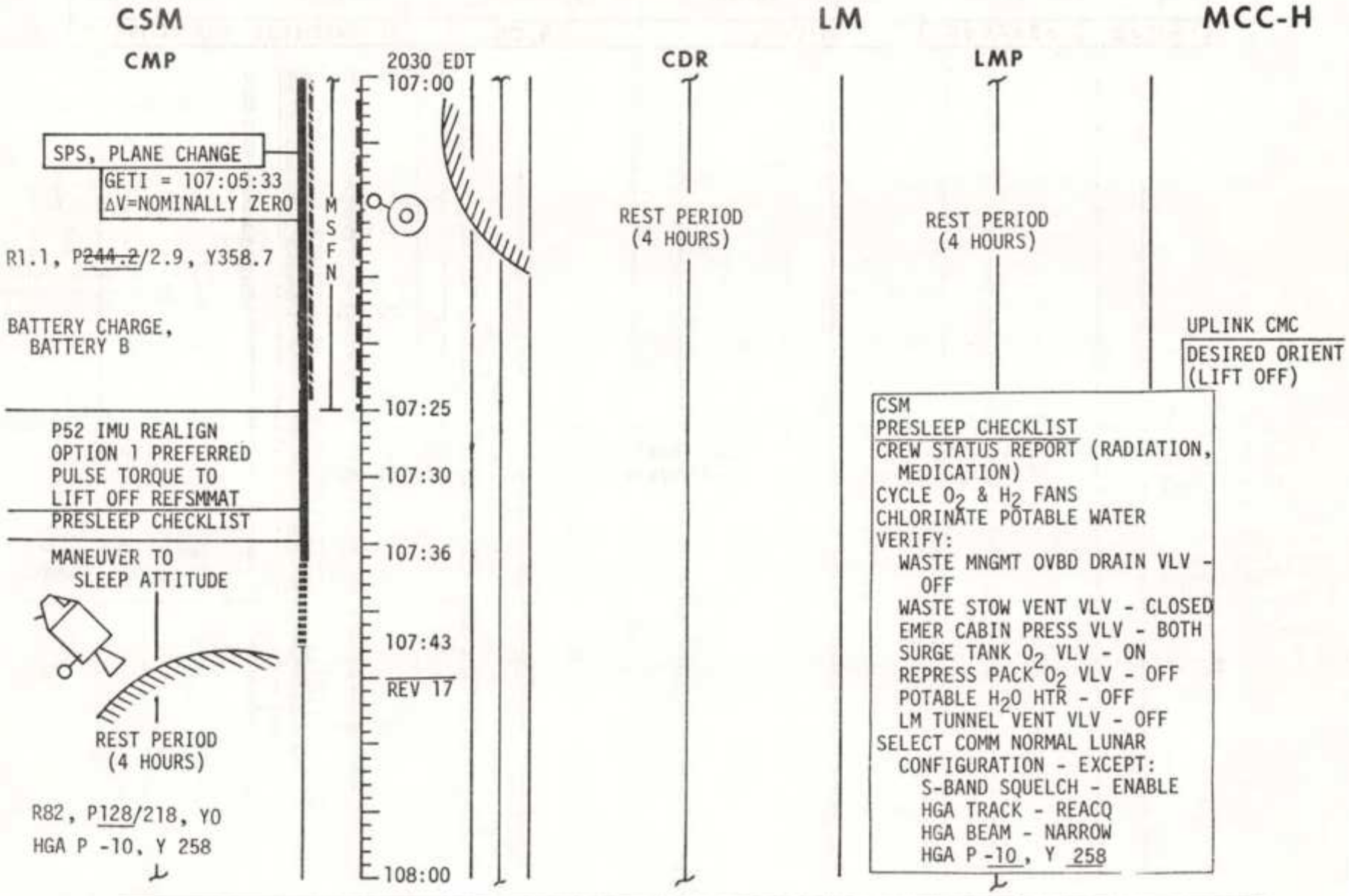
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	105:00 - 106:00	5/15-16	3-72

FLIGHT PLAN



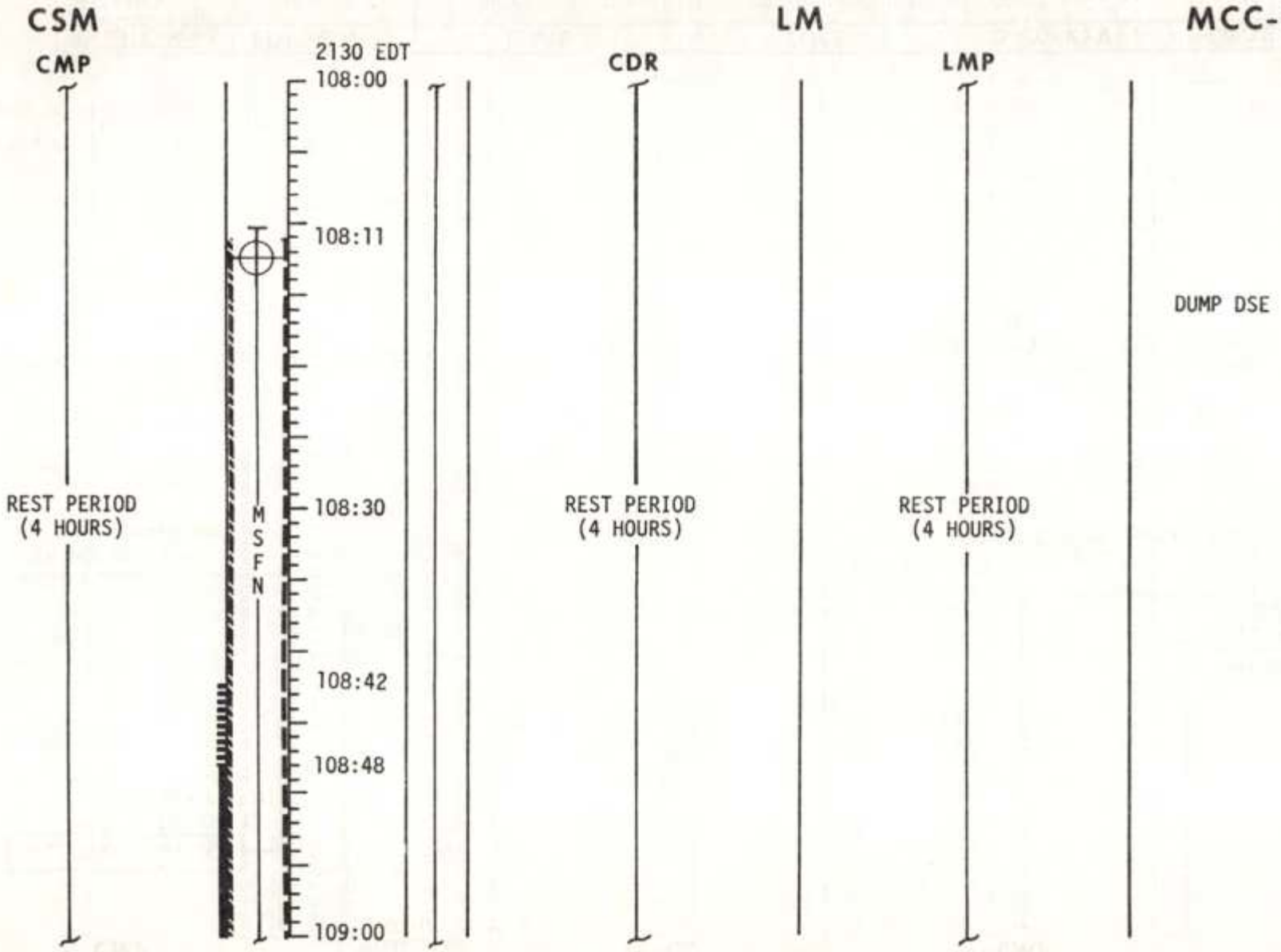
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	106:00 - 107:00	5/16	3-73

FLIGHT PLAN



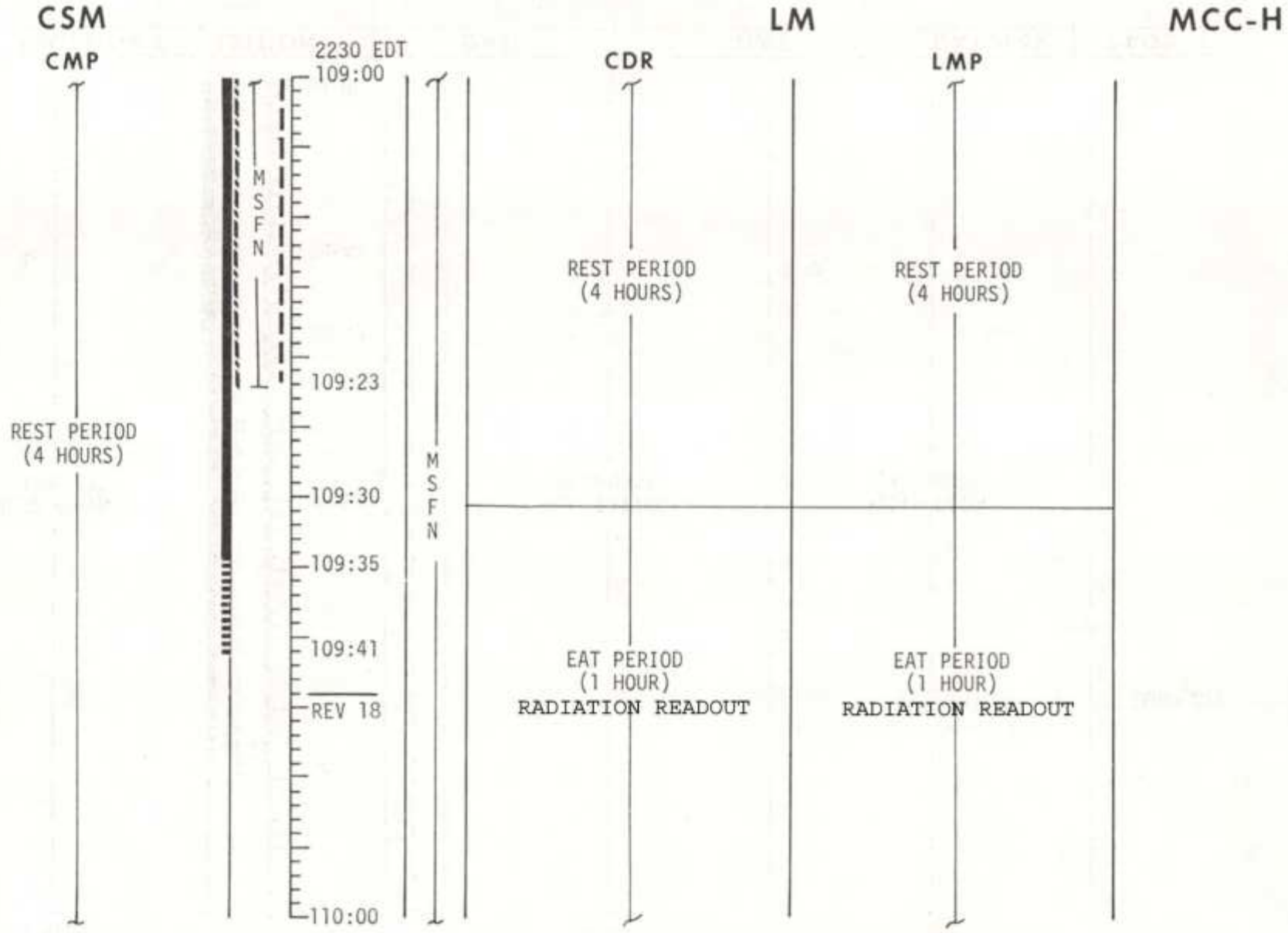
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	107:00 - 108:00	5/16-17	3-74

FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	108:00 - 109:00	5/17	3-75

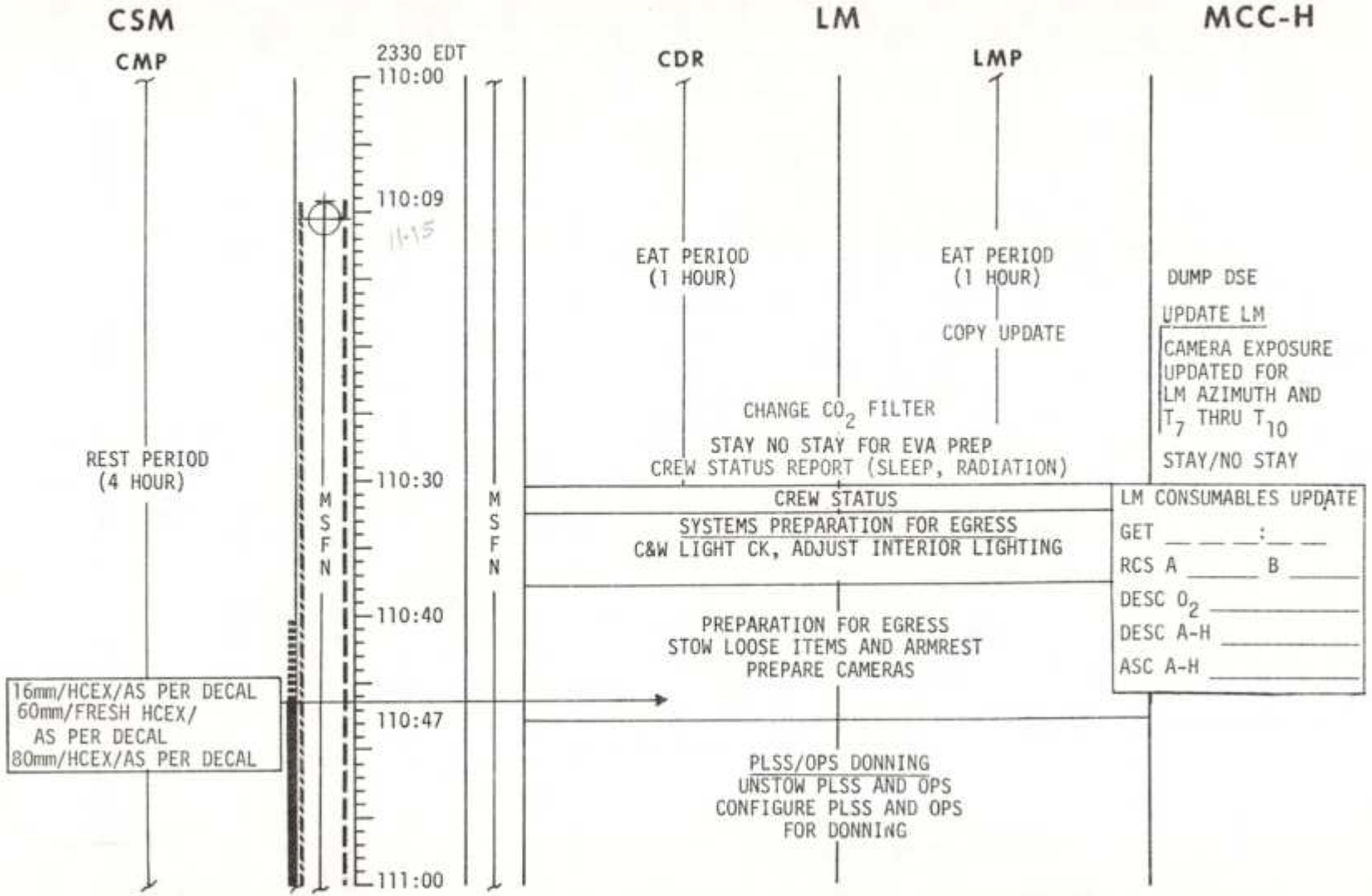
FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	109:00 - 110:00	5/17-18	3-76

FLIGHT PLANNING BRANCH

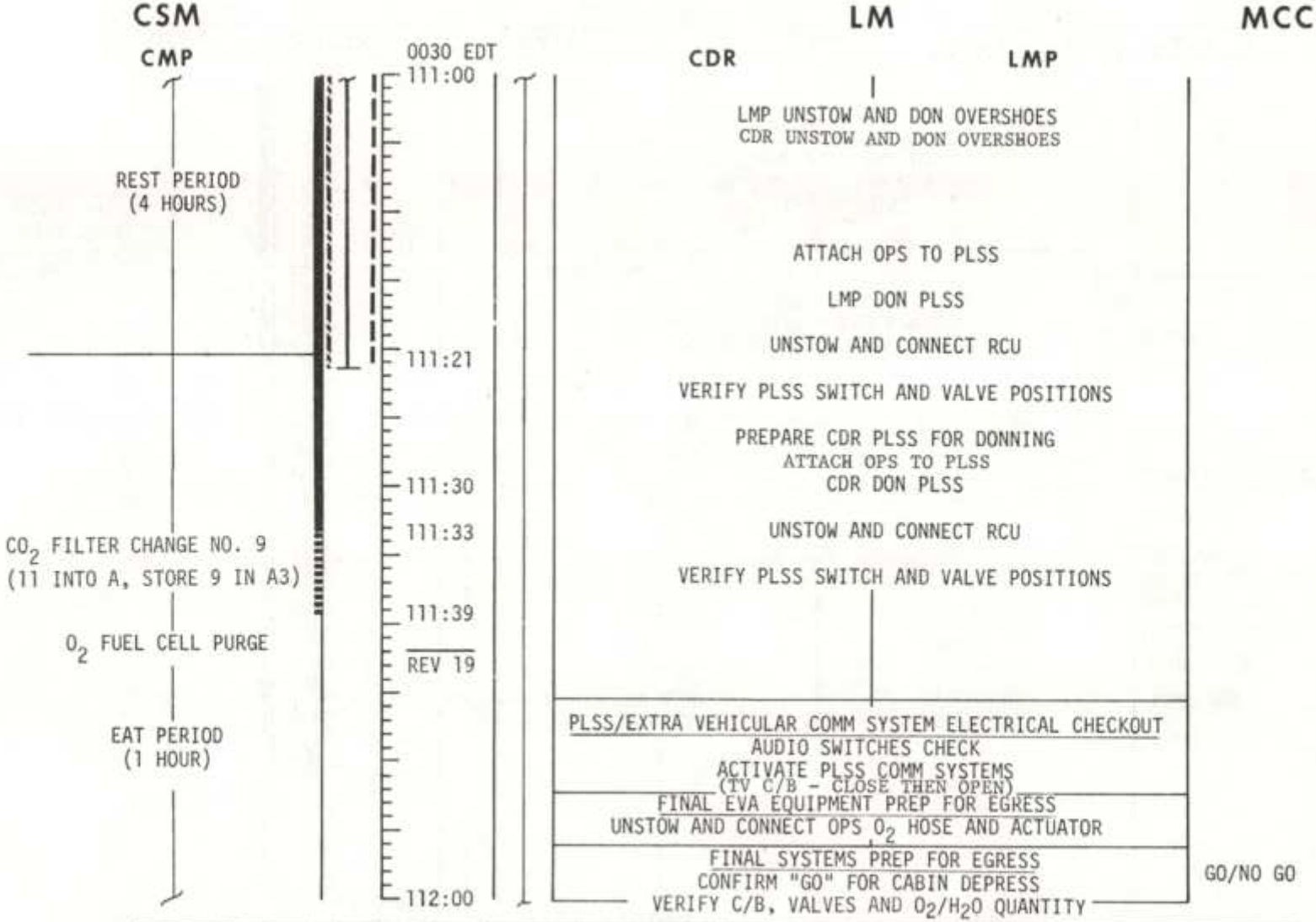
FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	110:00 - 111:00	5/18	3-77

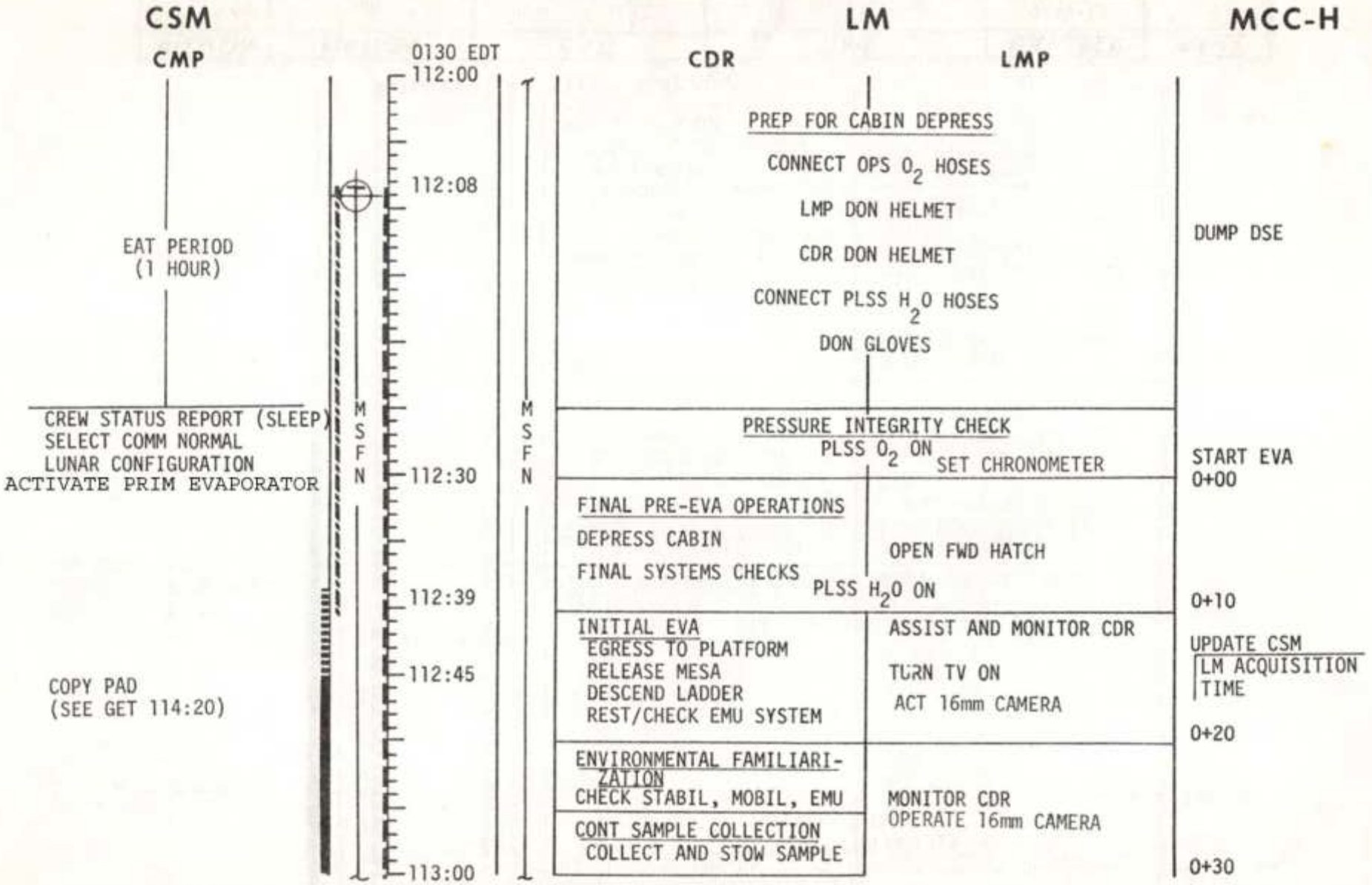
MSC Form 2189 (OT) (Nov 68) FLIGHT PLANNING BRANCH

FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	111:00 - 112:00	5/18-19	3-78

FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	112:00 - 113:00	5/19	3-79

FLIGHT PLAN

CSM

CMP

SET UP CAMERA FOR TRACKING
EL/250/BW-BRKT
INT (f5.6,250,INF)

PITCH DOWN 172° TO HEADS DOWN FOR LUNAR SURFACE OBSERVATION, ORB RATE

R180, P282/44, Y0

MSFN

MSFN

113:00
113:19
113:30
113:32
113:38
REV 20
114:00

LM

CDR

PRELIMINARY CHECKS
CK LM STATUS
CK LIGHTING VISIBILITY

REST
MONITOR AND PHOTOGRAPH
LMP EGRESS

TV DEPLOYMENT
CAMERA EQPT FROM MESA
CARRY TV TO SITE
MOUNT TRIPOD, PANORAMA, POSITION FOR EVA
PHOTOGRAPH SWC
PHOTO BULK SAMPLE AREA

BULK SAMPLE COLLECTION
CAMERA ON MESA
PREPARE SRC
COLLECT ROCK FRAGMENTS AND LOOSE MATERIAL
WEIGH SAMPLE
PACK AND SEAL SRC, CONNECT TO LEC
REST

LM INSPECTION
INSPECT QUAD IV, +Y GEAR
EVAL TERRAIN, VISIBILITY
INSPECT QUAD III, -Z GEAR
PHOTO QUAD II, EASEP OFF LOADING
INSPECT, PHOTO -Y GEAR
PHOTO PANORAMA
TAKE CLOSEUP PHOTOS
EASEP DEPLOYMENT

LMP

STILL-CAMERA TO SURFACE
FINAL LM CK
EVA GO

INITIAL EVA
EGRESS
DESCEND TO SURFACE

ENVIRONMENT FAMILIARIZATION
CK BALANCE, STABILITY, REACH, WALKING, EMU

SWC DEPLOYMENT
DEPLOY SWC IN SUN

EVA AND ENIRON EVAL
EVAL EVA CAPABILITY AND EFFECTS
EVAL LIGHTING/VISIBILITY AND SURFACE CHARACTERISTICS
PHOTO PANORAMA

LM INSPECTION
PHOTO QUAD I, +Z GEAR
PHOTO BULK SAMPLE AREA
DEPLOY ALSCC
PHOTO QUAD IV, +Y GEAR
PHOTO PANORAMA
PHOTO QUAD III, -Z GEAR
CAMERA TO CDR

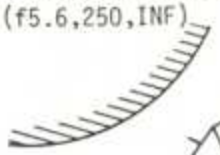
EASEP DEPLOYMENT
REMOVE EXPERIMENTS

MCC-H

0+30
EVA GO
0+40
0+50
1+00
1+10
1+20
1+30

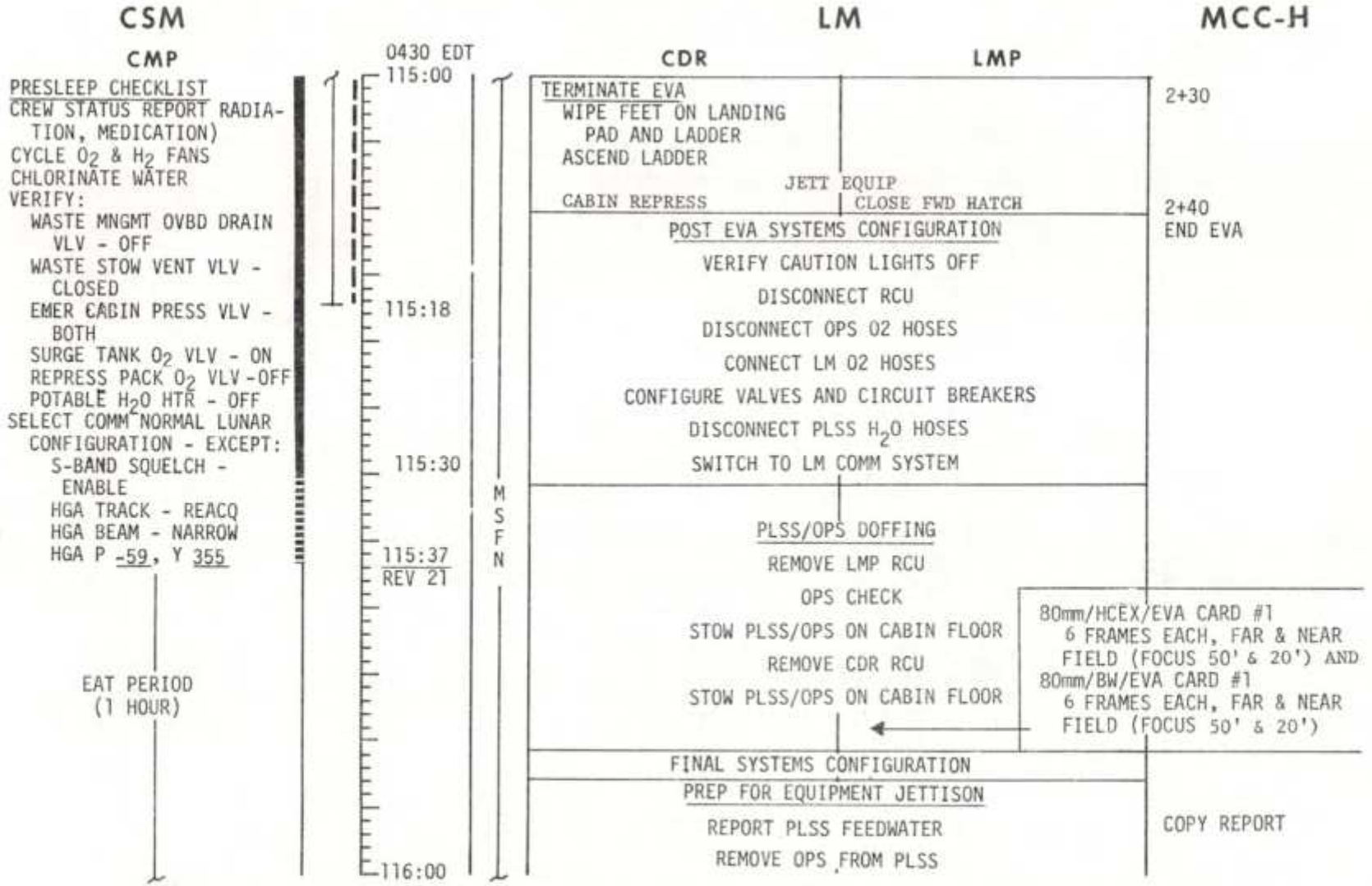
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	113:00 - 114:00	5/19-20	3-80

FLIGHT PLAN

CSM			LM		MCC-H
CMP	0330 EDT		CDR	LMP	
	114:00		SELECT DEPLOY SITE CARRY CAMERAS DEPLOY LR^3 EXPERIMENT PHOTO EXPERIMENTS	SELECT DEPLOY SITE CARRY EXPERIMENTS DEPLOY PSE TAKE CLOSEUP PHOTOS	1+30
	114:06				
EL/250/BW-BRKT, INT (f5.6,250,INF)  CHANGE SHUTTER TO 1/125 PITCH UP 38° ROLL 180° TO HEADS UP RO, P320/229,YO PITCH DOWN, PHOTOGRAPH LM WHILE TRACKING THROUGH COAS	MSFN	MSFN	DOCUMENTED SAMPLE COLLECTION REST/PHOTO LMP CLOSE-UP PHOTOS TETHER SAMPLE BAG TO LMP PHOTO SAMPLING UNSTOW GNOMON PHOTO DS AREA PHOTO SAMPLE COLLECTION STOW ALSCC FILM COLLECT ENVIRONMENTAL SAMPLES RETRIEVE AND STOW SWC PACK SRC CLOSE AND SEAL SRC	DOCUMENTED SAMPLE COLLECTION MOVE BULK SRC TO STRUTS OR FOOT PAD PREPARE DS SRC COLLECT CORE TUBE SAMPLE UNSTOW TOOLS COLLECT SAMPLES STOW ALSCC FILM COLLECT ENVIRONMENTAL SAMPLES COLLECT LOOSE MATERIAL CORE TUBE SAMPLE	1+40 DUMP DSE 1+50 LM ACQUISITION GET: __ __ __:__ __:__ 2+00
	114:30				
	114:37				
STOP PITCH AND ROLL 180° TO HEADS DOWN ATTITUDE FOR SURFACE OBSERVATIONS STOP CAMERA R180, P282/185, YO HGA P -7, Y 183			REST/PHOTO LMP	EVA TERMINATION WIPE SUIT AND EMU WIPE FEET ON LANDING PAD AND LADDER ASCEND LADDER INGRESS CABIN CHECK LM OPERATE SEQ CAMERA RECEIVE AND STOW SRC AND MAGAZINE RECEIVE AND STOW SRC	2+10 2+20 2+30
	114:43		SRC TRANSFER TRANSFER BULK SRC AND STILL CAMERA MAGAZINE REST CHECK EMU TRANSFER DS SRC		
	115:00				

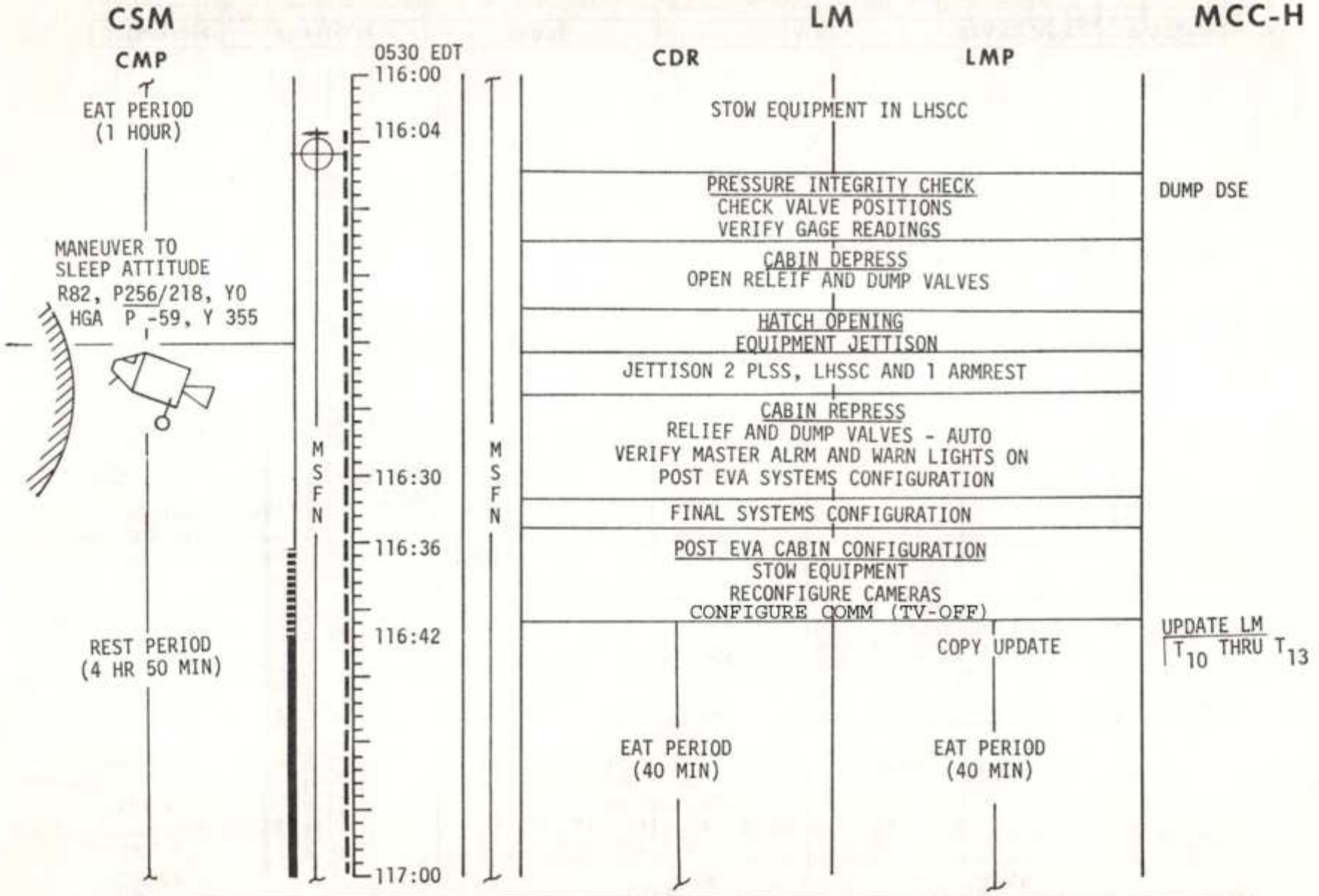
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	114:00-115:00	5/21	3-81

FLIGHT PLAN



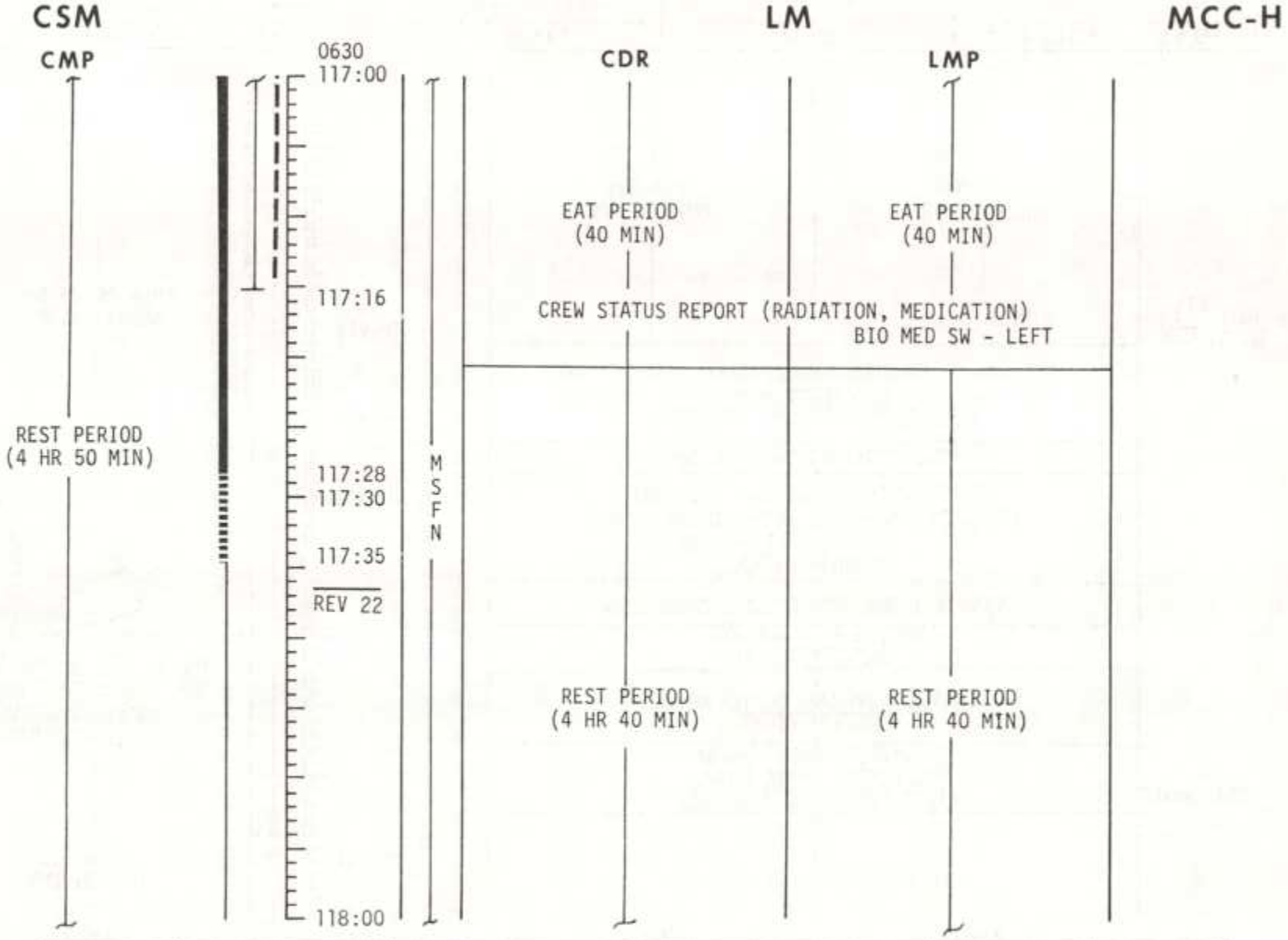
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	115:00-116:00	5/20-21	3-82

FLIGHT PLAN



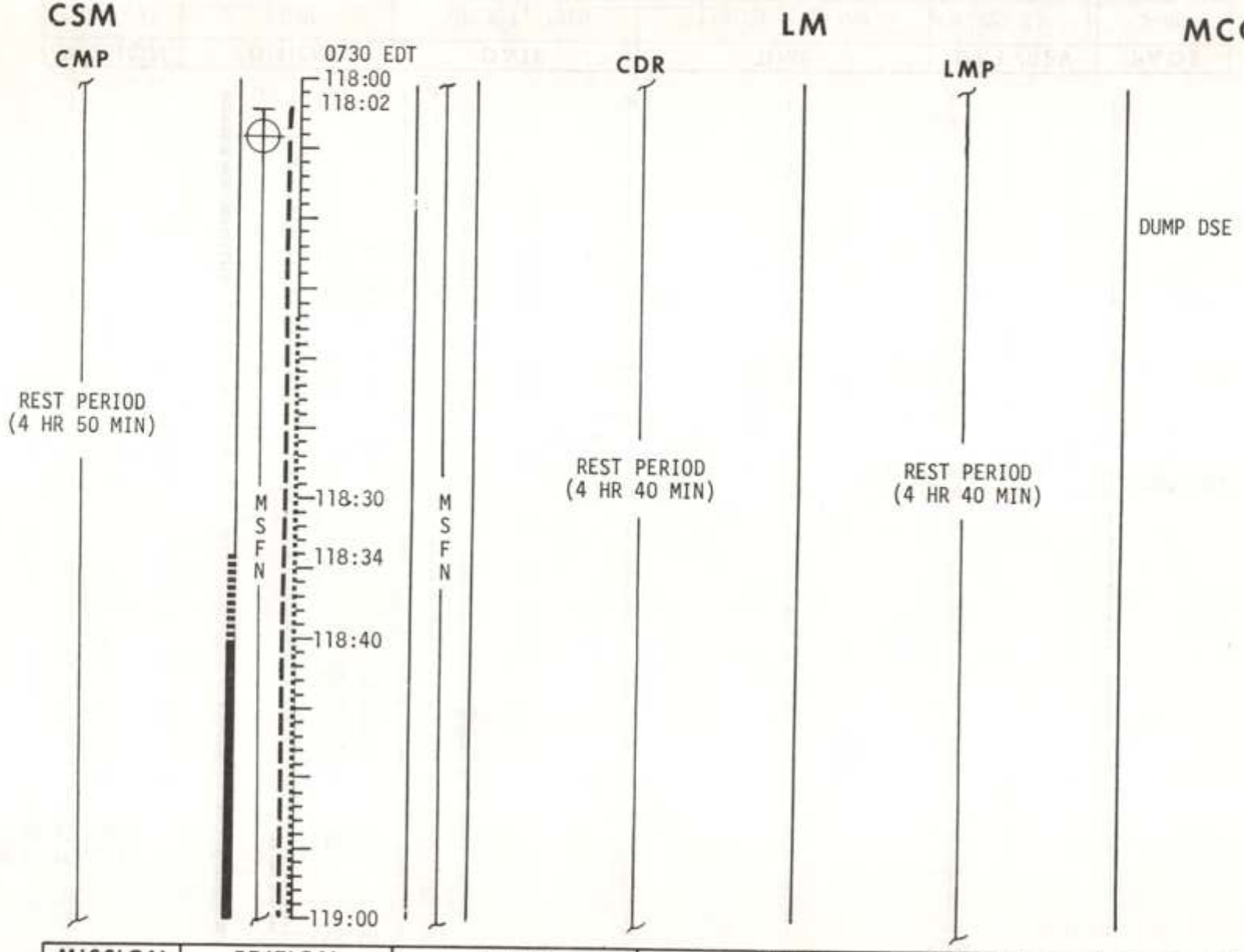
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	116:00 - 117:00	5/21	3-83

FLIGHT PLAN



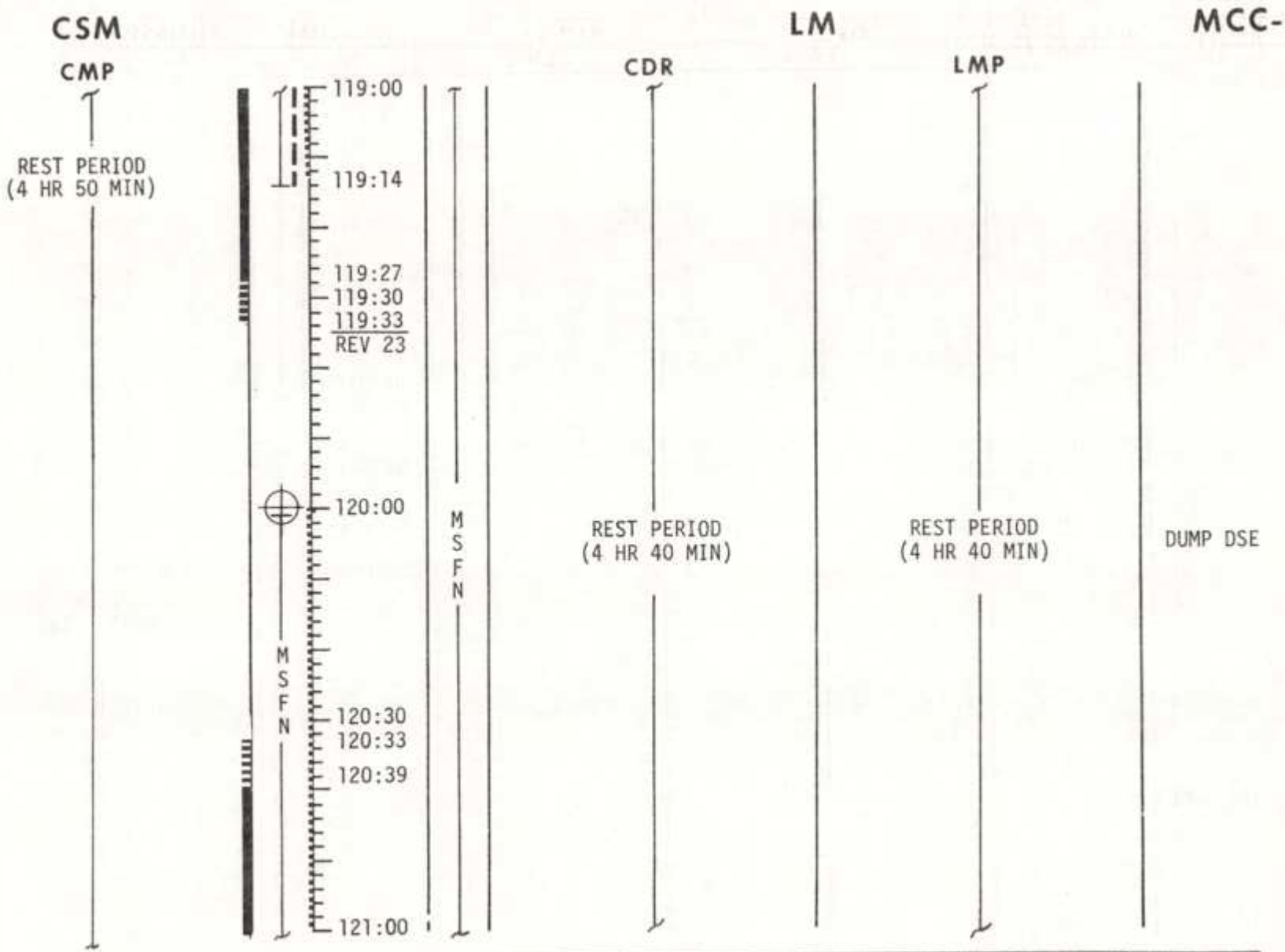
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	117:00 - 118:00	5/20-21	3-84

FLIGHT PLAN



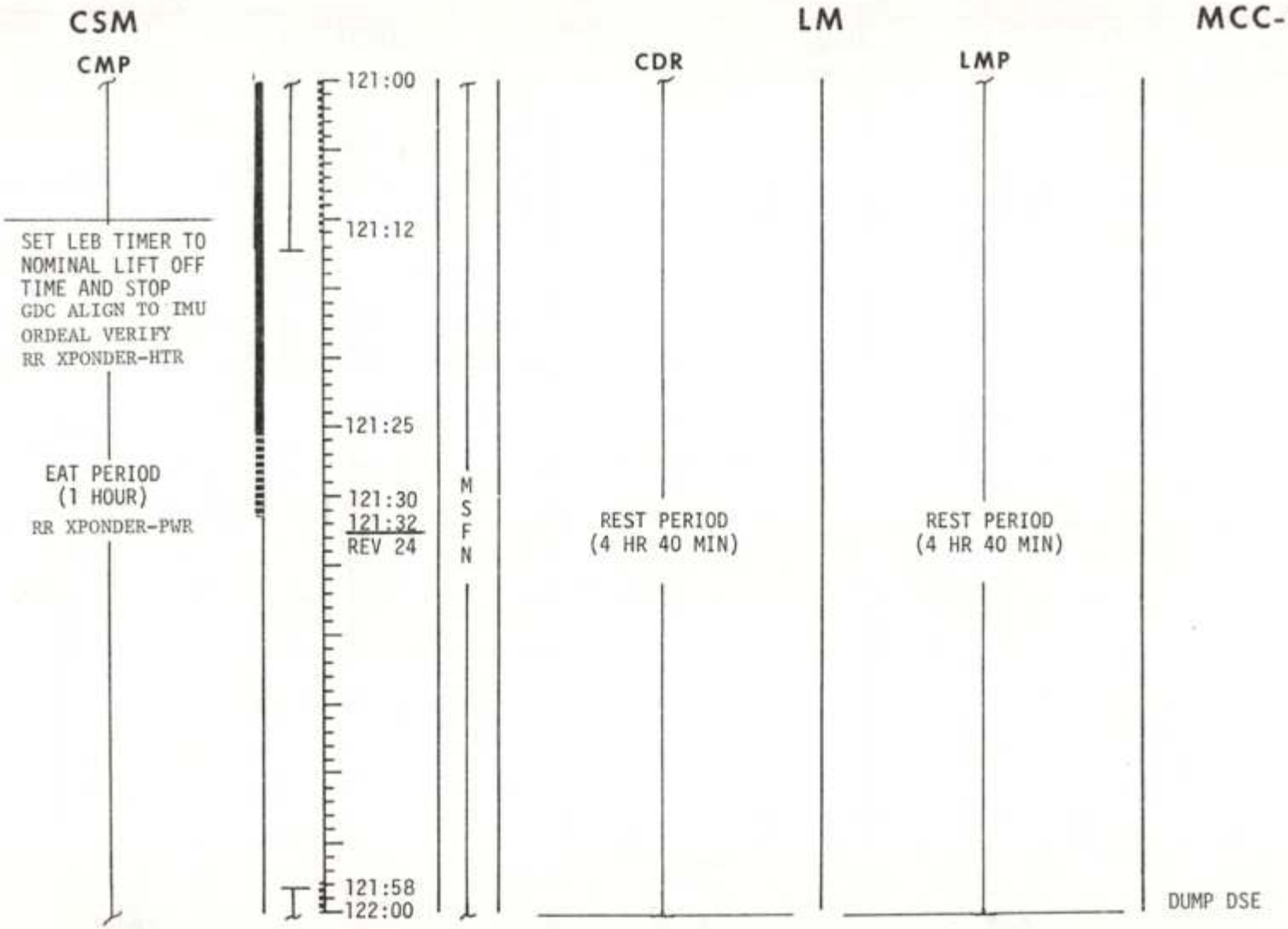
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	118:00 - 119:00	5/22-23	3-85

FLIGHT PLAN



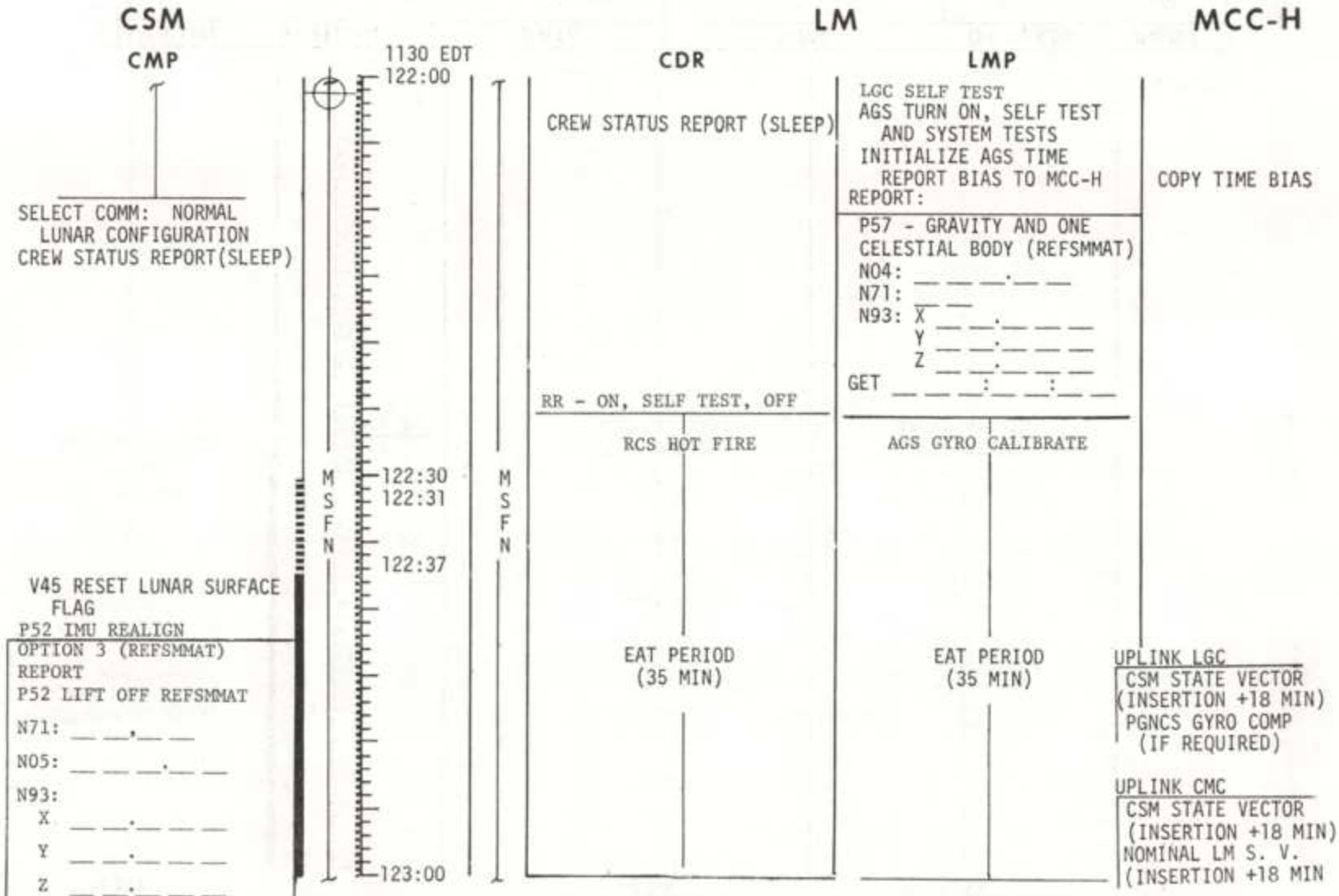
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	119:00 - 121:00	5/22-23	3-86

FLIGHT PLAN



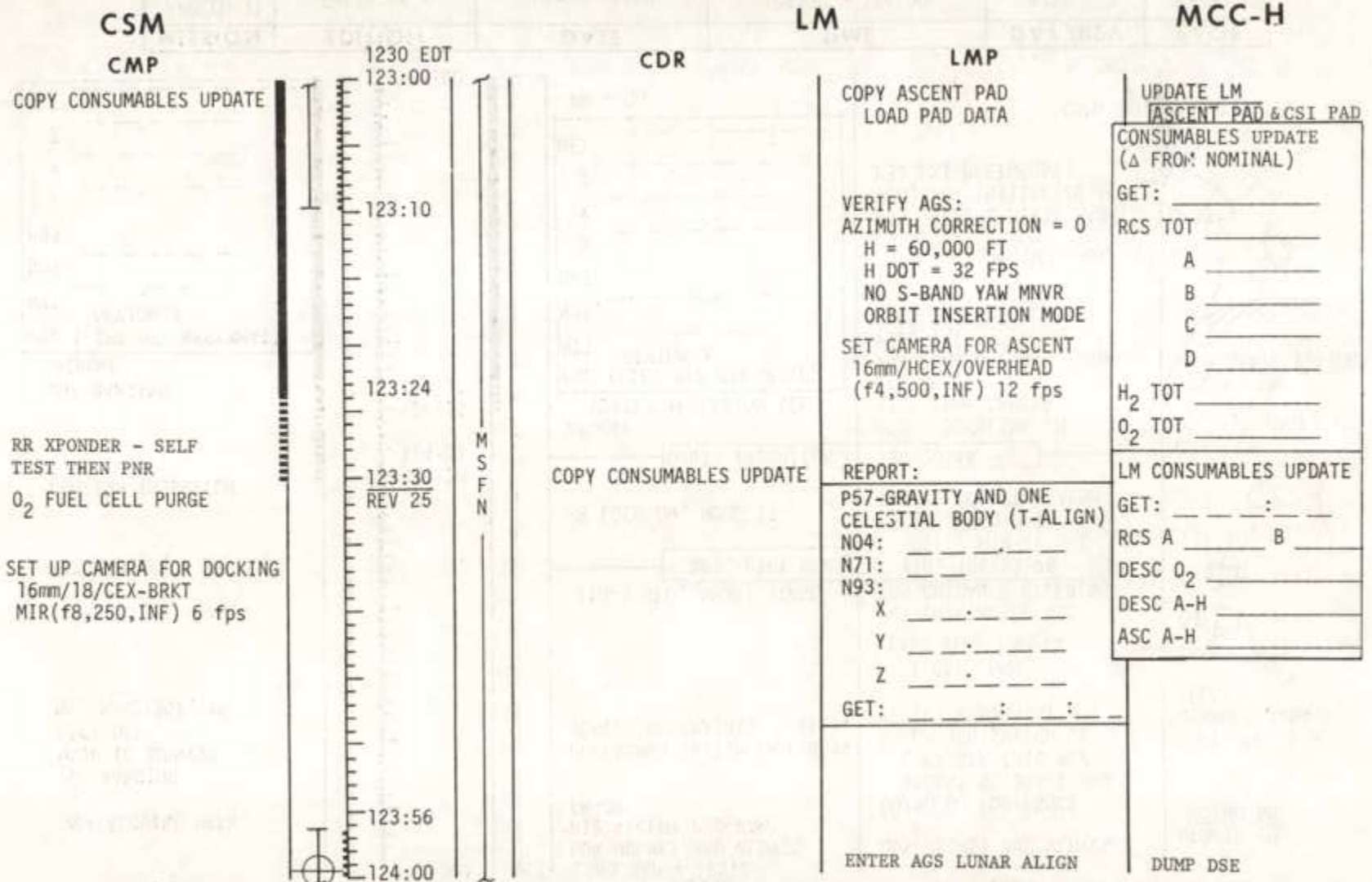
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	121:00 - 122:00	5/23-24	3-87

FLIGHT PLAN



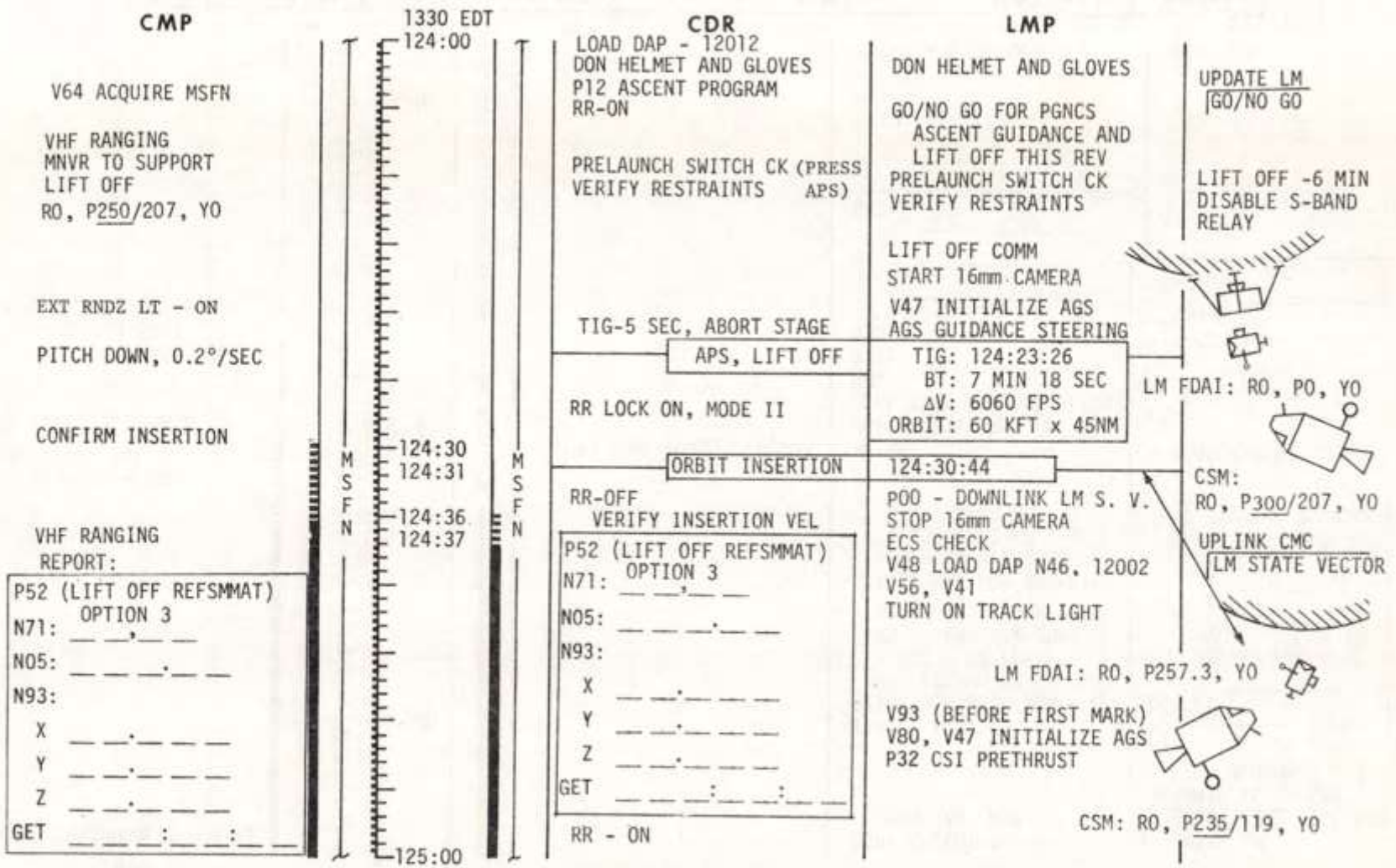
EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11 Revision A	July 8, 1969	122:00 - 123:00	6/24	3-88

FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	123:00-124:00	6/24-25	3-89

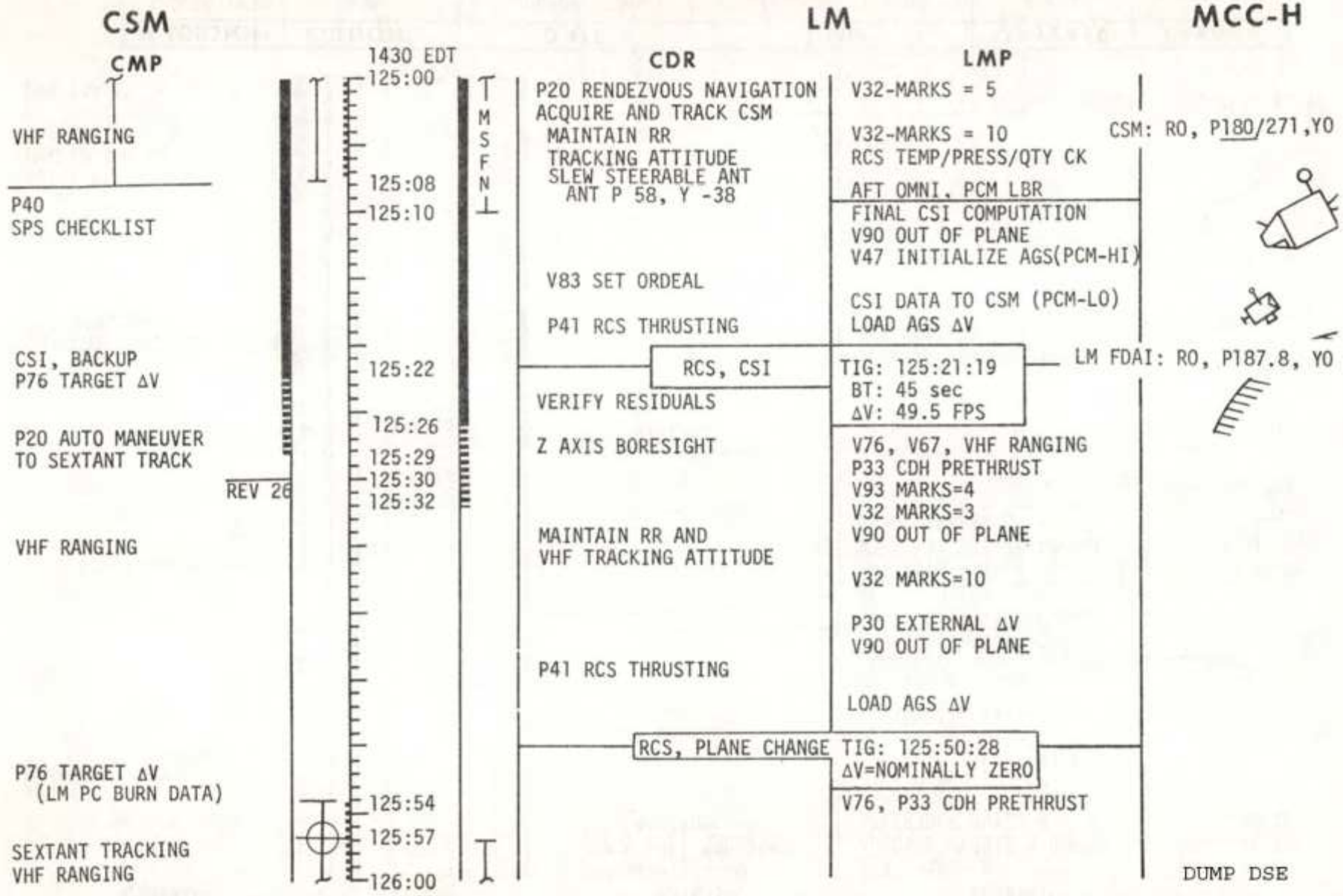
FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	124:00 - 125:00	6/25	3-90

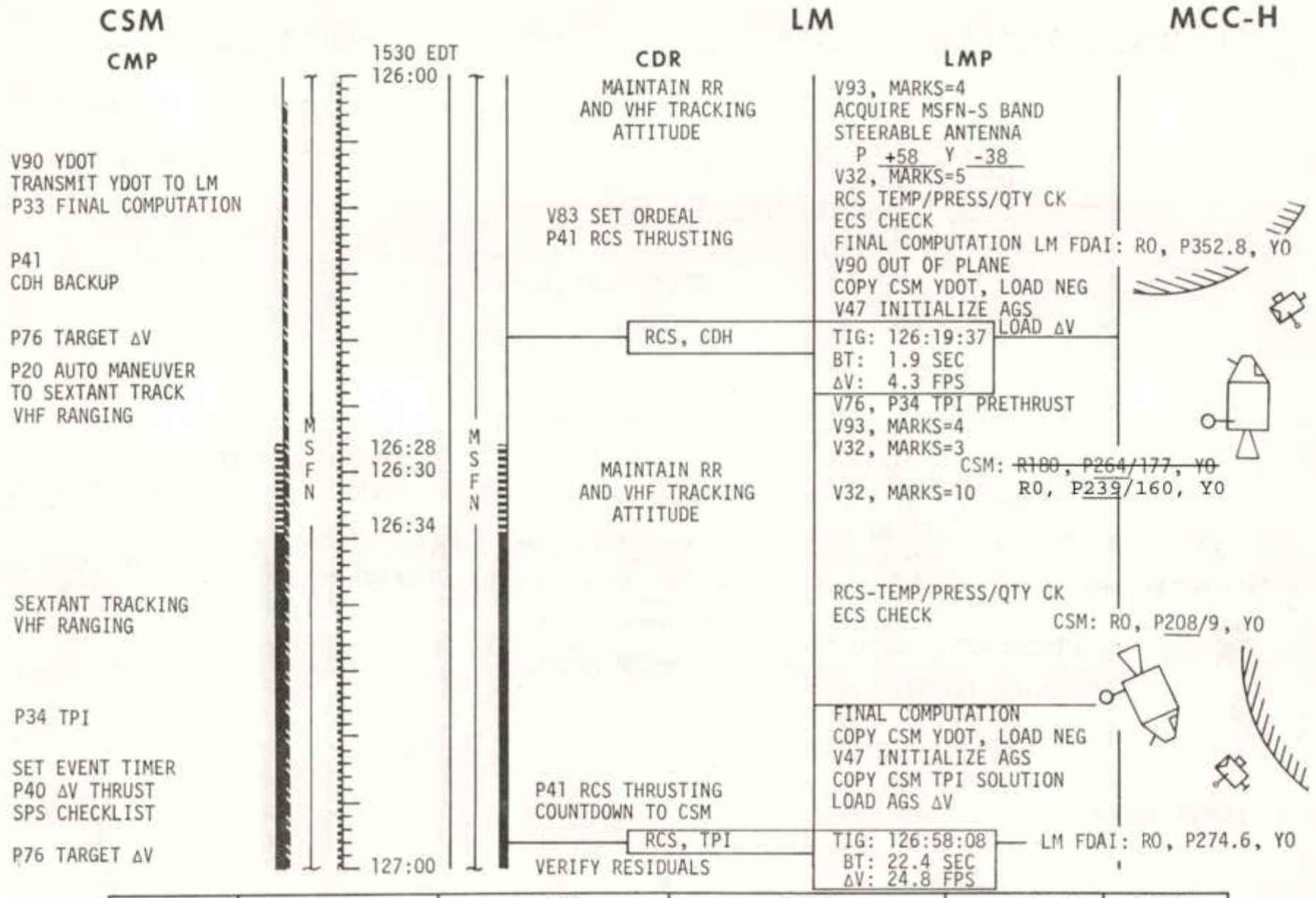
MSC Form 2189 (OT) (Nov 68) FLIGHT PLANNING BRANCH

FLIGHT PLAN



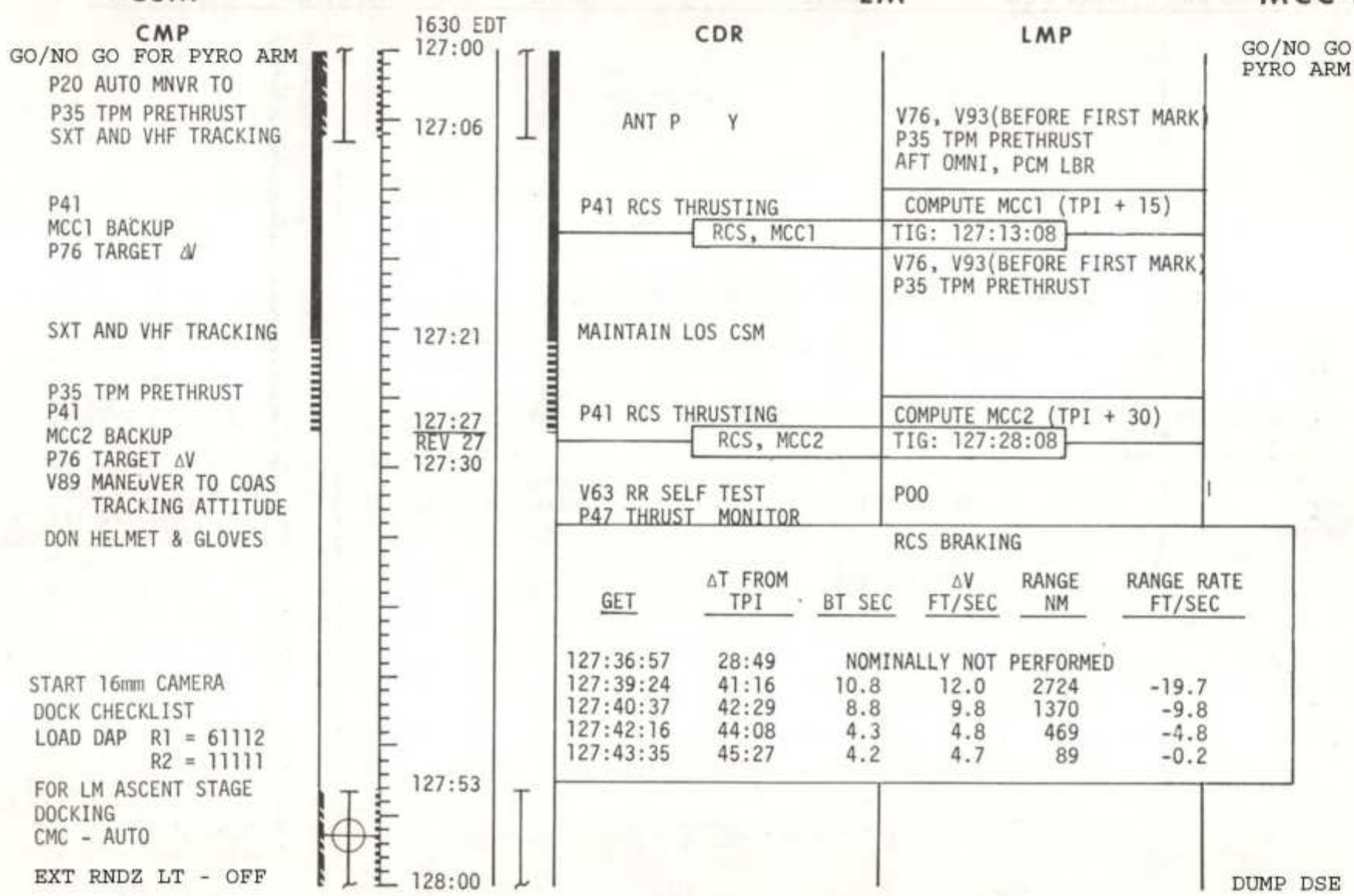
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	125:00 - 126:00	2/25-26	3-91

FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	126:00 - 127:00	6/26	3-92

FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	127:00 - 128:00	6/26-27	3-93

FLIGHT PLAN

CSM			LM		MCC-H
CMP	1730 EDT		CDR	LMP	
CONTACT: CMC - FREE NULL RATES ROLL RIGHT 60°, PITCH UP 53°	128:00		DOCKING 128:00		CSM: R300, P4/332, Y0
				LM FDAI: R0, P152, Y0	
GO INERTIAL PRESS CSM TO 5.5 PSIA DISABLE JETS B3 & C4 ADJUST O_2 FLOW TO .6 LBS/HR PRESS TUNNEL TO 3 PSID FOR LEAK CK, THEN EQUALIZE CM/LM ΔP INFORM LM WHEN PRESS EQUAL. REMOVE HATCH AND STOW VERIFY LATCHES PASS BAGS & BRUSH TO LM	128:26 128:30 128:32	MSFN	DOFF HELMET & GLOVES AND TEMPORARILY STOW OPEN LM HATCH REMOVE AND STOW PROBE & DROGUE RETRIEVE THE FOLLOWING ITEMS FROM CSM: HELMET STOWAGE BAGS (2) SRC (ROCK BOX) BAGS (2) CSC (GRAB SAMPLE) BAG(1) 70MM MAGAZINE BAG (1) CLOSEUP MAGAZINE BAG (1) VACUUM BRUSH & HOSE GLOVE BAGS (2)	DOFF HELMET & GLOVES AND TEMPORARILY STOW ASSIST CDR TEMPORARILY STOW BAGS AND BRUSH	CSM: R 0, P 102/25, Y 0 HGA: P -53, Y 3 LM FDAI: R 0, P 205, Y 60 ANT: P 173, Y 72
			CONFIGURE CDR SUIT LOOP FOR VACUUM CLEANING	ASSIST CDR	
	129:00				

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	128:00 - 129:00	6/27	3-94

FLIGHT PLAN

CSM		LM		MCC-H
CMP	1830 EDT	CDR	LMP	
	129:00			
V66 - TRANS CSM STATE VECTOR TO LM SLOT	129:05	VACUUM BRUSH FWD DUMP VALVE FILTER	UNSTOW AND HOLD SRC'S FOR CLEANING	UPLINK CSM STATE VECTOR
		VACUUM SRC'S		
RETRIEVE SRC'S FROM LM AND STOW IN B5 AND B6			BAG SRC'S AND TRANSFER TO CM	
		VACUUM:		
	129:19	CSC		
		70MM MAGAZINE		
		CLOSEUP MAGAZINE	HOLD EQUIPMENT FOR CLEANING	
	129:26 REV 28			
	129:30	HELMETS		
RETRIEVE BAGGED ITEMS FROM LM AND STOW: CSC - A5 CLOSEUP MAGAZINE - A5 70MM MAGAZINES - R13 HELMETS-FOOD CONTAINERS		GLOVES	BAG ITEMS AND TRANSFER TO CSM (GLOVES IN HELMETS)	
		VACUUM BRUSH LMP'S PGA		
			VACUUM BRUSH CDR'S PGA	
	129:51			
		VACUUM THE BRUSH AND STOW IN ISA		
	130:00			DUMP DSE

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	129:00 - 130:00	6/28	3-95

FLIGHT PLAN

CSM		LM		MCC-H
CMP		CDR	LMP	
	130:00	DISCONNECT FROM LM AND TRANSFER TO CM WITH ISA		
REMOVE ISA CONTENTS AND STOW. PLACE CM JETTISONABLE ITEMS INTO ISA AND TRANSFER ISA TO LM			RETRIEVE ISA AND INSTALL ON PANELS 1 & 2	
	130:25		CONFIGURE LM SYSTEMS FOR JETTISON	
	130:30		DISCONNECT FROM LM HOSES CLOSE LM HATCH	
	130:31			
UNSTOW AND INSTALL CSM HATCH			IVT TO CSM	
HATCH INTEGRITY CHECK				
DEPRESSURIZE TUNNEL				
	131:00			

MSFN

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	130:00 - 131:00	6/29	3-96

FLIGHT PLANNING BRANCH

FLIGHT PLAN

MCC-H

2030 EDT

131:00
131:03

EQUIPMENT STOWAGE

131:18

VACUUM PGAS

131:25
REV 29

SET UP CAMERA FOR LM JETTISON
16mm/18/CEX-BRKT,MIR
(f8,250,7) 12 fps

131:30

SM RCS CHECK
ENABLE JETS B-3 AND C-4
CONFIGURE DAP - R1=11102, R2=11111

GO/NO-GO FOR PYRO ARM
PYRO LOGIC ARM
THRUST MONITOR - P47
START CAMERA

GO/NO GO

131:50

DUMP DSE

LM JETTISON

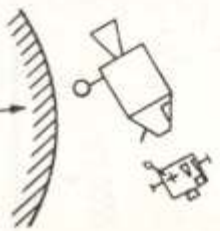
SM RCS CHECK

GETI = 131:53:05
BT =
ΔV = 1.5 FPS
RETROGRADE
ORBIT: 58.5 X 59.4

132:00

NOTES

DAP CONFIGURATION
FOR LM JETTISON
CSM, 0.5° DB, 0.5°/SEC
A/C ROLL, 4 JET



RO, P45/25, Y0
HGA P-53, Y3

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	131:00 - 132:00	6/29	3-97

FLIGHT PLAN

MCC-H

2130 EDT

132:00

UPDATE
TEI_{30} MNVR PAD

MNVR TO TEI BURN ATTITUDE (EXCEPT ROLL)

V66 - TRANS CSM STATE VECTOR TO LM SLOT
BURN STATUS REPORT
RECORD PRELIMINARY TEI_{30} MNVR PAD

GO INERTIAL

DOFF AND BAG PGA'S HELMETS
AND GLOVES

132:23

132:30

MSFN

EAT PERIOD

133:00

NOTES:

R1.1 P93.2/52.6 Y13.8
HGA P -79, Y 10

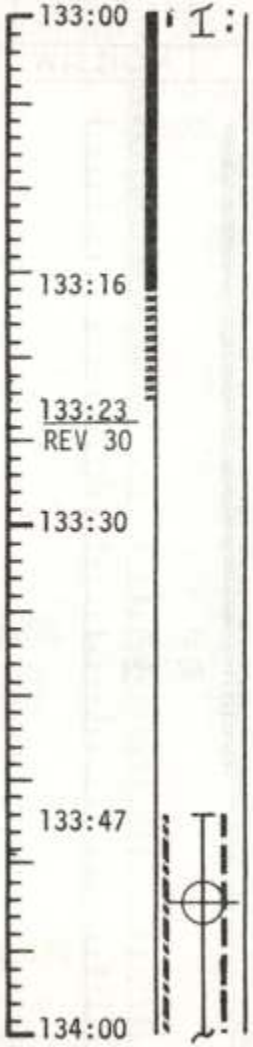
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	132:00 - 133:00	6/29	3-98

MSC Form 29 (May 69) FLIGHT PLANNING BRANCH

MCC-H

FLIGHT PLAN

NOTES



CO_2 FILTER CHANGE NO. 10
(12 INTO B, STORE 10 IN A3)

EAT PERIOD

O_2 FUEL CELL PRUGE

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	133:00 - 134:00	6/30	3-99

FLIGHT PLAN

MCC-H

2330 EDT

134:00

PRE TEI SYSTEMS CKS:
- C&W CK
- CM RCS MON CK
- SM RCS MON CK
- EPS MONITOR CK
- ~~ECS REDUNDANT COMPONENTS CK~~

RECORD FINAL TEI_{30} MNVR PAD AND BLOCK DATA
GO/NO GO FOR TEI_{31} THIS REV

UPDATE
- TEI_{30} MNVR PAD
- BLOCK DATA (TEI_{31})
- GO/NO GO

UPLINK CMC
- CSM STATE VECTOR
- TEI TGT LOAD

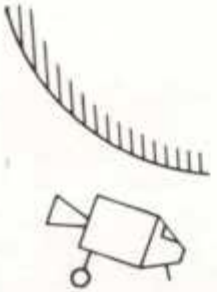
134:29

134:30

MSFN

V66 - TRANS CSM STATE VECTOR TO LM SLOT

IMU REALIGN - P52
OPTION 3 - REFSMMAT
AND DRIFT CHECK



R1.1 P201.4/52.6 Y13.8
HGA P -78, Y 5

P30 EXTERNAL ΔV

135:00

NOTES

REPORT:

P52 (LIFT OFF REFSMMAT)
N71: __ __,__ __
N05: __ __ __.__ __
N93:
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __
GET __ __ __:__ __:__

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	134:00 - 135:00	6/30	3-100

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TEI
BURN CHART

	P OR Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
TEI	10°/SEC TAKEOVER	+10° TAKEOVER	BT + 2 SEC & ΔV_C = -40 FPS	TRIM X AXIS TO 0.2 FPS

TEI ABORT MODES			
TEI V_{GO}	BT	TRAJECTORY	ABORT MODE
3292.7 - 1436.0	0-01:30	LUNAR ORBIT	MODE III - AFTER 1 REV
1436.0 - 1207.0	1:30- 1:40	UNSTABLE	MODE II - 2 SPS BURNS FOR ORBIT STABILIZATION AND WATER OR CLA LANDING.
1207.0 - 0	1:40-02:29	UNSTABLE/ HYPERBOLIC	MODE I - 1 BURN AT TEI + 2 HRS P37 AT SPHERE OF INFLUENCE HYPERBOLIC (ΔV 580 to 0, BT 02:05 - 02:29

FLIGHT PLAN

MCC-H

0030 EDT

135:00

135:02

SPS THRUST - P40
ROLL TO BURN ATT
DAP - R1=10101 R2=11111
R181.1, P280.4/52.6,Y13.8

SXT STAR CK

DAP CONFIGURATION FOR TEI
CSM, 5°DB, 0.2°/SEC
A/C ROLL, 4 JET

135:15

EMS ΔV TEST

~~DAP-R1=10111 R2=11111~~

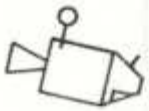
REV 31
135:21

SM RCS CK

GDC ALIGN TO IMU

TEI

GET: 135:24:34
ULLAGE = 2 JET 16 SEC
ΔV = 3292.7 FPS
BT = 2 MIN 29 SEC



135:30

V48-DAP UPDATE (S/C WT)
SM RCS MON CK
SPS MON CK
V66 - TRANS CSM STATE VECTOR TO LM SLOT
PITCH DOWN TO ACQ MSFN AND
VISUALLY ACQ MOON

R181.1, P355/52.6, Y13.8

UPLINK
DESIRED ORIENTATION
(PTC)

SET CAMERA FOR POST-TEI
EL/80/CEX(f5.6,250,INF)
OR
EL/80/BW(f4,250,INF)

MSFN

IMU REALIGN - P52
OPTION 1 - PREFERRED
AND STAR CK

136:00

NOTES

BURN STATUS REPORT

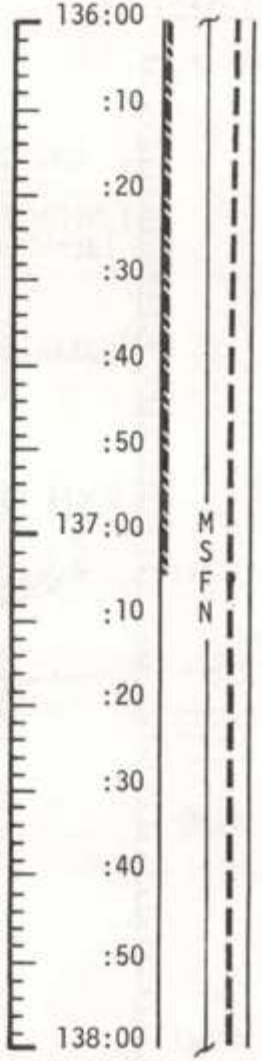
X	X			ΔTIG
X	X			BT
				V_{gx}
		TRIM		
X	X	X		R
X	X	X		P
X	X	X		Y
				V_{gx}
				V_{gy}
				V_{gz}
				ΔV_c
X	X	X		FUEL
X	X	X		OX
X	X	X		UNBAL

P52 - PULSE TORQUE
TO PTC REFSMMAT
PLATFORM ALIGN
CHECKED WITH
OPTICS

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	135:00 - 136:00	6/TEC	3-101

MCC-H

FLIGHT PLAN



WIPE EXCESSIVE
MOISTURE FROM
TUNNEL HATCH AREA

DEACTIVATE PRIM EVAPORATOR

START PTC
P 270° Y 0

EAT PERIOD-ALL

PRESLEEP CHECKLIST

CREW STATUS REPORT (RADIATION AND MEDICATION)
CYCLE O_2 & H_2 FANS
CHLORINATE POTABLE WATER
SELECT NORMAL LUNAR CONFIG
EXCEPT: (FOR COAST ASLEEP)
- S-BD NORMAL MODE VOICE - OFF
- S-BD AUX TAPE - OFF
- TAPE RCDR FWD - OFF
- GO TO HGA OR CONTINUE OMNI OPS PER MSFN

OMNI OPS
- S-BD ANT OMNI - OMNI
- S-BD ANT OMNI - B

HI GAIN OPS
- HI GAIN ANT BEAM - NARROW
- HI GAIN ANT TRACK - REACQ
- S-BD ANT-HI GAIN

VERIFY:
- WASTE MNGT OVBD DRAIN-OFF
- WASTE STOW VENT VLV-CLOSED
- EMERG CABIN PRESS VLV-BOTH
- SURGE TK O_2 VLV - OFF
- LM TUNNEL VENT VLV - OFF
- POT H_2O HTR - OFF

AUTO RCS JET SELECT (16) - OFF

REST PERIOD
(10 HOURS)

NOTES

PTC ESTABLISHED IN G&N P, Y +30° DB, R RATE OF 0.3°/SEC

P23-NO COMM, (5 SETS)
TEI + 30 MIN (136:00)
MENKENT (30), LNH
MENKENT (30), LNH
ATRIA (34), LNH
NUNKI (37), LFH
NUNKI (37), LFH

ONBOARD READOUT

BAT C _____
PYRO BAT A _____
PYRO BAT B _____
RCS A _____
B _____
C _____
D _____

DC IND SEL TO MNA OR MNB

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	136:00 - 138:00	6/TEC	3-102

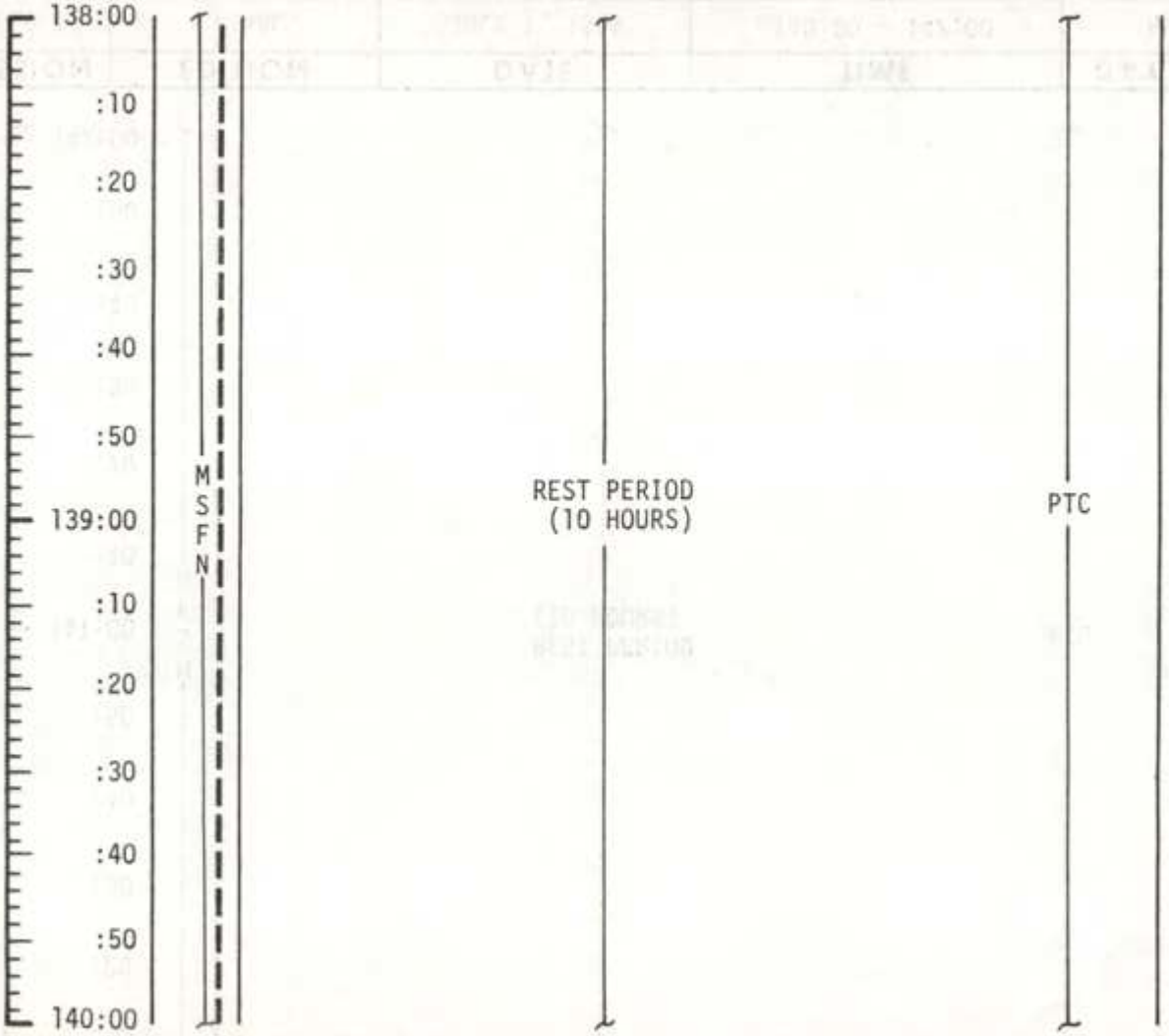
MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

MCC-H

FLIGHT PLAN

NOTES

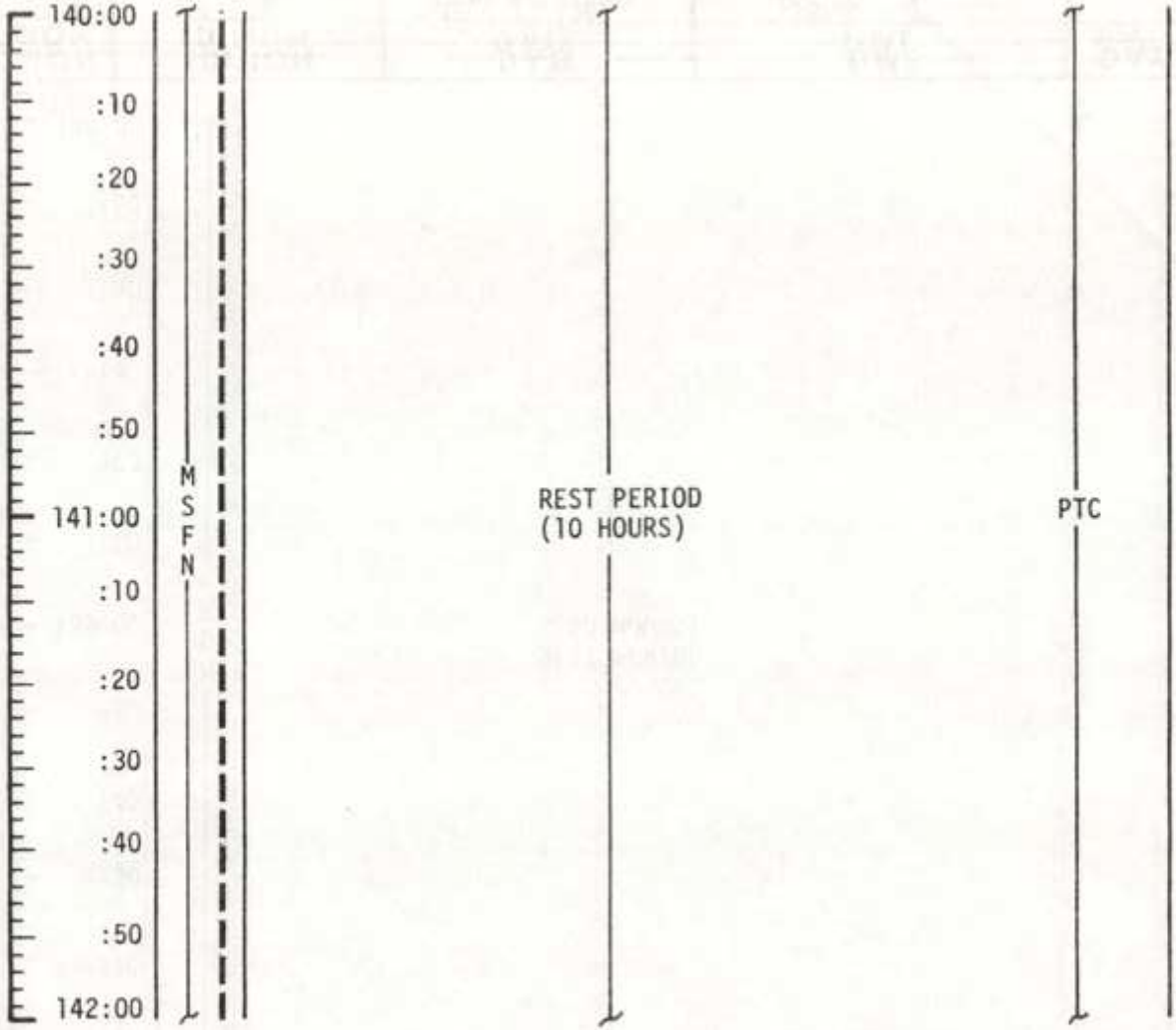


MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	138:00 - 140:00	6/TEC	3-103

MCC-H

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	140:00 - 142:00	6/TEC	3-104

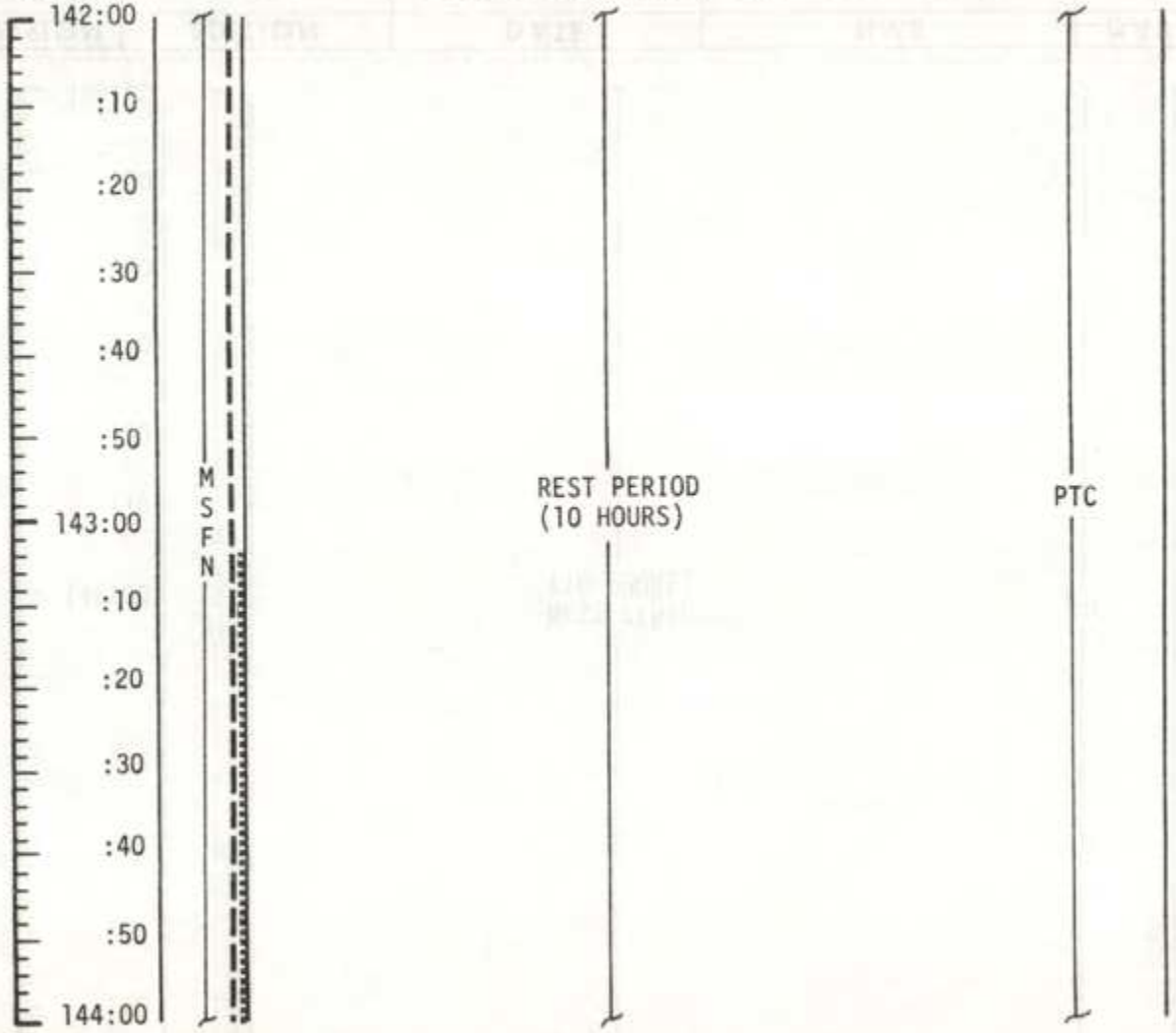
MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

MCC-H

FLIGHT PLAN

NOTES

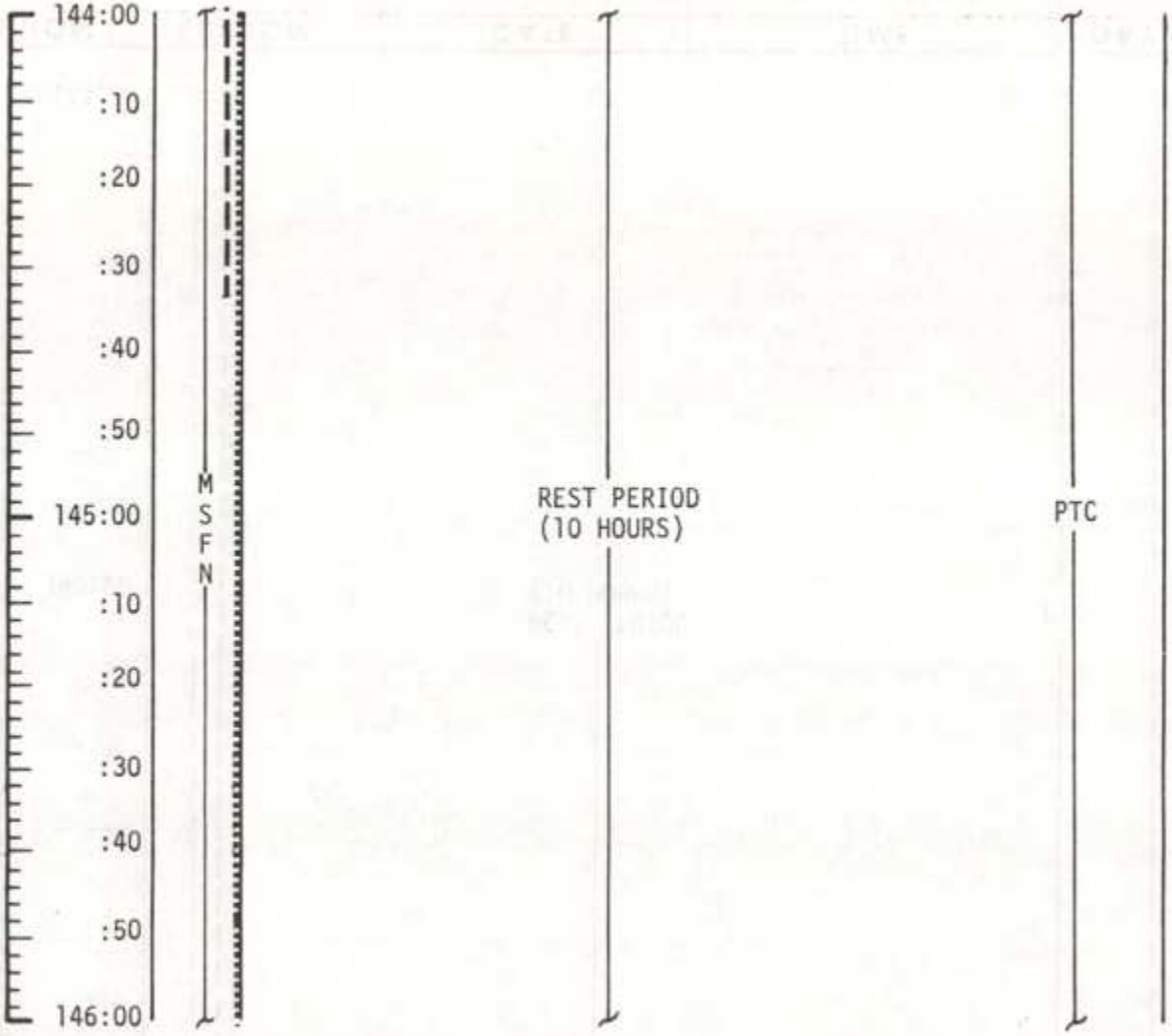


MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	142:00 - 144:00	6/TEC	3-105

MCC-H

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	144:00 - 146:00	6/TEC	3-106

MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

FLIGHT PLAN

MCC-H

1130 EDT

146:00

:10

:20

:30

:40

:50

147:00

:10

:20

:30

:40

:50

148:00

MSFN

REST PERIOD
(10 HOURS)

PTC

PHOTO AS CONVENIENT

EARTH:
EL/250/CEX-RING
(11,250,INF)

MOON:
EL/250/BW-RING
(5.6,250,INF)

CO_2 FILTER CHANGE NO. 11
(13 INTO A, STORE 11 IN A3)

POST SLEEP CHECKLIST

CREW STATUS REPORT(SLEEP)
CYCLE O_2 & H_2 FANS
GDC ALIGN TO IMU
CONSUMABLE UPDATES
SELECT NORMAL LUNAR
CONFIG EXCEPT:
S-BD AUX TAPE - OFF
TAPE RCDR FWD - OFF
POT H_2O HTR - ON

H_2 PURGE LINE HTR-ON

EAT PERIOD - ALL

O_2 & H_2 FUEL CELL PURGE

NOTES

P23-NO COMM, (5 SETS)
TEI + 11:30 (147:00)
SPICA (26), LNH
SPICA (26), LNH
MENKENT (30, LNH
NUNKI (37), LFH
NUNKI (37), LFH

CONSUMABLE UPDATE
(Δ FROM NOMINAL)

GET: _____

RCS TOT _____

A _____

B _____

C _____

D _____

H_2 TOT _____

O_2 TOT _____

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	146:00 - 148:00	7/TEC	3-107

FLIGHT PLAN

MCC-H

1330 EDT

148:00
:10
:20
:30
:40
:50
149:00
:10
:20
:30
:40
:50
150:00

MSFN

UPLINK
CSM STATE VECTOR
MCC5 TGT LOAD

UPDATE
MCC5 MNVR PAD

WIPE EXCESSIVE MOISTURE
FROM TUNNEL HATCH AREA

V66 - TRANS CSM STATE VECTOR TO
LM SLOT

RECORD MCC5 MNVR PAD

IMU REALIGN - P52
OPTION 3 - REFSMMAT

EXT ΔV - P30

SPS/RCS THRUST - P40/41

PTC

NOTES

P23-NO COMM, (5 SETS)
TEI + 13:00 (148:00)
ALPHERATZ (01), EFH
DIPHDA (02), EFH
MIRFAK (10), ENH
MIRFAK (10), ENH
ALDEBARAN (11), ENH

P52 (PTC REFSMMAT)
N71: __ __._ __
N05: __ __ __._ __
N93:
X __ __._ __ __
Y __ __._ __ __
Z __ __._ __ __
GET __ __ __:__ __:__ __

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	148:00 - 150:00	7/TEC	3-108

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MCC
BURN CHART

	P OR Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
MCC5	10°/SEC TAKEOVER	10° TAKEOVER	BT + 1 SEC	TRIM X AXIS ONLY

MCC-H

FLIGHT PLAN

1530 EDT
150:00

:10

:20

TEI +
15 HRS

:30

:40

:50

151:00

MSFN

MNVR TO BURN ATT

SXT STAR CK

EMS ΔV TEST

SM RCS MON CK

GDC ALIGN TO IMU

MCC5 ΔV=NOMINALLY ZERO

SM RCS MON CK
SPS MON CK

V66-TRANS CSM STATE VECTOR
TO LM SLOT
BURN STATUS REPORT

BATTERY CHARGE, BATTERY A

NOTES

PTC ESTABLISHED IN
G & N P,Y +30° DB
R RATE OF 0.3/SEC

START PTC
P 270° Y 0

PTC

BURN STATUS REPORT

X	X		•	ΔTIG
X	X		•	BT
			•	V_{gx}
		TRIM		
X	X	X		R
X	X	X		P
X	X	X		Y
			•	V_{gx}
			•	V_{gy}
			•	V_{gz}
			•	ΔV_c
X	X	X		FUEL
X	X	X		OX
X	X	X		UNBAL

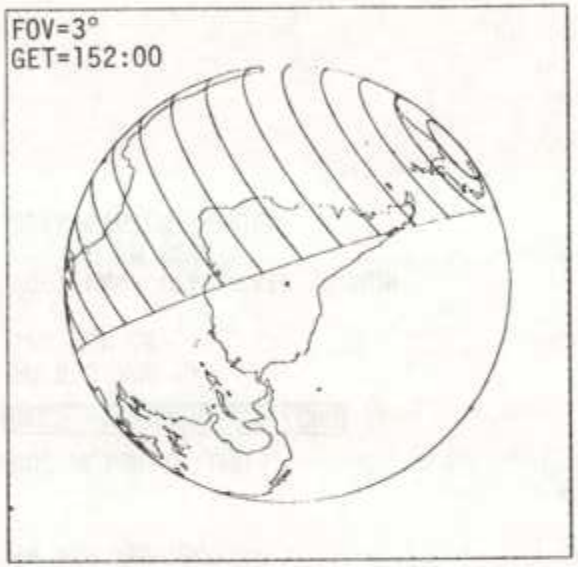
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	150:00 - 151:00	7/TEC	3-109

MCC-H

FLIGHT PLAN

1630 EDT
151:00
:10
:20
:30
:40
:50
152:00
:10
:20
:30
:40
:50
153:00

MSFN



PTC

NOTES

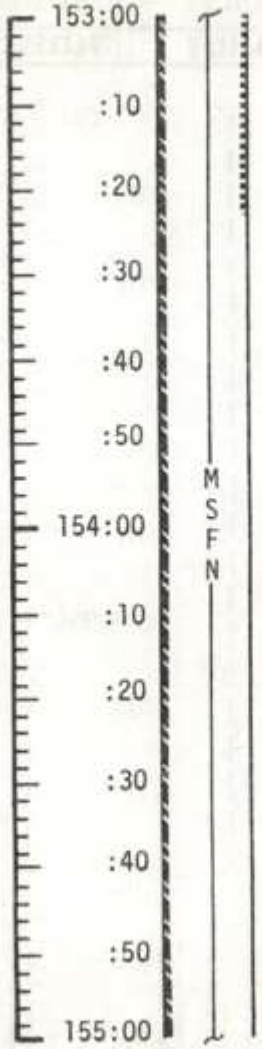
P23-NO COMM, (5 SETS)
TEI + 15:30 (152:00)
DIPHDA (02), EFH
DIPHDA (02), EFH
NAVI (03), ENH
MIRFAK (10), ENH
ALDEBARAN (11), ENH

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	151:00 - 153:00	7/TEC	3-110

MCC-H

FLIGHT PLAN

NOTES



MSFN

EAT PERIOD - ALL

PTC

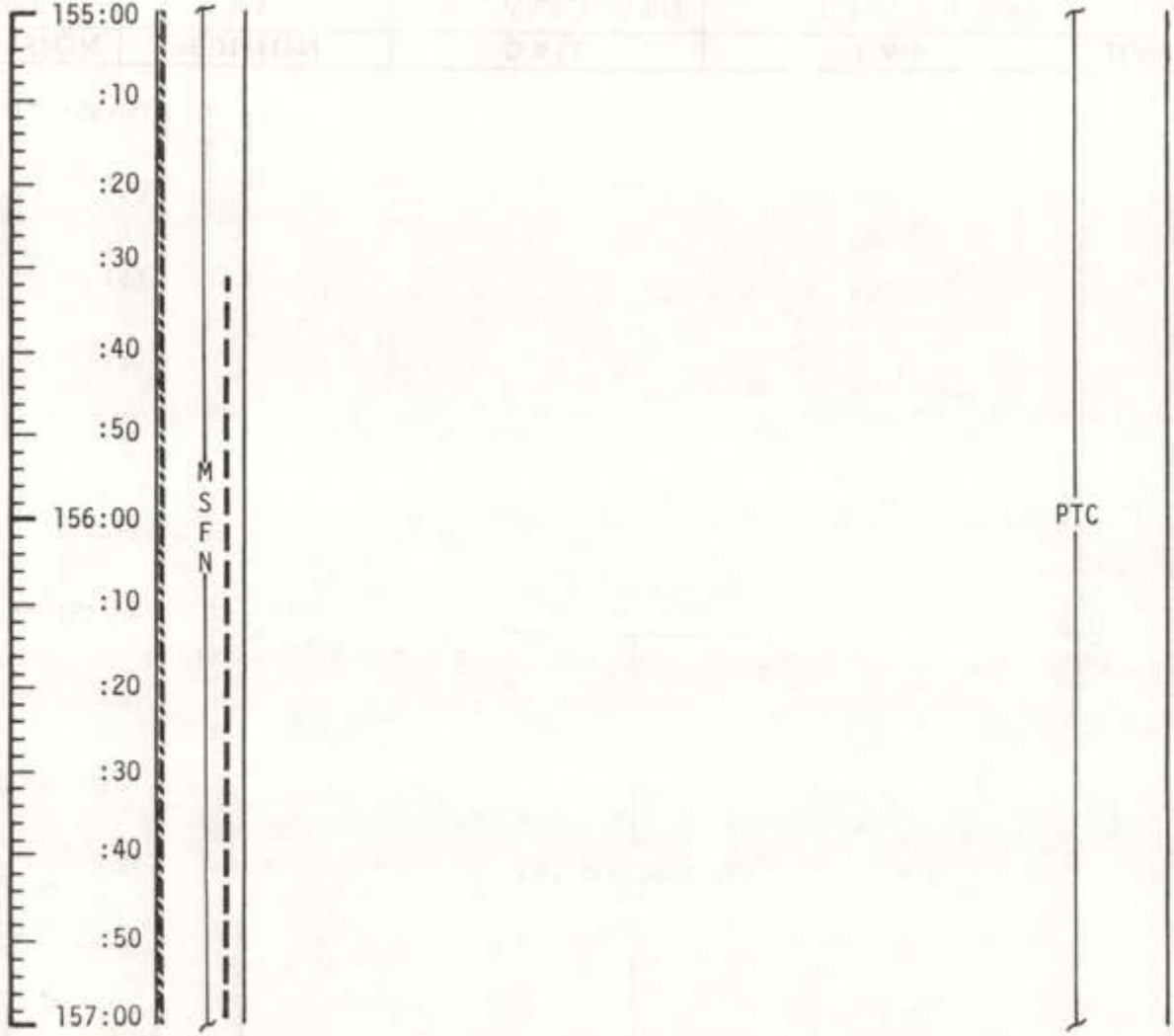
P23-NO COMM (3 SETS)
TEI + 19:00 (154:30)
SPICA (26), ENH
ANTARES (33), EFH
NUNKI (37), EFH

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	153:00 - 155:00	7/TEC	3-111

MCC-H

FLIGHT PLAN

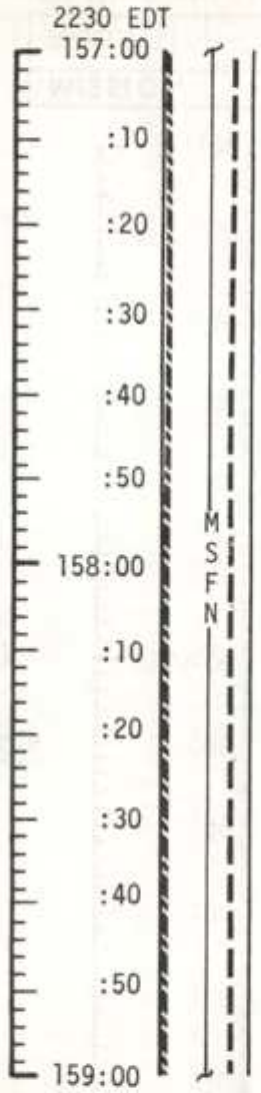
NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	155:00 - 157:00	7/TEC	3-112

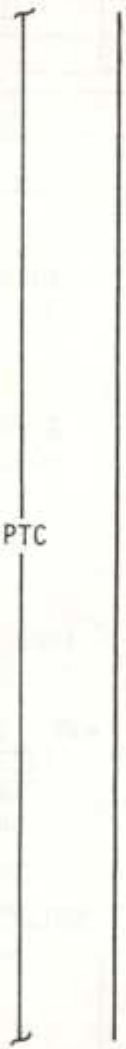
MCC-H

FLIGHT PLAN



WIPE EXCESSIVE MOISTURE
FROM TUNNEL HATCH AREA
O_2 FUEL CELL PURGE

CO_2 FILTER CHANGE NO. 12
(14 INTO B, STORE 12 IN A3)



NOTES

P23-NO COMM, (5 SETS)
TEI + 22:30 (158:00)
DIPHDA (02), EFH
DIPHDA (02), EFH
MENKAR (07), ENH
MIRFAK (10), ENH
ALDEBARAN (11), ENH

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	157:00 - 159:00	7/TEC	3-113

FLIGHT PLAN

MCC-H

NOTES

0030 EDT

159:00
:10
:20
:30
:40
:50
160:00
:10
:20
:30
:40
:50
161:00

MSFN

UPDATE

PRELIMINARY MCC6 MNVR PAD & ENTRY PAD (ASSUMES MCC6)

EAT PERIOD - ALL

EMS CK
RECORD MCC6 &
PRELIMINARY ENTRY PADS

REST PERIOD
(10 HOURS)

PRESLEEP CHECKLIST

CREW STATUS REPORT (RADIATION, MEDICATION)
CYCLE O_2 & H_2 FANS
CHLORINATE POTABLE WATER
SELECT NORMAL LUNAR CONFIG
EXCEPT: (FOR COAST ASLEEP)
- S-BD NORMAL MODE VOICE - OFF
- S-BD AUX TAPE - OFF
- TAPE RCDR FWD - OFF
- GO TO HGA OR CONTINUE OMNI OPS PER MSFN

OMNI OPS
- S-BD ANT OMNI - OMNI
- S-BD ANT OMNI - B

HI GAIN OPS
- HI GAIN ANT BEAM - NARROW
- HI GAIN ANT TRACK - REACQ
- S-BD ANT-HI GAIN

VERIFY:
- WASTE MNGT OVBD DRAIN-OFF
- WASTE STOW VENT VLV-CLOSED
- EMERG CABIN PRESS VLV-BOTH
- SURGE TK O_2 VLV-ON
- REPRESS PACK O_2 VLV-OFF
- LM TUNNEL VENT VLV - OFF
- POT H_2O HTR - OFF

AUTO RCS JET SELECT (16) - OFF

ON BOARD READOUT

BAT C _____
PYRO BAT A _____
PYRO BAT B _____
RCS A _____
B _____
C _____
D _____

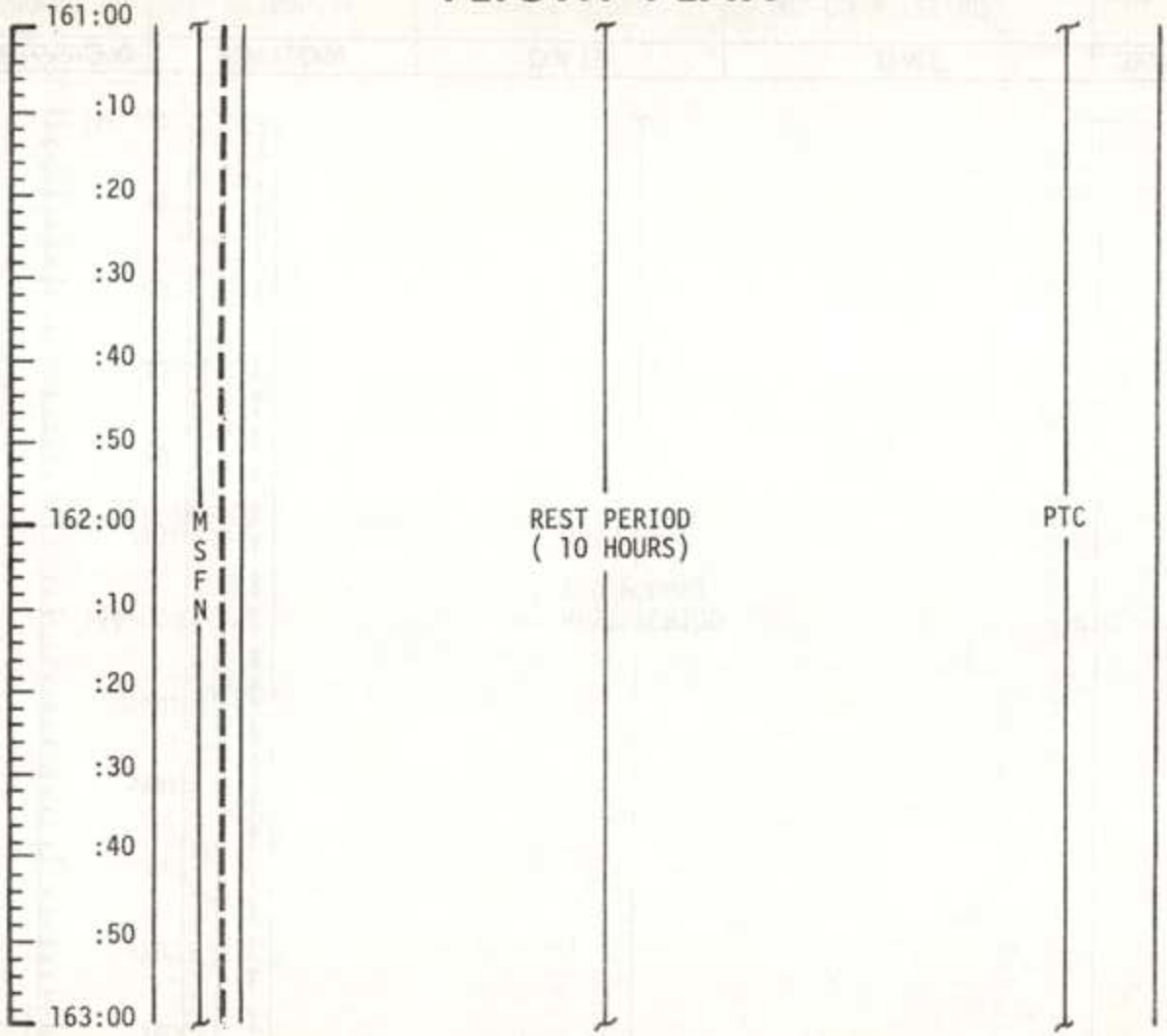
DC IND SEL TO MNA OR MNB

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	159:00 - 161:00	7/TEC	3-114

MCC-H

FLIGHT PLAN

NOTES

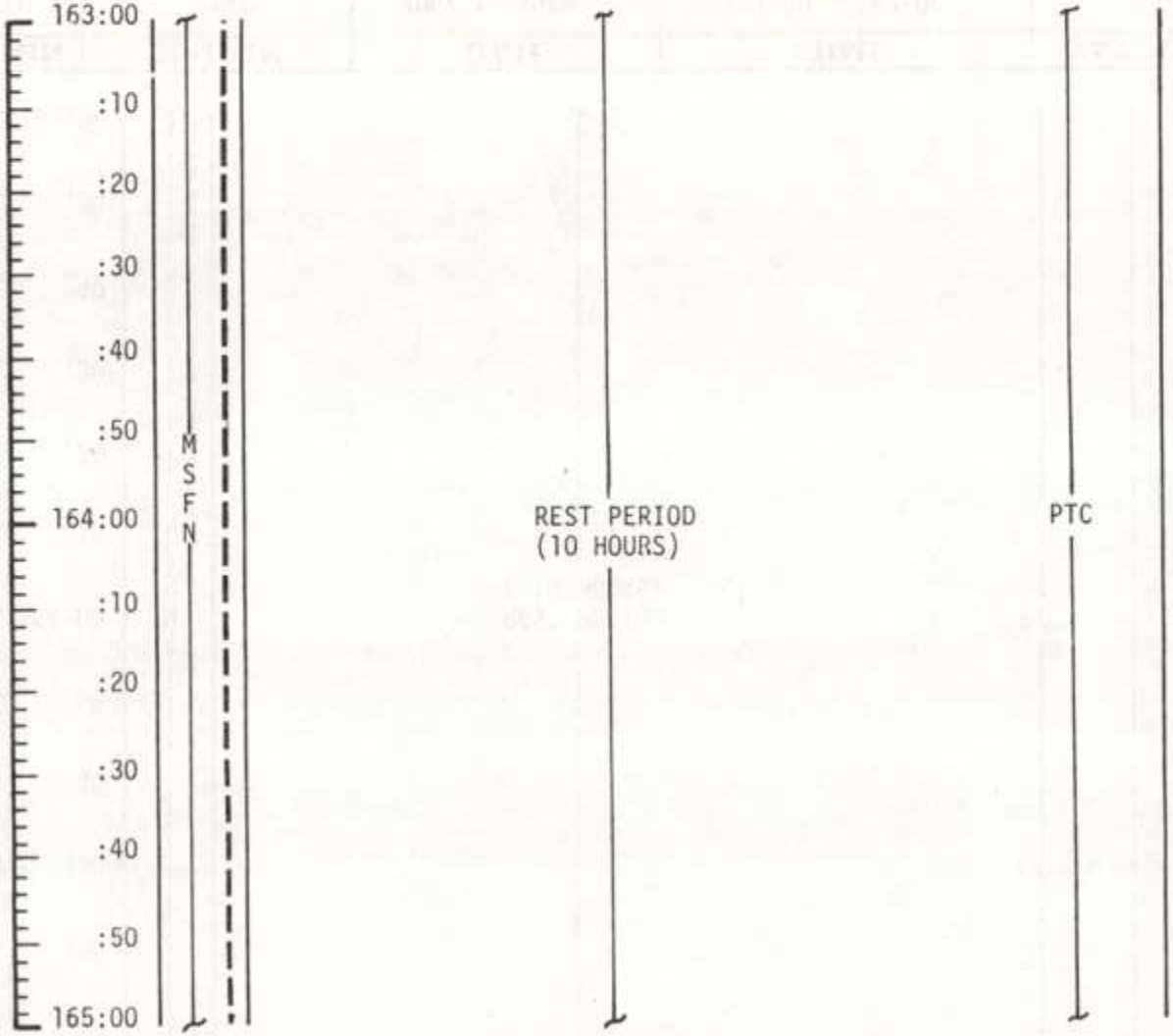


MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	161:00 - 163:00	7/TEC	3-115

MCC-H

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	163:00 - 165:00	7/TEC	3-116

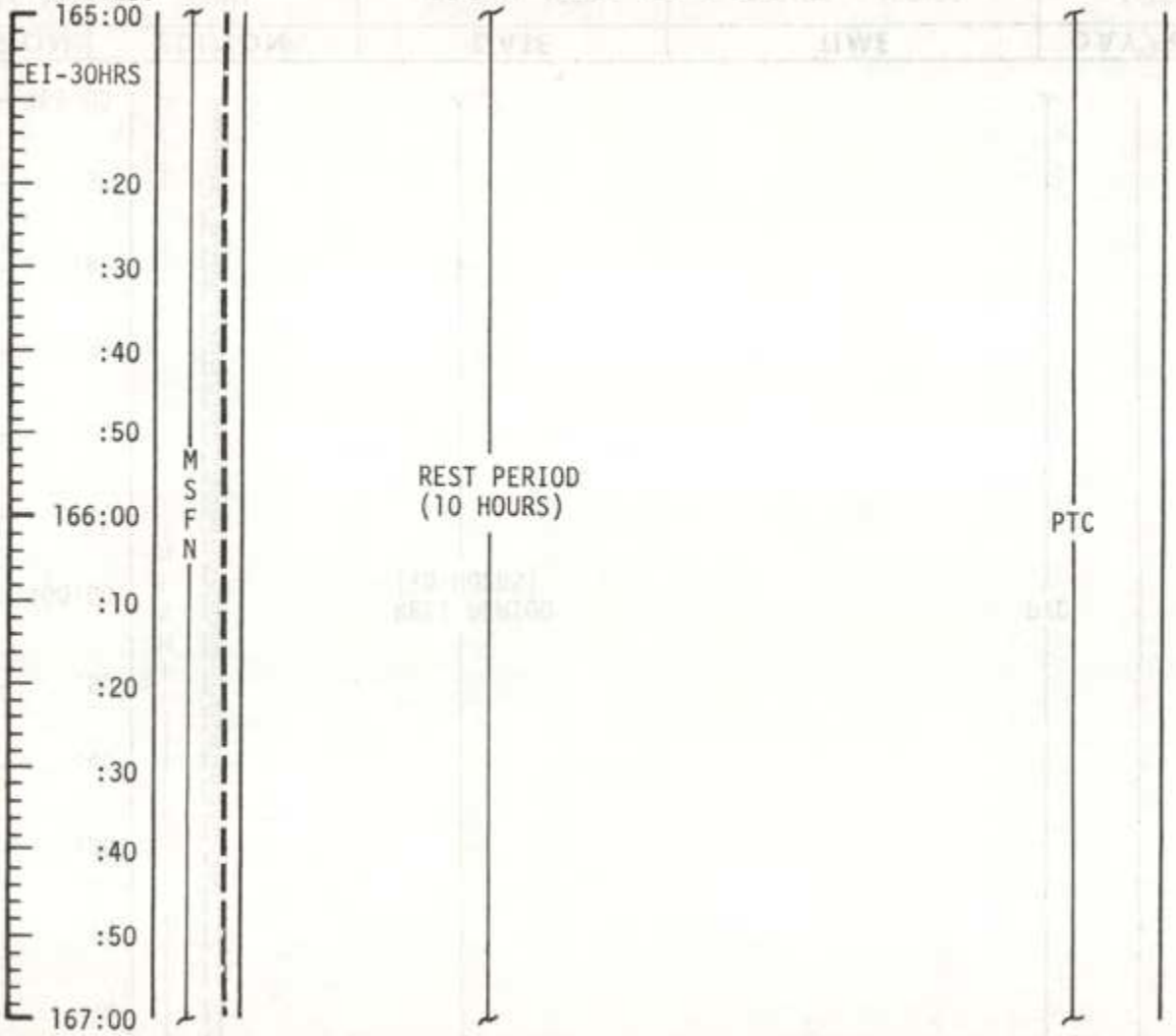
MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

MCC-H

FLIGHT PLAN

NOTES

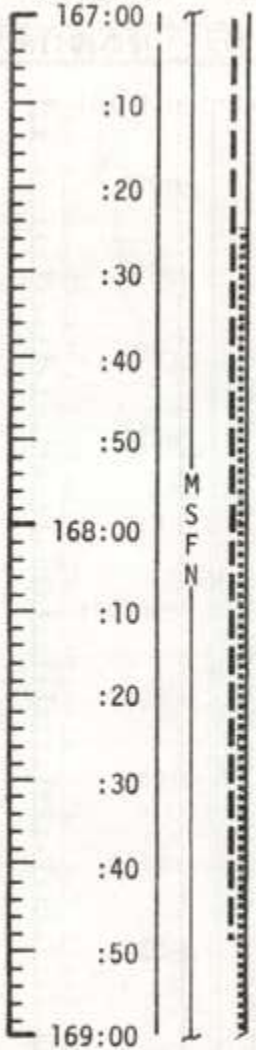


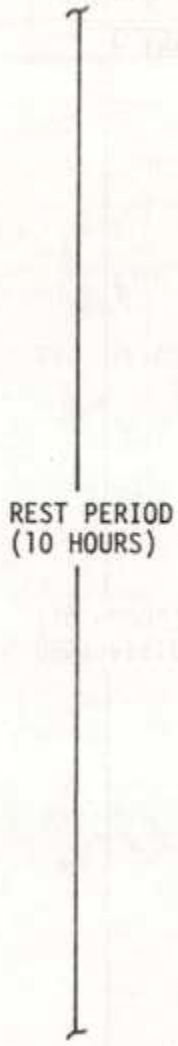
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	165:00 - 167:00	7/TEC	3-117

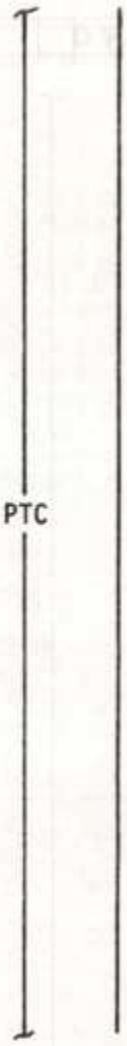
MCC-H

FLIGHT PLAN

NOTES







MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	167:00 - 169:00	7/TEC	3-118

FLIGHT PLAN

MCC-H

1030 EDT

169:00
:10
:20
:30
:40
:50
170:00
:10
:20
:30
:40
:50
171:00

MSFN

REST PERIOD
(10 HOURS)

CO_2 FILTER CHANGE NO. 13
(15 INTO A, STORE 13 IN A4)

REPORT CM RCS EJECTOR
VLV TEMP

EAT PERIOD - ALL

O_2 FUEL CELL PURGE

V66 - TRANSFER CSM STATE VECTOR TO LM SLOT

POST SLEEP CHECKLIST

CREW STATUS REPORT (SLEEP)
CYCLE O_2 & H_2 FANS
GDC ALIGN TO IMU
CONSUMABLES UPDATE
SELECT NORMAL LUNAR
CONFIGURATION EXCEPT:
  S-BD AUX TAPE - OFF
  TAPE RCDR FWD - OFF
POT H_2O HTR - ON
AUTO RCS JET SELECT (16) - ON

PTC

UPLINK CMC
CSM STATE VECTOR
MCC6 TGT LOAD

NOTES

P23-NO COMM (5 SETS)
TEI + 35:00 (170:30)
DIPHDA (02), EFH
DIPHDA (02), EFH
NAVI (03), ENH
MIRFAK (10), ENH
ALDEBARAN (11), ENH

CONSUMABLE UPDATE
(Δ FROM NOMINAL)

GET: _____

RCS TOT _____
A _____
B _____
C _____
D _____

H_2 TOT _____

O_2 TOT _____

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	169:00 - 171:00	8/TEC	3-119

MCC
BURN CHART

	P OR Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
MCC6	10°/SEC TAKEOVER	10° TAKEOVER	BT + 1 SEC	TRIM X AXIS ONLY

FLIGHT PLAN

MCC-H

1230 EDT

171:00

UPDATE
MCC6 PAD DATA
ENTRY PAD
(ASSUMES MCC6)

:10

:20

171:30

:40

:50

EI
-23 HRS

172:00

MSFN

WIPE EXCESSIVE MOISTURE
FROM TUNNEL HATCH AREA
RECORD MCC6 AND PRELIMINARY ENTRY PAD DATA

IMU REALIGN - P52

(OPTION 3 - REFSMMAT)

EXT ΔV - P30

SPS/RCS THRUST - P40/41

MNVR TO BURN ATT

SXT STAR CK

EMS ΔV TEST

SM RCS MON, CK

GDC ALIGN TO IMU

MCC6 ΔV=NOMINALLY ZERO

P52 - PTC REFSMMAT
N71: __,__ __
N05: __ __ __.__ __
N93:
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __
GET __ __ __:__ __:__ __

NOTES

BURN STATUS REPORT

X	X	□	•	ΔTIG
X	X		•	BT
□			•	V_{gx}
TRIM				
X	X	X		R
X	X	X		P
X	X	X		Y
□			•	V_{gx}
□			•	V_{gy}
□			•	V_{gz}
□			•	ΔV_c
X	X	X		FUEL
X	X	X		OX
X	X	X		UNBAL

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	171:00 - 172:00	8/TEC	3-120

MCC-H

FLIGHT PLAN

1330 EDT

172:00
:10
:20
:30
:40
:50
173:00
:10
:20
:30
:40
:50
174:00

MSFN

SM RCS MON CK
SPS MON CK
V66-TRANS CSM STATE VECTOR
TO LM SLOT
BURN STATUS REPORT
BATTERY CHARGE, BATTERY B

START PTC
P 270° Y 0

PTC

NOTES

PTC ESTABLISHED
IN G&N P, Y + 30°DB
R RATE OF 0.3°/SEC

P23-NO COMM (3 SETS)
TEI + 37:00 (172:30)
SPICA (26) LNH
ANTARES (33) LFH
NUNKI (37) LFH

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	172:00 - 174:00	8/TEC	3-121

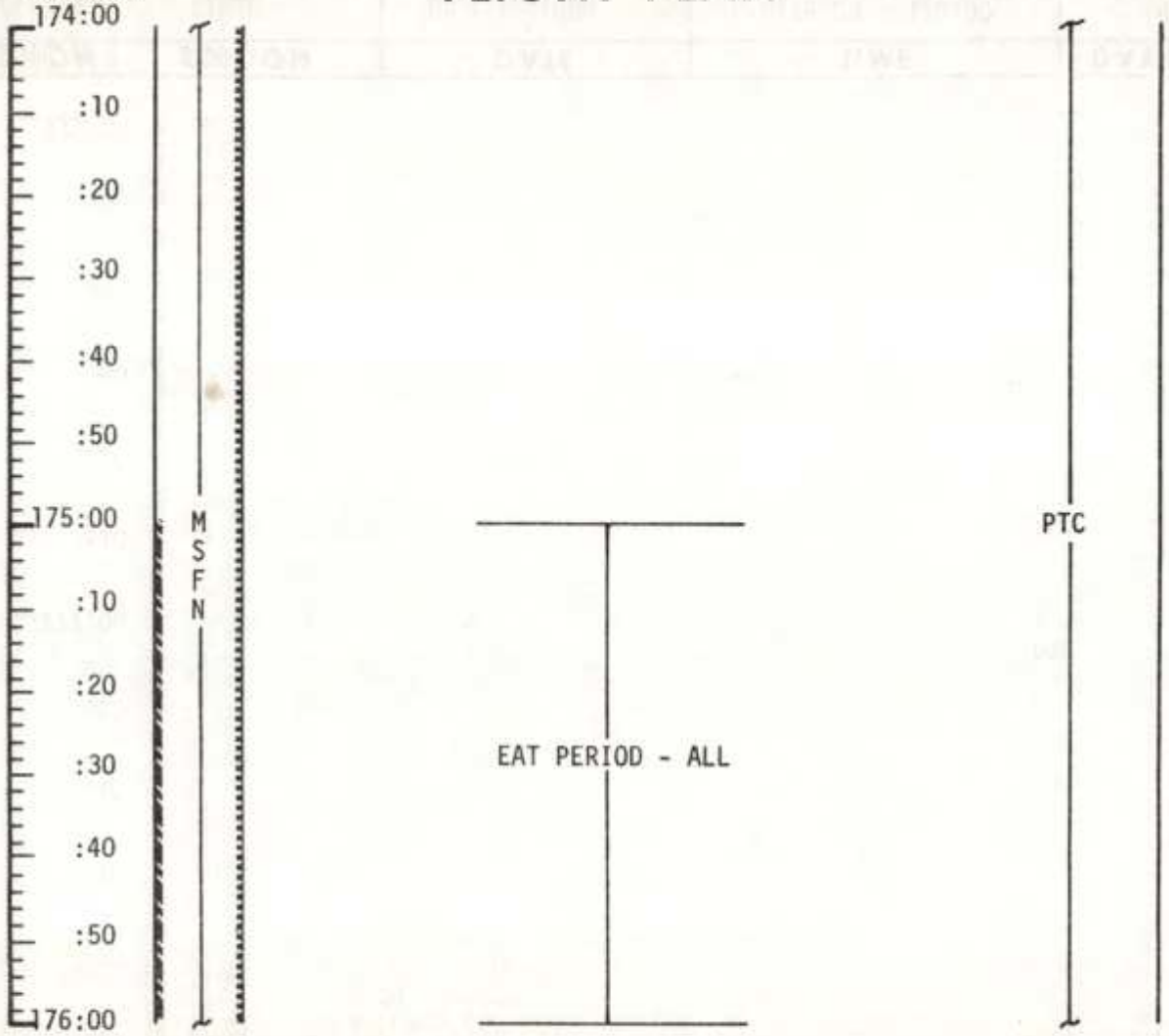
MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

MCC-H

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	174:00 - 176:00	8/TEC	3-122

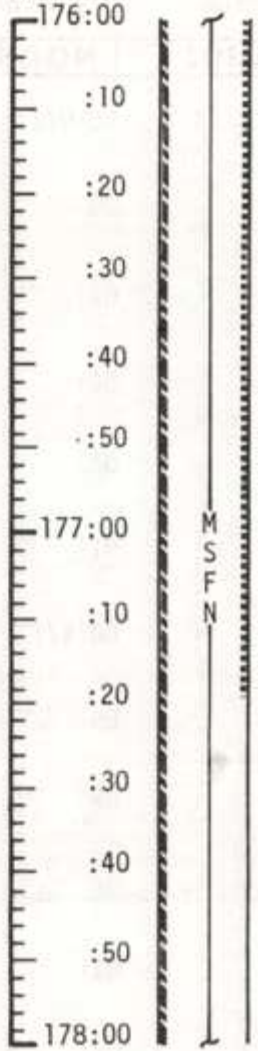
MSC Form 29 (May 69)

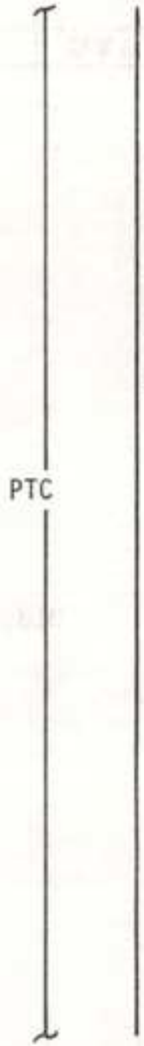
FLIGHT PLANNING BRANCH

MCC-H

FLIGHT PLAN

NOTES





P23-NO COMM (5 SETS)
TEI + 41:00 (176:30)
DIPHDA (02), EFH
MIRFAK (10), ENH
ALDEBARAN (11), ENH
CAPELLA (13), ENH
CAPELLA (13), ENH

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	176:00 - 178:00	8/TEC	3-123

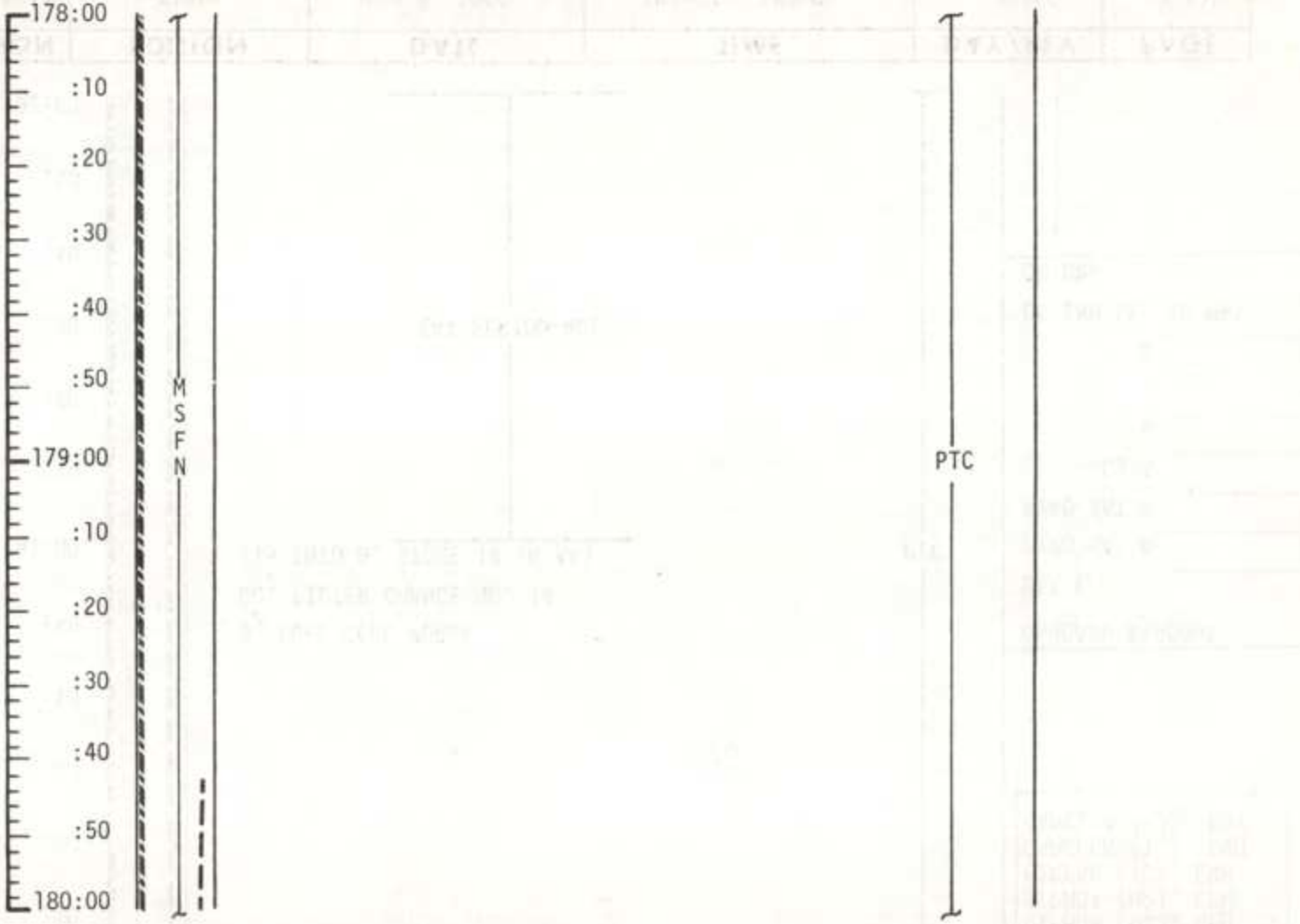
MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

MCC-H

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	178:00 - 180:00	8/TEC	3-124

MCC-H

FLIGHT PLAN

2130 EDT

180:00
:10
:20
:30
:40
:50
181:00
:10
:20
:30
:40
:50
182:00

M S F N

WIPE EXCESSIVE MOISTURE
FROM TUNNEL HATCH AREA

O_2 FUEL CELL PURGE
CO_2 FILTER CHANGE NO. 14
(16 INTO B, STORE 14 IN A4)

EAT PERIOD-ALL

PTC

NOTES

P23-NO COMM (5 SETS)
TEI + 44:30 (180:00)
DIPHDA (02), EFH
DIPHDA (02), EFH
MIRFAK (10), ENH
CAPELLA (13), ENH
CAPELLA (13), ENH

ONBOARD READOUT

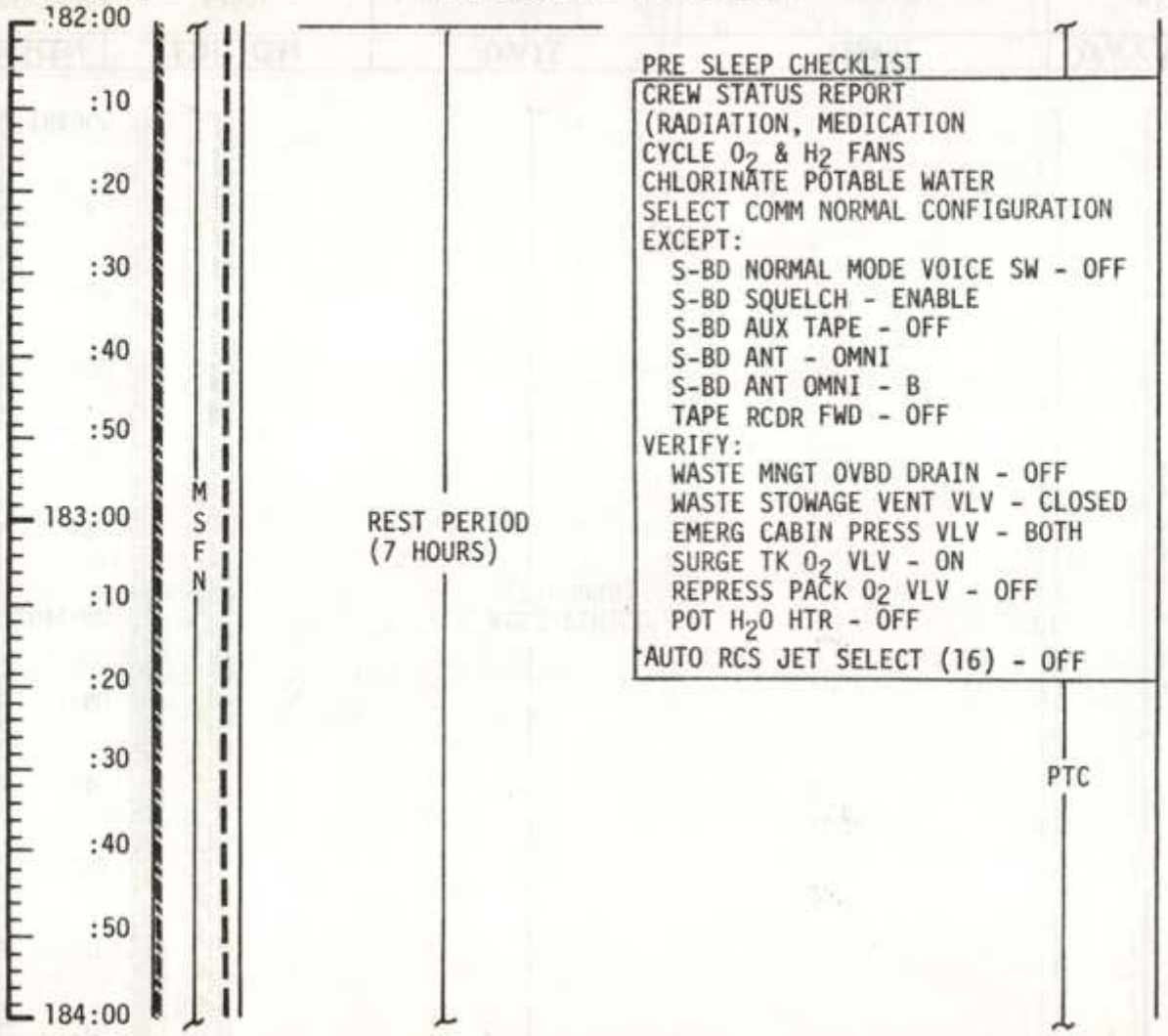
BAT C _____
PYRO BAT A _____
PYRO BAT B _____
RCS A _____
B _____
C _____
D _____
DC IND SEL TO MNA
OR MNB

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	180:00 - 182:00	8/TEC	3-125

MCC-H

FLIGHT PLAN

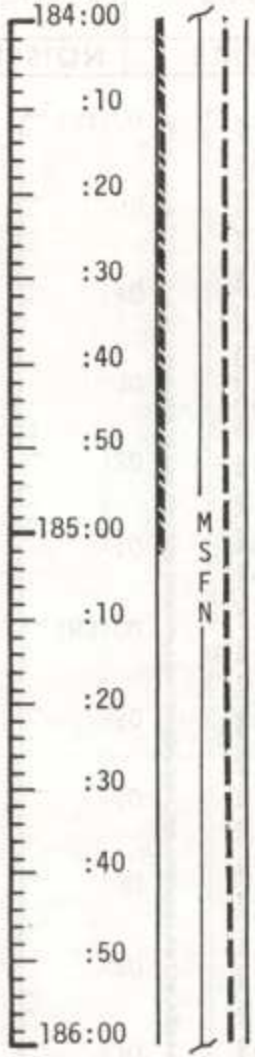
NOTES

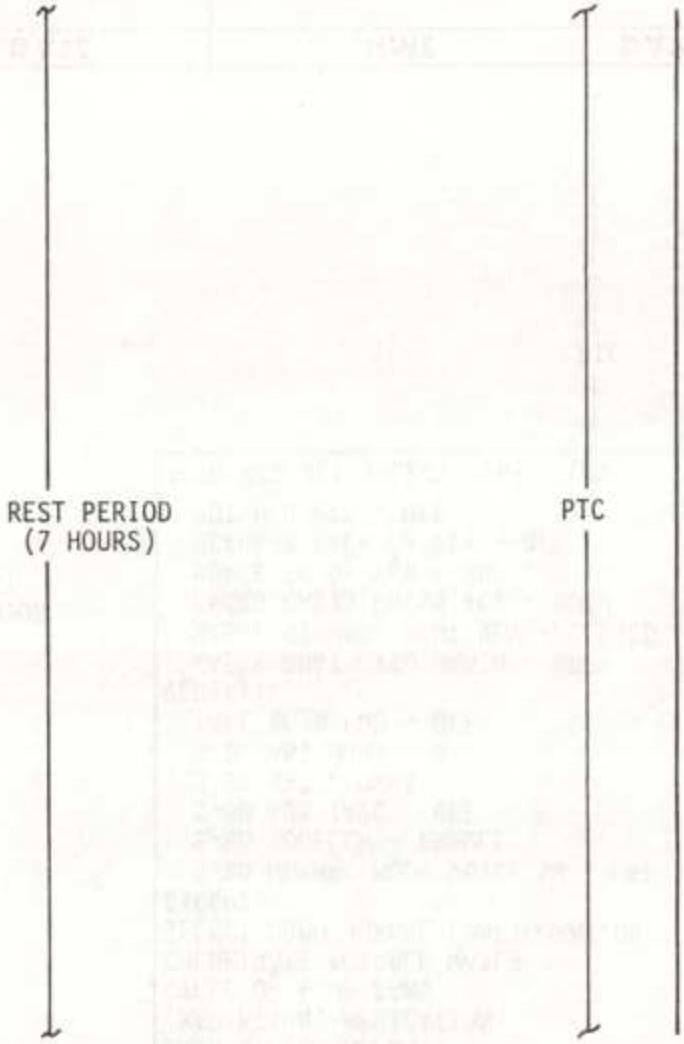


MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	182:00 - 184:00	8/TEC	3-126

MCC-H

FLIGHT PLAN





NOTES

P23-NO COMM (5 SETS)
TEI + 50 (185:30)
ALPHERATZ (01), EFH
MIRFAK (10), ENH
ALDEBARAN (11), ENH
CAPELLA (13), ENH
CAPELLA (13), ENH

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	184:00 - 186:00	8/TEC	3-127

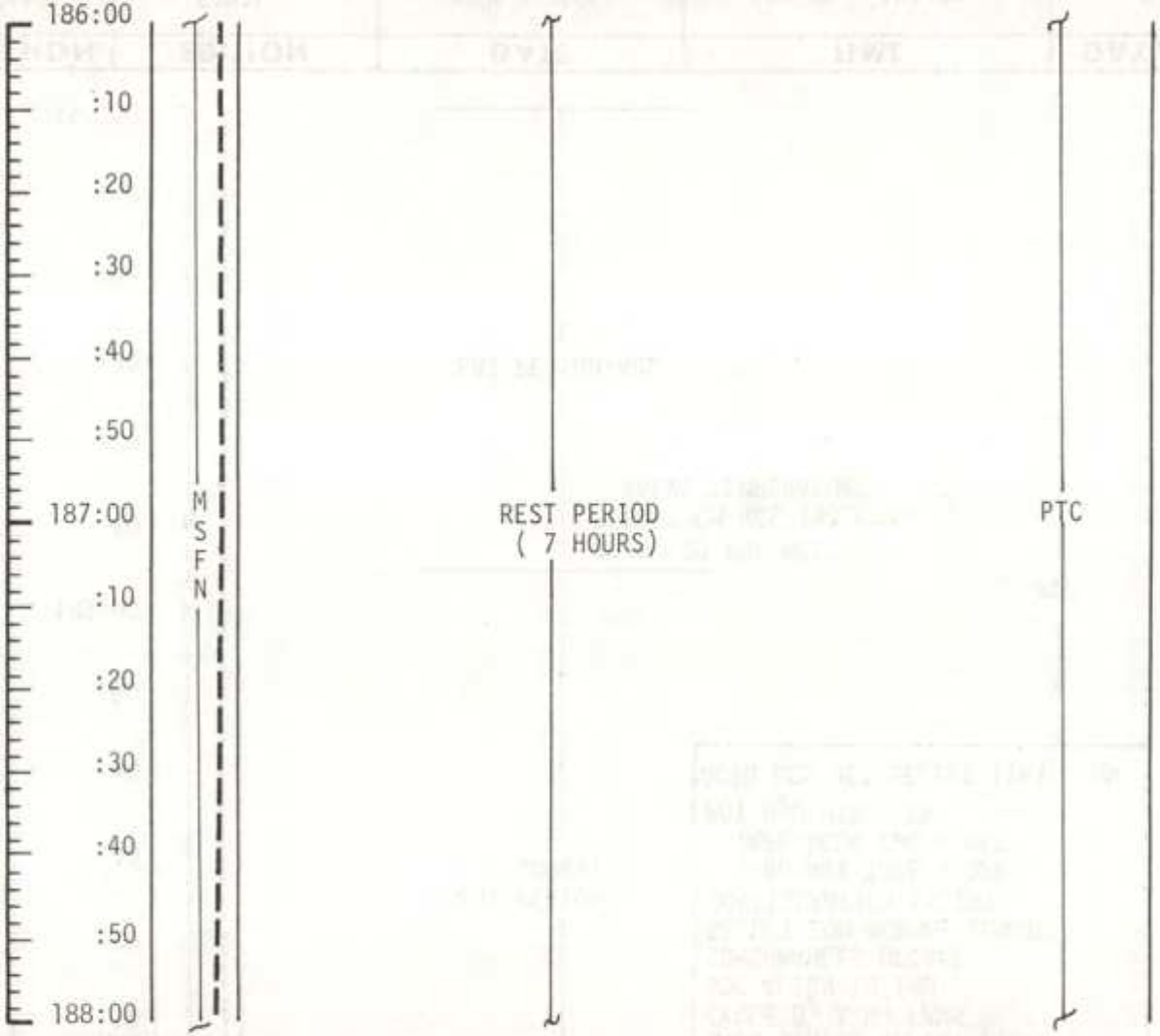
MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

MCC-H

FLIGHT PLAN

NOTES

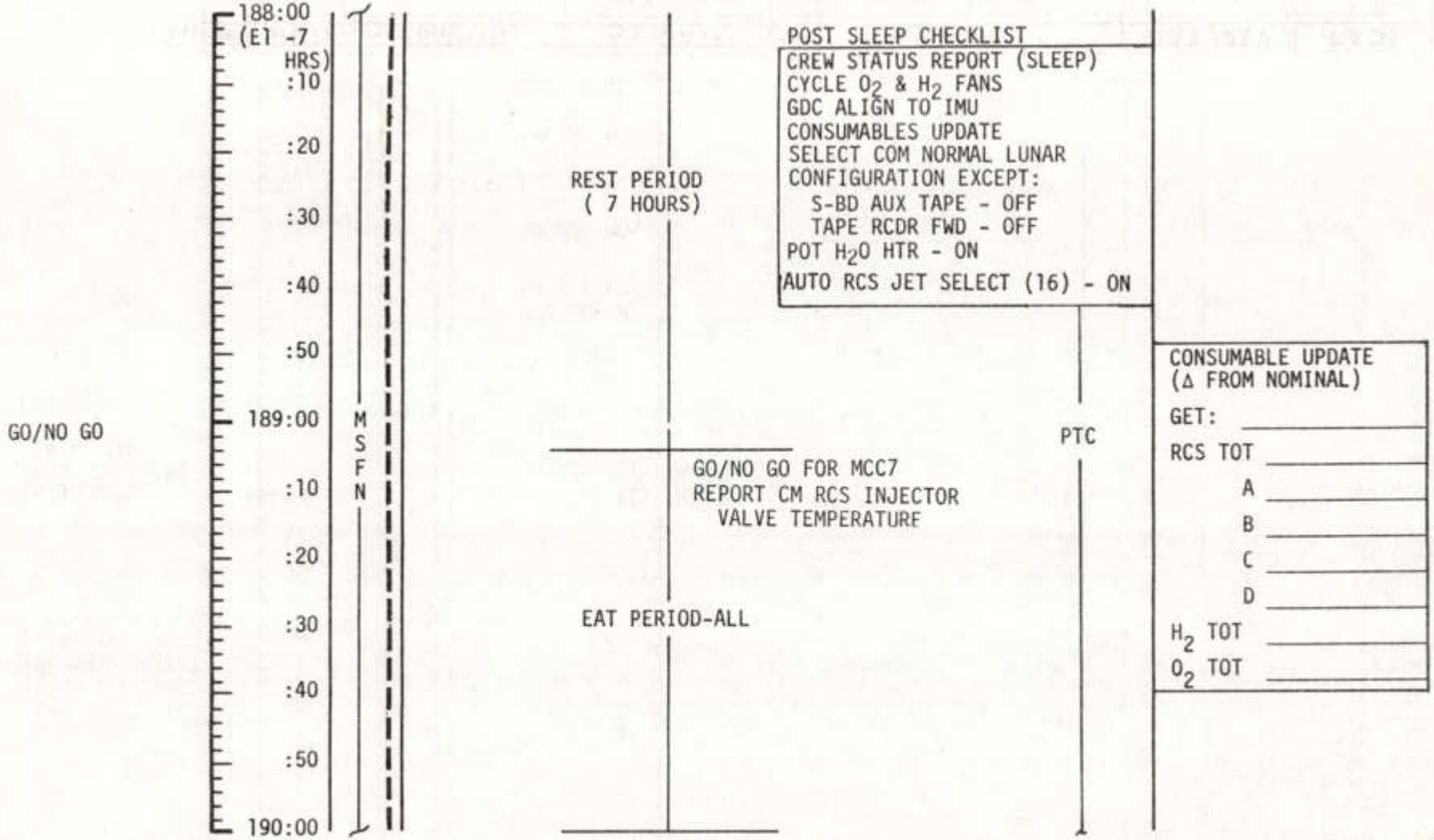


MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	186:00 - 188:00	8/TEC	3-128

MCC-H

FLIGHT PLAN

NOTES



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	188:00 - 190:00	8/TEC	3-129

FLIGHT PLAN

MCC-H

0730 EDT

190:00 (EI -5 HRS)

:10

:20

190:30

:40

:50

191:00

MSFN

DON MAE WEST & FOOT RESTRAINTS

V66 - TRANS CSM STATE VECTOR TO LM SLOT

RECORD MCC7 MNVR PAD & ENTRY PAD

EPS CHECK

SPS CHECK
CM RCS MON CK

SM RCS MON CK

C & W SYS CK

CMC SELF TEST

DSKY COND LT TEST

PTC

UPLINK CMC
- CSM STATE VECTOR
- MCC7 TGT LOAD
- DESIRED ORIENT (ENTRY)
- ENTRY LAT & LONG

UPLINK CMC
- MCC7 MNVR PAD
- ENTRY PAD

NOTES

P23-NO COMM, (5 SETS)
TEI + 54:30 (190:00)
ALPHERATZ (01), EFH
MIRFAK (10), ENH
ALDEBARAN (11), ENH
CAPELLA (13), ENH
CAPELLA (13), ENH

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	190:00 - 191:00	9 /TEC	3-130

MCC-H

FLIGHT PLAN

0830 EDT

191:00 (EI -4 HRS)

:10

:20

191:30

:40

:50

192:00

MSFN

IMU REALIGN - P52
OPTION 1 - PREFERRED

EXT ΔV - P30

SPS/RCS THRUST - P40/41

MNVR TO BURN ATT

SXT STAR CK

EMS ΔV TEST

SM RCS MON CK

GDC ALIGN TO IMU

NOTES

P52 (ENTRY REFSMMAT)
N71: __ __,__ __
N05: __ __ __.__ __
N93:
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __
GET __ __ __:__ __:__ __

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	191:00 - 192:00	9/TEC	3-131

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MCC
BURN CHART

	P OR Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
MCC7	10°/SEC TAKEOVER	10° TAKEOVER	BT + 1 SEC	TRIM X AXIS ONLY

FLIGHT PLAN

MCC-H

0930 EDT

192:00 (EI -3 HRS)

:10

:20

192:30

:40

:50

193:00

MSFN

MCC7 ΔV=NOMINALLY ZERO

SM RCS MON CK
SPS MON CK
BURN STATUS REPORT
V66 - TRANS CSM STATE VECTOR
TO LM SLOT

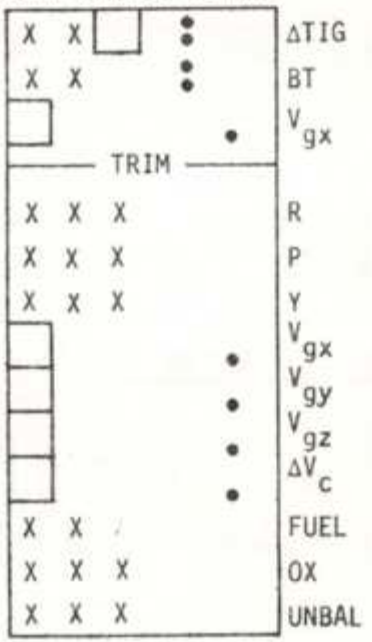
P23-NO COMM (3 SETS)
TEI + 57:00 (192:30)
MENKAR (07) ~~ENH~~ EFH
CAPELLA (13) ENH
CAPELLA (13) ENH

GO/NO GO

GO/NO GO FOR PYRO ARM SEQUENCE
VHF ACTIVATION

NOTES

BURN STATUS REPORT



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	192:00 - 193:00	9/TEC	3-132

MCC-H

FLIGHT PLAN

NOTES

LOGIC SEQUENCE CHECK

MNVR TO ENTRY ATTITUDE

COAS STAR CHECK

SXT STAR CHECK

IMU REALIGN - P52
(OPTION 3 - REFSMMAT)

GDC ALIGN TO IMU
CM RCS PREHEAT

P52 (ENTRY REFSMMAT)
N71: __ __,__ __
N05: __ __ __.__ __
N93:
X __ __.__ __ __
Y __ __.__ __ __
Z __ __.__ __ __
GET __ __ __:__ __:__ __

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	193:00 - 194:00	9/TEC	3-133

MSC Form 29 (May 69)

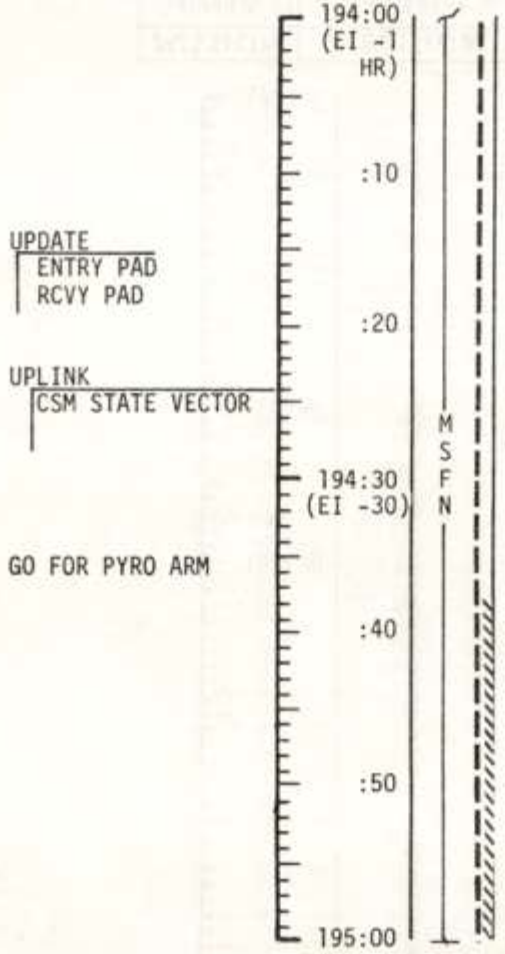
FLIGHT PLANNING BRANCH

FLIGHT PLAN

MCC-H

1130 EDT

NOTES



PRIMARY EVAPORATOR ACTIVATION

FINAL GDC DRIFT CHECK
TERM CM RCS PRFHEAT

SECONDARY EVAPORATOR ACTIVATION
PYRO BATTERY CHECK
RECORD ENTRY PAD AND RECOVERY DATA

P27 UPDATE - V66 - TRANS CSM STATE VECTOR TO LM SLOT

SET DET TO RRT
EMS INITIALIZATION
RSI ALIGN TO GDC

CM RCS CK
ENTRY BATTS - ON

SEPARATION CHECKLIST
GO FOR PYRO ARM

P61 ENTRY PREP
MNVR TO CM/SM SEP ATT

CM/SM SEP

P62 - ENTRY ATTITUDE
MNVR TO ENTRY ATT

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	194:00 - 195:00	9/TEC	3-134

MCC-H

FLIGHT PLAN

NOTES

P63 - ENTRY INITIATE

EI - GET = 195:05:04

P64 - ENTRY POST .05G

TRAJECTORY EVENT	TIME FROM ENTRY INTERFACE MIN:SEC
400,000 FEET (GET 195:05:04)	00:00
ENTER S-BAND BLACKOUT	00:18
0.05G	00:28
KA - INITIATE CONSTANT DRAG	00:52
RDOT = -700 FPS	01:18
PEAK G (6.6)	01:22
SUBCIRCULAR VELOCITY	02:08
P64 to P67	02:08
EXIT S-BAND BLACKOUT	03:24
GUIDANCE TERMINATION	07:14
DROGUE DEPLOYMENT	08:12
MAIN DEPLOYMENT	09:00
SPLASHDOWN	13:55

γ = -6.50°, L/D = 0.295, V = 36,195, & R = 1285

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	Revision A	July 8, 1969	195:00 - 196:00	9/TEC	3-135

MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

MSC - 1SC-70

SECTION IV - DETAILED TEST OBJECTIVES

SECTION 4

DETAILED OBJECTIVE ACTIVITIES

This section contains the activity summaries which reflect the test objectives for Mission G as described in "Mission Requirements G Type Mission", SPD9-R-038, Change A dated May 1, 1969. These activity summaries are presented in the approximate sequence in which they are planned to occur during the mission.

Each activity summary provides the following information:

A. TEST OBJECTIVES. This is the listing of the Functional Test Objectives (complete or partial) which relate to the particular activity;

B. TEST REQUIREMENTS. Here the special test prerequisites (and mission phase if necessary) are presented in addition to brief statements of the requirements for performing the activity;

C. TEST PROCEDURES/CHECKLISTS. These are the procedural references for the performance of the activity as far as the test objectives are concerned; and

D. DATA REQUIREMENTS. This part of the summary identifies the gross data which are needed for evaluation of test results in terms of flight crew and ground support requirements.

Cross references for relating Detailed and Functional Test Objectives with the activity summaries and relating activities to Functional Test Objectives, are provided as the initial part of this section.

The following ground rules are to be used in implementing data requirements:

A. The collection of highly desirable (HD) data should not constrain the timeline of the crew procedures.

B. Post-flight debriefing requirements which are fulfilled by real time transmission of data per the DATA REQUIREMENTS sections may be deleted from the post-flight debriefing.

All of the Test Requirements have not been totally implemented into the mission timeline. These items are identified in this section as "Not Implemented" or with the conditions by which they will be implemented.

TABLE 4-1
MISSION ACTIVITY AND
TEST OBJECTIVE CROSS REFERENCE

ACTIVITY	FTO
LM Descent	D-1, G-1, G-3, H-1, M-1
Lunar Surface Navigation	G-1, G-2, G-3, L-4, M-2
EVA Preparation and Egress	B-1, B-2, C-1, C-2, C-3, L-1
Surface Sample Collection	A-1, E-1, F-1, F-2, I-3, J-2, J-3, J-4, M-3
External LM Observations and Photography	D-1, D-2, D-3, D-4, L-2, M-3
Lunar Surface Observations and Photography	E-1, E-2, E-3, H-2, J-5, L-3, L-4, M-3
Experiment Deployment/Conduct	S-031, S-078, S-080
Post EVA Operations	B-1, C-1, C-2
Contamination Prevention	I-1, I-2

TABLE 4-2
TEST OBJECTIVE/MISSION ACTIVITY
CROSS REFERENCE

DTO/FTO NUMBER	TEST OBJECTIVE	MISSION ACTIVITY	SECTION PAGE NO.
A	Contingency Sample Collection		
A-1	Provide a Contingency Lunar Surface Sample	Surface Sample Collection	4-13
B	Lunar Surface EVA Operations		
B-1	Demonstrate Egress-To/Ingress-From the Lunar Surface	EVA Preparation and Egress	4-10
		Post EVA Operations	4-21
B-2	Evaluate Crew Lunar Surface EVA Capability	EVA Preparation and Egress	4-10
C	EMU Lunar Surface Operations		
C-1	EMU Capability to Provide a Habitable Environment	EVA Preparation and Egress	4-10
C-2	EMU Effects on Crew Mobility, Dexterity & Comfort	EVA Preparation and Egress	4-10
C-3	Demonstrate EVA Data/Voice Communications	EVA Preparation and Egress	4-10
D	Landing Effects on LM		
D-1	LM Landing Gear Performance Under Landing Conditions	LM Descent	4-6
		External LM Observ/Photo	4-15
D-2	Effects of Landing on LM Structure and Components	External LM Observ/Photo	4-15
D-3	Descent Engine Skirt Damage/Clearance After Landing	External LM Observ/Photo	4-15
D-4	Effects of RCS Plume Impingement on LM Structure & Components	External LM Observ/Photo	4-15
E	Lunar Surface Characteristics		
E-1	Data on Behavior/Characteristics of the Lunar Surface	Surface Sample Collection	4-13
		Lunar Surface Observ/Photo	4-17
E-2	Lunar Soil Erosion from DPS Plume Impingement	Lunar Surface Observ/Photo	4-17
E-3	Effect of DPS Venting on the Lunar Surface	Lunar Surface Observ/Photo	4-17
F	Bulk Sample Collection		
F-1	Collect Rock Samples and Fine Grained Material	Surface Sample Collection	4-13
F-2	Photograph Collection Area of Samples	Surface Sample Collection	4-13

TABLE 4-2
TEST OBJECTIVE/MISSION ACTIVITY
CROSS REFERENCE

DTO/FTO NUMBER	TEST OBJECTIVE	MISSION ACTIVITY	SECTION PAGE NO.
G	Landed LM Location		
G-1	Determine Location of Landed LM from LM Data	LM Descent	4-6
		Lunar Surface Navigation	4-8
G-2	Determine Location of Landed LM from CSM Data	Lunar Surface Navigation	4-8
G-3	Capability of Locating Landed LM in Real Time	LM Descent	4-6
		Lunar Surface Navigation	4-8
H	Lunar Environment Visibility		
H-1	Data on Landing Aids & Final Approach Visibility	LM Descent	4-6
H-2	Crew Performance of Visual Tasks on Lunar Surface	Lunar Surface Observ/Photo	4-17
I	Assessment of Contamination by Lunar Material		
I-1	Prevent Earth Contamination by Lunar Exposed Materials	Contamination Prevention	4-22
I-2	Minimize Crew/CM Contamination by Lunar Exposed Materials	Contamination Prevention	4-22
I-3	Lunar Sample for Quarantine Testing	Surface Sample Collection	4-13
J	Documented Sample Collection		
J-1	Obtain an Asceptic Sample of the Lunar Surface	Deleted	
J-2	Obtain a Core Sample of the Lunar Surface	Surface Sample Collection	4-13
J-3	Collect Lunar Geologic Samples	Surface Sample Collection	4-13
J-4	Collect a Lunar Environment Sample	Surface Sample Collection	4-13
J-5	Study and Describe Lunar Topography Features	Lunar Surface Observ/Photo	4-17
K	Lunar Surface Structure Photograph (Objective Deleted)	Deleted	

TABLE 4-2
TEST OBJECTIVE/MISSION ACTIVITY
CROSS REFERENCE

DTO/FTO NUMBER	TEST OBJECTIVE	MISSION ACTIVITY	SECTION PAGE NO.
L	Television Coverage		
L-1	TV Coverage of Astronaut Descending to the Lunar Surface	EVA Preparation and Egress	4-10
L-2	TV Coverage of External Landed LM	External LM Observ/Photo	4-15
L-3	TV Coverage of Lunar Surface Near LM	Lunar Surface Observ/Photo	4-17
L-4	TV Panoramic Coverage of Distant Terrain Features	Lunar Surface Navigation	4-8
		Lunar Surface Observ/Photo	4-17
L-5	TV Coverage of Astronaut Activities on the Lunar Surface	Lunar Surface Observ/Photo	4-17
M	Photographic Coverage		
M-1	Photograph Lunar Surface During LM Descent	LM Descent	4-6
M-2	Photograph Lunar Surface Post Touchdown/Pre EVA	Lunar Surface Navigation	4-8
M-3	Obtain Phtotgraphs During EVA	Surface Sample Collections	4-13
		External LM Observ/Photo	4-15
		Lunar Surface Observ/Photo	4-17
S-031	Lunar Passive Seismology	Experiment Deployment/Conduct	4-20
S-078	Laser Ranging Retro-Reflector	Experiment Deployment/Conduct	4-20
S-080	Solar Wind Composition	Experiment Deployment/Conduct	4-20

LM DESCENT

A. Test Objective

D-1 LM Landing Gear Performance Under Landing Conditions
G-1 Location of the Landed LM from LM Data
G-3 Capability of Locating the Landed LM in Real Time from LM/CSM/MSFN Data
H-1 Data on Landing Aids and Final Approach Visibility
M-1 Photograph Lunar Surface During LM Descent

B. Test Requirements

1. Determine landing site visibility, extent of washout and visibility of landing site landmarks. [H]

2. Photograph the landing site during the approach through the LM pilot's window with the data acquisition camera. [G, H, M]

3. Evaluate landing aids, i.e., Landing Point Designator, maps, photographs. [G, H]

4. Assess visual phenomena during LM landing which are significantly different from expected. [H]

5. Voice anotate location and identity of features during final descent. [G]

6. Determine landing location in real time by description of terrain features during descent. [G]

7. Assess LM landing conditions on the lunar surface. [D]

C. Procedures/Checklist

1. Photographic and Television Operations Plan.

2. Descent Procedures Document.

D. Data Requirements

1. Flight Crew Reports/Logs/Photographs

   a. LM crew comments on landing site visibility during final approach and landing phases and on effectiveness of the Landing Point Designator and landing site recognition aids. [H] (M)

   b. GET at start of data acquisition camera photographs during LM final approach. [H] (M)

   c. Voice track regarding observations of surface features during the descent phase. [G] (M)

   d. Photographs of the landing site and surrounding lunar surface features taken through a LM window during descent. [G, M] (M)

e. Data Acquisition Camera photographs of the landing site from high gate to touchdown. [H, M] (M)

f. Photographs of the landing site and surrounding lunar surface features taken through a LM window during descent. [G, M] (M)

g. Comments on any lunar dust observed during the final approach, the severity of the landing and vehicle stability after touchdown. [D] (M)

2. Ground Support

a. LM TM HBR. [D, G, H] (M)

b. LM TM LBR. [D, G] (M)

c. LM BET from DOI through touchdown. [G, H] (M)

d. MSFN tracking data of LM from acquisition of signal through touchdown. [G] (M)

LUNAR SURFACE NAVIGATION

A. Test Objectives

G-1 Determine the Location of the Landed LM from LM Data
G-2 Determine the Location of the Landed LM from CSM Data
G-3 Determine Capability of Locating the Landed LM in Real Time from LM/CSM/MSFN Data
L-4 Panoramic Coverage of Distant Terrain Features
M-2 Photograph Lunar Surface Post Touchdown/Pre EVA

B. Test Requirements

1. Correlate lunar surface features surrounding the landing site with photomaps and mark the LM location. [G, L, M]

2. Photograph terrain features thru the LM window to correlate LM location. [G, M]

3. Obtain two sets of LM IMU alignments after landing [G]

4. Provide TV coverage of prominent terrain features. [G, L]

5. Track the landed LM from the CSM during two orbital passes. Mark on a landmark near the landed LM. [G] - (Only one pass is implemented.)

6. Track the CSM with LM RR during one pass. [G] - (Not Implemented.)

7. Obtain 70 MM photographs of the landed LM or its shadow and the surrounding lunar features. [G]

8. Assist MCCH in determining the landing LM location in real time. [G, L]

C. Procedures/Checklist

1. Photographic and Television Operations Plan.

2. LM AOH, "PGNCS Lunar Surface Align Program (P57)".

3. LM AOH, "Lunar Surface Navigation Program (P22)".

4. CSM AOH, "Orbital Navigation (P22)".

D. Data Requirements

1. Flight Crew Reports/Logs/Photographs

   a. Estimate of the landed LM location on lunar photomaps. [G] (M)

b. Comments by LM crew regarding any difficulties encountered in estimating the location of the LM with respect to lunar surface features. [G] (HD)

c. Comments by LM crewman on location of landed LM with respect to prominent terrain features. [G] (M)

d. Obtain high resolution photographs of the landing area from the CSM. [G] (M)

e. Photographs of the landing site and surrounding lunar surface features taken through a LM window after landing. [G, M] (M)

f. Provide TV coverage of the lunar surface as viewed from the LM. [G, L] (M)

2. Ground Support

a. LM TLM HBR. [G] (M)

b. LM TLM LBR. [G] (M)

c. BET of CSM during the lunar surface phase. [G] (M)

d. BET of LM from DOI through touchdown. [G] (M)

e. Photographs of the landing area obtained during previous lunar missions. [G] (M)

f. Post-scan conversion video tape of all TV coverage. [L] (M)

g. Estimate solar illumination established by mission geometry. [L] (M)

h. Reflectivity and geometry of surfaces contributing to indirect illumination. [L] (HD)

EVA PREPARATION AND EGRESS

A. Test Objectives

B-1 Demonstrate Egress-to/Ingress-from the Lunar Surface
B-2 Evaluate Crew Lunar Surface EVA Capability
C-1 EMU Capability to Provide a Habitable Environment
C-2 EMU Effects on Crew Mobility/Dexterity/Comfort
C-3 Data/Voice Communications Capability During EVA
L-1 TV Coverage of an Astronaut Descending to the Lunar Surface

B. Test Requirements

1. Perform EVA preparations. [C]

2. Release the MESA pallet with pre-mounted TV camera and turn camera power on prior to descent to the lunar surface. [L]

3. Egress to the lunar surface. [B, C]

4. Deploy and set the TV camera to provide TV coverage of the lunar surface EVA. [L]

5. During EVA, communicate with MSFN via the EVA-LM-MSFN two way voice relay. [C]

6. Two-way voice communications to be performed between two EVA crewmen. [C]

7. EMU and biomedical data from two EVA crewmen will be simultaneously transmitted to MSFN via EVA-LM-MSFN one-way relay. [C]

C. Procedures/Checklist

1. EVA Procedures Document.

2. Lunar Surface Operations Plan.

D. Data Requirements

1. Flight Crew Reports/Logs/Photographs

   a. Notify MSFN of the initial and final positions of the PLSS water diverter valve, primary oxygen shutoff valve and water shutoff/ relief valve each time they are changed. [C] (M)

   b. Notify MSFN when PLSS; High O_2 flowrate, low vent flow, low feed water pressure or PGA pressure low remote control unit status indicators and audible warning tone come on. [C] (M)

c. Record EMU radiation dosimeter readings just prior to the EVA. [C] (M)

d. Notify MSFN if noxious odors occur or any condensation on the visor assembly. [C] (HD)

e. Comment on the adequacy of procedures and difficulties encountered during donning of EMU equipment. [C] (HD)

f. Comment on time required and adequacy of the EMU checkout procedures. [C] (HD)

g. Comment on the adequacy of EMU thermal environment when walking from a sunlit area to shadow and vice versa. [C] (M)

h. Comment on estimated energy expenditure and comfort as compared to simulation experience. [C] (HD)

i. Provide data on the adequacy of hardware and procedures, and the time required to perform the egress from the LM. [B] (M)

j. Comment on voice quality for EVA-EVA and EVA-LM-MSFN communications. [C] (M)

k. Provide sequence camera coverage and TV camera coverage of: [B, M] (M)

   1) A crew member descending to the lunar surface.

   2) A crew member walking on the lunar surface.

   3) A crew member performing lunar surface EVA operations.

2. Ground Support

a. LM TM FM. [B, C] (M)

b. Ground recorded TV signals. [B] (HD)

c. LM TM LBR. [L] (HD)

d. Post-scan conversion video tape of all TV coverage. [L] (M)

e. Record of S-band signal strength during video transmission. [L] (HD)

f. GET at beginning and end of TV transmission. [L]

g. Time period, if any, when LBR TM (in lieu of HBR TM) transmitted simultaneously with TV data. [L] (M)

h. Identity of ground station(s) used to record video transmission from LM. [L] (M)

i. Time period, if any, when erectable antenna used to transmit TV data. [L] (M)

j Estimate of incident illumination. [L] (M)

k. LM position on lunar surface. [H] (HD)

l. MSFN recording of EVA-LM-MSFN voice. [C] (M)

SURFACE SAMPLE COLLECTION

A. Test Objectives

A-1 Provide a Contingency Lunar Surface Sample
E-1 Behavior and Characteristics of the Lunar Surface
F-1 Collect Rock Samples and Fine Grained Material
F-2 Photograph Collection Area of Samples
I-3 Obtain a Lunar Sample for Quarantine Testing
J-2 Obtain a Core Sample of the Lunar Surface
J-3 Collect Lunar Geologic Samples
J-4 Collect a Lunar Environment Sample
M-3 Obtain Photographs of Geologic Inspection & Sampling

B. Test Requirements

1. Contingency Sample - Obtain upon first descending to the lunar surface. [A]

2. Bulk Material - Obtain 30 pounds consisting of 1/3 fragmentary and 2/3 loose samples. [F]

3. Core Sample - Obtain with the drive tube. [I, J]

4. Geologic Samples - Obtain using tools stowed in the MESA. Photograph sample areas. [J, M]

5. Lunar Environment Sample - Seal in gas analysis container. [J]

C. Procedures/Checklist

1. Lunar Landing Mission Flight Plan.

2. Lunar Surface Operations Plan.

3. Photographic and Television Operations Plan.

D. Data Requirements

1. Flight Crew Reports/Logs/Photographs

   a. Record areas in relation to LM where samples were collected. [A, F, J] (M)

   b. Record unusual lunar surface observations. [A, F, J] (M)

   c. Comment on soil behavior during collection of Bulk Sample. [E] (M)

   d. Comment on soil behavior during collection of Documented Sample. [E] (HD)

   e. Estimates of volume of fine grained material collected in one bag of the Documented Sample. [E] (HD)

f. Take photographs during sample collection. [A, F] (HD)

g. Photograph the lunar surface sample areas and of the samples as defined in the Photographic Operations Plan. [J] (M)

2. Ground Support

a. LM position on lunar surface. [J] (M)

b. MSFN recordings of all MSFN/EVA voice conferences. [J] (M)

EXTERNAL LM OBSERVATIONS AND PHOTOGRAPHY

A. Test Objectives

D-1 Effects of Landing on LM Landing Gear
D-2 Effects of Landing on LM Structure and Components
D-3 Descent Engine Skirt Damage and Clearance After Landing
D-4 Effects of RCS Plume Impingement on LM Structure and Components
L-2 TV Coverage of External Landed LM
M-3 Obtain Photographs of Landed LM

B. Test Requirements

1. Operate the TV camera to provide an external view of the LM. [L]

2. Photograph any observed LM external structural damage. [D, M]

3. Determine descent engine skirt ground clearance. [D, M]

4. Photograph any effects of RCS plume impingement observed. [D, M]

5. Obtain photographs of any lunar material collected on the LM. [D, M]

C. Procedures/Checklist

1. Mission G Photographic and Television Operations Plan.

D. Data Requirements

1. Flight Crew Reports/Logs/Photographs

   a. Comment on any LM component damage to include any visible discoloration or lunar soil accumulation. [D] (M)

   b. Comments describing any descent engine skirt damage and estimate of any skirt ground clearance. [D] (M)

   c. If the landing gear strut assembly photographs cannot be obtained, estimate the amount of stroking of each primary and secondary strut assembly. [D] (M)

   d. Photograph the landing gear to show the stroking of the primary and secondary strut assemblies. [D, M] (M)

   e. Photograph the LM exterior showing any structural damage. [D, M] (M)

   f. Photograph each landing gear assembly along the Z axis and the Y axis. [D, M] (HD)

g. Photograph the descent engine skirt. [D, M] (HD)

h. Photograph the LM base heat shield. [D, M] (HD)

i. Photograph the LM exterior, i.e., structure antenna, RCS jets, windows and foot pads. [D, M] (HD)

j. Photograph soil accumulation on the LM. [D, M] (HD)

k. Photographs by the close up stereo camera of lunar material adhering to LM surfaces. [M] (HD)

2. Ground Support

a. LM TM HBR. [D] (M), [L] (HD)

b. LM Mass, center of gravity and mass moment of inertia calculations. [E] (M)

c. Video tape of all TV coverage. [L] (M)

d. Record of S-band signal strength during TV coverage. [L] (HD)

e. GET at beginning and end of TV operations.

f. Time period of simultaneous LBR TM and TV transmission. [L] (M)

g. Identification of ground station(s) used to record video transmission. [L] (M)

h. Time period when erectable antenna was used to transmit from lunar surface. [L] (M)

LUNAR SURFACE OBSERVATIONS AND PHOTOGRAPHY

A. Test Objectives

E-1 Behavior and Characteristics of the Lunar Surface
E-2 Erosion of Lunar Surface by DPS Plume Impingement
E-3 Effect of Any DPS Venting on the Lunar Surface
H-2 Crew Performance of Visual Tasks on the Lunar Surface
J-5 Study and Description of Lunar Topography Features
L-3 TV Coverage of Lunar Surface Near LM
L-4 TV Panoramic Coverage of Distant Terrain Features
L-5 Coverage of Astronaut Activities on the Lunar Surface
M-3 Obtain Photographs During EVA

B. Test Requirements

1. Provide TV coverage of the lunar surface in the vicinity of the LM and panoramic scenes of distant terrain features. [L]

2. Photograph the lunar terrain at various azimuths with respect to the sun including 9, 90 and 180 degrees. Comment on ability to see terrain features in these areas. [H, M]

3. Estimate the distance to prominent terrain features within the field of view of photographs taken. [H]

4. Observe lunar surface characteristics including texture, consistency, compressibility, cohesiveness, adhesiveness, density and color. [E]

5. Study and photograph the mechanical behavior of the lunar surface from interactions of astronauts boots and equipment with the lunar soil, erosion by DPS plume impingement and DPS venting. [E, M]

6. Describe and photograph field relationships such as shape, size, range, pattern of alignment or distribution of all accessible types of lunar topographic features. [J,M]

7. Photograph the structure of lunar surface material in its natural state. [M]

C. Procedures/Checklist

1. Mission G Photographic and Television Operations Plan.

D. Data Requirements

1. Flight Crew Report/Logs/Photographs

   a. Report condition of the temperature indicator viewing ports on the TV camera at the beginning and the end of the TV operations. [L] (M)

b. Position of the TV camera scan rate switch at start of TV operation. [L] (M)

c. Comments describing the interaction between astronaut boots and lunar surface while walking. [E] (M)

d. Comments on slope and roughness characteristics of the landing terrain to include descriptions of craters, depressions, embankments or other obstacles. [E] (M)

e. Comments on the color and texture of both undisturbed and mechanically disturbed areas of the lunar surface. [E] (M)

f. Comments on lunar soil conditions adjacent to DPS vents to include any discoloration. [E] (M)

g. Comments describing the lunar surface penetration by the Solar Wind Composition Staff and core sample tool under their own weight and the estimated force. [E] (Mandatory for either the staff or the core sample tool: highly desirable for the other.)

h. Comments on lunar soil erosion as caused by the DPS plume impingement during landing. [E] (M)

i. Record vent valves opened. [E] (M)

j. Photograph the lunar surface showing DPS plume impingement erosive effects. [E, M] (M)

k. Photograph the lunar surface adjacent to DPS vents if soil discoloration is observed. [E, M] (M)

l. Photograph an astronaut footprint showing interaction between astronaut boots and lunar surface. [E, M] (M)

m. Photograph the Solar Wind Composition Experiment Staff and core sampling tool after being inserted to their maximum depth as penetrometers. [E, M] (HD)

n. Photograph the natural slopes, crater walls and embankments in the vicinity of the landing site. [E,M] (M)

o. Photograph from the CSM of the lunar surface surrounding the LM. [E, M] (HD)

p. Comments on the visibility of the lunar terrain as a function of the sun/viewing angle and on their ability to perform visual tasks while on the lunar surface. [H] (M)

q. Comments on color/contrast perception. [H] (M)

r. Comments on and significant unexpected visual phenomena. [H] (M)

s. Estimate of distance to at least one prominent terrain feature within the field of view of the photographs in item t below. [H] (M)

t. Photograph the lunar terrain at various sun azimuths to include 0 degrees, 90 degrees and 180 degrees. [H, M] (M)

u. Photograph any unexpected visual phenomena. [H, M] (HD)

v. Photograph a representative depression caused by use of the scoop in collecting fine grained fragmental material. [E, M] (M)

w. Photograph one scoop of fine grained fragmental material placed in one of the pre-numbered bags. [E, M] (HD)

x. Photograph of each LM foot pad and surrounding lunar soil exhibiting evidence of LM foot pad-lunar soil interaction. [M] (HD)

2. Ground Support

a. LM TM HBR. [E, L] (HD)

b. Estimate of incident illumination. [D] (M)

c. Video tape of all TV coverage. [L] (M)

d. Record of S-band signal strength during TV transmission. [L] (M)

e. GET at beginning and end of TV transmission. [L] (M)

f. Time period when LBR TM was transmitted simultaneously with TV. [L] (M)

g. Identity of ground station(s) used to record LM video transmission. [L] (M)

h. Time period when erectable antenna was used to transmit from the lunar surface. [L] (M)

EXPERIMENT DEPLOYMENT/CONDUCT

A. Test Objectives

S-031 Deploy the Passive Seismic Experiment Package
S-078 Deploy the Laser Ranging Retro-Reflector Experiment
S-080 Conduct the Solar Wind Composition Experiment

B. Test Requirements

1. Emplace, level and orient the Passive Seismic Experiment Package (PSEP). Deploy the solar panels and aim the antenna at the earth. [S-031]

2. Photograph the deployed PSEP and deployment area. [S-031]

3. Remove the Laser Ranging Retro-Reflector (LRRR) from the descent stage and carry it to the deployment site. [S-078]

4. Emplace, level and orient the LRRR to the alignment marks corresponding to the landing site. [S-078]

5. Remove the Solar Wind Composition Experiment from the LM MESA and deploy it on the lunar surface. [S-080]

6. After one hour operation, disassemble the Solar Wind Composition Experiment, place the reel and foil in a teflon bag and store in a sample return container. [S-080]

C. Procedures/Checklist

None

D. Data Requirements

1. Flight Crew Reports/Logs/Photographs

   a. Comment on deployment of experiment. [S-031] (M)

   b. Photograph deployment area. [S-031, S-078, S-080] (HD)

   c. Comment on location of deployed experiment with respect to the LM, attitude of deployed foil with respect to the sun and total time foil was deployed. [S-080] (M)

   d. Retrieve reel and foil from the Solar Wind Composition Experiment. [S-080] (M)

   e. Comments on orientation and elevation setting used for deployment. [S-078] (HD)

2. Ground Support

   a. Experiment TLM Data [S-031] (M)

POST EVA OPERATIONS

A. Test Objectives

B-1 Demonstrate Egress-to/Ingress-from the Lunar Surface
C-1 EMU Capability to Provide a Habitable Environment
C-2 EMU Effects on Crew Mobility, Dexterity/Comfort

B. Test Requirements

1. Perform post EVA preparations and ingress. [B]

2. Perform PLSS shutdown. [C]

C. Procedures/Checklist

1. EVA Procedures Document.

D. Data Requirements

1. Flight Crew Reports/Logs/Photographs

a. Notify MSFN of the initial and final positions of the PLSS water diverter valve, primary oxygen shutoff valve and water shutoff/relief valve each time they are changed. [C] (M)

b. Notify MSFN when PLSS; High O_2 flowrate, low vent flow, low feed water pressure or PGA pressure low remote control unit status indicators and audible warning tone come on. [C] (M)

c. Provide data on the adequacy of hardware and procedures, and the time required to perform the ingress to the LM. [B] (M)

d. Comment on the adequacy of procedures and difficulties encountered during doffing of EMU equipment. [C] (HD)

e. Record quantity of water drained from PLSS at end of EVA period. [C] (M)

f. Record EMU radiation dosimeter readings after completion of the EVA. [C] (M)

g. Provide sequence camera coverage and TV camera coverage of a crew member ascending the LM ladder. [B] (M)

Contamination Prevention

A. Test Objectives

I-1 Prevent Earth Contamination by Lunar Exposed Materials
I-2 Minimize Crew/CM Contamination by Lunar Exposed Materials

B. Test Requirements

1. All contamination related operations from the initial astronaut egress to the lunar surface until postflight crew/cm quarantine will be completed per procedures contained in the documents listed below. [I]

C. Procedures/Checklist

1. Lunar Surface Operations Plan

2. EVA Procedures Document

3. Quarantine Procedures

D. Data Requirements

1. Flight Crew Reports/Logs/Photographs

   a. Crew comments on the adequacy of Biological Isolation Garment, sample return containers, Mobile Quarantine Facility and related equipment and procedures used to prevent back contamination. [I] (M)

   b. Photograph boots, clothing and equipment showing adhesion of particles. [I, M] (HD)

2. Ground Support

   a. Deliver samples, CM and Mobile Quarantine Facility to the Lunar Receiving Laboratory. [I] (M)

   b. Comment on ground procedures and hardware used for retrieval, biological isolation and CM transfer to the Lunar Receiving Laboratory. [I] (M)

   c. Report on the existence of contamination of the crew on CM. [I] (M)

SECTION V - CONSUMABLES ANALYSIS

NOTE

Acknowledgement is made to the Consumables Analysis Section (CAS) of the Mission Planning and Analysis Division (MPAD) for their work in the preparation of the consumable analysis presented herein and to the Crew Systems Division for the PLSS Consumables.

CSM-107/LM5 PROPELLANT BUDGET

The results of the Propellant Budget Analysis are summarized in the following Tables and Figures:

TABLE 5-1 SM RCS Propellant Loading And Usage Summary

TABLE 5-2 SM RCS Budget

TABLE 5-3 CM RCS Propellant Summary

TABLE 5-4 SPS Propellant Summary

TABLE 5-5 SPS Assumptions

TABLE 5-6 LM RCS Propellant Loading And Usage Summary

TABLE 5-7 LM RCS Budget

TABLE 5-8 DPS Propellant Summary

TABLE 5-9 DPS Assumptions

TABLE 5-10 APS Propellant Summary

TABLE 5-11 APS Assumptions

FIGURE 5-1 Total SM RCS Propellant Profile

FIGURE 5-2 Quad A SM RCS Propellant Profile

FIGURE 5-3 Quad B SM RCS Propellant Profile

FIGURE 5-4 Quad C SM RCS Propellant Profile

FIGURE 5-5 Quad D SM RCS Propellant Profile

FIGURE 5-6 Total LM RCS Propellant Profile

SM-RCS BUDGET

GROUND RULES and ASSUMPTIONS

1. The transposition and docking phase of the mission includes an SPS evasive maneuver.

2. The first and third midcourse corrections (translunar) are executed as SPS burns with the third MCC followed by an RCS trim.

3. No SM RCS propellant is required during PTC or lunar orbit coast.

4. The sixth midcourse correction (transearth) is executed as an RCS burn of 5 fps.

5. The individual quad plots are included for reference only as quad management is determined by the flight controllers during the mission.

TABLE 5-1

SM RCS PROPELLANT LOADING AND USAGE SUMMARY

Nominal loaded		1342.4 lb
Initial outage due to loaded mixture ratio		15.6
Total trapped		26.4
Gauging inaccuracy		80.4
Deliverable SM-RCS propellant		1220.0
Nominal usage		590
Translunar phase (through LOI-2)	204	
Lunar orbit phase	311	
Transearth Phase (includes TEI)	75	
Nominal remaining		630 lb

TABLE 5-2

SM-RCS PROPELLANT BUDGET

TIME (HR)	EVENT	S/C WT (LBS) (a)	SM-RCS USED (LBS)	SM-RCS LEFT (LBS) (b)	SM-RCS LEFT (%) (b)
.0	MISSION G	63457.	.0	1220.0	100.
.0	INITIALIZE PROP LOADING	63457.	.0	1220.0	100.
1.7	SM RCS CHECKOUT	63451.	5.8	1214.2	100.
3.2	TRANSPOSITION AND DOCKING +X 0.8 FPS	63445.	6.1	1208.1	99.
3.2	-X 0.3 FPS	63443.	2.4	1205.7	99.
3.2	PITCH TO ACQUIRE SIVB PITCH 180 DEG AT 1.5 DEG/SEC	63440.	2.3	1203.4	99.
3.2	ROLL CSM 60 DEG 2 DEG/SEC	63439.	1.3	1202.1	99.
3.2	NULL RELATIVE DEL V 0.5 FPS	63435.	4.0	1198.1	98.
3.5	INDEX AND DOCK	63409.	26.0	1172.1	96.
4.2	LM EJECTION -X 5 SEC 4 JET	96717.	7.4	1164.6	95.
4.5	SPS BURN TO EVADE SIVB ORIENT AT 0.2 DEG/SEC	96712.	4.4	1160.2	95.
4.5	ATTITUDE HOLD 0.5 DEG DB PGNCS	96712.	.8	1159.4	95.
4.5	START TRANSIENT CONTROL	96710.	1.3	1158.1	95.
4.5	SPS BURN BUILD UP	96707.	.0	1158.1	95.
4.5	STEADY STATE BURN	96508.	.3	1157.8	95.

(a) Spacecraft weights are approximate and are included for reference only.

(b) Note: These refer to usable SM RCS propellant.

TABLE 5-2 (CONT'D)

SM-RCS PROPELLANT BUDGET

TIME (HR)	EVENT	S/C WT (LBS)	SM-RCS USED (LBS)	SM-RCS LEFT (LBS)	SM-RCS LEFT (%)
4.5	TAILOFF	96467.	.7	1157.2	95.
4.5	DAMP SHUTDOWN TRANSIENT	96466.	1.1	1156.1	95.
5.5	P52 IMU ALIGN	96466.	.2	1155.9	95.
5.9	NAVIGATION SIGHTINGS ORIENT AT 0.2 DEG/SEC	96461.	4.4	1151.5	94.
6.1	NAVIGATION SIGHTINGS ORIENT AT 0.2 DEG/SEC	96457.	4.4	1147.1	94.
7.0	ORIENT FOR PTC 3AXIS 0.2 DEG/SEC	96453.	4.1	1143.0	94.
7.0	ATTITUDE HOLD 0.5 DEG DB PGNCS	96452.	.8	1142.2	94.
7.0	ROLL 0.3 DEG/SEC	96451.	.4	1141.8	94.
10.6	TERMINATE PTC DAMP RATES	96447.	4.4	1137.4	93.
10.7	P52 IMU ALIGN	96447.	.2	1137.1	93.
11.5	MIDCOURSE CORRECTION NO 1 3 AXIS ORIENT PGNCS	96442.	4.4	1132.7	93.
11.5	ATTITUDE HOLD 0.5 DEG DB PGNCS	96442.	.8	1131.9	93.
11.5	START TRANSIENT CONTROL	96440.	1.3	1130.6	93.
11.5	SPS BURN BUILD UP	96437.	.0	1130.6	93.
11.5	STEADY STATE BURN 3 FPS PGNCS	96402.	.1	1130.5	93.

TABLE 5-2 (CONT'D)

SM-RCS PROPELLANT BUDGET

TIME (HR)	EVENT	S/C WT (LBS)	SM-RCS USED (LBS)	SM-RCS LEFT (LBS)	SM-RCS LEFT (%)
11.5	TAILOFF	96361.	.8	1129.7	93.
11.5	DAMP SHUT-DOWN TRANSIENT	96359.	1.1	1128.6	93.
12.0	P52 IMU ALIGN	96359.	.2	1128.4	92.
12.5	ORIENT FOR PTC 3AXIS 0.2 DEG/SEC	96355.	4.1	1124.3	92.
12.5	ATTITUDE HOLD 0.5 DEG DB PGNCS	96354.	.8	1123.5	92.
12.5	ROLL 0.3 DEG/SEC	96354.	.4	1123.1	92.
24.2	TERMINATE PTC DAMP RATES	96349.	4.4	1118.7	92.
24.3	P52 IMU ALIGN	96349.	.2	1118.5	92.
24.5	CISLUNAR NAVIGATION STAR/EARTH HORIZON ORIENT	96345.	4.4	1114.2	91.
24.7	NAVIGATION SIGHTINGS ORIENT AT 0.2 DEG/SEC	96341.	4.4	1109.8	91.
26.6	MIDCOURSE CORRECTION NO 2 MNVR TO BURN ATT	96336.	4.4	1105.4	91.
26.6	ATTITUDE HOLD 0.5 DEG DB PGNCS	96335.	.8	1104.7	91.
26.7	DELTA VEL = NOMINALLY ZERO	96335.	.0	1104.7	91.
27.0	ORIENT FOR PTC 3AXIS 0.2 DEG/SEC	96331.	4.2	1100.5	90.
27.0	ATTITUDE HOLD 0.5 DEG DB PGNCS	96330.	.8	1099.7	90.

TABLE 5-2 (CONT'D)

SM-RCS PROPELLANT BUDGET

TIME (HR)	EVENT	S/C WT (LBS)	SM-RCS USED (LBS)	SM-RCS LEFT (LBS)	SM-RCS LEFT (%)
27.0	ROLL 0.3 DEG/SEC	96330.	.4	1099.3	90.
52.8	TERMINATE PTC DAMP RATES	96326.	4.4	1094.9	90.
53.0	PS2 IMU ALIGN	96325.	.2	1094.7	90.
53.6	MIDCOURSE CORRECTION NO 3 MNVR TO BURN ATT	96321.	4.4	1090.3	89.
53.6	ATTITUDE HOLD 0.5 DEG DB PGNCS	96320.	.8	1089.5	89.
53.6	START TRANSIENT CONTROL	96319.	1.3	1088.2	89.
53.6	SPS BURN BUILD UP	96316.	.0	1088.2	89.
53.6	STEADY STATE BURN 3 FPS	96281.	.1	1088.1	89.
53.6	TAILOFF	96239.	.8	1087.3	89.
53.6	DAMP SHUT-DOWN TRANSIENT	96238.	1.1	1086.2	89.
53.6	RCS TRIM 1 FPS	96227.	11.2	1075.0	88.
54.0	ORIENT FOR PTC 3AXIS 0.2 DEG/SEC	96223.	4.1	1070.9	88.
54.0	ATTITUDE HOLD 0.5 DEG DB PGNCS	96222.	.8	1070.1	88.
54.0	ROLL 0.3 DEG/SEC	96222.	.4	1069.8	88.
69.5	TERMINATE PTC DAMP RATES	96217.	4.4	1065.3	87.

TABLE 5-2 (CONT'D)

SM-RCS PROPELLANT BUDGET

TIME (HR)	EVENT	S/C WT (LBS)	SM-RCS USED (LBS)	SM-RCS LEFT (LBS)	SM-RCS LEFT (%)
70.0	PS2 IMU ALIGN	96217.	.2	1065.1	87.
70.5	MIDCOURSE CORRECTION NO 4 MNVR TO BURN ATT	96213.	4.4	1060.7	87.
70.5	ATTITUDE HOLD 0.5 DEG DB PGNCS	96212.	.8	1059.9	87.
70.5	DEL VEL = NOM ZERO	96212.	.0	1059.9	87.
72.7	PS2 IMU ALIGN	96212.	.2	1059.7	87.
74.0	ORIENT AND SXT STAR CHECK	96207.	4.4	1055.2	86.
74.5	ORIENT AND OBSERVE LUNAR SURFACE	96203.	4.4	1050.8	86.
75.5	LUNAR ORBIT INSERTION BURN 1 3-AXIS ORIENT PGNCS	96198.	4.4	1046.5	86.
75.5	ATTITUDE HOLD 0.5 DEG DB PGNCS	96198.	.8	1045.7	86.
75.5	START TRANSIENT CONTROL	96196.	1.3	1044.4	86.
75.9	LOI BURN BUILD UP	96193.	.0	1044.4	86.
75.9	STEADY STATE BURN	72357.	.5	1043.9	86.
75.9	TAILOFF	72316.	.0	1043.9	86.
75.9	DAMP SHUT DOWN TRANSIENT	72315.	1.1	1042.8	85.
76.2	REV 1 ATTITUDE HOLD WIDE DEADBAND	72312.	3.0	1039.8	85.

TABLE 5-2 (CONT'D)

SM-RCS PROPELLANT BUDGET

TIME (HR)	EVENT	S/C WT (LBS)	SM-RCS USED (LBS)	SM-RCS LEFT (LBS)	SM-RCS LEFT (%)
77.5	P52 IMU ALIGN	72312.	.1	1039.6	85.
78.2	REV 2 ATTITUDE HOLD	72309.	3.0	1036.6	85.
79.2	P52 IMU ALIGN	72309.	.1	1036.5	85.
80.0	LOI 2 LPO CIRC MNVR TO BURN ATT	72306.	3.5	1033.0	85.
80.0	ATTITUDE HOLD 0.5 DEG DB PGNCS	72305.	.8	1032.2	85.
80.0	B-D ULLAGE	72290.	15.1	1017.1	83.
80.1	SPS BURN BUILD UP	72287.	.0	1017.1	83.
80.1	STEADY STATE BURN	71316.	.2	1017.0	83.
80.1	TAILOFF	71276.	.0	1017.0	83.
80.1	DAMP SHUTDOWN TRANSIENT	71275.	1.1	1015.9	83.
80.2	REV 3 ATTITUDE HOLD	71272.	3.0	1012.9	83.
80.4	REACQUIRE MSFN ROLL 0.2 DEG/SEC	71272.	.1	1012.8	83.
82.2	REV 4 ATTITUDE HOLD	71269.	3.0	1009.8	83.
82.3	MNVR TO LDG SITE OBS ATT	71265.	3.5	1006.3	82.
82.3	LDG SITE OBSERVATION	71265.	.4	1005.8	82.

TABLE 5-2 (CONT'D)

SM-RCS PROPELLANT BUDGET

TIME (HR)	EVENT	S/C WT (LBS)	SM-RCS USED (LBS)	SM-RCS LEFT (LBS)	SM-RCS LEFT (%)
82.3	REORIENT	71261.	3.5	1002.3	82.
82.3	REACQUIRE MSFN	71261.	.2	1002.1	82.
84.2	MANEUVER TO SLEEP ATTITUDE 3 AXIS 0.2 DEG/SEC	71258.	3.5	998.6	82.
94.4	DAMP RATES	71254.	3.5	995.0	82.
94.5	REACQUIRE MSFN	71254.	.1	994.9	82.
95.1	MNVR TO ALIGN ATT	71250.	3.5	991.4	81.
96.2	REV 11 ATTITUDE HOLD	71247.	3.0	988.4	81.
98.2	REV 12 ATTITUDE HOLD	71244.	3.0	985.4	81.
98.5	MNVR TO LOS SITE OBS ATT	71241.	3.5	981.8	80.
98.5	LOS SITE OBSERVATION	71240.	.4	981.4	80.
98.9	REACQUIRE MSFN ROLL 0.2 DEG/SEC	71240.	.2	981.3	80.
99.8	MANEUVER TO AGS CAL ATTITUDE	71237.	3.5	977.7	80.
100.0	PRE UNDOCKING ALLOCATION	71213.	24.0	953.7	78.
100.0	ORIENT TO UNDOCKING ATTITUDE ROLL 0.2 DEG/SEC	71212.	.2	953.6	78.
100.2	CSM ACTIVE UNDOCK SEP AND NULL VEL 0.5 FPS	37893.	4.5	949.0	78.

TABLE 5-2 (CONT'D)

SM-RCS PROPELLANT BUDGET

TIME (HR)	EVENT	S/C WT (LBS)	SM-RCS USED (LBS)	SM-RCS LEFT (LBS)	SM-RCS LEFT (%)
100.2	FORMATION FLYING	37883.	10.0	939.0	77.
100.2	REACQUIRE MSFN	37883.	.1	938.9	77.
100.6	ORIENT FOR SEP BURN	37880.	3.1	935.8	77.
100.7	RCS SEPARATION BURN 2.5 FPS	37868.	11.2	924.6	76.
100.7	REV 13 ATTITUDE HOLD	37865.	3.0	921.6	76.
101.5	MANEUVER TO SXT TRACKING	37862.	3.1	918.6	75.
102.6	MANEUVER TO SXT TRACKING	37859.	3.1	915.5	75.
104.4	REACQUIRE MSFN ROLL 0.5 DEG/SEC	37859.	.3	915.3	75.
104.5	MANEUVER TO SXT TRACKING	37856.	3.1	912.2	75.
104.6	REV 14 ATTITUDE HOLD	37853.	3.0	909.2	75.
104.6	MNVR TO LDG SITE OBS ATT	37850.	3.1	906.1	74.
104.6	LDG SITE OBS	37850.	.4	905.7	74.
104.7	TRACK LM	37846.	3.1	902.6	74.
104.9	REACQUIRE MSFN ROLL 0.5 DEG/SEC	37846.	.3	902.3	74.
105.0	REV 15 ATTITUDE HOLD	37843.	3.0	899.3	74.

TABLE 5-2 (CONT'D)

SM-RCS PROPELLANT BUDGET

TIME (HR)	EVENT	S/C WT (LBS)	SM-RCS USED (LBS)	SM-RCS LEFT (LBS)	SM-RCS LEFT (%)
105.0	REACQUIRE MSFN ROLL 0.5 DEG/SEC	37843.	.3	899.1	74.
107.0	PLANE CHANGE MNVR TO BURN ATT	37840.	3.1	896.0	73.
107.0	ATTITUDE HOLD 0.5 DEG DB PGNCS	37839.	.8	895.2	73.
107.0	ULLAGE	37825.	14.3	880.9	72.
107.0	SPS BURN BUILD UP	37822.	.0	880.9	72.
107.0	STEADY STATE	37754.	.1	880.8	72.
107.0	TAILOFF	37713.	1.0	879.8	72.
107.0	DAMP SHUTDOWN TRANSIENT	37712.	1.1	878.7	72.
107.2	PS2 IMU ALIGN	37712.	.1	878.6	72.
107.2	MANEUVER TO SLEEP ATTITUDE	37710.	1.7	876.9	72.
111.5	DAMP RATES	37707.	3.1	873.9	72.
112.2	REV 19 ATTITUDE HOLD	37704.	3.0	870.9	71.
114.2	REV 20 ATTITUDE HOLD	37701.	3.0	867.9	71.
114.3	ORIENT FOR SEXTANT TRACKING	37698.	3.1	864.8	71.
115.0	MANEUVER TO SLEEP ATT	37697.	.7	864.1	71.

TABLE 5-2 (CONT'D)

SM-RCS PROPELLANT BUDGET

TIME (HR)	EVENT	S/C WT (LBS)	SM-RCS USED (LBS)	SM-RCS LEFT (LBS)	SM-RCS LEFT (%)
120.0	DAMP RATES	37697.	.7	863.5	71.
120.0	SEXTANT TRACKING	37695.	1.3	862.2	71.
120.0	REACQUIRE MSFN	37695.	.1	862.1	71.
120.2	REV 23 ATTITUDE HOLD	37692.	3.0	859.1	70.
122.2	REV 24 ATTITUDE HOLD NARROW DEADBAND	37687.	5.2	853.9	70.
124.5	SUPPORT LM LIFT OFF	37669.	18.0	835.9	69.
124.6	MANEUVER TO TRACK LM POST LIFTOFF	37666.	3.1	832.8	68.
125.5	MANEUVER TO SUPPORT LM CSI BURN	37663.	3.1	829.7	68.
125.6	MANEUVER TO TRACK LM POST CSI	37660.	3.1	826.6	68.
125.6	REV 25 ATTITUDE HOLD NARROW DEADBAND	37654.	5.2	821.4	67.
126.5	MANEUVER TO SUPPORT LM CDH BURN	37651.	3.0	818.4	67.
126.6	MANEUVER TO TRACK LM POST CDH	37648.	3.1	815.3	67.
126.6	RNDZ NAV	37645.	3.1	812.2	67.
126.6	REINITIATE RNDZ NAV	37642.	3.1	809.1	66.
127.0	MANEUVER TO SUPPORT LM TPI BURN	37639.	3.1	806.1	66.

TABLE 5-2 (CONT'D)

SM-RCS PROPELLANT BUDGET

TIME (HR)	EVENT	S/C WT (LBS)	SM-RCS USED (LBS)	SM-RCS LEFT (LBS)	SM-RCS LEFT (%)
127.1	MANEUVER TO TRACK LM POST TPI	37636.	3.1	803.0	66.
127.1	MANEUVER TO COAS TRACK	37633.	3.1	799.9	66.
127.1	MANEUVER TO SXT TRACKING	37630.	3.1	796.9	65.
127.2	MANEUVER TO SUPPORT LM MCC1 BURN	37627.	3.1	793.8	65.
127.2	MANEUVER TO SXT TRACKING	37624.	3.1	790.8	65.
127.5	MANEUVER TO SUPPORT LM MCC2 BURN	37621.	3.1	787.7	65.
127.5	MANEUVER TO SUPPORT LM TPF BURN	37618.	3.0	784.7	64.
127.5	MANEUVER TO SXT TRACKING	37615.	3.1	781.6	64.
127.8	ORIENT TO DOCKING ATTITUDE	37612.	3.1	778.5	64.
127.8	ALLOCATION FOR TERMINAL RDZ USAGE FROM POSTFLIGHT	37577.	35.0	743.5	61.
127.9	MAINTAIN BORESIGHT	37574.	3.1	740.5	61.
128.0	DOCKING	43212.	26.0	714.5	59.
131.5	MNVR TO JETTISON ATT	43210.	1.1	713.3	58.
132.0	JETTISON LM 1 FPS	37542.	4.7	708.6	58.
132.0	ORIENT TO TRACKING ATT	37540.	1.6	707.0	58.

TABLE 5-2 (CONT'D)

SM-RCS PROPELLANT BUDGET

TIME (HR)	EVENT	S/C WT (LBS)	SM-RCS USED (LBS)	SM-RCS LEFT (LBS)	SM-RCS LEFT (%)
132.0	TRACK LM	37540.	.4	706.6	58.
132.6	HOLD INERTIAL ATT	37539.	.4	706.1	58.
132.6	P52 IMU ALIGN	37539.	.7	705.5	58.
134.5	P52 IMU ALIGN	37538.	.7	704.8	58.
134.5	SXT STAR CHECK	37537.	.4	704.4	58.
135.0	TRANS-EARTH INJECTION MNVR TO BURN ATT	37536.	1.6	702.7	58.
135.0	ATTITUDE HOLD 0.5 DEG DB PGNCS	37535.	.8	702.0	58.
135.0	ULLAGE	37521.	14.3	687.6	56.
135.5	SPS BURN BUILD UP	37518.	.0	687.6	56.
135.5	STEADY STATE SPS BURN	27478.	.2	687.4	56.
135.5	TAILOFF	27437.	.0	687.4	56.
135.5	DAMP SHUTDOWN TRANSIENT	27436.	1.1	686.3	56.
136.0	P52 IMU ALIGN	27436.	.6	685.7	56.
136.0	ORIENT FOR PTC	27435.	1.1	684.6	56.
136.0	ATTITUDE HOLD 0.5 DEG DB PGNCS	27434.	.8	683.8	56.

TABLE 5-2 (CONT'D)

SM-RCS PROPELLANT BUDGET

TIME (HR)	EVENT	S/C WT (LBS)	SM-RCS USED (LBS)	SM-RCS LEFT (LBS)	SM-RCS LEFT (%)
136.0	ROLL 0.3 DEG/SEC	27434.	.1	683.7	56.
147.5	TERMINATE PTC DAMP RATES	27432.	1.3	682.3	56.
147.6	PS2 IMU ALIGN	27432.	.6	681.8	56.
150.0	MIDCOURSE CORRECTION NO 5 MNVR TO BURN ATT	27430.	1.3	680.5	56.
150.0	ATTITUDE HOLD 0.5 DEG DB PGNCS	27430.	.8	679.7	56.
150.0	DEL VEL = NOM ZERO	27430.	.0	679.7	56.
150.5	ORIENT FOR PTC	27428.	1.1	678.5	56.
150.5	ATTITUDE HOLD 0.5 DEG DB PGNCS	27428.	.8	677.8	56.
150.5	ROLL 0.3 DEG/SEC	27428.	.1	677.6	56.
171.0	TERMINATE PTC	27426.	1.3	676.3	55.
172.0	PS2 IMU ALIGN	27426.	.6	675.8	55.
172.5	MIDCOURSE CORRECTION NO 6 MNVR TO BURN ATT	27424.	1.3	674.5	55.
172.5	ATTITUDE HOLD 0.5 DEG DB PGNCS	27424.	.8	673.7	55.
172.5	RCS -X TRANS 5 FPS	27408.	15.9	657.8	54.
173.0	ORIENT FOR PTC	27407.	1.1	656.6	54.

TABLE 5-2 (CONT'D)

SM-RCS PROPELLANT BUDGET

TIME (HR)	EVENT	S/C WT (LBS)	SM-RCS USED (LBS)	SM-RCS LEFT (LBS)	SM-RCS LEFT (%)
173.0	ATTITUDE HOLD 0.5 DEG DB PGNCS	27406.	.8	655.8	54.
173.0	ROLL 0.3 DEG/SEC	27406.	.1	655.7	54.
190.0	TERMINATE PTC	27404.	1.3	654.4	54.
191.2	P52 IMU ALIGN	27404.	.6	653.8	54.
192.0	MIDCOURSE CORRECTION NO 7 MNVR TO BURN ATT	27402.	1.3	652.5	53.
192.0	ATTITUDE HOLD 0.5 DEG DB PGNCS	27402.	.8	651.7	53.
192.0	DEL VEL = NOM ZERO	27402.	.0	651.7	53.
192.0	STAR CHECK MIN IMPULSE	27401.	.4	651.3	53.
193.0	MANEUVER TO REENTRY ATTITUDE	27399.	2.6	648.7	53.
193.0	ATTITUDE HOLD 0.5 DEG DB PGNCS	27390.	8.6	640.1	52.
194.8	MANEUVER TO SEP ATTITUDE	27387.	2.6	637.4	52.
194.8	CM/SM SEPARATION DELTA VEL=3 FPS	15001.	7.9	629.6	52.

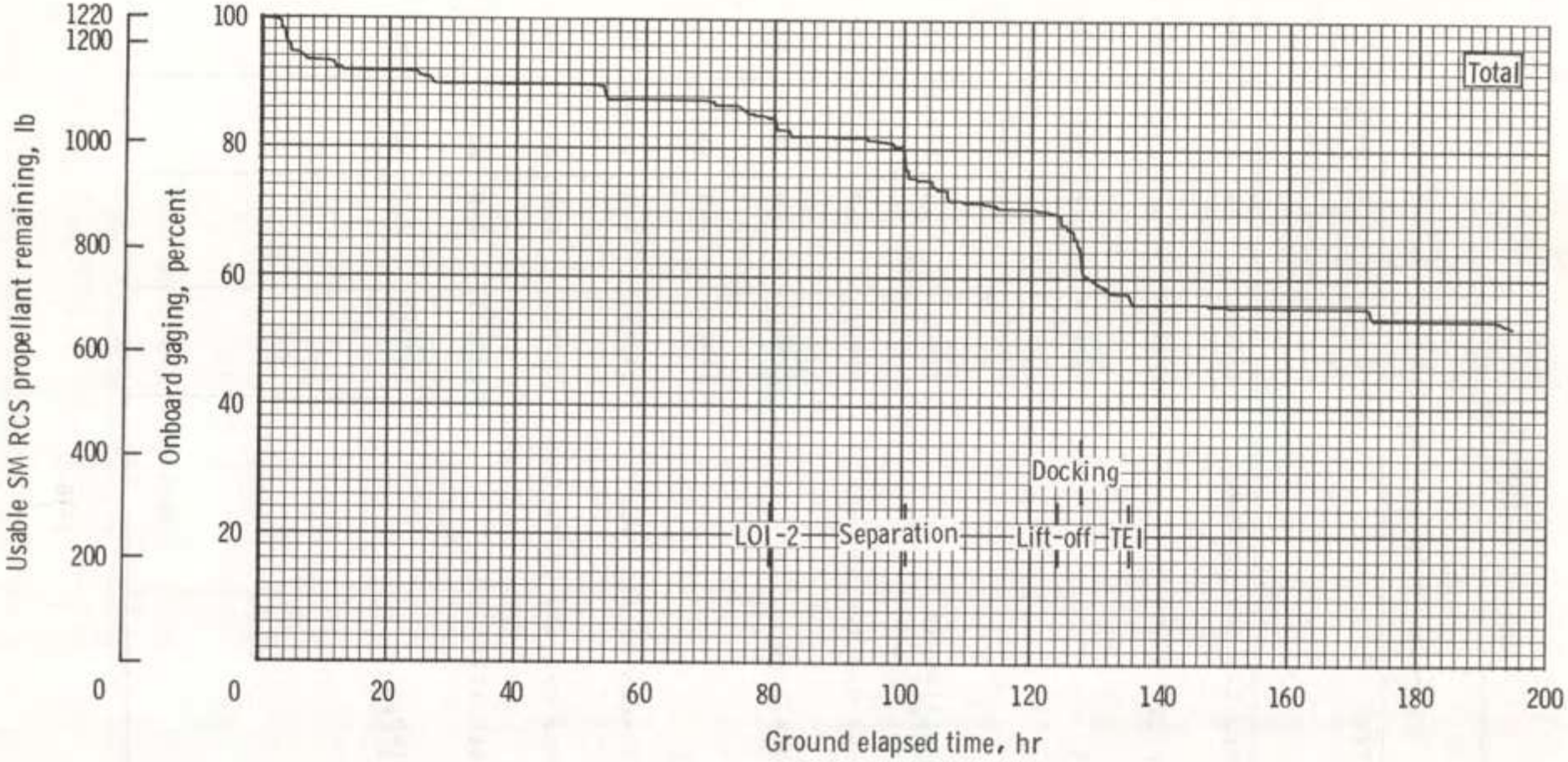


FIGURE 5-1
SM RCS propellant profile - total.

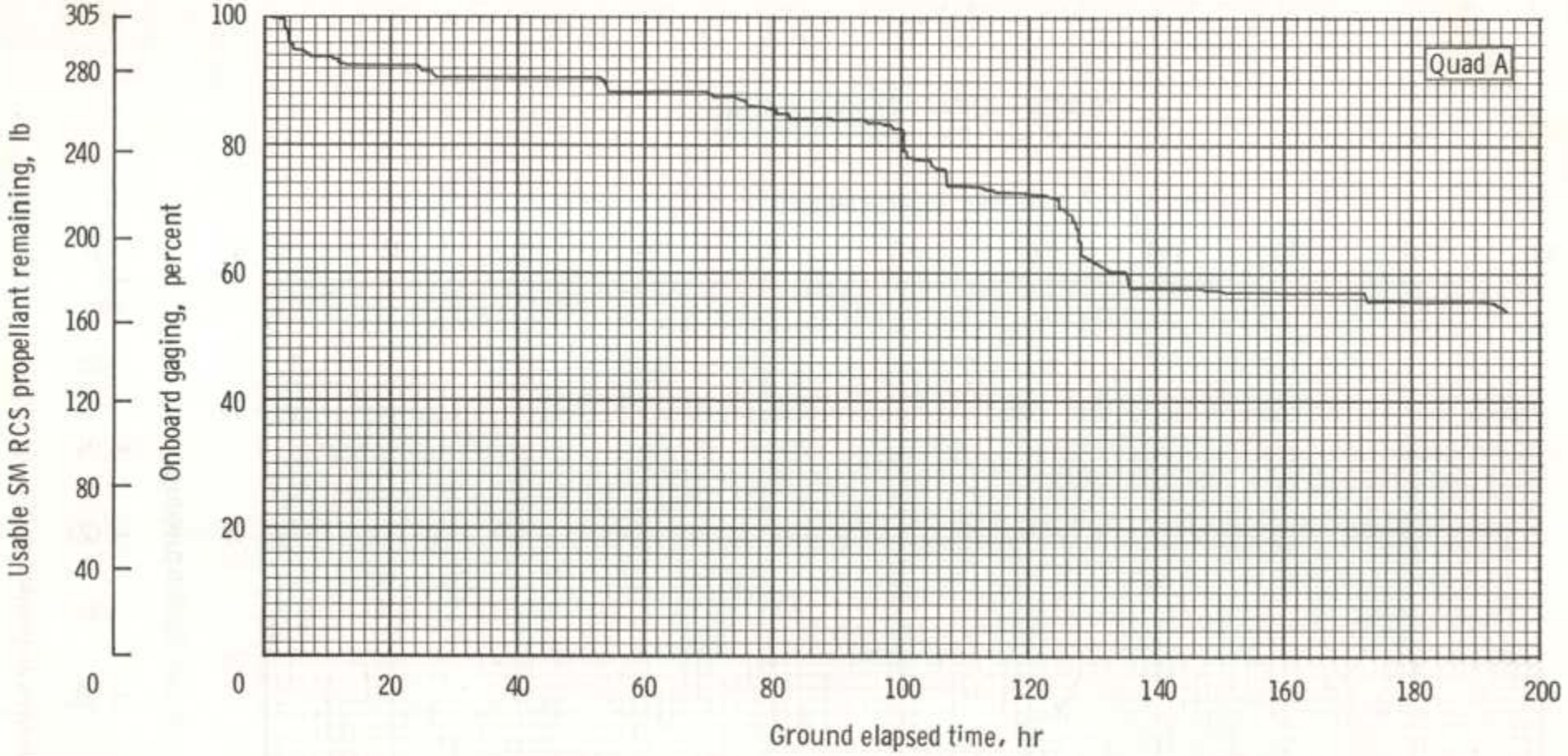


FIGURE 5-2
SM RCS propellant profile - quad A.

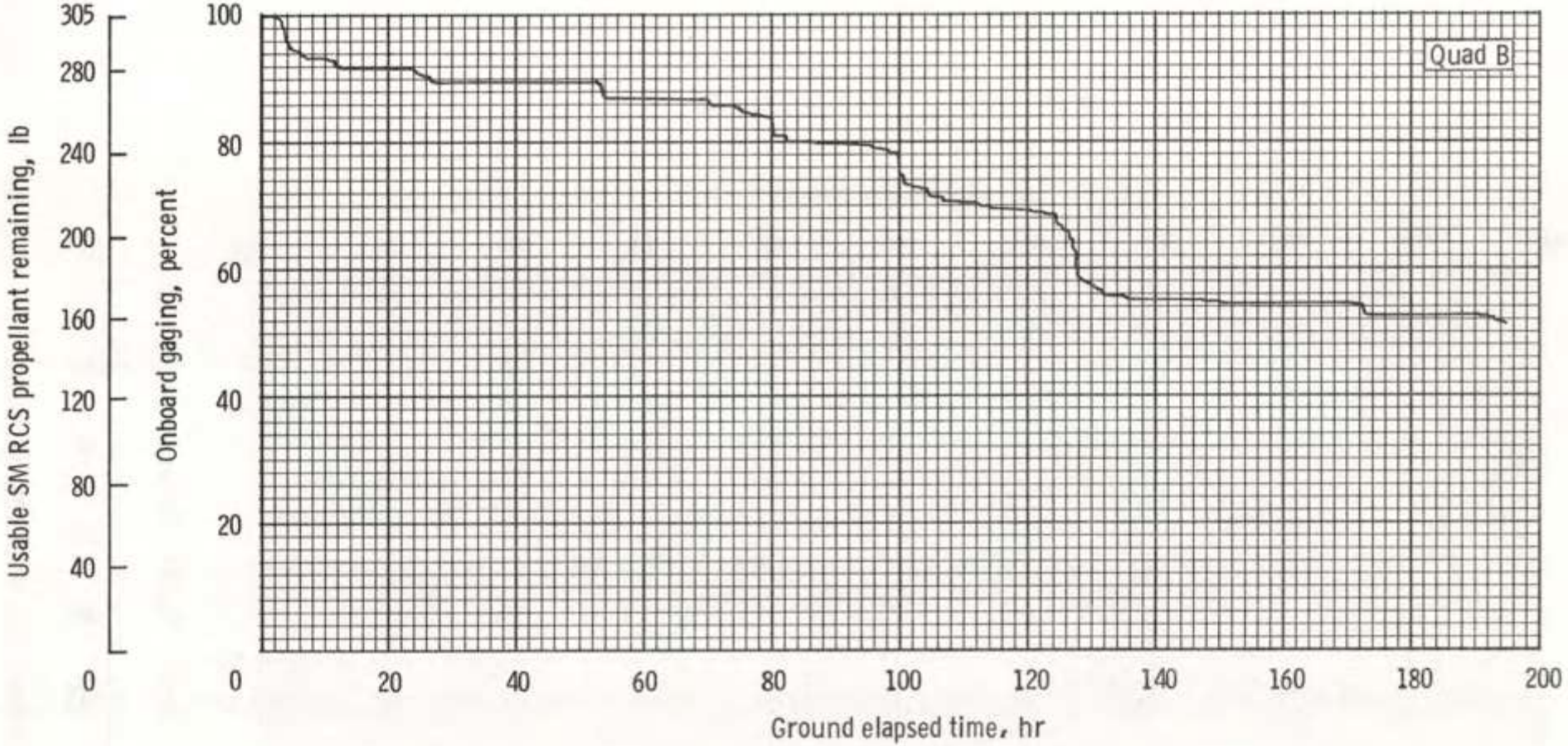


FIGURE 5-3
SM RCS propellant profile - quad B.

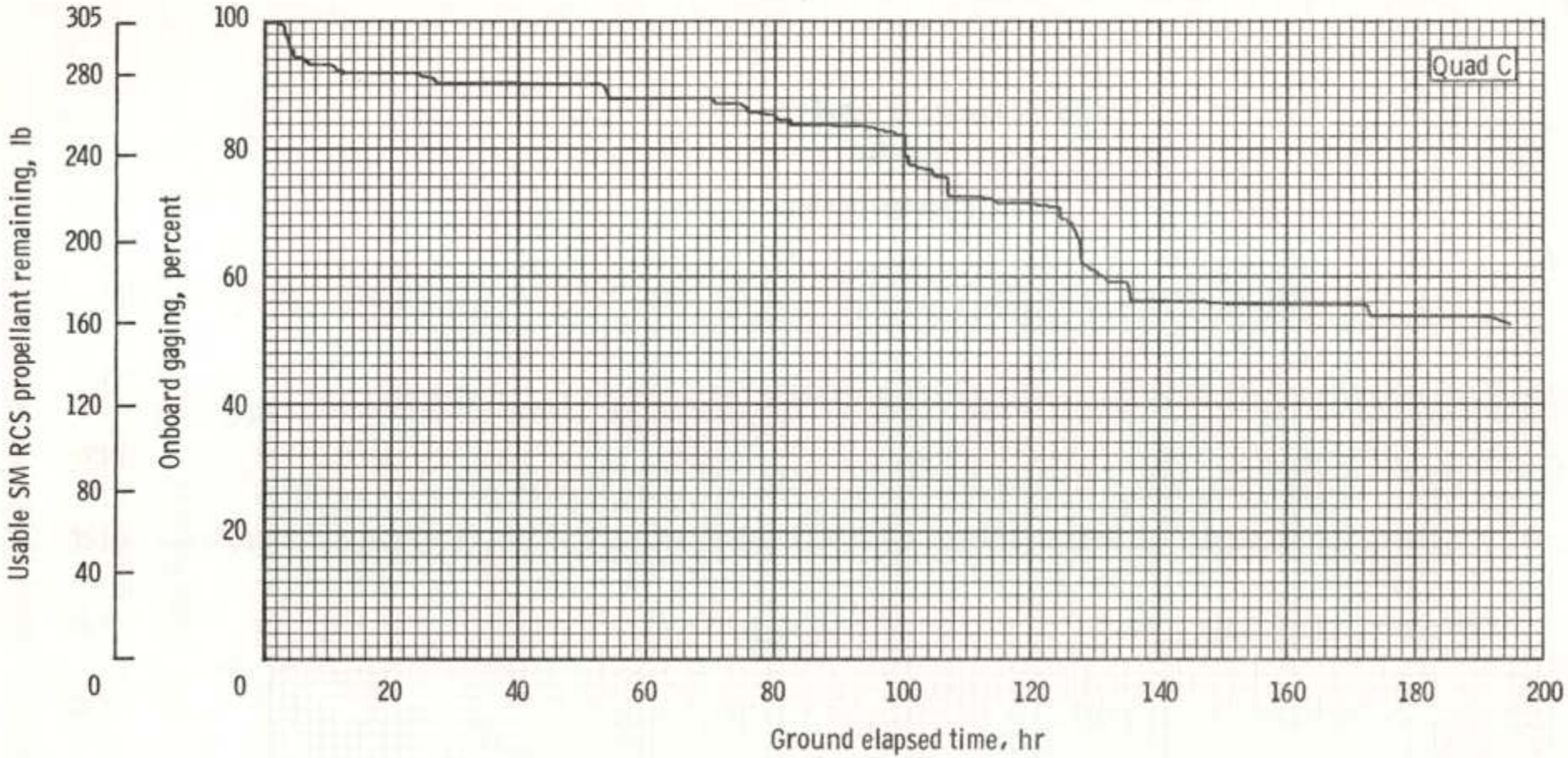


FIGURE 5-4
SM RCS propellant profile - quad C.

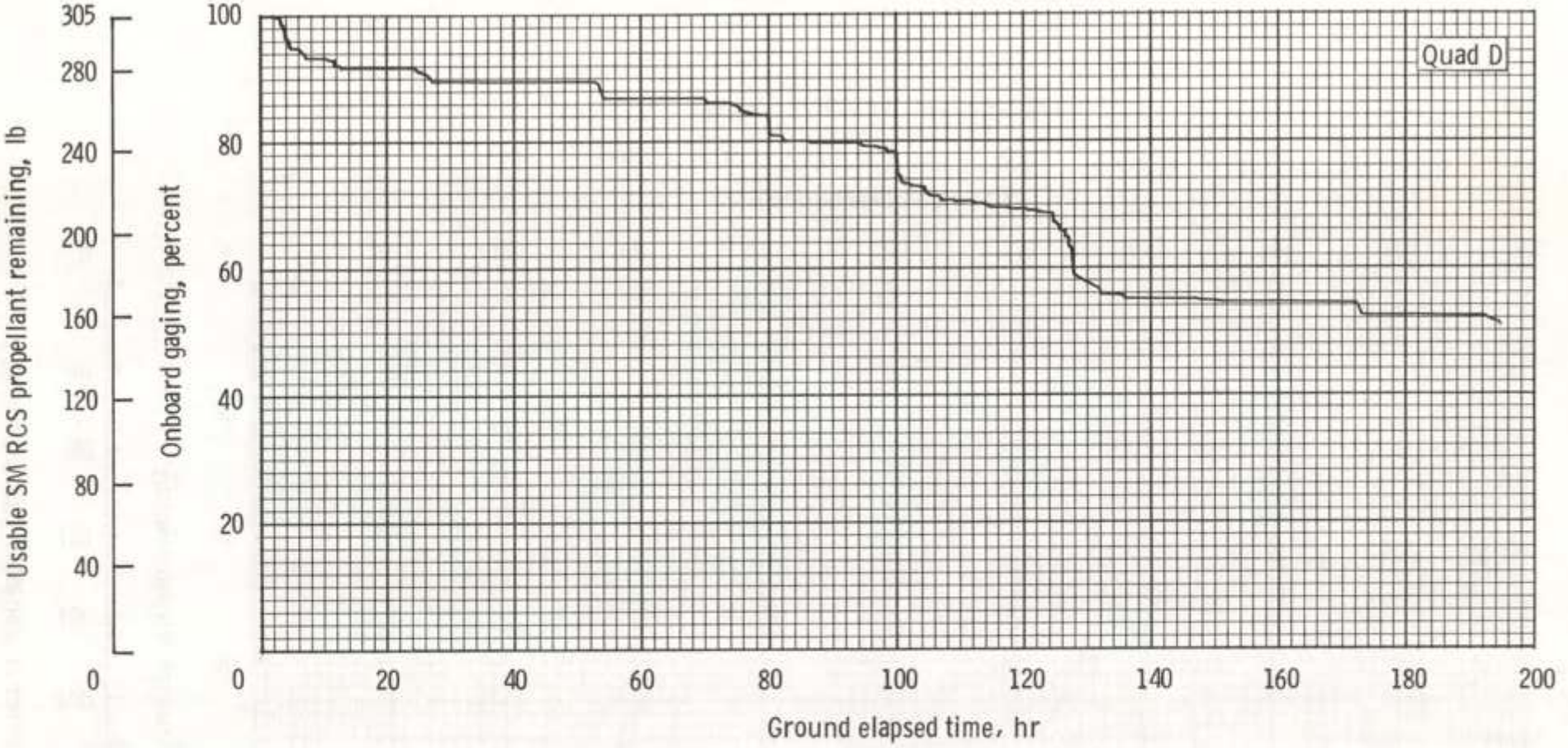


FIGURE 5-5

SM RCS propellant profile - quad D.

TABLE 5-3

CM RCS Propellant Summary

Item	Propellant required, lb.	Propellant remaining, lb.
Loaded	--	245.0
Trapped	36.4	208.6
Available for mission planning	--	208.6
Nominal usage	39.3	169.3
Nominal remaining	--	169.3

SERVICE PROPULSION SYSTEM (SPS). - The budget presented in table 5-4 is for a July 16 launch, 72 degree launch azimuth, first opportunity injection, 59.5 hour lunar parking orbit, and fast earth return. The assumptions used in preparing this budget are presented in table 5-5. ΔV requirements were coordinated with IMAB in MPAD.

It should be noted that the mission flexibility allowance of 900 fps has been used in addition to the fast return. In real time however, it is highly likely that a slower earth return would be performed in the mission flexibility ΔV had already been used (e.g., for LM rescue). Table 5-4 shows 3906 lbs of propellant remaining nominally and a total propellant margin (accounting both for the flexibility ΔV and the fast return) of 1268 lb.

TABLE 5-4 - APOLLO 11 SPS PROPELLANT SUMMARY

ITEM	PROPELLANT REQUIRED, LB	PROPELLANT REMAINING, LB
Loaded [a]	--	40803.0
Trapped and unavailable	441.4	40361.6
Outage	59.5	40302.1
Unbalance meter	100.0	40202.1
Available for ΔV	--	40202.1
Required for ΔV		
TLMC (120 fps)[b]	1166.4	39035.7
LOI-1 (2924 fps, 5 min. 59 sec.)	23862.4	15173.3
LOI-2 (157.8 fps, 16.4 sec.)	1115.4	14057.9
LOPC (16.6 fps, .9 sec.)	73.8	13984.1
TEI (3292.7 fps, 149 sec.)	10077.8	3906.3
Nominal remaining	--	3906.3
Mission flexibility (900 fps)	2212.4	1693.9
Dispersions (-3 σ)	426.0	1267.9
Propellant margin	--	1267.9

[a] 15712.0 lb of fuel and 25091.0 lb of oxidizer; this is loaded on CSM-107.

[b] Includes 19.7 fps for evasive maneuver

TABLE 5-5 - ASSUMPTIONS FOR THE APOLLO 11 SPS PROPELLANT BUDGET

1. There is a non-propulsive propellant loss of 14.4 lb for each engine start. LM rescue assumed three engine starts.

2. A mission flexibility ΔV of 900 fps has been included in the SPS budget to provide the capability to perform a worst case LM rescue, or to handle several other contingencies (such as loss of PGNCS), or to perform a quicker earth return.

3. Spacecraft weight:

CM	12 280.0 lb
SM	10 551.3 lb
SLA Ring	98.0 lb
Tanked SPS	40 600.7 lb
LM (unmanned)	33 278.3 lb
Total	96 808.3 lb

4. Lunar Orbit Activity
   Total weight transfer (CSM to LM) = 436.7 lb
   Total weight transfer (LM to CSM) = 284.0 lb

5. SM RCS, EPS, and ECS weight losses:

Mission Period	Incremental Weight Loss, lb
EL to TLMC	151.8
TLMC to LOI-1	327.1
LOI-1 to LOI-2	32.0
LOI-2 to LOPC	146.5
LOPC to TEI	216.1

6. SM RCS usage (above nominal rendezvous requirement) for LM rescue was 216 lb.

LM RCS BUDGET

Ground Rules and Assumptions

1. Data for the LM RCS engine performance and propellant requirements were obtained from the Spacecraft Operational Data Book and postflight analysis from Apollo 9 and Apollo 10.

2. All orientation maneuvers were assumed to be made at 2.0 °/sec.

3. All orientation maneuvers were assumed to be three-axis maneuvers.

TABLE 5-6

LM RCS Propellant Loading and Usage Summary

Loaded	633.0
Trapped	40.6
Nominal deliverable	592.4
Gaging Inaccuracy and loading tolerance	39.5
Mixture ratio uncertainty	17.0
Usable	535.9
Nominal mission requirement	252.7
Nominal remaining	283.2

TABLE 5-7

LM - RCS PROPELLANT BUDGET PAGE 1

TIME HRS M	EVENT TITLE	[a] S/C WT (LBS)	LM RCS USED (LBS)	[b] LM RCS LEFT (LBS)	[b] LM RCS LEFT (%)
0 0	OUTPUT PROPELLANT LOADINGS	33714.	.0	633.0	100.0
99 25	RCS HOT FIRE	33709.	5.0	628.0	99.2
100 15	UNDOCKING	33709.	.0	628.0	99.2
100 15	NULL UNDOCKING VELOCITY	33707.	1.9	626.1	98.9
100 20	LM MNVR FOR INSPECTION YAW	33705.	1.7	624.4	98.6
100 20	LM MNVR FOR INSPECTION PITCH	33703.	2.0	622.4	98.3
100 25	LM MNVR FOR INSPECTION YAW	33702.	.8	621.6	98.2
100 25	FORMATION FLYING	33690.	2.0	619.6	97.9
100 50	RR LOCK ON MNVR	33687.	3.6	616.0	97.3
101 0	IMU REALIGN STAR 1	33683.	3.6	612.4	96.7
101 0	IMU REALIGN STAR 2	33680.	3.6	608.8	96.2
101 0	IMU REALIGN STAR 3	33676.	3.6	605.2	95.6
101 32	MNVR TO DOI BURN ATTITUDE	33672.	3.6	601.6	95.0
101 32	ATTITUDE HOLD	33672.	.1	601.5	95.0
101 36	2 JET ULLAGE	33667.	5.9	595.6	94.1
101 38	DOI BURN	33419.	.0	595.6	94.1
101 38	MOMENT CONTROL DOI BURN	33414.	5.0	590.6	93.3
101 38	TRIM HORIZONTAL RESIDUAL	33407.	7.6	583.0	92.1
101 38	ATTITUDE HOLD	33407.	.3	582.8	92.1
101 38	PITCH DOWN	33406.	1.0	581.8	91.9
101 42	RR LOCK ON MNVR	33402.	3.6	578.2	91.3
101 55	PITCH DOWN	33401.	.6	577.6	91.3
101 55	YAW LEFT	33401.	.6	577.0	91.2
102 0	ALIGNMENT CHECK	33400.	1.2	575.8	91.0
102 10	RR LOCK ON MNVR	33396.	3.6	572.2	90.4

[a] These weights were used for analysis only and do not reflect the actual weight after consumables loading.

[b] RCS propellant remaining of total loaded

TABLE 5-7 (CONT'D)

LM - RCS PROPELLANT BUDGET PAGE 2

TIME HRS M	EVENT TITLE	S/C WT[a] (LBS)	LM RCS USED (LBS)	LM[b] RCS LEFT (LBS)	LM[b] RCS LEFT (%)
102 14	MNVR TO PDI ATTITUDE	33392.	3.6	568.6	89.8
102 14	MAINTAIN LOS	33391.	1.0	567.6	89.7
102 29	ATTITUDE HOLD	33391.	.1	567.5	89.7
102 35	2 JET ULLAGE	33385.	5.9	561.7	88.7
102 35	PDI BURN	16753.	.0	561.7	88.7
102 35	POWERED DESCENT	16710.	34.1	527.5	83.3
102 47	TOUCHDOWN	16710.	.0	527.5	83.3
112 40	ADD LUNAR SAMPLES	16580.	.0	527.5	83.3
124 23	LUNAR LIFT OFF	10840.	.0	527.5	83.3
124 23	POWERED ASCENT PHASE WITH RCS/APS INTERCONNECT	6087.	.0	527.5	83.3
124 23	POWERED ASCENT PHASE WITHOUT RCS/APS INTERCONNECT	5969.	.9	526.7	83.2
124 25	RR LOCK ON MNVR	5969.	.4	526.2	83.1
124 30	INSERTION BURN CONTROL	5967.	1.8	524.4	82.8
124 30	TRIM OUT OF PLANE ERROR	5964.	3.3	521.2	82.3
124 30	ATTITUDE HOLD	5962.	1.3	519.9	82.1
124 37	IMU REALIGN STAR 1	5962.	.4	519.5	82.1
124 37	IMU REALIGN STAR2	5961.	.4	519.0	82.0
124 37	IMU REALIGN STAR3	5961.	.4	518.6	81.9
124 55	RR LOCK ON MNVR	5961.	.4	518.1	81.9
124 55	MAINTAIN LOS	5958.	2.7	515.5	81.4
125 15	ATTITUDE HOLD	5957.	1.3	514.2	81.2
125 21	CSI BURN RCS +Z	5923.	33.6	480.6	75.9
125 26	MAINTAIN LOS	5920.	3.3	477.2	75.4
125 44	MNVR TO PLANE CHANGE ATTITUDE	5919.	.4	476.8	75.3

[a] These weights were used for analysis only and do not reflect the actual weight after consumables loading.

[b] RCS propellant remaining of total loaded

TABLE 5-7 (CONT'D)

LM - RCS PROPELLANT BUDGET PAGE 3

TIME HRS M	EVENT TITLE	S/C WT[a] (LBS)	LM RCS USED (LBS)	LM[b] RCS LEFT (LBS)	LM[b] RCS LEFT (%)
125 45	ATTITUDE HOLD	5918.	1.3	475.5	75.1
125 50	RCS PLANE CHANGE BURN	5914.	4.1	471.4	74.5
126 0	RR LOCK ON MNVR	5913.	.4	471.0	74.4
126 0	MAINTAIN LOS	5911.	2.0	469.0	74.1
126 15	ATTITUDE HOLD	5910.	1.3	467.7	73.9
126 19	CDH RCS BURN	5906.	4.0	463.7	73.3
126 19	MAINTAIN LOS	5902.	4.0	459.7	72.6
126 53	ATTITUDE HOLD	5901.	1.3	458.4	72.4
126 58	RCS TPI BURN	5884.	17.0	441.4	69.7
126 58	MAINTAIN LOS	5883.	1.3	440.1	69.5
127 36	PCC AND BRAKING	5849.	33.9	406.3	64.2
127 36	ATTITUDE AND LOS CONTROL	5833.	16.0	390.3	61.7
128 00	LM CONTROL CSM ACTIVE DOCKING	5823.	10.0	380.3	60.1

[a] These weights were used for analysis only and do not reflect the actual weight after consumables loading.

[b] RCS propellant remaining of total loaded

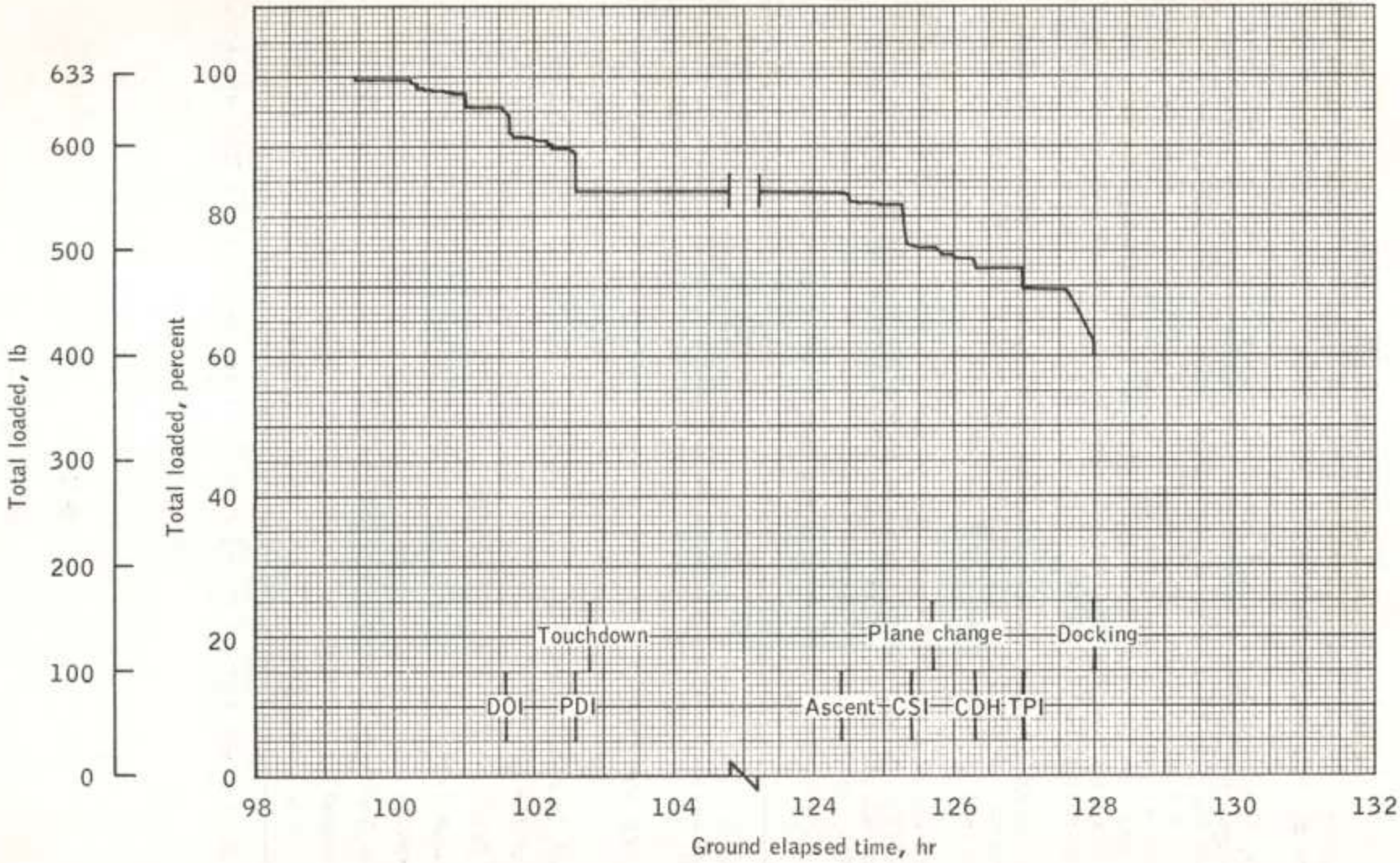


Figure 5-6.- LM RCS propellant profile.

DESCENT PROPULSION SYSTEM PROPELLANT BUDGET

Descent Propulsion Subsystem (DPS) - The DPS budget is shown in table 5-8 and the ground rules and assumptions in table 5-9.

Previously, the uncertainty in the low-level sensor (68.7 lb) has been shown as a contingency allowance. This is now included as part of the unusables. Also, there has previously been a contingency allowance for manual hover to allow for 2 minutes of burn time from 500 feet to touchdown. The present budget shows a nominal ΔV which includes a manual allowance of 477 fps (90 sec) from 500 feet to touchdown. Any additional hover time will be used from the propellant margin (unassigned capability). The rate of use for hover is approximately 9.1 lb/sec.

Propellant loads are those actually loaded on LM-5, and trapped and residual propellants are from Volume III, SODB. Engine performance data and ΔV requirements have been coordinated with LAB in MPAD.

Three sigma dispersions represent total propellant cost due to 3 σ uncertainties in propellant loading, trapped, I_{SP}, ΔV, separation weight, non-ΔV consumables weight, and mixture ratio. There is a total propellant margin of 669 lb or approximately 73 seconds of hover time.

TABLE 5-8 - APOLLO 11 DPS PROPELLANT SUMMARY

ITEM	PROPELLANT REQUIRED, LB	PROPELLANT REMAINING, LB
Loaded [a]	--	18184.2
Trapped and unavailable	223.5	17960.7
Outage	14.0	17946.7
Low-Level Sensor Uncertainty	68.7	17878.0
Available for ΔV	--	17878.0
Nominal Required for ΔV of 6728.6 fps	16799.7	1078.3
Dispersions (-3σ)	224.7	853.6
Contingencies		
Engine Valve-Pair Malfunction ($\Delta MR=\pm.016$)	81.1	772.5
Redesignation (60 fps)	104.0	668.5
Margin (73 sec. hover)	--	668.5

[a] 6974.8 lb fuel and 11209.4 lb oxidizer; this is loaded on the LM-5 spacecraft.

TABLE 5-9 - ASSUMPTIONS FOR THE APOLLO 11 DPS PROPELLANT BUDGET

1. Integrated average I_{sp} = 301.9 ± 3.54 seconds
2. LM separation weight = 33746. lb
3. Mixture ratio = 1.596 ± 0.0108
4. Nominal ΔV = 6728.6 ± 96 fps
5. Non-ΔV consumables of 47.4 lb from separation to DOI and 106.1 lb from DOI to touchdown

ASCENT PROPULSION SYSTEM PROPELLANT BUDGET

Ascent Propulsion Subsystem (APS) - Tables 5-10 and 5-11 present the ascent propellant budget for the current mission. Propellant loads are those actually on LM-5. Mission ΔV was coordinated with LAB in MPAD. The budget shown in table 5-10 accounts for an engine valve-pair malfunction, a PGNCS to AGS switchover, and a touchdown abort. There is a total propellant margin of 68 lb or about 6 seconds of burn time.

TABLE 5-10 - APOLLO 11 APS PROPELLANT SUMMARY

ITEM	PROPELLANT REQUIRED, LB	PROPELLANT REMAINING, LB
Loaded [a]	--	5238.4
Trapped and Unavailable	48.9	5189.5
Outage	17.5	5172.0
Available for ΔV	--	5172.0
Nominal Required for ΔV of 6072.5 fps	4965.8	206.2
Dispersions (-3 σ)	57.8	148.4
Contingencies		
Engine Valve-pair Malfunction ($\Delta MR=\pm.016$)	19.6	128.8
PGNCS to AGS Switchover (40 fps)	23.8	105.0
Touchdown Abort (ΔW=+99.9 lb, $\Delta\Delta V$=-15fps)	36.8	68.2
Margin (6 seconds)	--	68.2

[a] Includes 2019.9 lb fuel and 3218.5 lb oxidizer; this is loaded on the LM-5 spacecraft.

TABLE 5-11 - ASSUMPTIONS FOR THE APOLLO 11 APS PROPELLANT BUDGET

1. I_{sp} = 308.97 $\pm$ 3.553 seconds
2. Mixture ratio = 1.602 $\pm$ 0.0225
3. Nominal ΔV = 6072.5 $\pm$ 33.5 fps
4. Ascent stage lift-off weight = 10873.6 lb

CSM-107/LM5 CRYOGENIC/EPS AND ECS BUDGET

The results of the Cryogenic, EPS, and ECS analysis are summarized in the following tables and figures:

TABLE 5-11 CSM Cryogenic Loading And Usage Summary

TABLE 5-13 LM EPS Summary

TABLE 5-14 LM ECS Summary

FIGURE 5-7 CSM O_2 PROFILE

FIGURE 5-8 CSM H_2 PROFILE

FIGURE 5-9 CSM POWER PROFILE

FIGURE 5-10 CSM BUS VOLTAGE VS TIME

FIGURE 5-11 LM DESCENT POWER PROFILE

FIGURE 5-12 LM ASCENT POWER PROFILE

FIGURE 5-13 LM TOTAL CURRENT PROFILE

FIGURE 5-14 LM DESCENT O_2 PROFILE

FIGURE 5-15 LM ASCENT O_2 PROFILE

FIGURE 5-16 LM DESCENT H_2O PROFILE

FIGURE 5-17 LM ASCENT H_2O PROFILE

CSM EPS BUDGET

ASSUMPTIONS AND GROUND RULES

1. The system was assumed to operate with three fuel cells and two inverters.

2. Fuel cell purging is included in the EPS requirements.

3. 100% fill for both H_2 and O_2.

4. Three entry and postlanding batteries were considered available to supply the total spacecraft power required for entry, parachute descent, and postlanding time. Each battery was assumed to have a 40 A-h capacity until splashdown, at which time the capacity was uprated to 45 A-h.

5. Two batteries were considered to be in parallel with the fuel cells during ascent and for each SPS maneuver.

6. No cryogenic venting was assumed in flight.

7. The EPS hydrogen consumption rate (lb/hr) = $0.00257 \times I_{fc}$

8. The EPS oxygen consumption rate (lb/hr) = $7.936 \times \dot{H}_2$

9. Six battery charges were assumed: three on battery A and three on battery B.

TABLE 5-12

APOLLO 11 CRYOGENIC SUMMARY

		H_2, lb	O_2, lb
I.	Planning Allowance		
	A. Total Loaded	58.60	660.20
	B. Less Residual	2.32	13.00
	C. Less Instrumentation Error	1.50	17.50
	Available for Mission Planning	54.78	629.70
II.	Predicted Usages		
	A. Prelaunch [1]		
	1. Inline HTR + Pressure Relief (T-28 to T-3 (Incl 12.5 hr hold))	1.61	18.60
	2. Power Production (plus ECS O_2) (T-3 to liftoff)	.57	6.96
	Total Prelaunch requirements	2.18	25.50
	B. Flight		
	1. EPS Requirements (Incl FC Purge)	36.60	288.33
	2. CM ECS (Incl Cabin Purge)	-	72.40
	3. LM Pressurizations	-	10.35
	Total Flight Requirements	36.60	371.08
III.	Nominal Reserves (RSS)		
	EPS Uncertainty (5 percent)	1.83	14.42
	ECS Uncertainty (.08 lb/hr)	-	15.60
	Tank Unbalance (AOH)	.80	12.90
	Launch Window	.86	10.20
	RSS Subtotal	2.17	26.87
IV.	Operational Reserves		
	A. Available for Mission Planning	54.78	629.70
	B. Less Nominal Predicted Usage	38.78	396.58
	C. Less Nominal Reserves	2.17	26.87
	Operational Reserve	13.83	206.25

[1] KSC Supplied Data

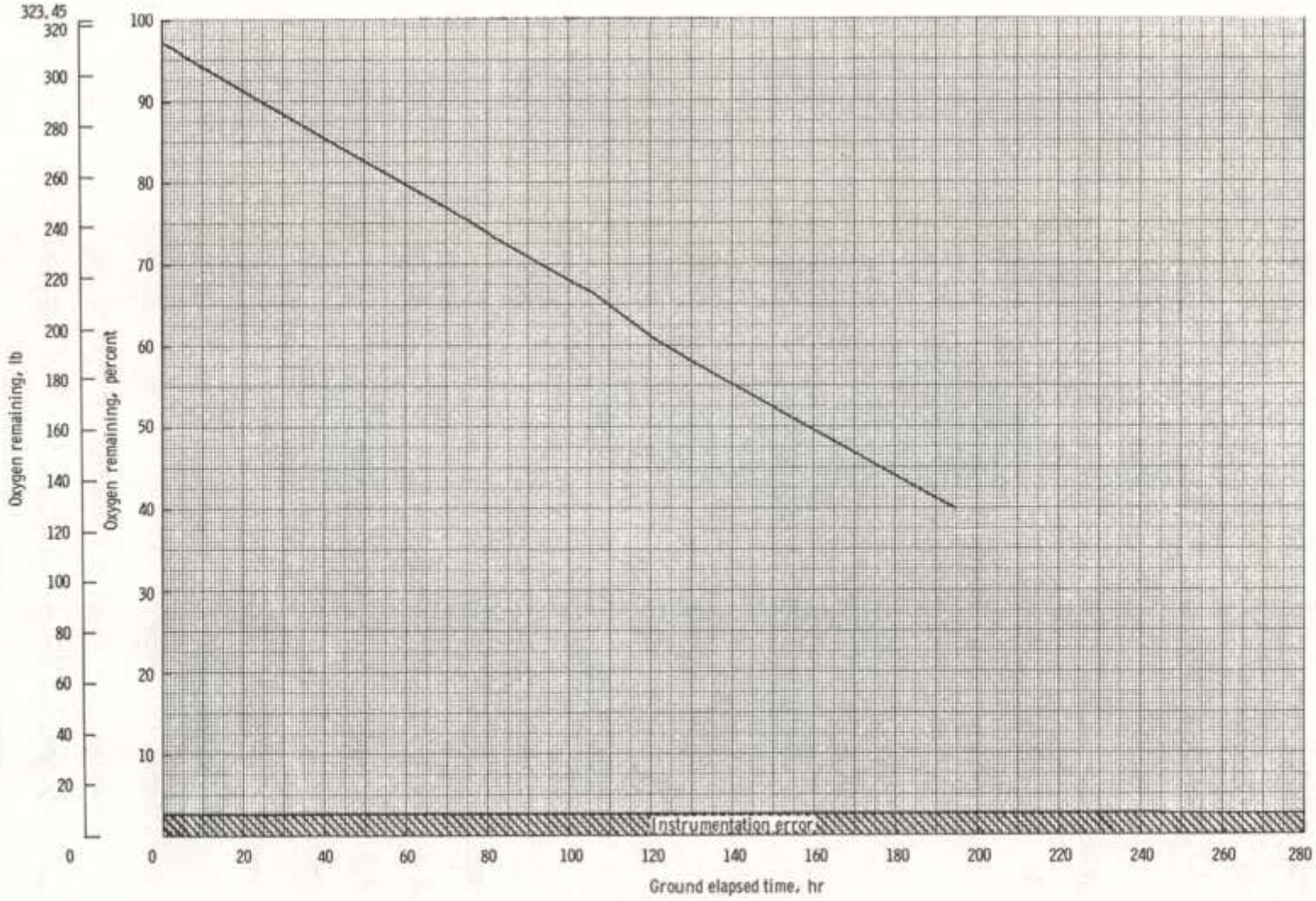


Figure 5-7.- Oxygen remaining for mission for one tank versus time.

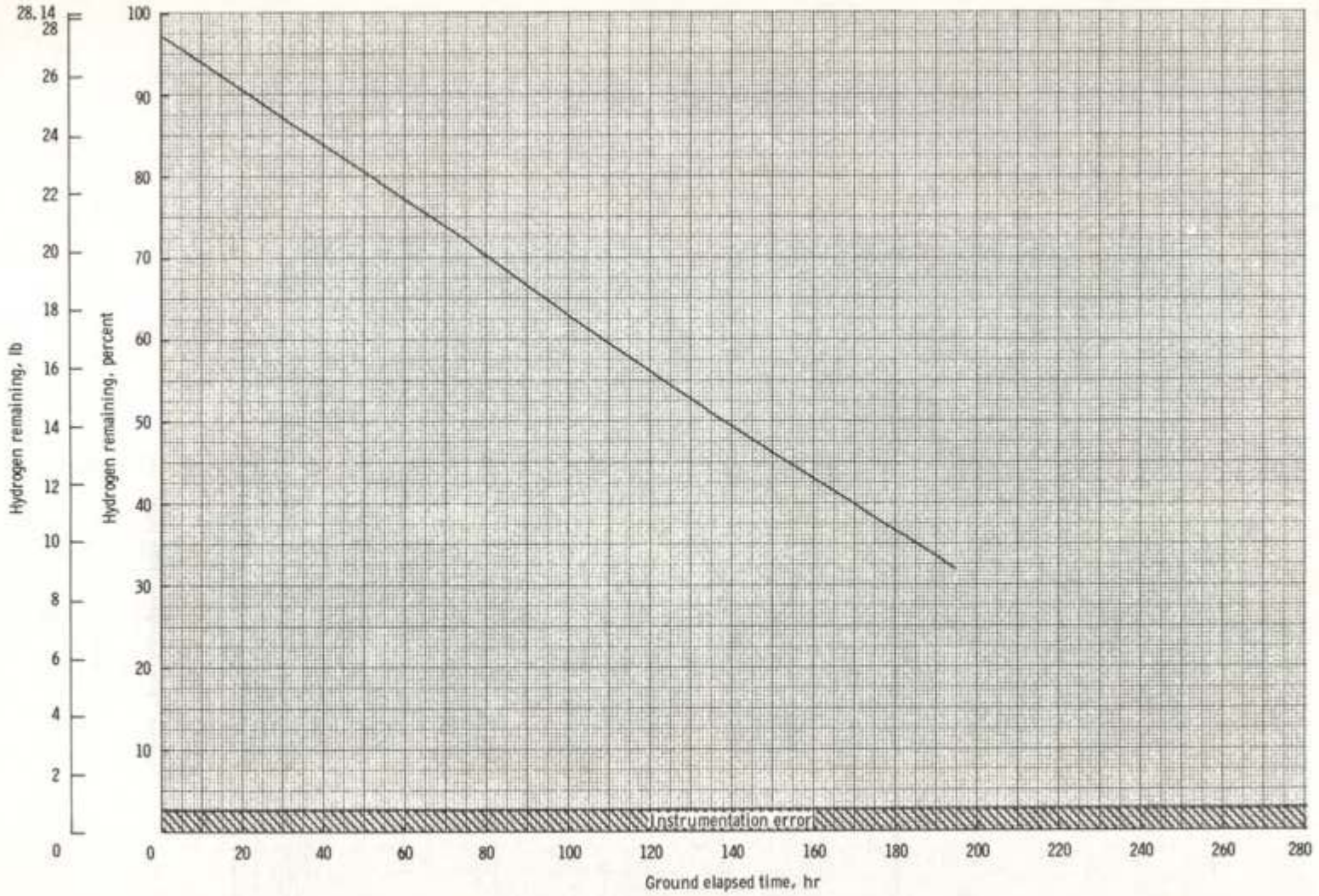


Figure 5-8.- Hydrogen remaining for mission for one tank versus time.

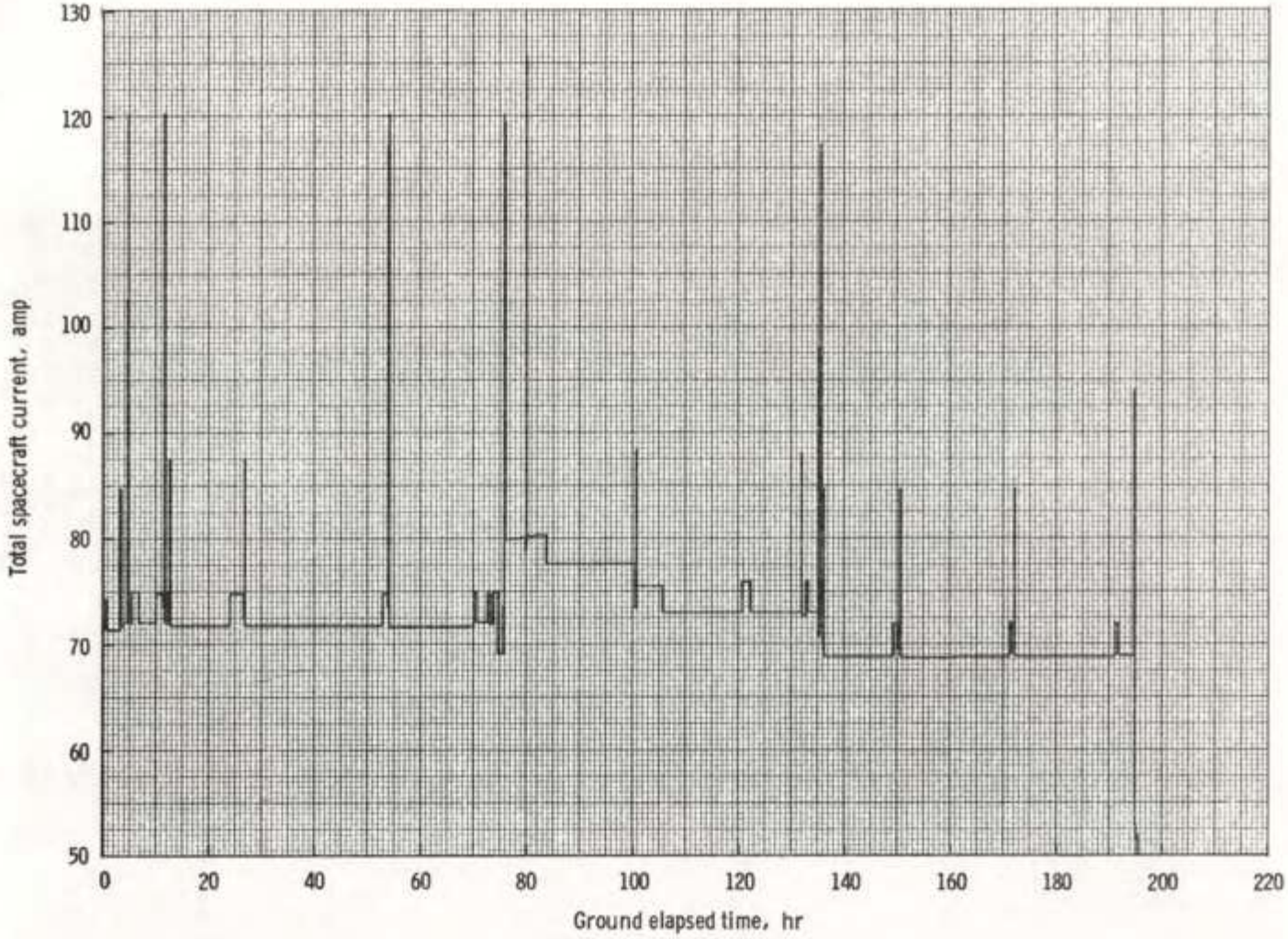


Figure5-9.- CSM total spacecraft current profile.

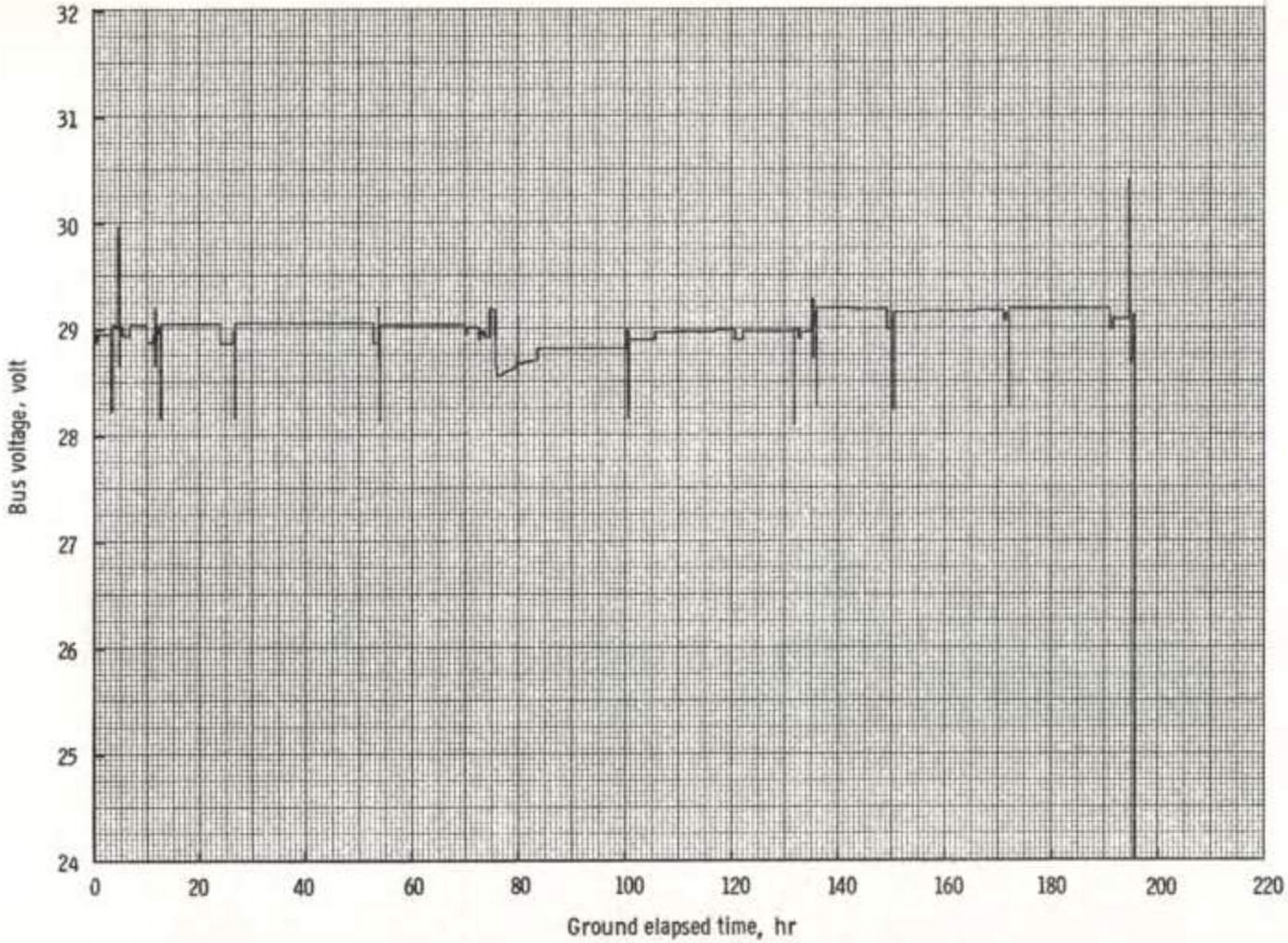


Figure 5-10. - CSM bus voltage versus time.

LM EPS ANALYSIS

GROUND RULES AND ASSUMPTIONS

1. The descent stage batteries go on the line 30 minutes prior to earth liftoff.

2. A 3.8 hour checkout was assumed for lunar orbit.

3. Ascent and descent batteries were paralleled for the powered descent burn and prior to liftoff from the lunar surface.

4. The S-band equipment was assumed on 100 percent from initial activation in lunar orbit until completion of the mission.

5. The rendezvous radar electronics was assumed to be operational for the period of time dictated by the current G Mission flight plan.

6. The primary navigation and guidance subsystem (PGNCS) was left in the operate mode for the entire lunar stay.

7. The forward window heaters were left off for the entire mission.

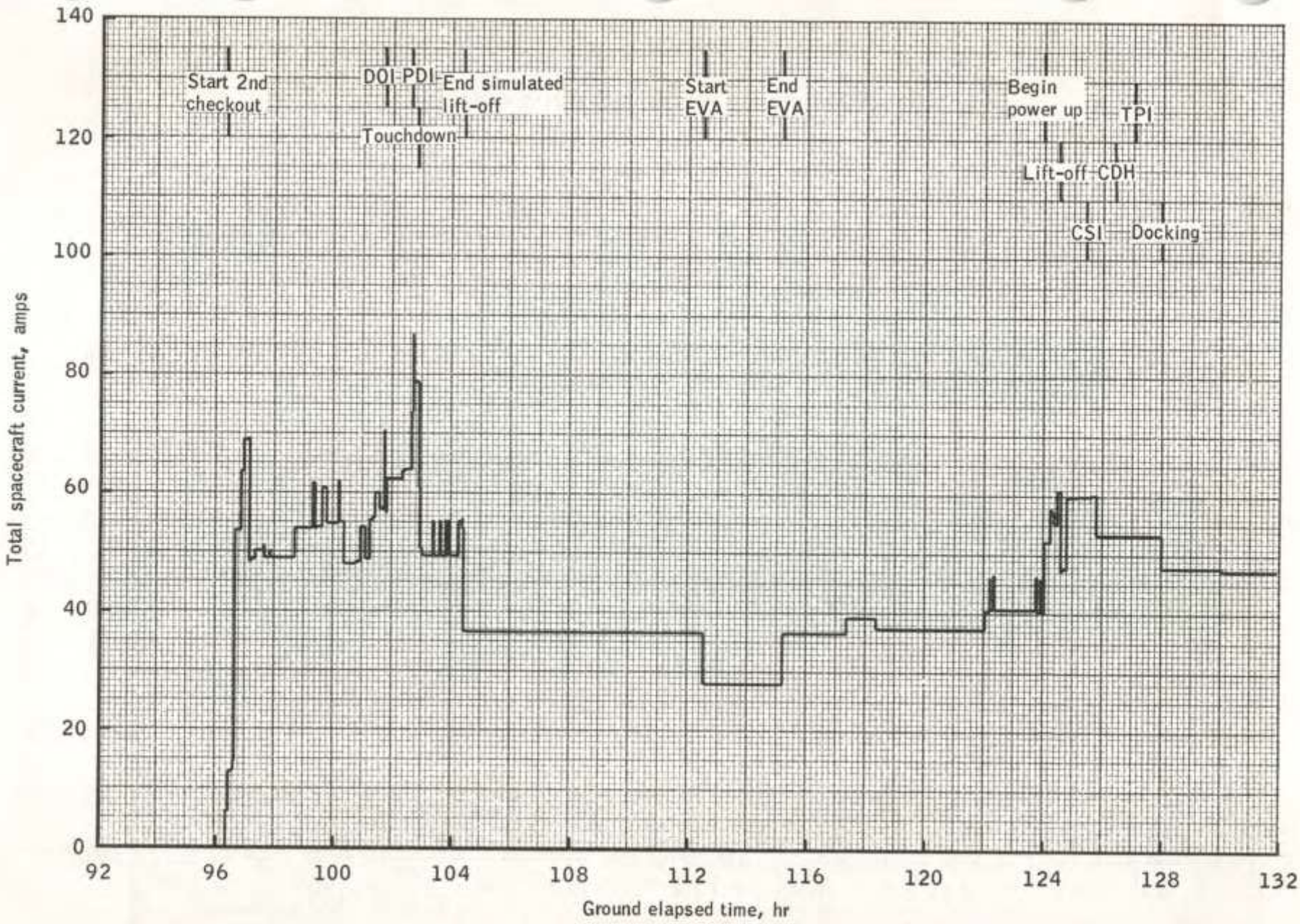


Figure 5-11.- LM-5 total spacecraft current.

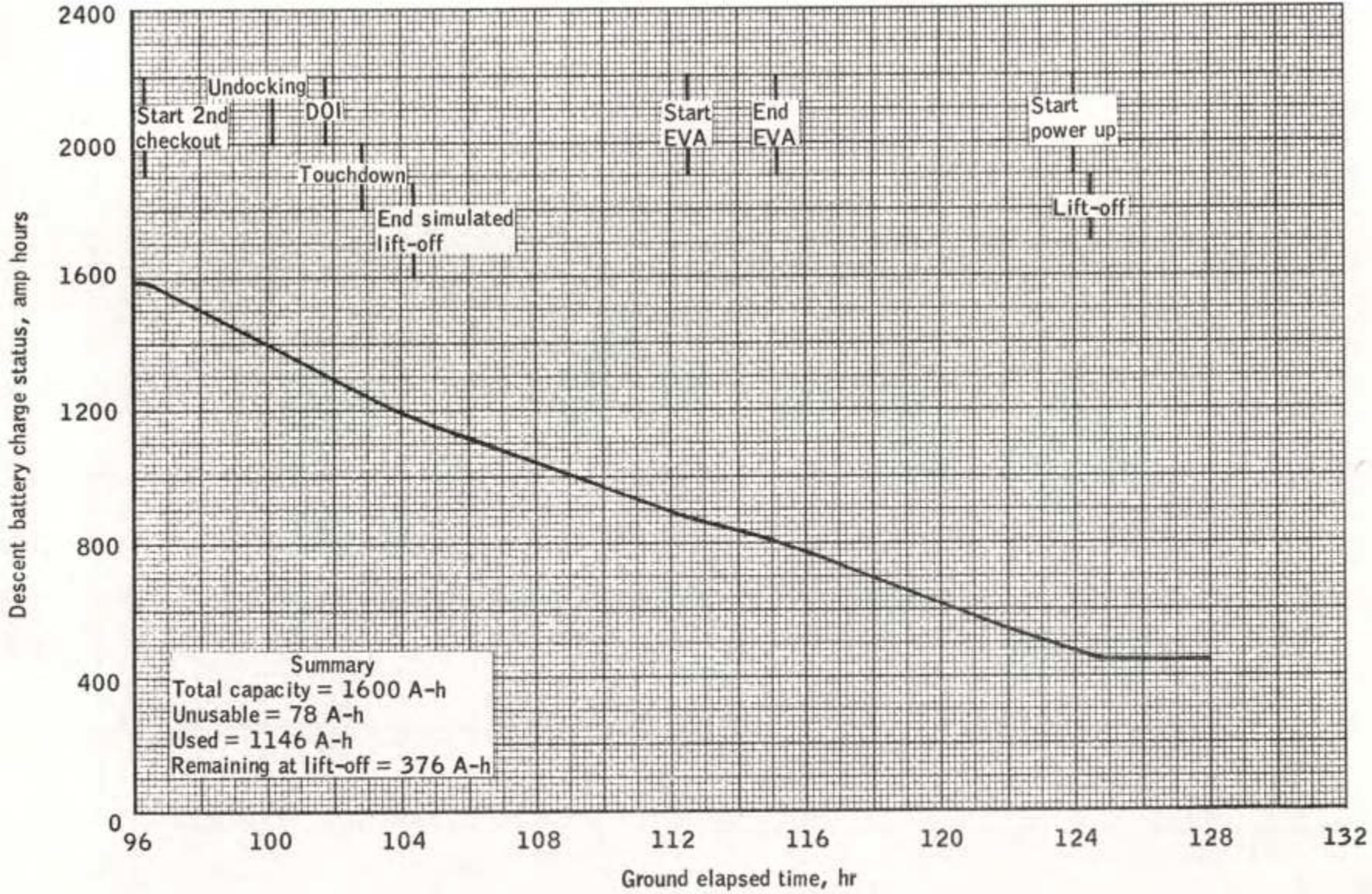


Figure 5-12.- Descent stage amp hours remaining.

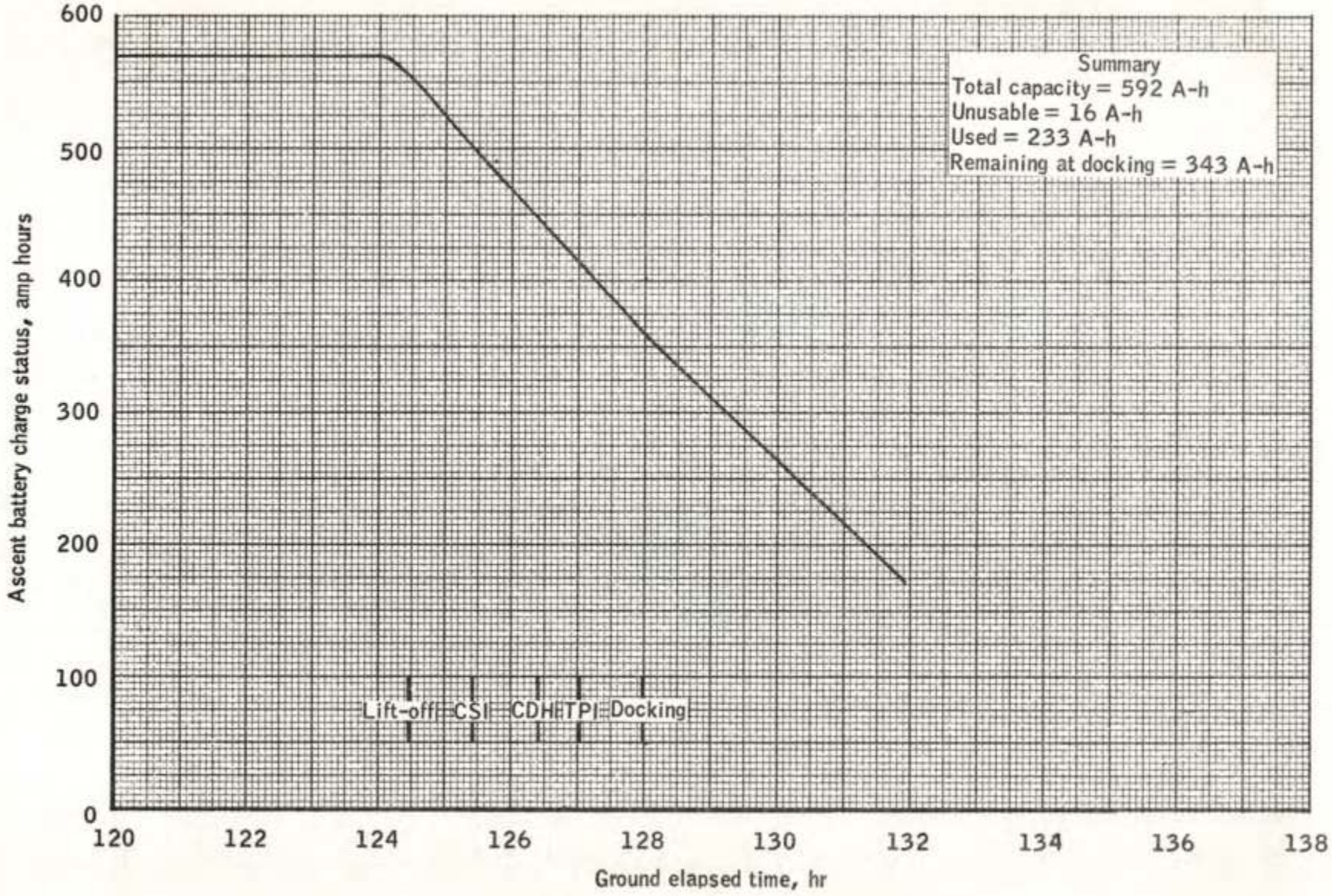


Figure 5-13.- Ascent stage amp hours remaining.

LM ECS BUDGET

GROUND RULES AND ASSUMPTIONS

1. Cabin O_2 leakage rate was 0.2 lb/hr while pressurized
2. Metabolic rates were varied according to Volume 2 of the Spacecraft Operational Data Book
3. Metabolic O_2 consumed was $(1.643 \times 10^{-4}) \times$ (metabolic rate)
4. LM pressurization requires 6.62 lb of O_2
5. Cabin pressure regulator check requires 2.65 lb of O_2
6. H_2O consumed because of sublimator cooling was total heat removed divided by 1040 (btu per lb) of H_2O
7. H_2O lost due to urination was 0.11 lb/hr per man
8. Cabin temperature control was set to 72° F
9. Average glycol flow rate was 250 lb/hr
10. Budget was performed on the operational trajectory and may change when the revision 1 is analyzed.

TABLE 5-13
LM ECS Summary

(a) Descent Stage

Description	O_2, lb	H_2O, lb
Loaded	48.00	210.6
Unusable	3.40	16.4
Available for mission	44.60	194.2
Required for mission	26.17	142.4
Usable remaining in tanks	18.43	51.8

(b) Ascent Stage

Loaded	4.86	85.00
Unusable	.74	4.20
Available for mission	4.12	80.80
Required for mission	1.95	45.48
Usable remaining in tanks	2.17	35.32

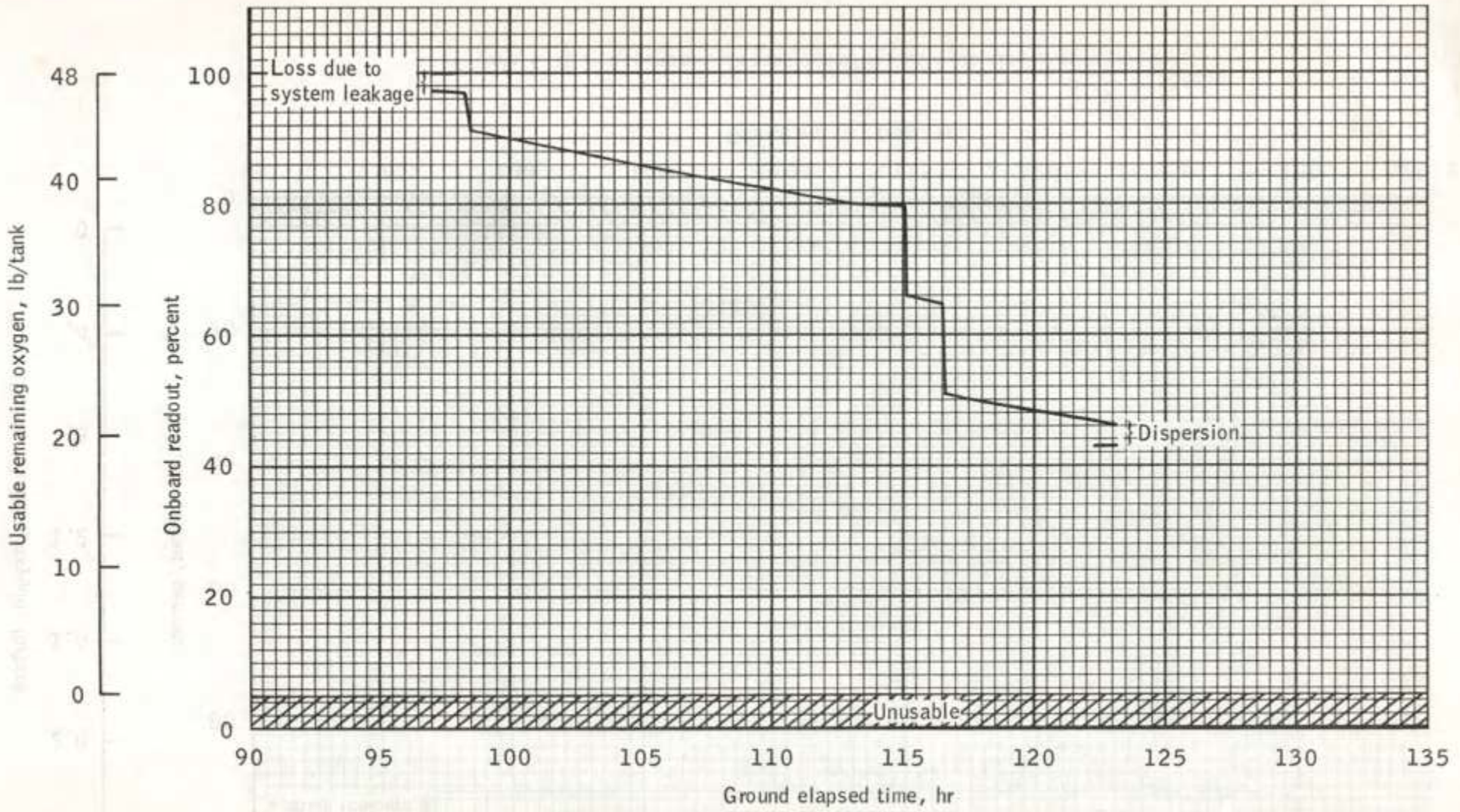


Figure 5-14 .- Descent oxygen tank quantities as a function of mission time.

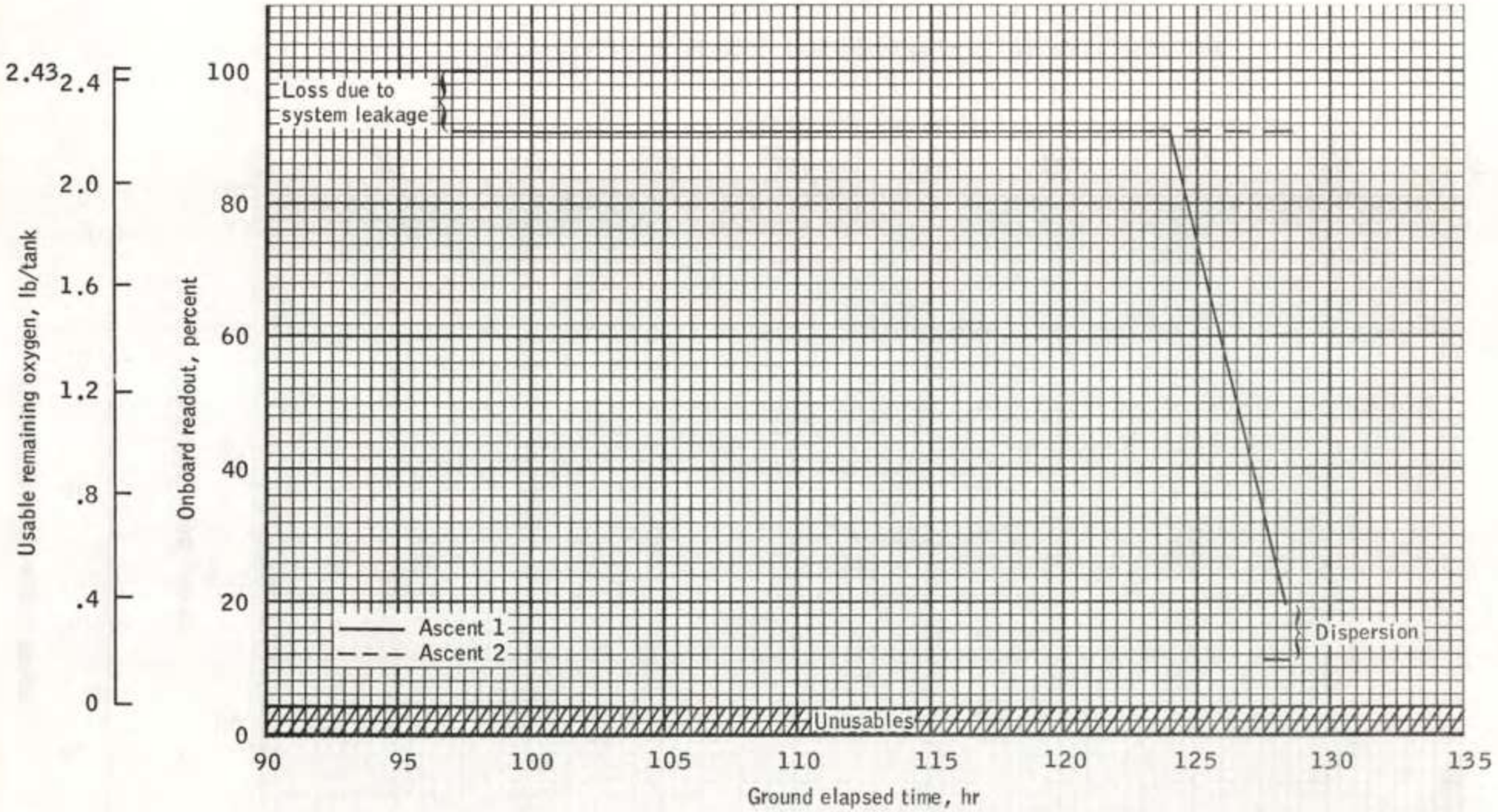


Figure 5-15 .- Ascent oxygen tank quantities as a function of mission time.

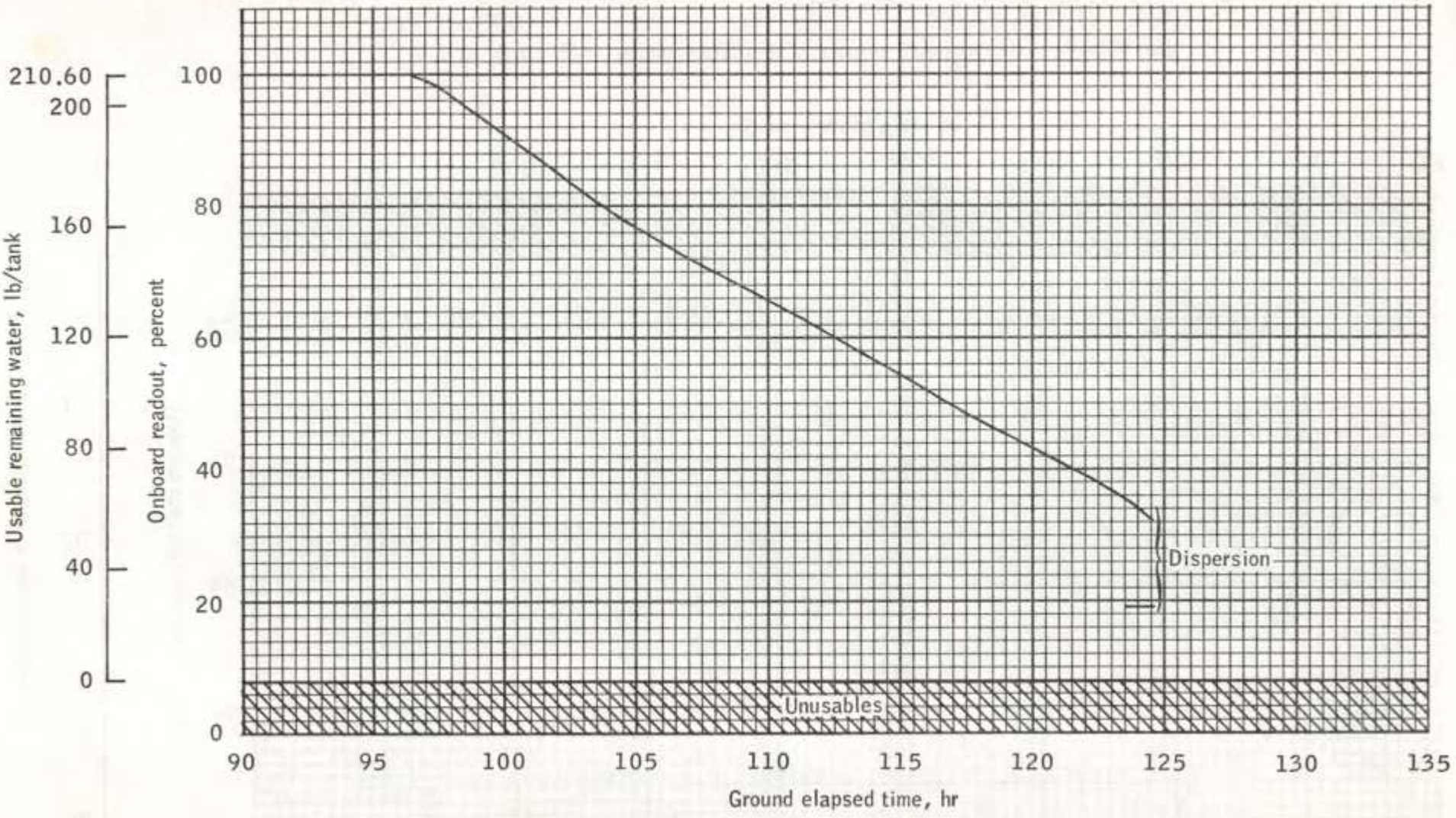


Figure 5-16 .- Descent water tank quantities as a function of mission time.

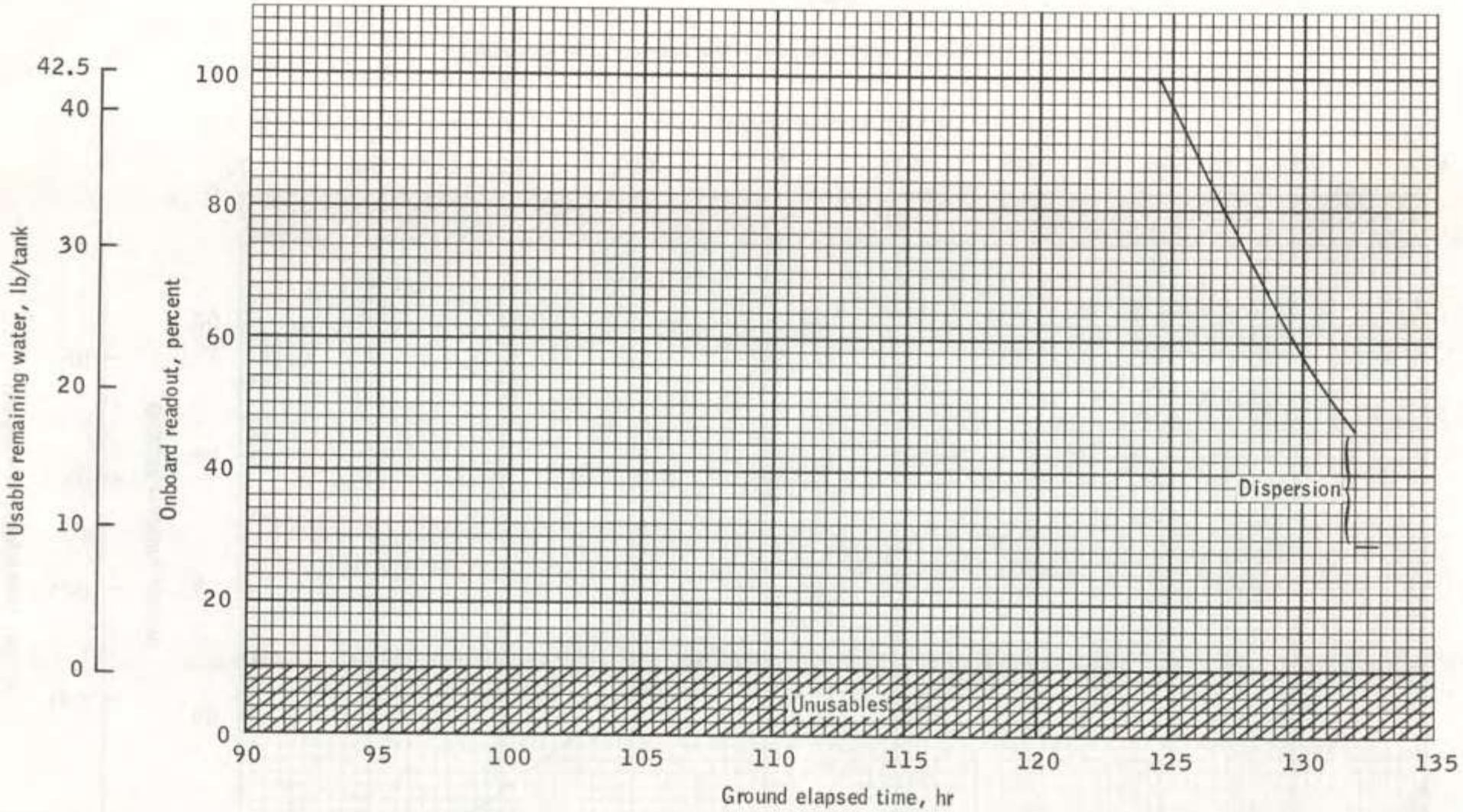


Figure 5-17.- Ascent water tank quantities as a function of mission time.

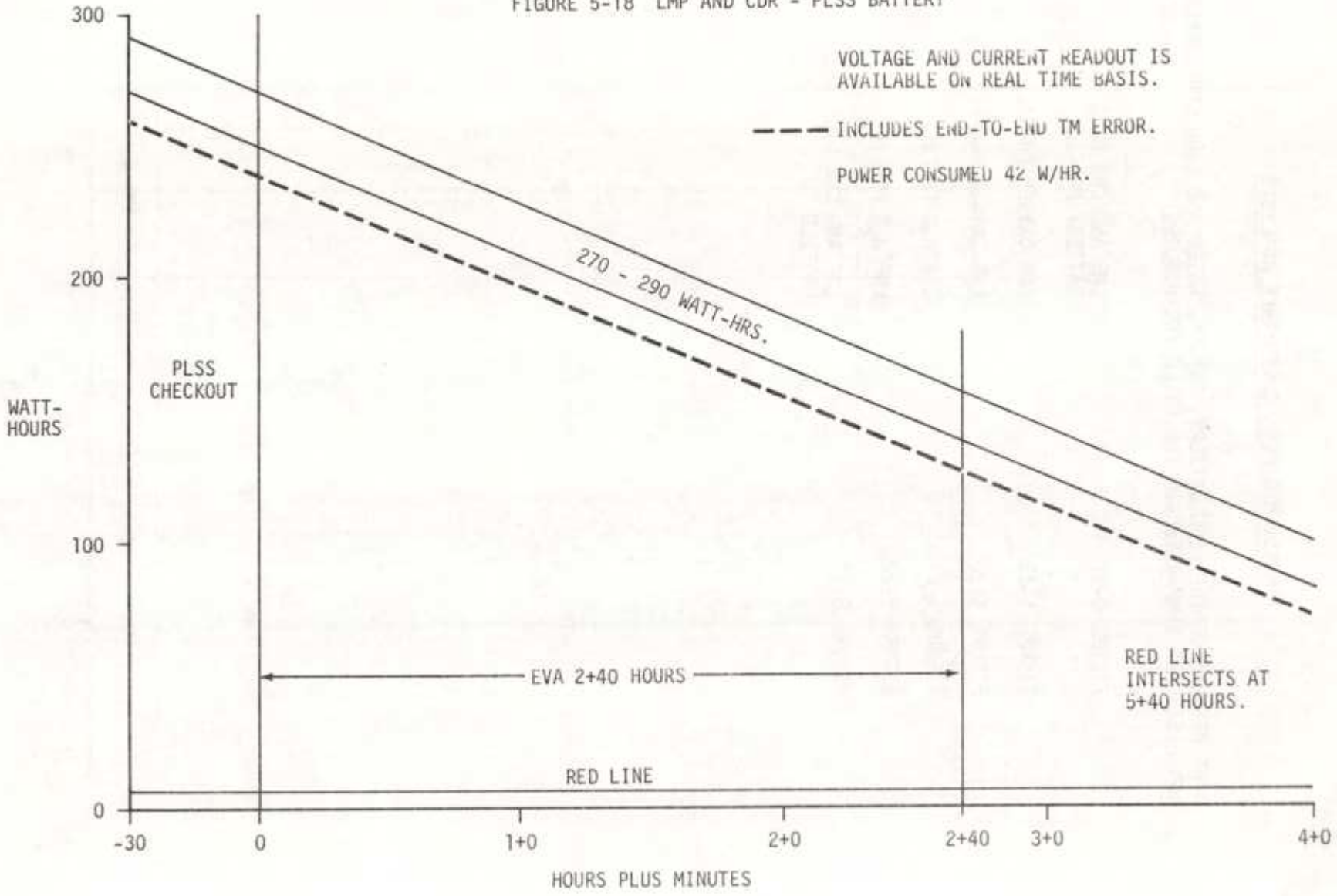
MISSION G PLSS CONSUMABLE ANALYSIS

THE RESULTS OF THE PLSS BATTERY, OXYGEN, WATER AND LiOH CONSUMABLE ANALYSIS ARE SUMMARIZED IN THE FOLLOWING FIGURES:

FIGURE 5-18	LMP AND CDR PLSS BATTERY PROFILE
FIGURE 5-19	CDR OXYGEN PROFILE
FIGURE 5-20	LMP OXYGEN PROFILE
FIGURE 5-21	CDR H_2O PROFILE
FIGURE 5-22	LMP H_2O PROFILE
FIGURE 5-23	LMP AND CDR LiOH CO_2 PROFILE

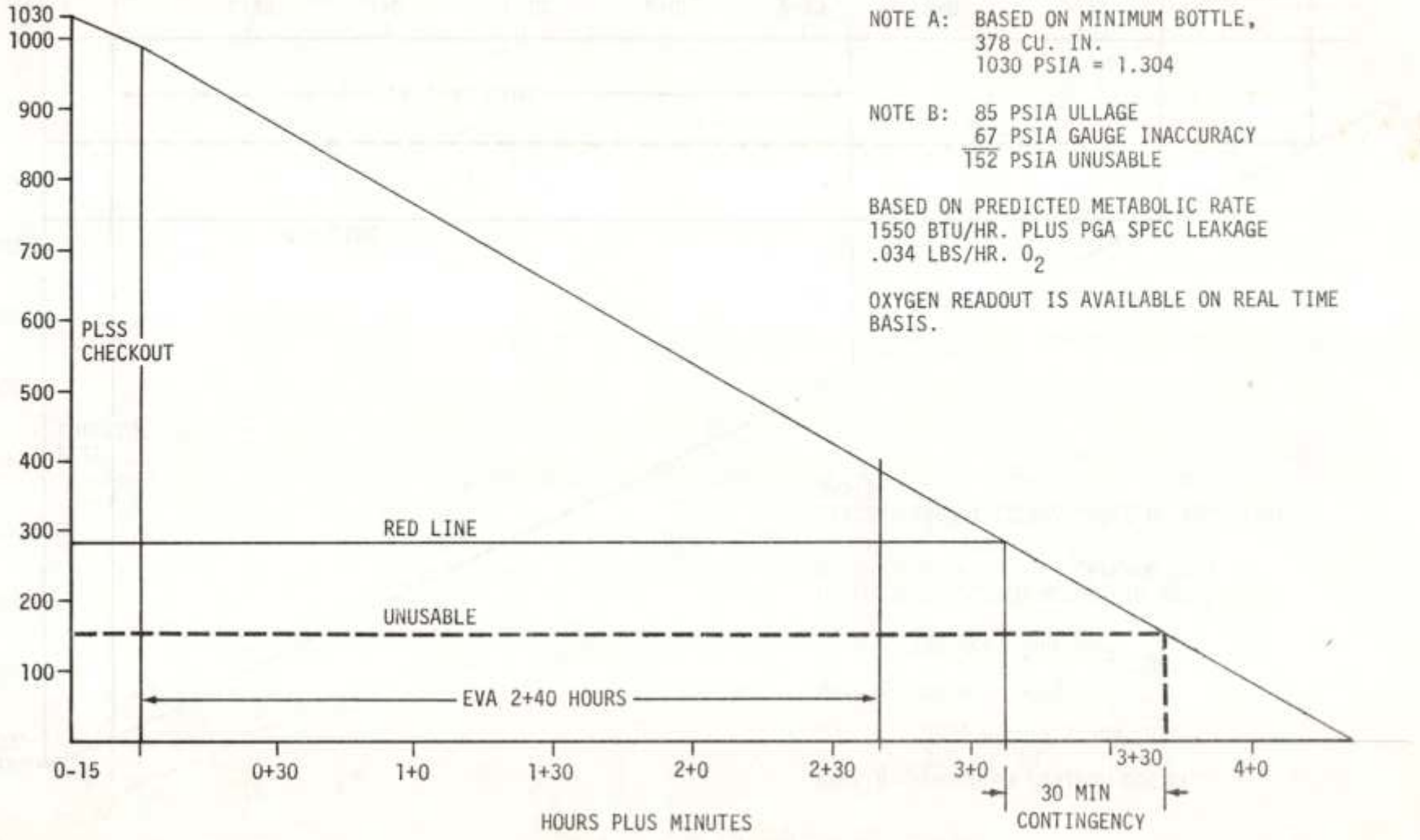
NOMINAL LUNAR SURFACE EVA

FIGURE 5-18 LMP AND CDR - PLSS BATTERY



NOMINAL LUNAR SURFACE EVA

FIGURE 5-19 CDR - OXYGEN



NOTE A: BASED ON MINIMUM BOTTLE,
378 CU. IN.
1030 PSIA = 1.304

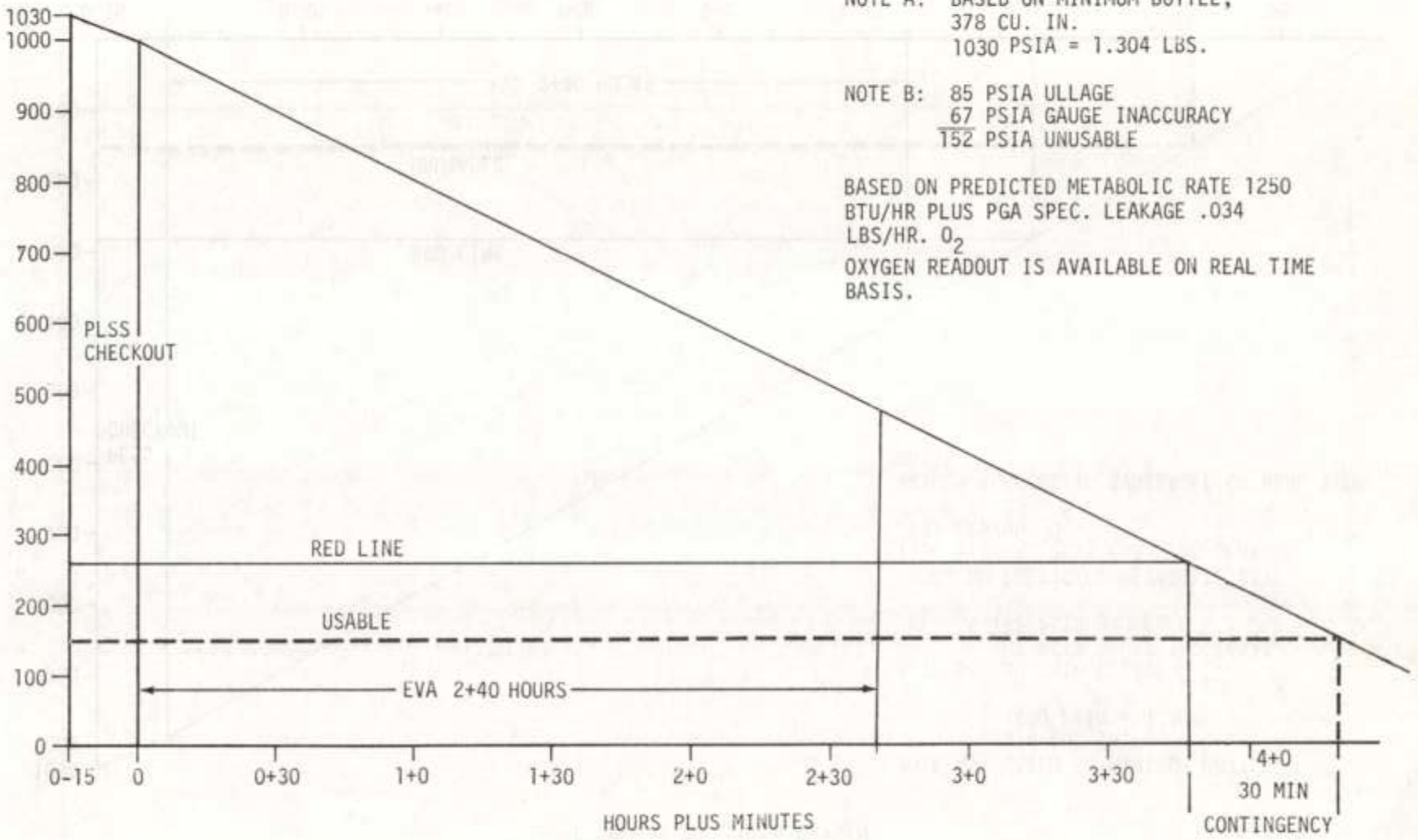
NOTE B: 85 PSIA ULLAGE
67 PSIA GAUGE INACCURACY
152 PSIA UNUSABLE

BASED ON PREDICTED METABOLIC RATE
1550 BTU/HR. PLUS PGA SPEC LEAKAGE
.034 LBS/HR. O_2

OXYGEN READOUT IS AVAILABLE ON REAL TIME BASIS.

NOMINAL LUNAR SURFACE EVA

FIGURE 5-20 LMP - OXYGEN

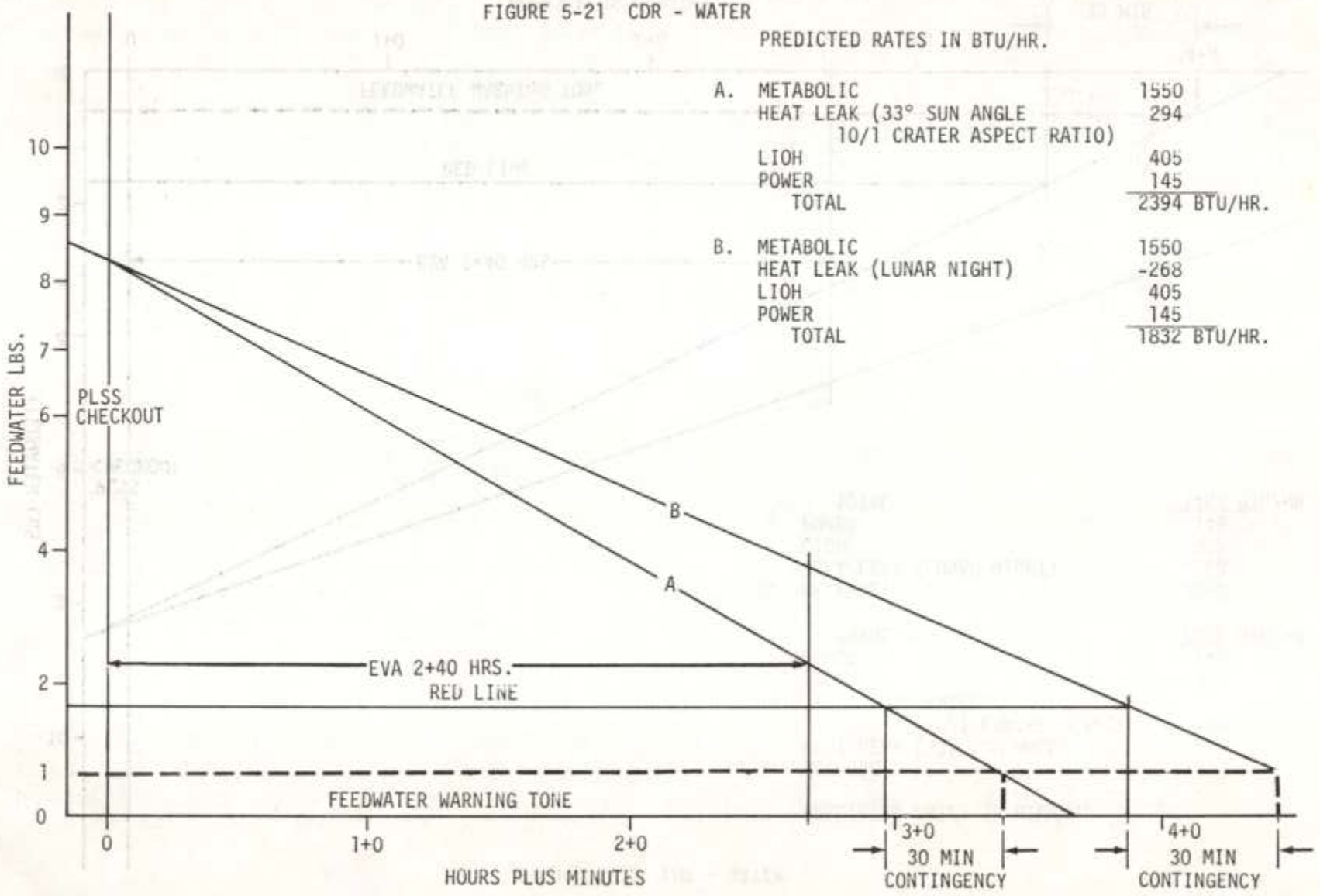


NOMINAL LUNAR SURFACE EVA

FIGURE 5-21 CDR - WATER

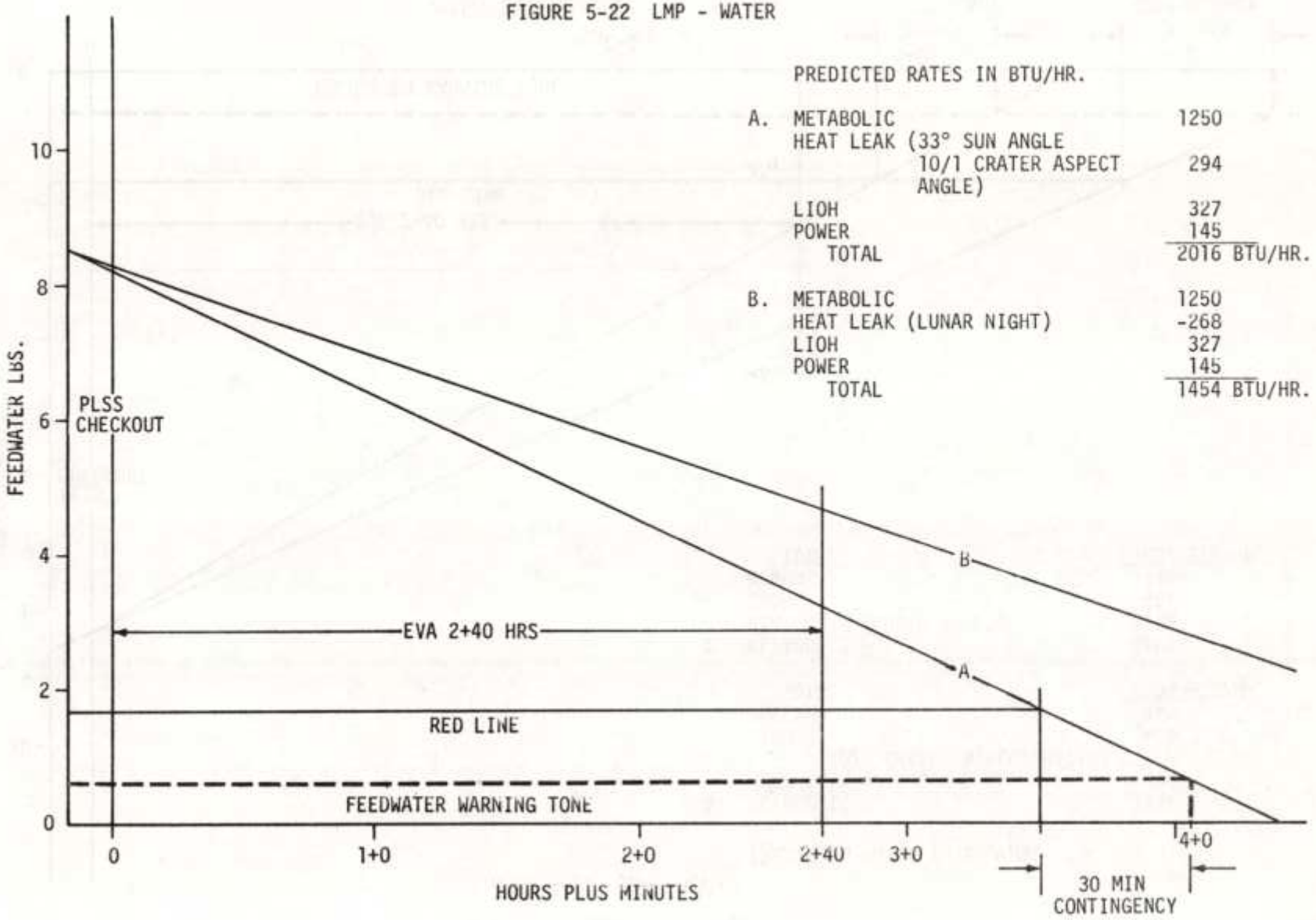
PREDICTED RATES IN BTU/HR.

A.	METABOLIC	1550
	HEAT LEAK (33° SUN ANGLE 10/1 CRATER ASPECT RATIO)	294
	LIOH	405
	POWER	145
	TOTAL	2394 BTU/HR.
B.	METABOLIC	1550
	HEAT LEAK (LUNAR NIGHT)	-268
	LIOH	405
	POWER	145
	TOTAL	1832 BTU/HR.



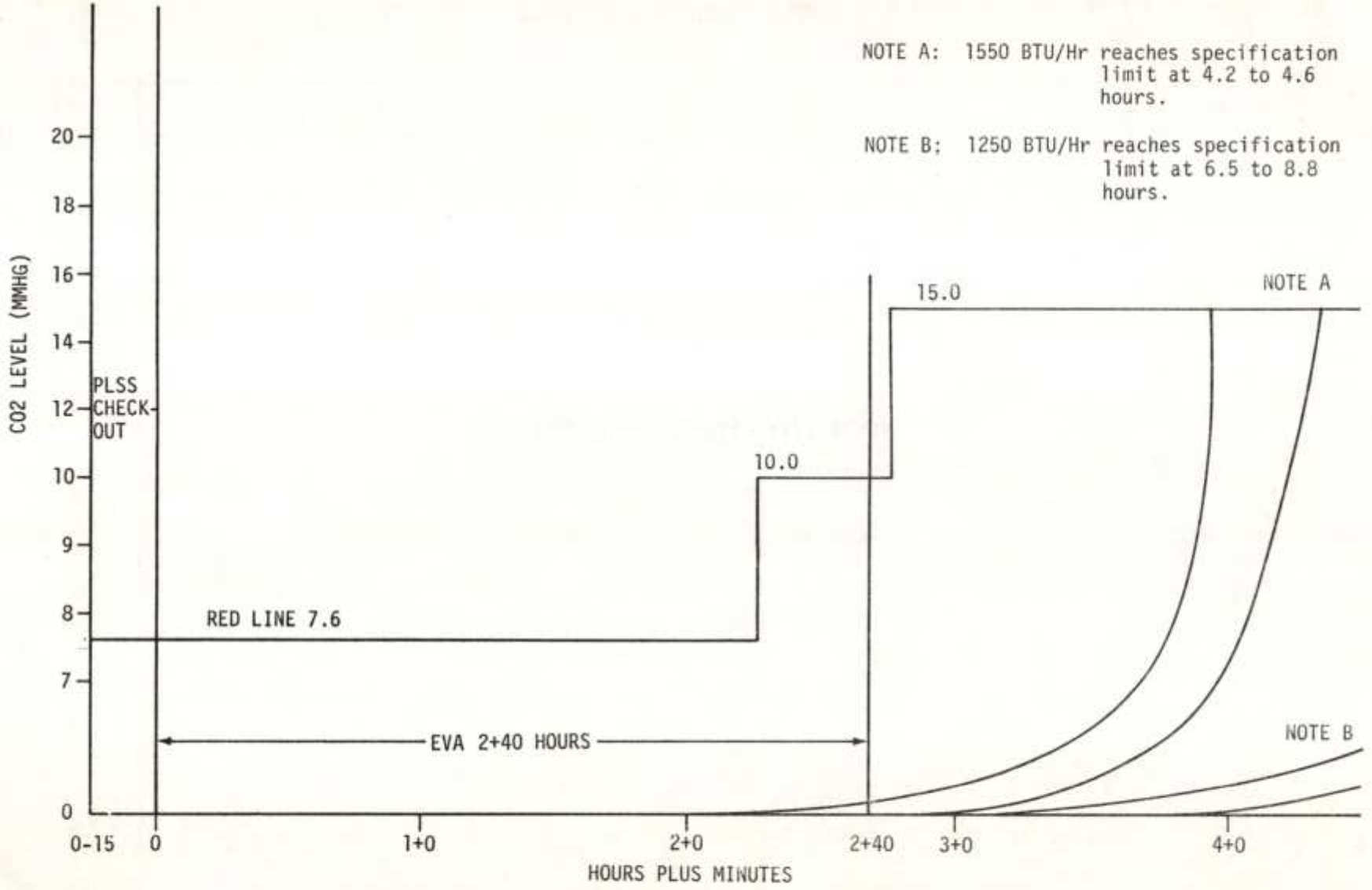
NOMINAL LUNAR SURFACE EVA

FIGURE 5-22 LMP - WATER



NOMINAL LUNAR SURFACE EVA

FIGURE 5-23 LMP & CDR LiOH



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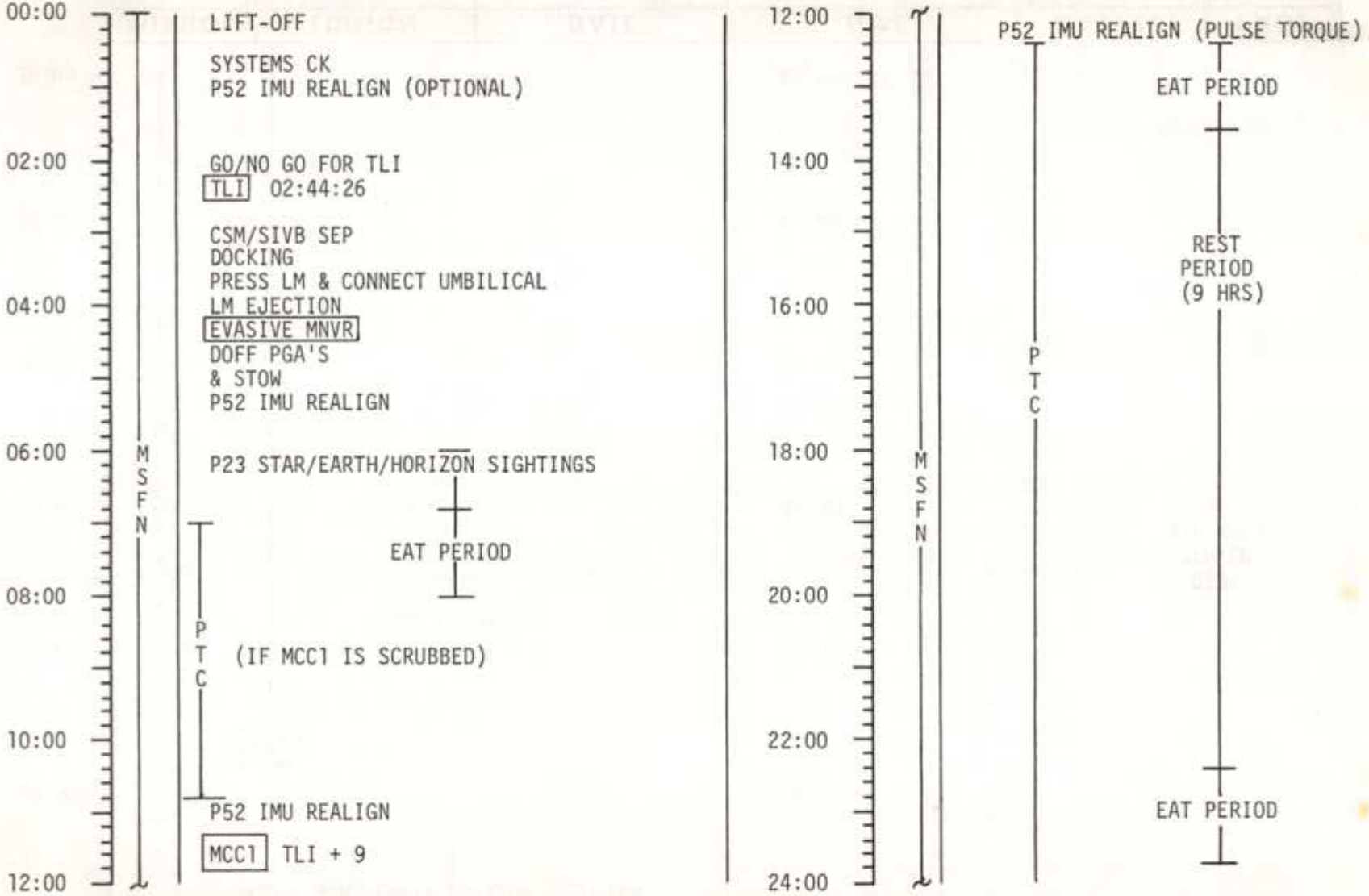
SECTION VI - SUMMARY FLIGHT PLAN

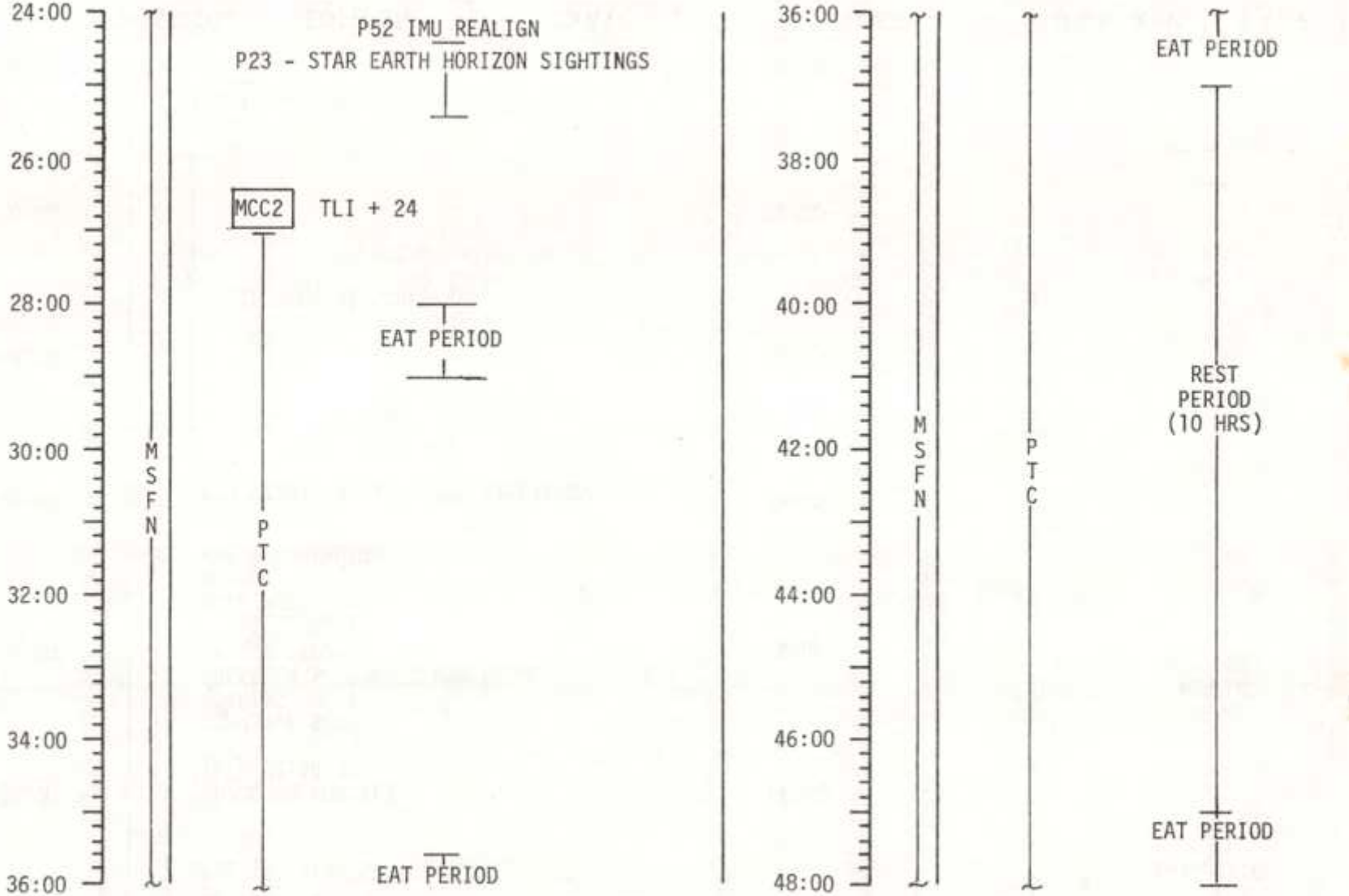
FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	00:00 - 24:00	1/TLC	6-1

MSC Form 8450 (Jan 69) FLIGHT PLANNING BRANCH

FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
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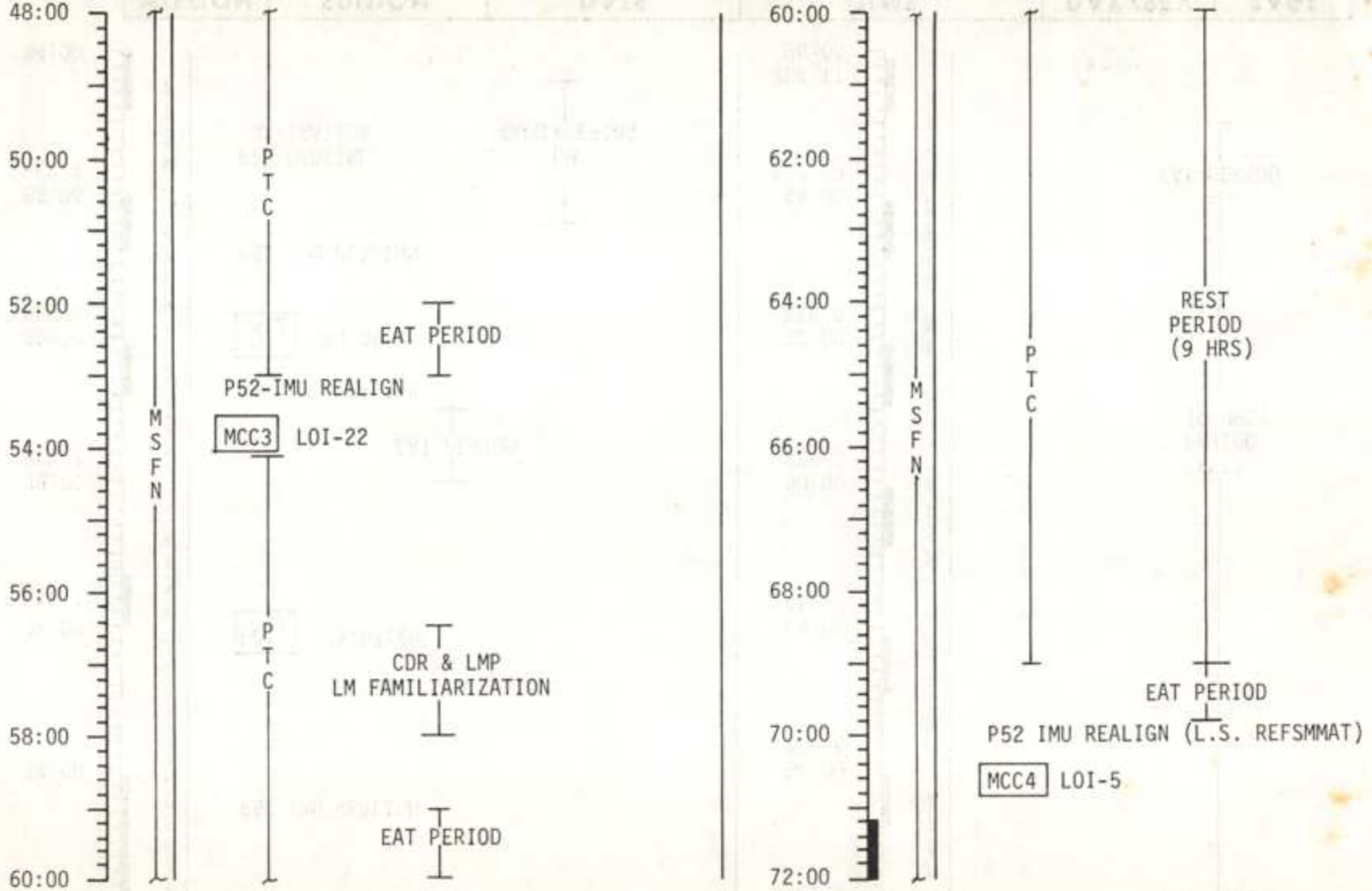
MSC Form 8450 (Jan 69) FLIGHT PLANNING BRANCH

FLIGHT PLAN



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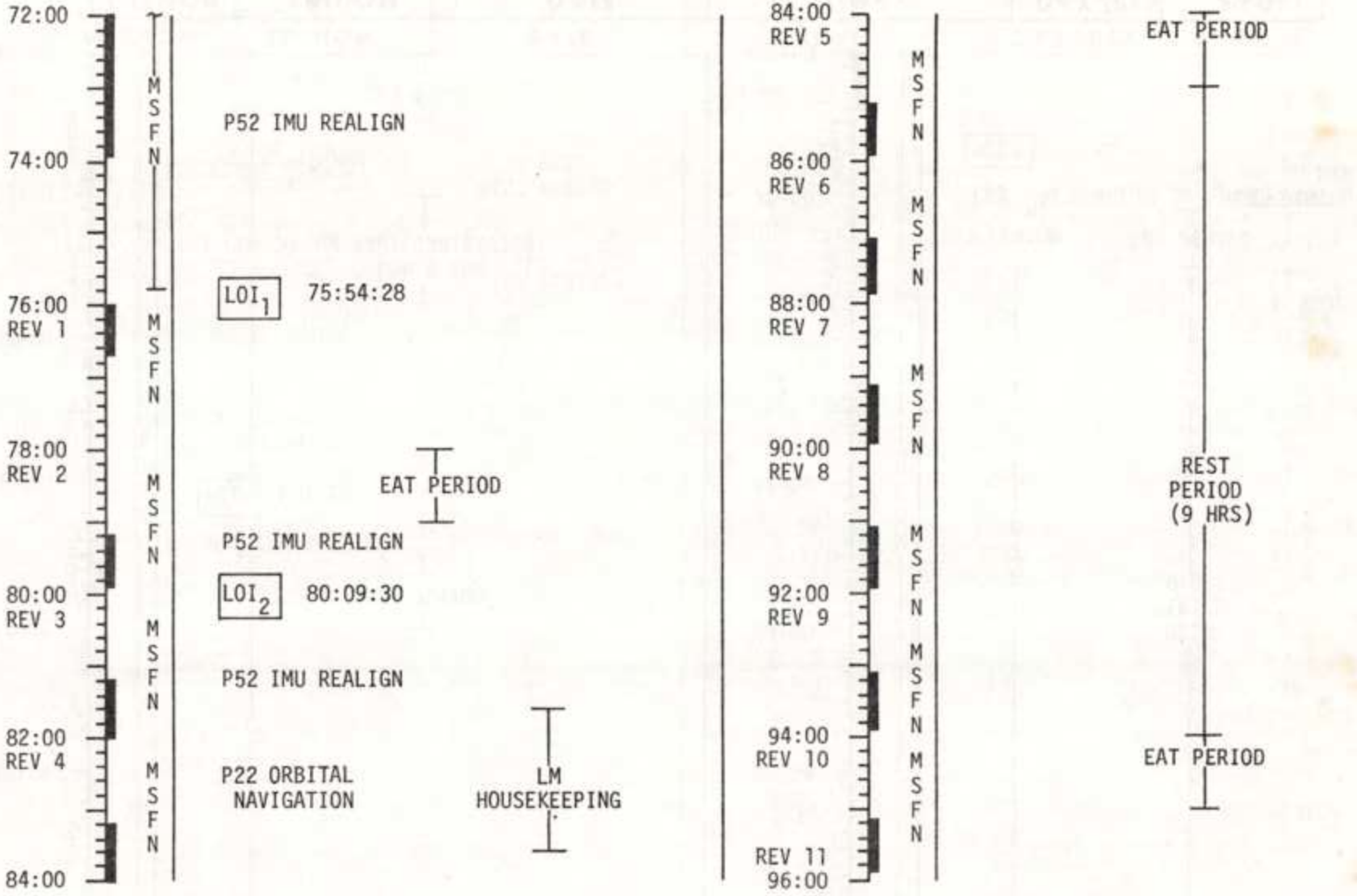
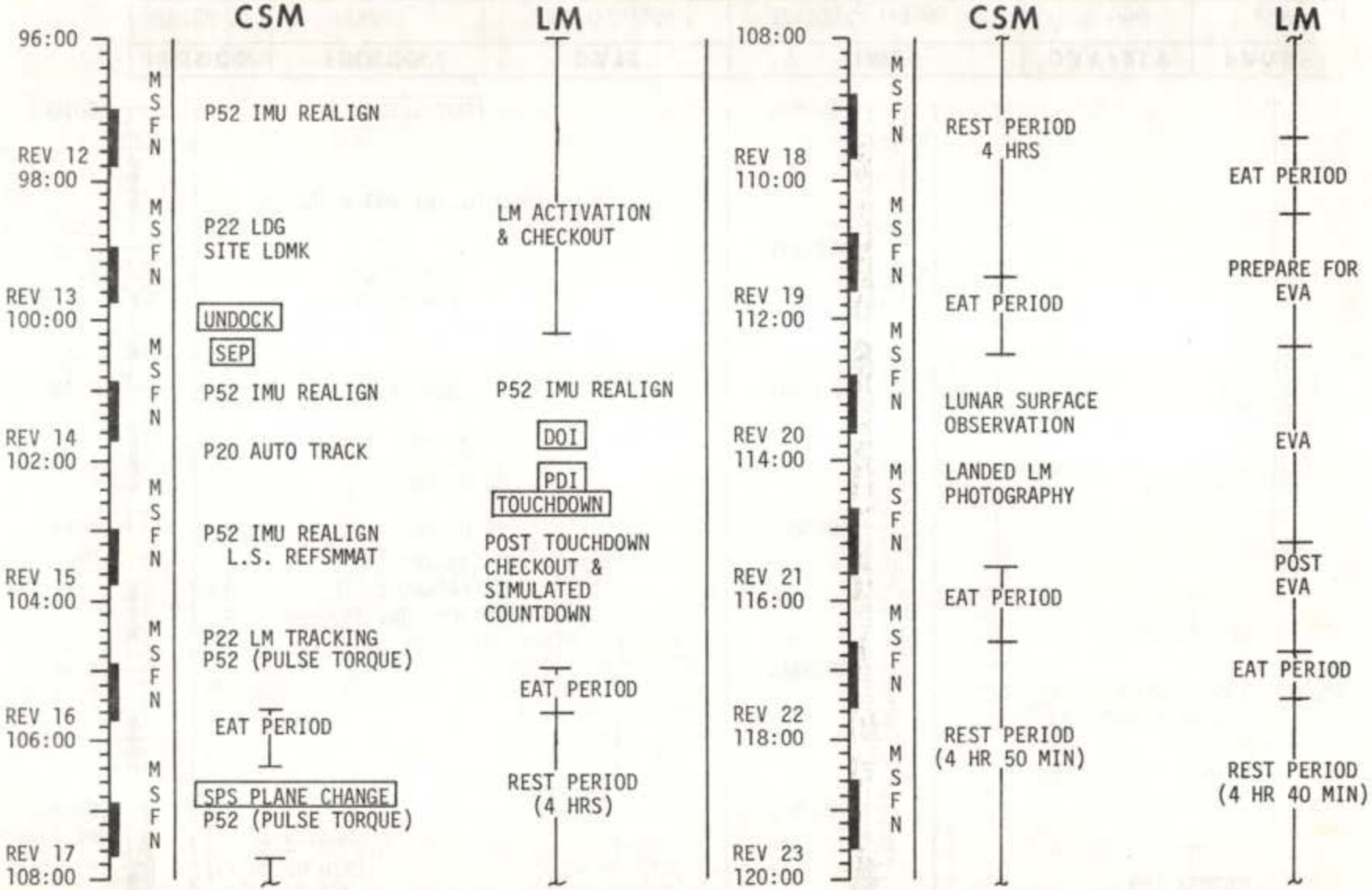
FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	72:00 - 96:00	4/1 THRU 10	6-4

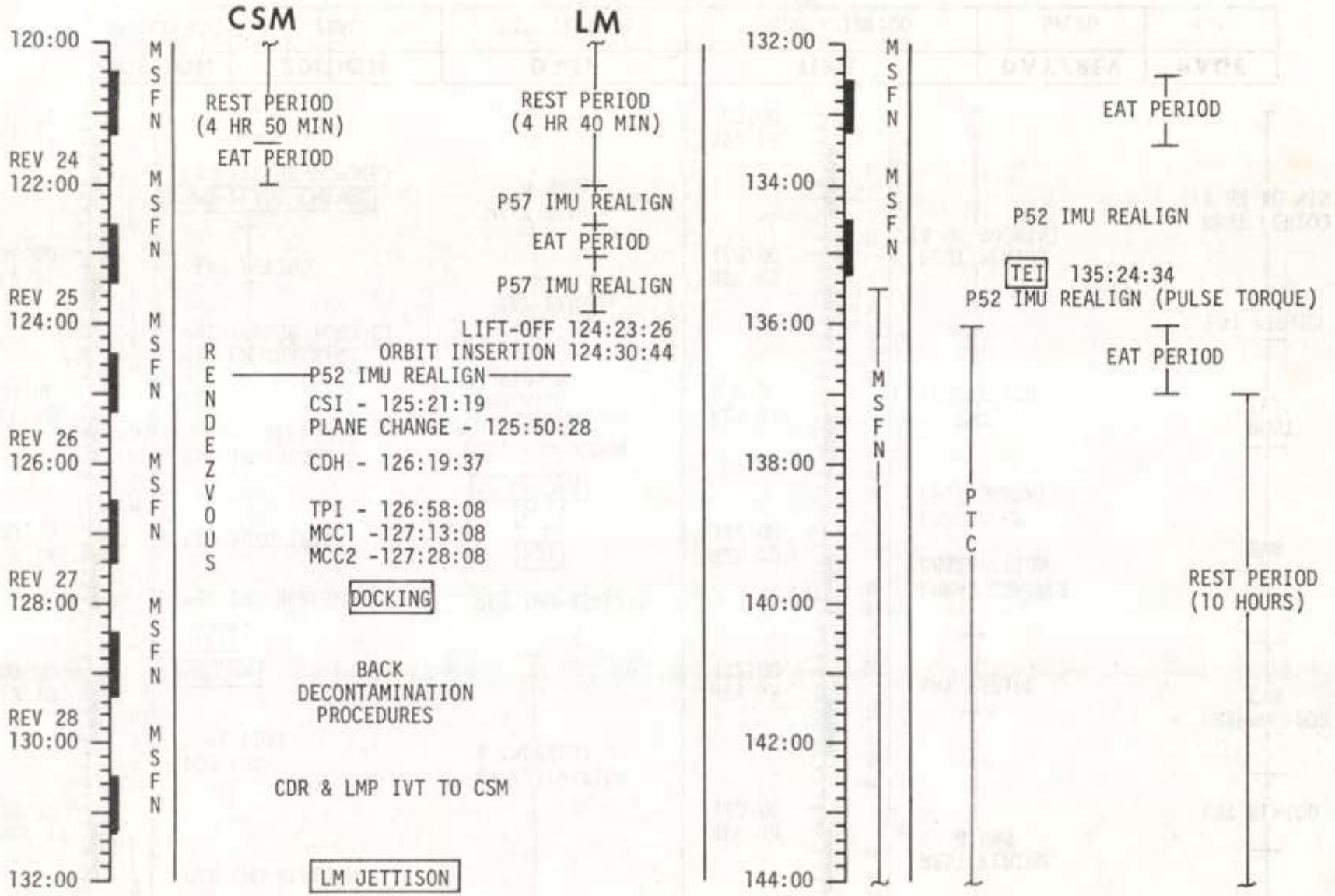
FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	96:00 - 120:00	5/LPO	6-5

MSC Form 845D (Jan 69) FLIGHT PLANNING BRANCH

FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
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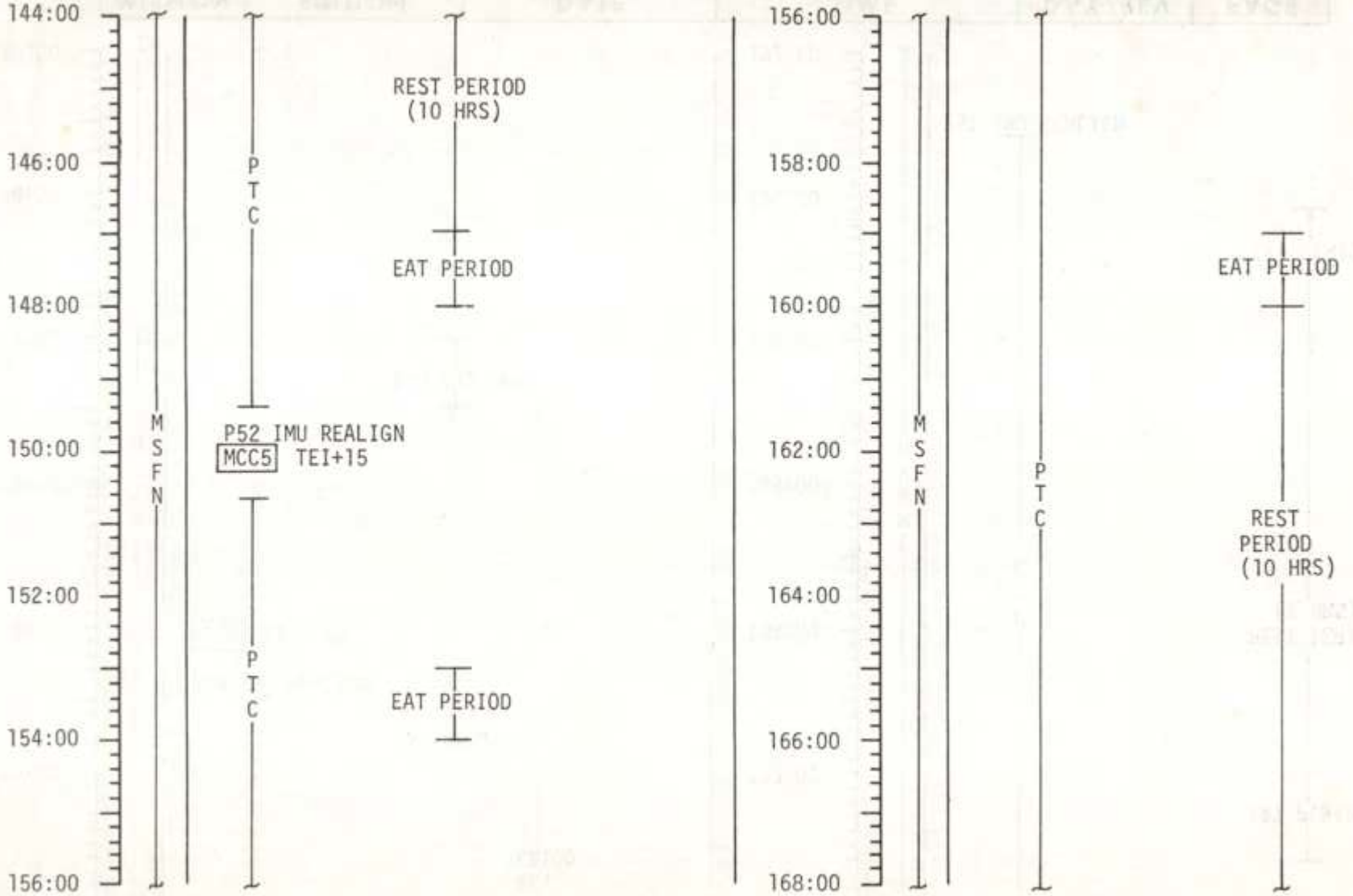
MSC Form 845D (Jan 69) FLIGHT PLANNING BRANCH

FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
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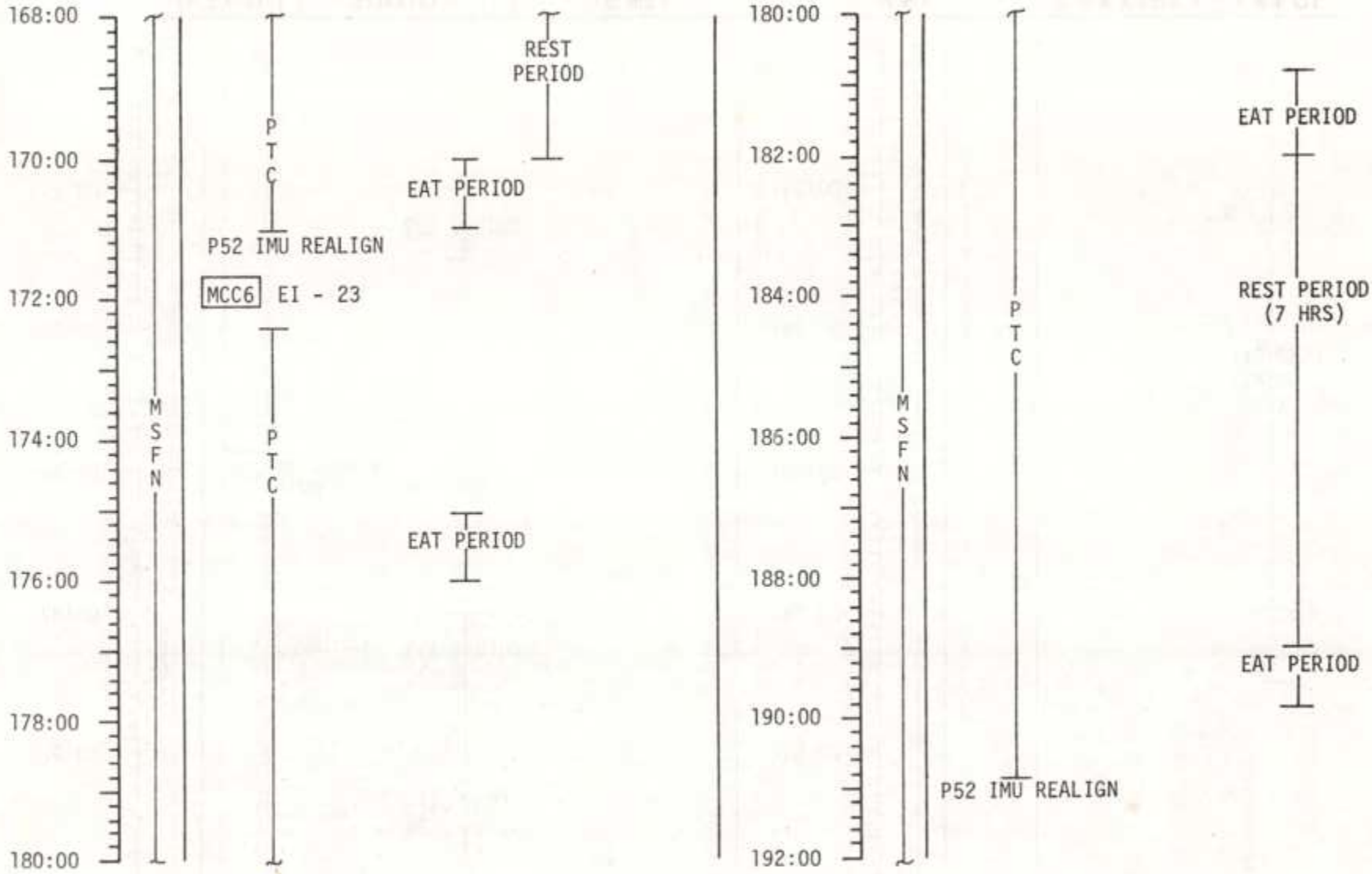
FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
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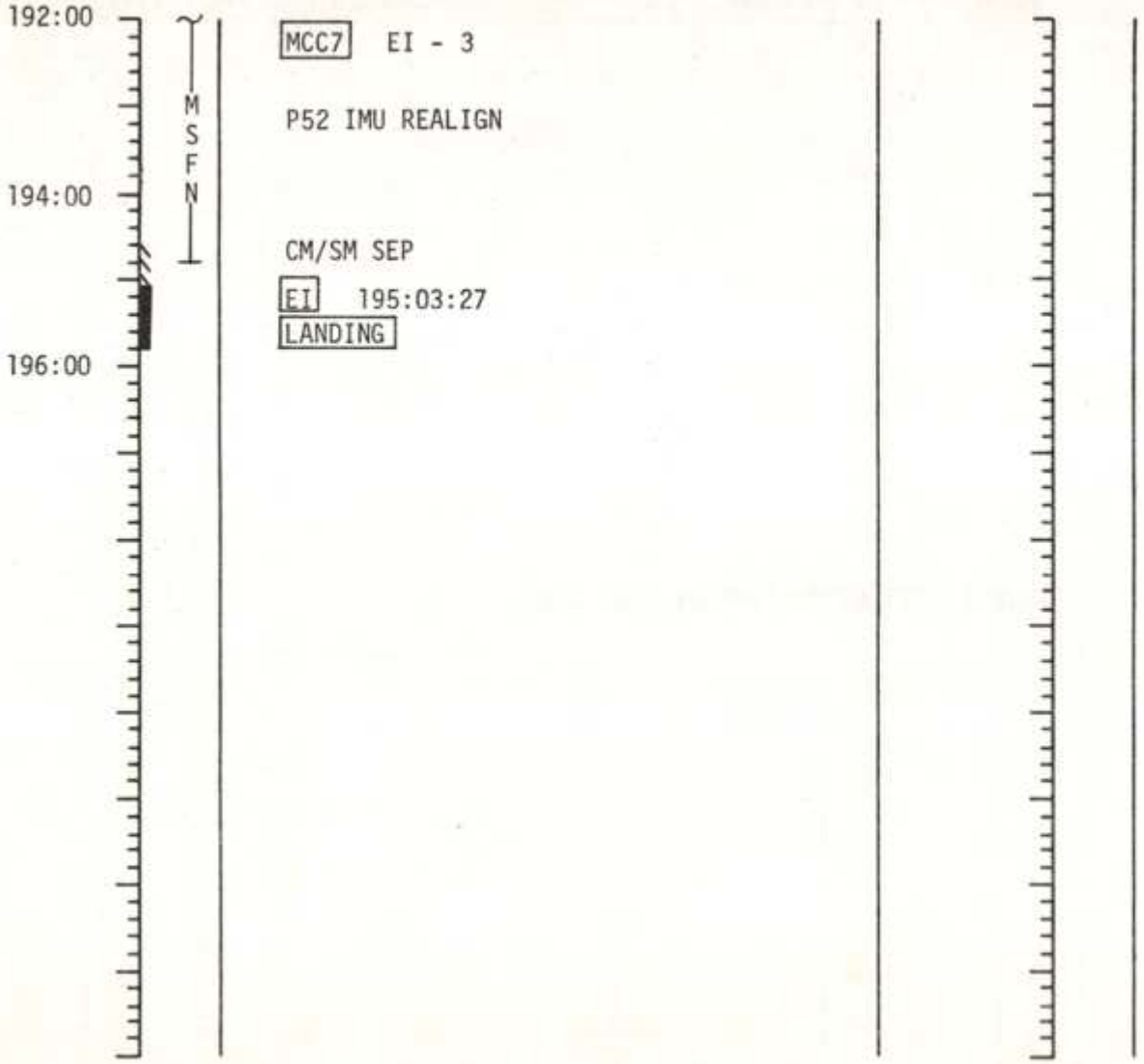
MSC Form 845D (Jan 69) FLIGHT PLANNING BRANCH

FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 11	FINAL	JULY 1, 1969	192:00 - 195:00	9	6-9

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